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**STUDY OF RESISTANCE MECHANISM OF TEM-TYPE
EXTENDED SPECTRUM β -LACTAMASES (ESBLs)
BY MASS SPECTROMETRY**

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Study of Resistance Mechanism of TEM-type

Extended Spectrum β -Lactamases (ESBLs)

by Mass Spectrometry

Wong Tsz Fung

A Thesis Submitted in Partial Fulfillment of
the Requirements of the Degree of Doctor of Philosophy

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Wong Tsz Fung

Abstract

Antibiotics resistance has long been a big worldwide health concern, and expression of β -lactamases is the most common strategy for bacteria to elude the attack of antibiotics. Since the discovery of the first TEM-type β -lactamase, its various variants have been found. G238S, E104K and M182T, which are three point mutations commonly found in variants, either with single or in combination with other point mutation(s), are responsible for conferring the extra hydrolytic power towards new generation antibiotics, e.g., cefotaxime. Substantial efforts have been made in the past few decades to study the properties and structures of these extended spectrum β -lactamases (ESBLs). X-ray crystallography has been extensively utilized to investigate ESBLs and it can provide valuable information of static structures for β -lactamases, particularly their intra-molecular interactions. However, there is still a lack of comprehensive study for the conformational dynamics of TEM-type ESBLs. Therefore, in chapters 1 - 3 of this thesis, individual and interactive effects of the three point mutations on the conformational dynamics of TEM-type β -lactamases were investigated with the help of hydrogen/deuterium exchange-mass spectrometry (HDX-MS). Six TEM-type β -lactamases, TEM-1, TEM-19 (G238S), TEM-17 (E104K), TEM-15 (G238S/E104K), TEM-20 (G238S/M182T) and TEM-52 (G238S/E104K/M182T), were included in this study. Conformational changes among the apo-TEM-type β -lactamases, cefotaxime-bound TEM-type β -lactamases and those due to cefotaxime binding were comprehensively compared. The comparison between apo-enzymes reflected that G238S caused more significant enhancement in flexibility than E104K. Despite the catalytic efficiency enhancement towards cefotaxime due to E104K, the conformation did not undergo any significant changes compared with the wild type TEM-1. However, the synergistic effect of G238S and E104K caused an acute enhancement in flexibility. Apart from key mutations like G238S and E104K, the reported global suppressor

M182T, a secondary point mutation, optimized the flexibility of mutants in order to compensate for the severe conformational changes caused by key mutations. Upon the cefotaxime binding, key regions such as the region near S70, Ω -loop, SDN loop and β -3 strand underwent significant conformational changes and became rigid, which could be attributed to the direct interaction of these regions to cefotaxime. Two allosteric regions were first discovered and discussed in this thesis. H5-H6 region, which was far away from the active site and not directly involved in the cefotaxime hydrolysis, was found to relate to the M182T stabilization during binding. The flexibility enhancement of Ω -loop was compensated by a more compact H5-H6 in M182T-containing mutants and hence thermostabilities of mutants were increased. Therefore, this region could be a potential allosteric region as a target for novel antibiotics and inhibitors. Moreover, one edge of the β -sheet, which was also not directly involved in the antibiotics binding, was found to become more flexible particularly upon cefotaxime binding to G238S-containing mutants. However, the edge had insignificant flexibility changes or became slightly compact than TEM-1 after mutants incorporating M182T. This could be a result of compensating for destabilization effect of the flexibility enhancement for the active site related β -3 strand to accommodate the bulky side chain of cefotaxime and promote the effective interaction to the cefotaxime in the binding site. These findings on conformational dynamics changes of TEM-type ESBLs upon single or combined point mutations, particularly in the two allosteric regions, provided new insights into the resistance mechanism of ESBLs and design of new inhibitors targeting non-active site regions.

The occurrence of coronavirus disease in 2019 (COVID-19) has caused the unprecedented global health threat. Rapid detection of infected cases is an important step to combat the disease. Currently, nucleic acid detection approach is regarded as the golden method for detection of severe acute respiratory syndrome coronavirus 2

(SARS-CoV-2), the causative agent of the COVID-19, with serological detection as the alternative approach. However, these approaches suffer from certain drawbacks including time-consuming and labor-intensive steps, false positive or negative results, etc. In chapter 4, a targeted proteomics-based approach has been attempted for detection of SARS-CoV-2 based on the detection of specific peptides from the SARS-CoV-2 nucleocapsid (N) protein. Four specific peptides were obtained with a 3-minute tryptic digestion of the N protein, and successfully eluted within 3 minutes by liquid chromatography and detected by mass spectrometry using multiple reaction monitoring, with a limit of detection of the specific peptides down to low ppb level in saliva, demonstrating the possibility of the targeted proteomics approach for rapid detection of SARS-CoV-2.

Research Publications

Journal paper

Wong, T. F.; So, P. K.; Yao, Z. P. Rapid Detection of SARS-CoV-2 by Mass Spectrometry: A Review. *Trends Anal. Chem.* **2022** (submitted).

Conference papers

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Wong, T. F.; So, P. K.; Huang, L. W.; Yao, Z. P. Study of Dynamics and Kinetics of β -Lactamases towards Antibiotics by Hydrogen/Deuterium Exchange Mass Spectrometry. The 26th Symposium on Chemistry Postgraduate Research in Hong Kong. Hong Kong. 4 May **2019**. (Poster)

Wong, T. F.; So, P. K.; Huang, L. W.; Yao, Z. P. Conformational Dynamics of β -Lactamase Mutants as Revealed by Hydrogen/Deuterium Exchange Mass Spectrometry. HKSMS Symposium 2019. Hong Kong. 22 June **2019**. (Poster)

Wong, T. F.; So, P. K.; Yao, Z. P. Study of Conformational Dynamics of TEM-type β -Lactamases by Hydrogen/Deuterium Exchange Mass Spectrometry. 8th Asia-Oceania Mass Spectrometry Conference 2020. Macau. 3-7 January **2020**. (Oral for the Young Scientist Forum and poster for the main conference)

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Table of Content

Abstract.....	i
Research Publications.....	iv
Acknowledgements	vi
Table of Content.....	viii
List of Abbreviations.....	xii
List of Figures.....	xv
List of Tables.....	xxi
Chapter 1. Introduction.....	1
1.1. The history of antibiotics	1
1.2. Mechanism of action of β -lactam antibiotics.....	3
1.3. Antibiotics resistance	5
1.3.1. How bacteria defend against antibiotics	5
1.3.2. General reaction scheme and classification of β -lactamases	9
1.4. TEM-type β -lactamases	12
1.4.1. Kinetics of β -lactam antibiotics hydrolysis by TEM-type β -lactamases .	13
1.4.2. Structure and mechanism of TEM-type β -lactamases	14
1.4.3. TEM-type extended spectrum β -lactamases (ESBLs)	20
1.4.3.1. Effect of G238S	21
1.4.3.2. Effect of E104K	22
1.4.3.3. Effect of M182T	22
1.5. Hydrogen/deuterium exchange-mass spectrometry (HDX-MS) for dynamic	

properties of protein.....	24
1.6. Objectives	30
Chapter 2. Experimental.....	31
2.1. Protein expression.....	31
2.1.1. Expression and purification of TEM-type β -lactamases.....	31
2.1.2. Expression and purification of E166A TEM-type β -lactamases.....	32
2.2. Study of the completeness of binding between TEM-17 and cefotaxime	32
2.3. Preparation of phosphate buffer and deuterated phosphate buffer	32
2.4. Preparation of the E166A-TEM-type β -lactamase-cefotaxime complexes	33
2.5. Hydrogen/deuterium exchange (HDX) reaction.....	33
2.6. Global HDX-MS analysis	33
2.7. Optimization of digestion enzyme of local HDX-MS	34
2.8. Local HDX-MS analysis.....	35
2.9. Data analysis	35
Chapter 3. Results and discussions.....	37
3.1. Mass spectrometric characterization of the TEM-type β -lactamases	37
3.2. Global HDX-MS of apo-TEM-type β -lactamases	41
3.3. Local HDX-MS analysis of free TEM-type β -lactamases	44
3.3.1. Choice of digestion enzyme for local HDX-MS.....	44
3.3.2. Study of digestion patterns of apo-TEM-type β -lactamases.....	48
3.3.3. Local HDX-MS results of apo-TEM-type β -lactamases	53
3.4. Study of binding between the six β -lactamases with cefotaxime	76
3.4.1. Completeness of binding between apo-TEM-type β -lactamases and	

cefotaxime.....	76
3.4.2. Expression of the six TEM-type β -lactamases with E166A point mutations	79
3.4.3. Complete binding between E166A-TEM-type β -lactamases and cefotaxime.....	82
3.4.4. Global HDX-MS of cefotaxime-bound E166A-TEM-type β -lactamases .	84
3.4.5. Local HDX-MS of cefotaxime-bound E166A-TEM-type β -lactamase complexes	85
3.4.6. Comparison of conformational changes in localized regions of six E166A- TEM-type β -lactamases upon cefotaxime binding	114
3.4.7. Insights into cefotaxime catalytic efficiencies of TEM-type ESBLs from conformational changes due to incorporation of single or combination of point mutations.....	140
Chapter 4. Rapid detection of SARS-CoV-2 by mass spectrometry	145
4.1. Introduction.....	145
4.1.1. SARS-CoV-2 background.....	145
4.1.1.1. Features	145
4.1.1.2. Variants of SARS-CoV-2	146
4.1.2. Detection of SARS-CoV-2.....	147
4.1.2.1. Nucleic acid detection.....	147
4.1.2.2. Serological detection.....	151
4.1.2.3. Detection of SARS-CoV-2 by targeted proteomic approach using mass spectrometry.....	153
4.1.3. Objectives	156
4.2. Experimental.....	157
4.2.1. Nucleocapsid (N) protein.....	157

4.2.2. Buffer exchange	157
4.2.3. Trypsin digestion.....	158
4.2.4. LC-MS analysis for searching unique peptides of SARS-CoV-2 N-protein	158
4.2.5. Database search by PLGS and protein-protein basic local alignment search tool (blastp)	159
4.2.6. MRM transitions of N-protein model peptides	159
4.2.7. LC-MS analysis of model peptides in MRM mode	160
4.3. Results and discussions.....	162
4.3.1. Strategy of detection of SARS-CoV-2-specific viral peptides.....	162
4.3.2. Sample pretreatment procedures and the searching of peptides specific to N protein of SARS-CoV-2	163
4.3.3. MRM transitions development	168
4.3.4. LC separation	169
4.3.5. Linearities of the four selected peptides	171
4.3.6. Limit of detection of the four peptides in saliva matrix.....	172
4.3.7. Limitations and improvements of this approach.....	173
Chapter 5. Conclusions and future prospects	174
References	178

List of Abbreviations

Abbreviation	Full form
2xTY	2x yeast extract tryptone
ACN	Acetonitrile
ARDS	Acute respiratory distress syndrome
ES*	Acylated-enzyme-substrate complex
k_2	Acylation rate
ACE2	Angiotensin-converting enzyme 2
Å	Angstrom
BLDB	Beta-Lactamase DataBase
Cef	Cefotaxime
CDC	Centers for Disease Control and Prevention
CLIA	Chemiluminescence assay
CRISPR	Clustered regularly interspaced short palindromic repeats
COVID-19	Coronavirus disease 2019
D-Ala-D-Ala	D-Alanine-D-alanine
Da	Dalton
k_3	Deacylation rate
DNA	Deoxyribonucleic acid
D ₂ O	Deuterium oxide
K _d	Dissociation equilibrium constant
E	Envelope
ELISA	Enzyme-linked immunosorbent assay
ES	Enzyme-substrate complex
ESBLs	Extended spectrum β -lactamases
FA	Formic acid

H1	Helix-1
HDX	Hydrogen/deuterium exchange
HDX-MS	Hydrogen/deuterium exchange-mass spectrometry
IgG	Immunoglobulin G
IgM	Immunoglobulin M
IPTG	Isopropyl β -d-1-thiogalactopyranoside
LOD	Limit of detection
LC-MS	Liquid chromatography mass spectrometry
MS	Mass spectrometry
MALDI-MS	Matrix-assisted laser desorption ionization mass spectrometry
M	Membrane
m/z	Mass-to-charge ratio
K_M	Michalis constant
MERS-CoV	Middle East respiratory syndrome coronavirus
MRM	Multiple reaction monitoring
NAG	N-acetylglucosamine
NAM	N-Acetylmuramic acid
(-)ssRNA	Negative-sense single stranded ribonucleic acid
NDM-1	New Delhi metallo- β -lactamase 1
NMR	Nuclear magnetic resonance
N	Nucleocapsid
ppb	Parts per billion
PBP	Penicillin binding protein
PCR	Polymerase chain reaction
(+)ssRNA	Positive-sense single stranded RNA

PLGS	ProteinLynx Global Server
qRT-PCR	Real-time reverse transcription polymerase chain reaction
RAS	Renin–angiotensin system
RT-LAMP	Reverse transcription-loop-mediated isothermal amplification
RNA	Ribonucleic acid
RdRps	RNA-dependent RNA polymerases
SIM	Selected ion monitoring
SARS-CoV	Severe acute respiratory syndrome coronavirus
SARS-CoV-2	Severe acute respiratory syndrome coronavirus 2
SDS-PAGE	Sodium dodecyl sulphate–polyacrylamide gel electrophoresis
S	Spike
sgRNAs	Subgenomic RNAs
MS/MS	Tandem mass spectrometry
TEM-1	Temoniera-1
TIC	Total ion chromatogram
LC-MS	Ultra-performance liquid chromatography mass spectrometry
VOCs	Variants of concern
WB	Western blot
WHO	World Health Organization

List of Figures

Figure 1.1 Core structures of a) penicillin, b) cephalosporin, c) carbapenem and d) monobactam.....	3
Figure 1.2 Structures of Gram-positive and Gram-negative bacteria. The zoom-in figure depicts the peptidoglycan structure.	4
Figure 1.3 Structures of penicillin and D-Ala-D-Ala.	4
Figure 1.4 Hydrogen network of binding of vancomycin with (a) D-Ala-D-Ala or (b) D-Ala-D-Lactate of the peptidoglycan terminus (modified from reference 22).	7
Figure 1.5 General reaction scheme of hydrolysis of β -lactam antibiotic by β -lactamase.	9
Figure 1.6 Classification of β -lactamases under Ambler scheme.	10
Figure 1.7 Molecular and functional features of the major groups of β -lactamases...	11
Figure 1.8 General reaction mechanism of TEM-type β -lactamases. (a) S70 is activated indirectly by the general base E166 through WAT297 (b) Glu166 deprotonates S70 through K73.	19
Figure 1.9 The structure of peptide Gly-Gln-Glu-Ser-Pro. Three types of hydrogen, exchangeable backbone amide hydrogens (red), side-chain hydrogens (blue), alkyl hydrogens (purple with rounded rectangle, indicated here only for part of the peptide backbone), are shown here.....	25
Figure 1.10 Procedures of continuous labeling local HDX-MS.	26
Figure 1.11 Normal process of HDX. Either (a) continuous labeling or (b) pulse labeling can be used for deuteration, followed by either (c) local exchange analysis or (d) global exchange analysis.....	29

Figure 3.1 (a-g) ESI-MS spectra and (h-l) deconvoluted mass spectra of six TEM-type β -lactamases.	39
Figure 3.2 Total ion chromatogram of TEM-1 as a representative. The single dominant peak of TEM-1 indicated the high sample purity.	40
Figure 3.3 SDS-PAGE of the six TEM-type β -lactamases. Lane M: Ladder; Lane 1: TEM-1; Lane 2: TEM-15; Lane 3: TEM-17; Lane 4: TEM-19; Lane 5: TEM-20; Lane 6: TEM-52.	40
Figure 3.4 Isotopic distribution shift of TEM-1 at various HDX time intervals.	41
Figure 3.5 Deuterium uptake plot of the six apo-TEM-type β -lactamases.	42
Figure 3.6 Coverage map of apo- (a) TEM-1, (b) TEM-15, (c) TEM-17, (d) TEM-19, (e) TEM-20 and (f) TEM-52.	52
Figure 3.7 Comparisons of TEM-type β -lactamases that had one different point mutation.	54
Figure 3.8 Different plot of comparison between (a) TEM-1 and TEM-19, (b) TEM-1 and TEM-17, (c) TEM-19 and TEM-20, (d) TEM-19 and TEM-15, (e) TEM-17 and TEM-15, (f) TEM-20 and TEM-52 and (g) TEM-15 and TEM-52.	61
Figure 3.9 Deuterium uptake plot of residues 59-69 for apo-TEM-type β -lactamases.	62
Figure 3.10 Deuterium uptake plot of residues (a) 163-169 and (b) 170-179 for apo-TEM-type β -lactamases.	64
Figure 3.11 Deuterium uptake plot of residues (a) 130-138, (b) 127-137 and (c) 127-138 for apo-TEM-type β -lactamases.	67
Figure 3.12 Deuterium uptake plot of residues 230-246 for apo-TEM-type β -lactamases.	68
Figure 3.13 Deuterium uptake plot of residues (a) 103-117 and (b) 103-118 for apo-TEM-type β -lactamases.	70

Figure 3.14 Deuterium uptake plot of residues 178-186 for apo-TEM-type β -lactamases.	72
Figure 3.15 Deuterium uptake plot of residues 140-150 for apo-TEM-type β -lactamases.	73
Figure 3.16 Deuterium uptake plot of residues (a) 49-59, (b) 222-230 and (c) 250-260 for apo-TEM-type β -lactamases.	75
Figure 3.17 (a) ESI-MS and (b) deconvoluted mass spectra of reaction mixture of cefotaxime hydrolysis by TEM-17. Peak A and B represented free TEM-17 and the acylated complex of TEM-17 and cefotaxime, respectively.	77
Figure 3.18 Plot of $[ES^*]/[E_{total}]$ against time of TEM-17 and cefotaxime reaction. .	78
Figure 3.19 SDS-PAGE of the six E166A-TEM-type β -lactamases. Lane M: Ladder; Lane 1: TEM-1A; Lane 2: TEM-15A; Lane 3: TEM-17A; Lane 4: TEM-19A; Lane 5: TEM-20A; Lane 6: TEM-52A.	79
Figure 3.20 (a-g) ESI-MS spectra and (h-l) deconvoluted mass spectra of six E166A-TEM-type β -lactamases.	80
Figure 3.21 Total ion chromatogram of TEM-1A as a representative. The single dominant peak of TEM-1 indicated the high sample purity.....	81
Figure 3.22 (a-g) ESI-MS spectra and (h-l) deconvoluted mass spectra of six cefotaxime-bound E166A-TEM-type β -lactamases.	83
Figure 3.23 Deuterium uptake plot of the six cefotaxime-bound E166A-TEM-type β -lactamases.	85
Figure 3.24 Coverage map of (a) TEM-1A-Cef, (b) TEM-15A-Cef, (c) TEM-17A-Cef, (d) TEM-19A-Cef, (e) TEM-20A-Cef and (f) TEM-52A-Cef.....	89
Figure 3.25 Different plot of comparison between (a) TEM-1A-Cef and TEM-19A-Cef, (b) TEM-1A-Cef and TEM-17A-Cef, (c) TEM-19A-Cef and TEM-20A-Cef, (d) TEM-19A-Cef and TEM-15A-Cef, (e) TEM-17A-Cef and TEM-15A-Cef, (f) TEM-20A-Cef and TEM-52A-Cef and (g) TEM-15A-Cef	

and TEM-52A-Cef.	97
Figure 3.26 Aligned X-ray crystal structures of TEM-1 (green) and TEM-52 (orange). Sticks represent the S70 in TEM-1 (green) and TEM-52 (orange) respectively.....	98
Figure 3.27 Deuterium uptake plot of residues (a) 59-69 and (b) 60-69 for cefotaxime- bound TEM-type β -lactamases.	99
Figure 3.28 Deuterium uptake plot of residues (a) 170-179 and (b) 163-169 for cefotaxime-bound TEM-type β -lactamases.	101
Figure 3.29 Deuterium uptake plot of residues (a) 130-138 and (b) 127-138 for cefotaxime-bound TEM-type β -lactamases.	103
Figure 3.30 Deuterium uptake plot of residues 231-248 for cefotaxime-bound TEM- type β -lactamases.	105
Figure 3.31 Deuterium uptake plot of residues (a) 103-117 and (b) 103-118 for cefotaxime-bound TEM-type β -lactamases.	106
Figure 3.32 Deuterium uptake plot of residues (a) 170-186, (b) 178-186 and (c) 180- 186 for cefotaxime-bound TEM-type β -lactamases.....	109
Figure 3.33 Deuterium uptake plot of residues 140-150 for cefotaxime-bound TEM- type β -lactamases.	111
Figure 3.34 Deuterium uptake plot of residues (a) 52-59, (b) 222-230 and (c) 250-260 for cefotaxime-bound TEM-type β -lactamases.....	113
Figure 3.35 Coverage map of comparison between (a) TEM-1A and TEM-1A-Cef, (b) TEM-15A and TEM-15A-Cef, (c) TEM-17A and TEM-17A-Cef, (d) TEM-19A and TEM-19A-Cef, (e) TEM-20A and TEM-20A-Cef and (f) TEM-52A and TEM-52A-Cef.....	117
Figure 3.36 Different plot of comparison between (a) TEM-1A and TEM-1A-Cef (b) TEM-15A and TEM-15A-Cef, (c) TEM-17A and TEM-17A-Cef, (d) TEM-19A and TEM-19A-Cef, (e) TEM-20A and TEM-20A-Cef and (f)	

TEM-52A and TEM-52A-Cef.....	124
Figure 3.37 Plot of change of deuterium uptake of residues 59-69 upon cefotaxime binding.	125
Figure 3.38 Plot of change of deuterium uptake of residues (a) 163-169 and 163-170 (TEM-52A only), (b) 163-169 (TEM-20A only) and 163-170 (TEM-20A and TEM-52A), (c) 170-179 upon cefotaxime binding.	128
Figure 3.39 Plot of change of deuterium uptake of residues 130-138 upon cefotaxime binding.	129
Figure 3.40 Plot of change of deuterium uptake of residues 230-246 upon cefotaxime binding.	130
Figure 3.41 Plot of change of deuterium uptake of residues (a) 113-117 and (b) 113-118 upon cefotaxime binding.	132
Figure 3.42 Plot of change of deuterium uptake of residues (a) 170-186, (b) 170-177, (c) 180-186 and (d) 178-186 upon cefotaxime binding.	135
Figure 3.43 Plot of change of deuterium uptake of residues 140-150 upon cefotaxime binding.	136
Figure 3.44 Plot of change of deuterium uptake of residues (a) 49-59, (b) 250-260, (c) 222-230 and (d) 214-225 and 214-230 (TEM-52A only) upon cefotaxime binding.	139
Figure 4.1 Schematic diagram of the SARS-CoV-2 structure	146
Figure 4.2 SDS-Page of the expressed N-protein. Lane M and 1 represent protein marker and induced strain respectively. This result is provided by the Guangdong Laboratory Animals Monitoring Institute.....	157
Figure 4.3 Sequences alignment of SARS-CoV-2 and SARS-CoV. Residues highlighted in yellow indicate differences between the protein sequences.	162

- Figure 4.4** An example of blastp search result of a peptide specific to N protein of SARS-CoV-2. The red rectangular box indicates all results belonging to SARS-CoV-2 have 100% percent identity. 166
- Figure 4.5** An example of blastp search result of a peptide not specific to N protein of SARS-CoV-2. The peptide sequence subjected to blastp searching also matched the sequence of SARS-CoV..... 167
- Figure 4.6** An example of blastp search result of a peptide having almost all results belonging to SARS-CoV-2 with 100% percent identity, but one result also perfectly matched the bat coronavirus RaTG13. This is not included in the specific peptide list..... 167
- Figure 4.7** A chromatogram of the LC separation of the four selected peptides. Only one MRM transition from each peptide was selected for demonstration of the separation efficiency..... 169
- Figure 4.8** All three MRM transitions of (a-d) peptide 1-4 were eluted at the same time which indicated the positive detection of these peptides. 171
- Figure 4.9** An example of the confirmation of detection for peptide 1. Two out of three channels had their S/N greater than 3. 172

List of Tables

Table 3.1 Amino acid at position 104, 182 and 238, theoretical and measured molecular masses of the six TEM-type β -lactamases.	38
Table 3.2 Comparison of three digestion methods, pepsin column digestion, protease type XVIII digestion and combination of protease type XVIII and pepsin column.	47
Table 3.3 Number of peptides generated, sequence coverage, redundancies and number of common peptides resulted from pepsin digestion of the six apo-TEM-type β -lactamases.	52
Table 3.4 Amino acid at position 104, 166, 182 and 238, theoretical and measured molecular masses of the six E166A mutated TEM-type β -lactamases.	81
Table 3.5 Mass differences between the cefotaxime-bound and the apo E166A-mutated β -lactamases.	84
Table 3.6 Number of peptides generated, sequence coverage, redundancies and number of common peptides resulted from pepsin digestion of the six cefotaxime-bound E166A-TEM-type β -lactamases.	90
Table 3.7 Number of peptides generated, sequence coverage, redundancies and number of common peptides resulted from pepsin digestion of each comparison between E166A-TEM-type β -lactamases and their cefotaxime-bound complexes.	118

Table 4.1 MRM transition setting for the four model N-protein peptides. The product ions in bold are the quantifier ions.	161
Table 4.2 Four selected peptides used for SARS-CoV-2 detection. Peptides 2 and 3 were missed cleaved and contained the sequence of peptide 1, which are highlighted in red.....	165
Table 4.3 Linearities and linear ranges of the four selected peptides.....	171
Table 4.4 LOD of the four selected peptides in saliva matrix.	173

Chapter 1. Introduction

1.1. The history of antibiotics

The history of the use of antibiotics can be traced back to ancient time when people utilized filamentous fungi, mold and plant extracts to treat bacterial infections.^{1,2} However, such unsystematic use of various oddities to combat bacteria was unreliable and unsafe and this aroused the interest of many scientists to the need of developing reliable antimicrobial reagents.

The discovery of a synthetic dye-derived antibiotics is the first milestone of the modern use of antibiotics. Paul Ehrlich (a German-Jewish scientist) and Sahachiro Hata (a Japanese bacteriologist) synthesized and discovered arsphenamine, an organoarsenic compound, in 1907.^{3,4} Then in 1932, another German scientist, Gerhard Domagk, developed the first sulfonamide antibacterial drug Prontosil.^{3,5,6} These achievements significantly spurred and contributed to the development of antibiotics. Paul Ehrlich and Gerhard Domagk received the Nobel Prize in Physiology or Medicine in 1908 and 1939, respectively.

Penicillin is the most renowned revolutionized antibiotic. Various significant observations by scientists led to the discovery of penicillin. One of them was made in 1897 by Ernest Duchesne, a doctoral student at École du Service de Santé Militaire. He published his thesis⁷, one of the very first scholarly works that discussed his discovery of the healing properties of a *Penicillium glaucum* mold that could even cure typhoid-infected guinea pigs. He also proposed that the battle between molds and bacteria was endless. However, the work of Duchesne received hardly any attention by the scientific community at that time.

The revolutionary discovery of penicillin happened in 1928 when it was found out by Scottish biologist Alexander Fleming. He observed that his *Staphylococcus* plate culture was contaminated by certain blue-green mold, and surprisingly bacteria growth around the mold was inhibited. He then postulated that certain antimicrobial substance must have been secreted from the mold and inhibited the bacterial growth. This substance was later named penicillin. However, the failure to purify penicillin hindered its clinical development until 1942 when the first penicillin, penicillin G or benzylpenicillin, was successfully purified by Ernst Chain, Howard Florey, and Edward Abraham. Abraham also proposed the chemical structure of penicillin in the same year⁸ and finally Dorothy Crowfoot Hodgkin confirmed its structure by X-ray crystallography in 1945.⁹⁻¹¹ Fleming, Chain, and Florey shared the Nobel Prize in Physiology or Medicine in 1945 for the discovery of penicillin and its curative effect in various infectious diseases.

The golden age of antibiotic discovery started after the clinical use of penicillin and various types and classes of antimicrobial drugs were discovered from the 1940s to the 1970s. One of the most common types of antibiotics is β -lactam. Antibiotics of this type share the 4-membered 2-azetidinone ring, or commonly called the β -lactam ring, as their core structure.¹² The first discovery of the ancestor in each class of antibiotics is originated from naturally occurring sources. Currently there are four major classes of β -lactam antibiotics, which are penicillin, cephalosporin, carbapenem and monobactam (Figure 1.1).

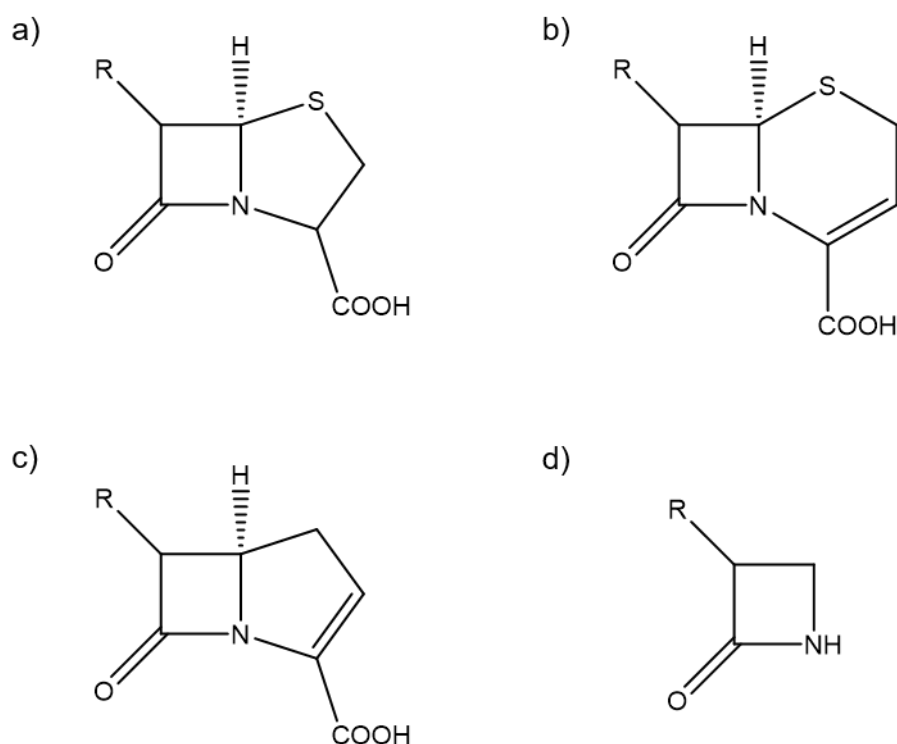


Figure 1.1 Core structures of a) penicillin, b) cephalosporin, c) carbapenem and d) monobactam.

1.2. Mechanism of action of β -lactam antibiotics

β -Lactam antibiotics inhibit the growth of bacteria by interfering with the synthesis of the peptidoglycan layer, an important component of bacterial cell wall that maintains the integrity and shape of bacteria. It is the outermost layer of Gram-positive bacteria. A simplified monomer of peptidoglycan consists of N-acetylglucosamine (NAG) and N-Acetylmuramic acid (NAM), and these two sugars are connected by β -(1,4)-glycosidic linkage. A four or five amino acid residue chain is also attached on NAM, which the last two amino acids are D-Ala-D-Ala. As peptidoglycan monomers connect to form a linear chain, a peptidoglycan layer is established by crosslinking linear peptidoglycan chains. The peptidoglycan crosslinking is facilitated by DD-peptidase, which is also called penicillin binding protein (PBP), and it connects the short amino acid chain attached on NAM (Figure 1.2). β -Lactam antibiotics is the analog of the terminal D-Ala-D-Ala of the crosslinking chain and hence they are able to bind with PBP

(Figure 1.3).¹³ The binding between β -lactam antibiotic and PBP is irreversible and the acyl-complex is highly stable, rendering the inactivation of the PBP. Thus, the bacterial cell wall cannot acquire a stiff and rigid structure because of the interference of the peptidoglycan crosslinking, which directly causes bacterial cell death.¹⁴⁻¹⁶

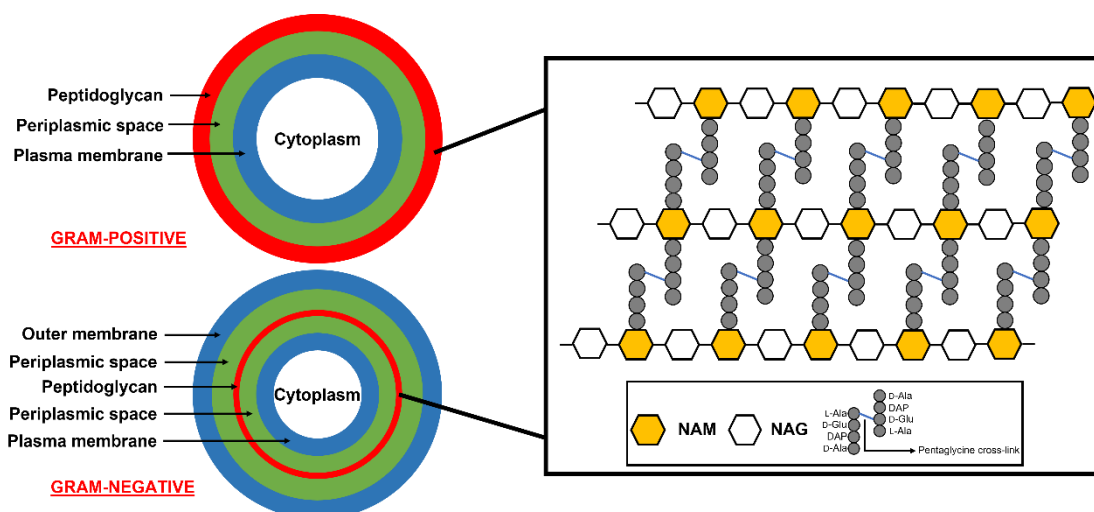


Figure 1.2 Structures of Gram-positive and Gram-negative bacteria. The zoom-in figure depicts the peptidoglycan structure.

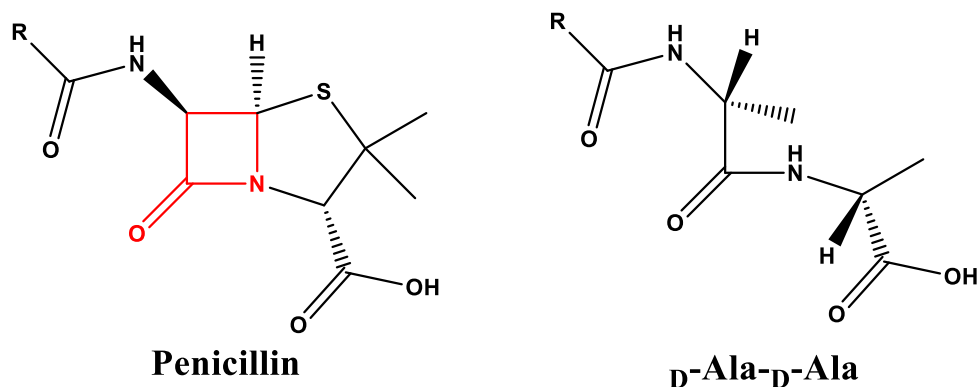


Figure 1.3 Structures of penicillin and D-Ala-D-Ala.¹⁷

1.3. Antibiotics resistance

The pioneer of penicillin, Fleming, had warned about the misuse of the drug which would finally lead to bacterial resistance to antibiotics in as early as 1945 in his Nobel Lecture. In 2019, the World Health Organization (WHO) listed antibiotic resistance as one of the ten threats to global health. As predicted by Fleming, over-prescription and misuse of antibiotics in different sectors render and aggravate the problem of antibiotic resistance. In the clinical sector, especially at the beginning of the use of penicillin, doctors totally relied on this wonder drug to treat all medical problems. This increases the chance for bacteria to adapt to and develop resistance towards antibiotics.

Various social and economic consequences are followed by the development and worsened situation of antibiotics resistance. Even worse, ineffectiveness of antibiotics leads to the post-antibiotics era, in which even a small bleeding wound cause death because of incurable bacterial infection. The curing time of patients in hospital is prolonged, which heightens the risk of hospital-acquired bacterial infection. In the USA, over 63,000 patients pass away every year because of the nosocomial infection.¹⁸ This is accompanied by extra healthcare cost and productivity losses.¹⁸⁻²⁰ In the European Union, Iceland and Norway, the annual loss is about £1.5 billion, and this figure is predicted to increase. It is the responsibility of all governments and related agencies to establish immediate and effective measures to halt and control the misuse of antibiotics, as well as preventive measures to avoid the easy transmission of resistance genes among bacteria all over the world.^{20,21}

1.3.1. How bacteria defend against antibiotics

The molecular mechanisms of antibiotics resistances have been extensively investigated. Four major approaches, target modification, molecular bypass, efflux and

chemical modification, are utilized by bacteria to escape from the attack of antibiotics.²²

Target modification is a straightforward approach to resist antibiotics. PBPs are the targets of β -lactam antibiotics. Once PBPs are altered, the affinity of β -lactam antibiotics towards PBPs would be reduced and hence resistance could be developed.²³ The modification could be achieved by mutation, although extensive alteration is not feasible as it may also affect the affinity between PBPs and the original substrates. Previous works have revealed that there is more than just a single point mutation involved in target modification.²⁴ It is not limited to PBPs as the target to be modified. Single mutation of *gryA* gene, which causes the modification of GryA protein and hence render the ciprofloxacin (a synthetic fluoroquinolone antibiotics) to be ineffective, is one of the examples.²⁵

Resistance to vancomycin is a representative example of molecular bypass. Vancomycin acts as a bactericidal agent by binding to the peptidoglycan terminus D-Ala-D-Ala, and hence hinders the complete bacterial cell wall synthesis. The binding between vancomycin and D-Ala-D-Ala involves five hydrogen bonds (Figure 1.4a). Resistance to vancomycin originates from the substitution of the amide linkage between the two alanine to an ester linkage, resulting in a D-Ala-D-Lactate terminus. This depsipeptide eliminates one of the five hydrogen bonds between the vancomycin and peptidoglycan (Figure 1.4b), and introduces electronic repulsion which significantly reduces the binding affinity to about 1000-fold. D-Ala-D-Ser is another possible mutation for the peptidoglycan. Although only 6-fold reduction in affinity is observed, the extra steric hinderance introduced can effectively prevent the binding with vancomycin.²⁶

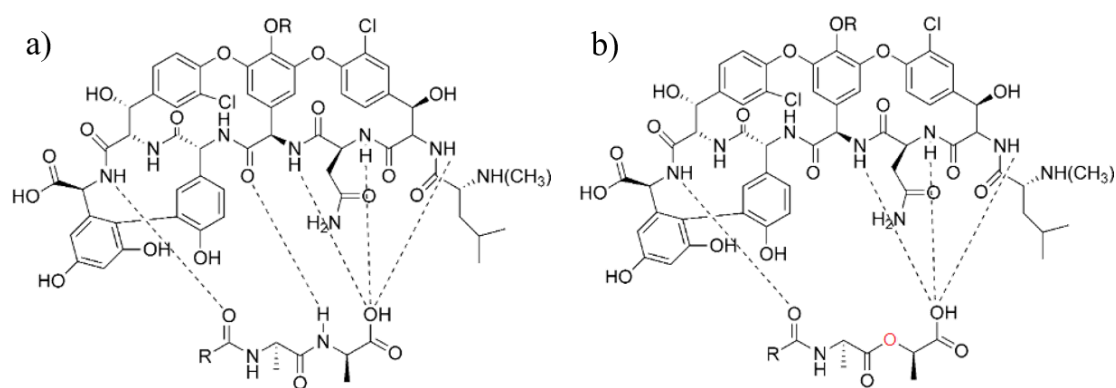


Figure 1.4 Hydrogen network of binding of vancomycin with (a) D-Ala-D-Ala or (b) D-Ala-D-Lactate of the peptidoglycan terminus (modified from reference 22).

Efflux of antibiotics is another mechanism of antibiotic resistance. Bacteria develop their own specific set of efflux pump to actively transport the foreign antibiotics out of the cell. This mechanism contributes to the intrinsic resistance of Gram-negative bacteria, especially to antibiotics that treat Gram-positive bacteria.²² Efflux pump is a set of proteins for transportation within the cell, and overexpression of efflux pumps aggravates resistance to antibiotics. These pumps have different substrate specificities. Those having a broad efflux spectrum are regarded as multidrug resistance efflux pumps.²⁷ *Citrobacter freundii*, which can express New Delhi metallo- β -lactamase 1 (NDM-1), carries genes that encode the tripartite resistance nodulation division pump.²⁸ Other examples include MdeA in *Streptococcus mutans*, FuaABC in *Stenotrophomonas maltophilia*, KexD in *K. pneumoniae* and LmrS in *S. aureus*.^{27,29-32}

Perhaps chemical modification is the most commonly heard resistance mechanism as it involves the expression of β -lactamases to change the chemical structure of β -lactam antibiotics (i.e., hydrolysis of β -lactam ring). It is not only limited to β -lactam antibiotics; numerous enzymes have also been discovered to degrade and alter the chemical structure of different types of antibiotics including aminoglycosides,

phenicols and macrolides.²⁷ In this thesis, antibiotics resistance by β -lactamases would be mainly discussed, as this mechanism is regarded as the most clinically prevalent and has been encountered since the discovery of penicillin.³³ In fact, even though extensive research studies on β -lactamases have been performed for more than 50 years, this antibiotics resistance mechanism is still prevalent nowadays, and the number of β -lactamases discovered is still continuously and exponentially surging, rendering it a more complicated and serious situation.^{34,35}

As discussed above, PBPs are the target for β -lactam antibiotics. The alteration of PBPs to reduce the affinity towards antibiotics and the maintaining of the original PBP activity, is a straightforward defense strategy.³⁶ This mechanism, indeed, is the most effective way for Gram-positive bacteria to defend. However, this resistance mechanism required 20 years after the use of penicillin to be emerged and confirmed by isolation of penicillin-resistance *S. pneumoniae*^{37,38} and methicillin-resistance *S. aureus*.³⁹⁻⁴¹ Instead, the producing of β -lactamases was the first identified resistance mechanism for Gram-positive bacteria.⁴² Later, “trans-Gram gene transfer”, which means β -lactamase genes transfer between Gram-positive and Gram-negative bacteria, was also observed.⁴³

Gram-negative bacteria that possess β -lactamases are more clinically significant. The first β -lactamase discovered in 1940 was isolated from *Bacillus coli*, which is categorized as a class C β -lactamase, and that is the AmpC chromosomal cephalosporinase from *Escherichia coli*.⁴⁴ Later, more Gram-negative bacteria including various enteric species and *P. aeruginosa* were found to produce specific chromosomal β -lactamases.^{45,46} But those plasmid-mediated β -lactamases are more difficult to deal with because they can be widely disseminated. Some plasmids even

carry multiple resistance genes, and dissemination of these plasmids poses more serious threats to the community.

1.3.2. General reaction scheme and classification of β -lactamases

In general, β -lactamases resemble the initial target of β -lactam antibiotics, i.e., PBPs. PBPs bind with β -lactam antibiotics to form an irreversible acyl-complex and this complex undergoes deacylation very slowly. In contrast, β -lactamases have higher affinity towards β -lactam antibiotics than PBPs, and the deacylation rate is significantly higher. The β -lactam antibiotic therefore undergoes chemical alteration and hence inactivation. The binding between serine β -lactamases and β -lactam antibiotics shares similar molecular mechanism of the binding between β -lactam antibiotics and PBPs. The active site serine of β -lactamases attacks the carbonyl group in the β -lactam ring, forming the acyl-complexes. The deacylation and release of the final product involves the opening of the β -lactam ring (Figure 1.5), which inactivates the β -lactam antibiotics, and β -lactamases are regenerated for another round of enzymatic reaction.

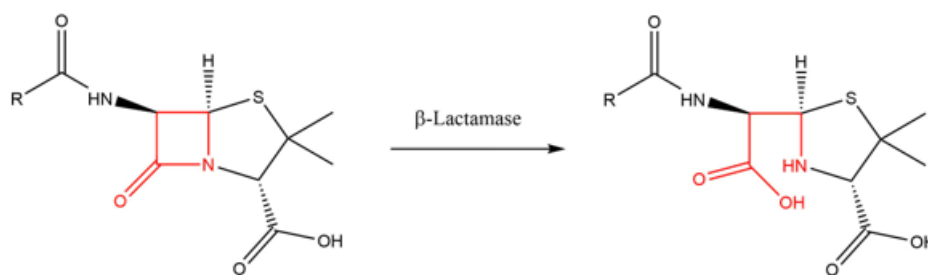


Figure 1.5 General reaction scheme of hydrolysis of β -lactam antibiotic by β -lactamase.

The first β -lactamase was discovered in 1940.⁴⁴ Nowadays, over 7100 β -lactamases have been discovered and identified. These numerous β -lactamases are classified by two systems. The rationale of the first classification is according to the function of antibiotics. Bush-Jacoby-Medeiros system is a sophisticated and common functional

classification system established in 1995.⁴⁷ Four groups (groups 1-4) and several subgroups are utilized to classify cephalosporinases, penicillinase and metallo- β -lactamases.^{47,48} This system is commonly used clinically and mainly beneficial for physicians.

Another classification system, the Ambler scheme, categorizes β -lactamases according to their amino acid sequence. This scheme is continuously revised and nowadays four classes, A-D, are used to classify the β -lactamases.⁴⁹⁻⁵¹ Classes A, C and D are serine β -lactamases, while class B is metallo- β -lactamases. There are various subclasses under each major class (Figure 1.6). A comprehensive, weekly updated web-based database, Beta-Lactamase DataBase (BLDB), was established in 2017. It provides a lot of useful information related to β -lactamases such as sequences, structural and kinetic information.⁵² The relationships of these two systems are integrated in Figure 1.7.

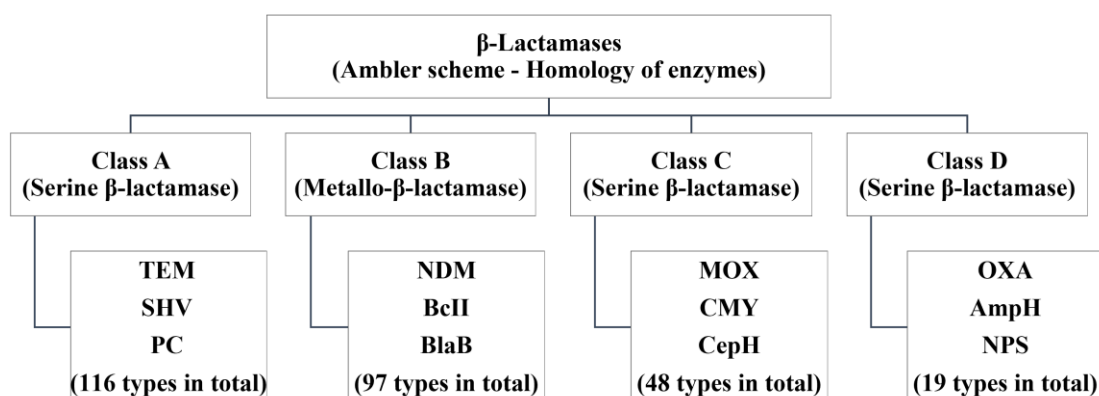


Figure 1.6 Classification of β -lactamases under Ambler scheme.

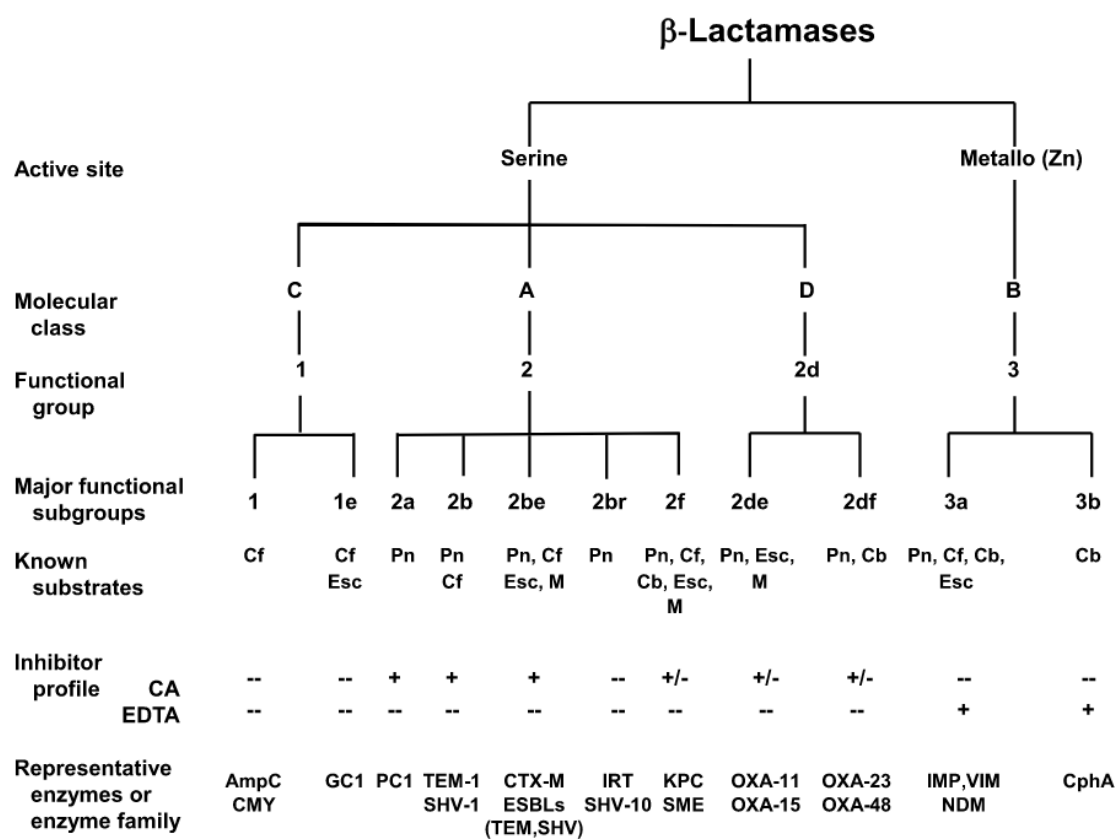


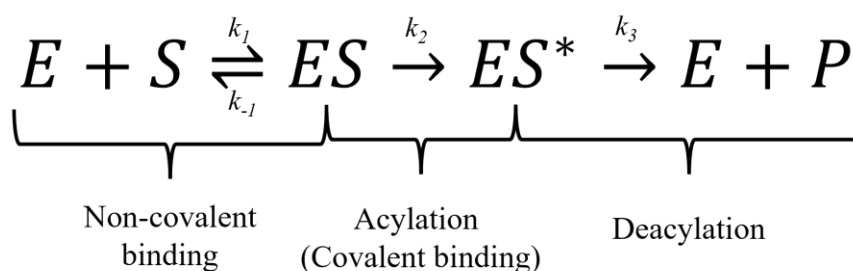
Figure 1.7 Molecular and functional features of the major groups of β-lactamases.³⁴

1.4. TEM-type β -lactamases

TEM-type β -lactamases belong to Ambler class A, and they are the most widely spread and intensively investigated β -lactamases because of their high clinical relevance. TEM-1, the ancestor in this class A sub-group, was isolated from an Athenian patient called Temoniera in 1963.⁵³⁻⁵⁵ It is the first plasmid-borne β -lactamase identified in Gram-negative bacteria and it is effective towards aminopenicillins and old generation cephalosporins. Since the discovery of β -lactamases, various new antibiotics have been developed and introduced to evade the attack from β -lactamases. Unfortunately, that initiates and triggers the endless competition between human and β -lactamase-producing bacteria, i.e., introduction of novel antibiotics or inhibitors will naturally trigger the production of new or mutated β -lactamases by the bacteria which conferring resistance to those newly developed antibiotics. The problem of mutation happens frequently for TEM-type β -lactamases, and more than 200 variants have been discovered nowadays. Sequence alignment have revealed that these variants usually incorporate one to five point mutations.^{56,57} Single or a combination of point mutation(s) confers β -lactamase variants different antibiotics phenotypes and various antibiotics catalytic properties compared with TEM-1, rendering a serious consequence that later generations of β -lactam antibiotics and β -lactamase inhibitors become ineffective. It is worth noting that even various TEM-type variants have been identified nowadays, the ancestral TEM-1 and TEM-2 have never disappeared, they are still ubiquitously found in contemporary clinical and environmental isolates (these isolates are able to carry multi-resistance genes). This could be attributed to their high hydrolyzing ability for old generation penicillins which are still commonly used.

1.4.1. Kinetics of β -lactam antibiotics hydrolysis by TEM-type β -lactamases

TEM-type β -lactamases belong to class A which contain an active site serine, therefore their catalytic mechanism reasonably follows that of the serine proteases.⁵⁸ Generally, a three-step kinetic model is used to describe the antibiotics hydrolysis by β -lactamases (Scheme 1).



Scheme 1. The three-step model of antibiotic hydrolysis by β -lactamases.

There are three steps in the reaction. The first step is non-covalent binding for forming a non-covalent Michaelis enzyme-substrate complex (ES). The binding affinity is represented by dissociation equilibrium constant, K_d . The constant is the ratio of dissociate rate constant to the ES formation rate constant, k_{-1}/k_1 . The second step is acylation, and the acylated-enzyme-substrate complex (ES^*) is formed. The acylation rate constant is represented by k_2 . The final step is deacylation. In this step, a hydrolyzed product (P) is released and the apo-enzyme (E) is restored for the next round of hydrolysis.^{59,60} The deacylation rate constant is represented by k_3 . For the reaction between PBP and antibiotics, since antibiotics act like the PBP inhibitor, the deacylation rate (k_3) is very low, and is 100-1000 fold lower than the acylation rate (k_2), so as to maintain the ES^* state for PBP inhibition.⁶¹ However, since β -lactamases act as a “regenerating machine” to keep hydrolyzing the antibiotics, it is expected and reported that the binding affinity ($1/K_d$), acylation (k_2) and deacylation rate (k_3) are much higher than those of PBP- β -lactam antibiotics reaction.⁶²

Michaelis kinetic parameters are also commonly used to describe and compare the catalytic efficiencies between different β -lactamases and antibiotics. These parameters are related to the kinetic rate constants of the three individual steps, and the following equations summarize these relationships.

$$k_{cat} = \frac{k_2 k_3}{(k_2 + k_3)} \quad (1)$$

$$K_M = \frac{k_3(k_{-1} + k_2)}{k_1(k_2 + k_3)} \quad (2)$$

$$\frac{K_M}{k_{cat}} = \frac{k_1 k_2}{(k_{-1} + k_2)} \quad (3)$$

Where the Michaelis constant, K_M , is conceptually similar to K_d , but k_2 and k_3 also contribute to K_M . Turnover number, k_{cat} , reflects the speed of the decomposition of the ES^* to release the hydrolyzed antibiotics. K_M/k_{cat} indicates the ES^* formation efficiency, as k_3 does not contribute to this parameter.⁶³

1.4.2. Structure and mechanism of TEM-type β -lactamases

The structure of TEM-1 has been well studied by X-ray crystallography (PDB: 1BTL).⁶⁴ It consists of two structural domains, an α -domain and an α/β domain, with 11 helices and 5 β -strands in total. The α -domain consists of the H2 – H9, while the α/β domain is constructed by the five-strand β -sheet, H1, H10 and H11. The inner hydrophobic core is the interface of the two domains. Two hinge regions, residues 60-68 and 212-222, are responsible for depression generation for the two domains at the interface in order to form the enzymatic active site. To maintain this cavity, hinge regions are constrained by several hydrogen bonds (e.g., E37-V44, E37-R61, WAT294-L221, WAT294-L222) and salt bridges (R43-E64, R61-E64, R222-D233) to minimize the occurrence of significant conformational changes. TEM-1 also contains one

disulfide bond (C77-C123) which links H2 and H4. This linker is not critical for the catalytic efficiency of TEM-1 towards antibiotics, but thermostability is undermined after the removal of the disulfide bond, as reflected by the C77S mutant of TEM-1.⁶⁵

The active site cavity is constructed by several regions of TEM-1. The most catalytically important S70 is located in one turn 3_{10} helix containing residues 69-71, and interacts with K73, which is located in the continued α -helix H2 through a hydrogen bond. Side chains of both residues are directed towards the active site cavity and are directly involved in the antibiotics hydrolysis to result in the highly conserved property of the Ser-Xaa-Xaa-Lys motif in TEM-type β -lactamases.

S130, D131 and N132, commonly termed as the SDN loop, are another segment of the binding pocket. This loop links H4 and H5. SDN confers certain constraints to the protein conformation, particularly to the N-terminals of H3 and H5, through the hydrogen bond between the buried side chain D131 and four residues, V108, T109, N132, T133 and A134. These interactions also assist the proper orientation of S130, which is also involved in the hydrolysis.⁶⁴

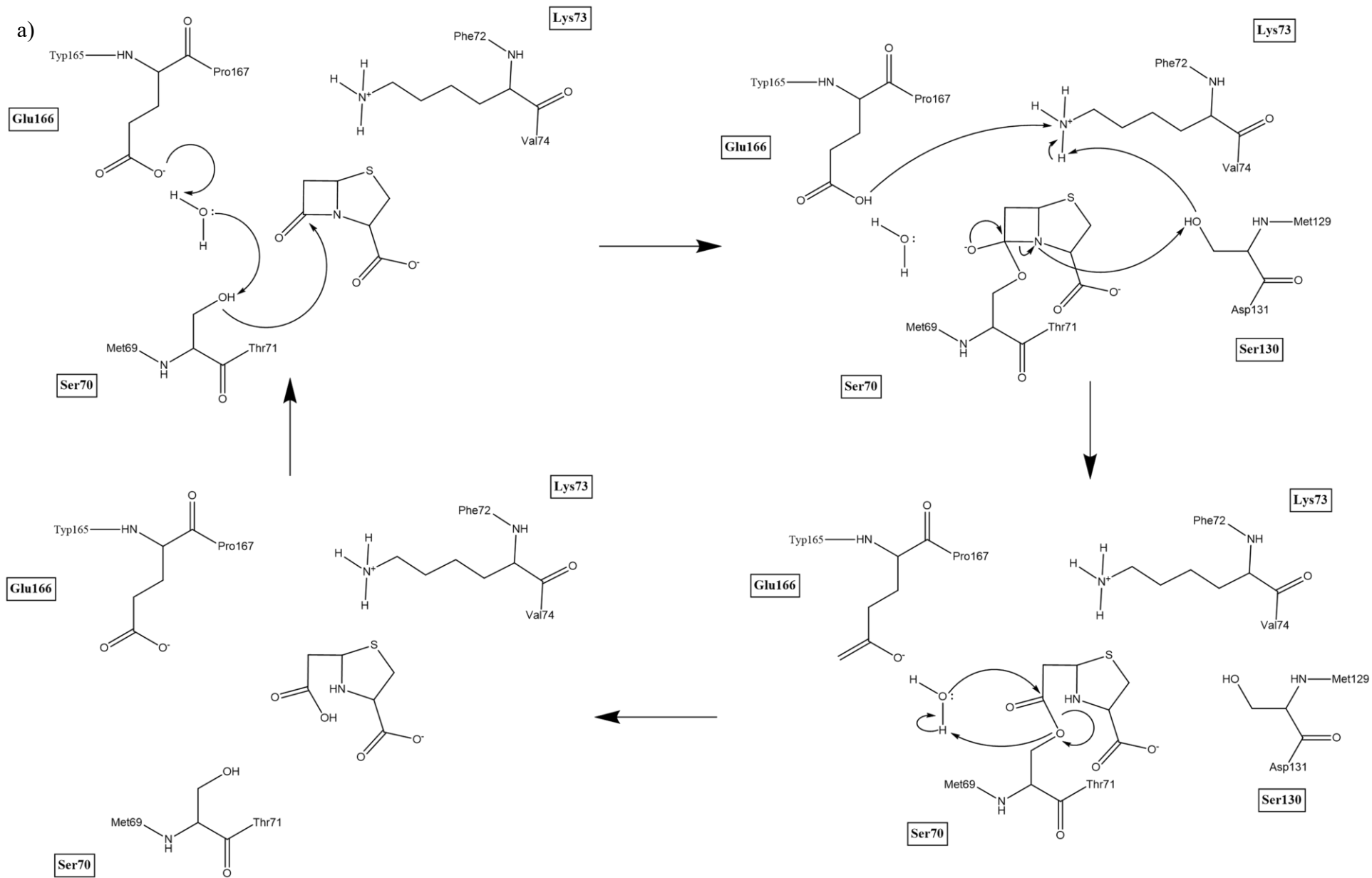
β -3 and β -4 strands (residues 230-237 and 244-251, respectively) are located at the opposite side of the active site with respect to the SDN loop, forming another side of the binding pocket. Residues in these two strands significantly affect the catalytic efficiencies and stability of antibiotics. For example, the nitrogen atom in A237 is part of the oxyanion hole that stabilizes the charged intermediate state of the antibiotics.⁶⁶ Alteration of the position of the side chain of this residue may create significant steric hindrance for substrate binding, or cause the substrate incapable to bind if the side chain orients to the inner part of the active site. R244 in the β -4 strand interacts with N276,

G236, and the sulphate ion in the TEM-1 structure, rendering R244 a valuable residue to be investigated.⁶⁷ Interestingly, R244 is not highly conserved in TEM-type β -lactamases but still deemed as a catalytically related residue. Mutations occurring at this point can change the phenotype of the β -lactamases, providing insights into the resistance mechanism of TEM-type β -lactamases.

Ω -loop (residues 161-179) is another edge of the active site. This solvent-exposed loop is directly involved in the enzymatic reaction, and it strongly interacts with nearby regions. Maintaining the shape of the Ω -loop depends on the strictly conserved, 3 Å salt bridge between R164 and D179. Loss of this interaction is detrimental to the β -lactamase stability as it causes significant distortion of the Ω -loop conformation, and leads to improper orientation of several catalytically important residues.⁶⁸ In addition, four salt bridges are found in TEM-1, hence its Ω -loop is believed to be more constrained than that of PC-1⁶⁴, another class A β -lactamase, as only the essential R164 and D179 salt bridge exists in the Ω -loop of PC-1. E166 can be regarded as the second most important residue for the catalysis that has been extensively investigated. It interacts with several important residues such as K73, N132, N136 and WAT297, which is predicted that this water molecule acts as the bridge in the proton transferring pathway during antibiotics hydrolysis. Proper orientation of E166 plays a decisive role in the catalysis. Generally, peptide bonds in protein are in trans orientation, but cis-peptide bond can be formed between proline and the residues that precede it, however, with very low possibility.^{69,70} In TEM-1, E166 is followed by P167 and linked by a cis-peptide bond. P167 also hydrogen bonds to N170 side chain and interacts with L169 through Van der Waals force to stabilize the cis-peptide bond. The side chain of P167 reduces the steric hindrance and therefore the cis-peptide bond is beneficial for the side chain of E166 to perfectly orient towards the core of the binding pocket.^{68,71}

The solvent-exposed H2/H3 connecting loop, particularly the C-terminal of loop, is located at the entrance of the active site. Residue E104 and Y105 are two residues closest to the entrance in the connecting loop. Even though their side chains are not directly involved in the hydrolysis, since E104 is close to P167, it can also interact with the penicillin C6 side chain.⁶⁴

The hydrolysis reaction mechanism has been studied extensively. Although certain details are still controversial, a simple and general mechanism has been depicted and is commonly accepted. As previously discussed, the reaction involves both acylation and deacylation. The acylation is triggered by the nucleophilic attack by the deprotonated side chain of S70 to the carbonyl group in the β -lactam ring. One of the controversies comes from the deprotonation of the hydroxyl group in S70. One hypothesis suggests that the S70 is activated by the general base E166, which deprotonates S70 indirectly through WAT297 as a bridge (Figure 1.8a). The second prediction does not involve any water molecules, instead, one hydrogen of the charged K73 is captured by the general base E166, and the uncharged K73 deprotonates S70, followed by the subsequent nucleophilic attack steps similar to the first hypothesis (Figure 1.8b). S130 is the proton donor that assists the β -lactam ring opening process, followed by the deacylation step.⁷²⁻⁷⁶



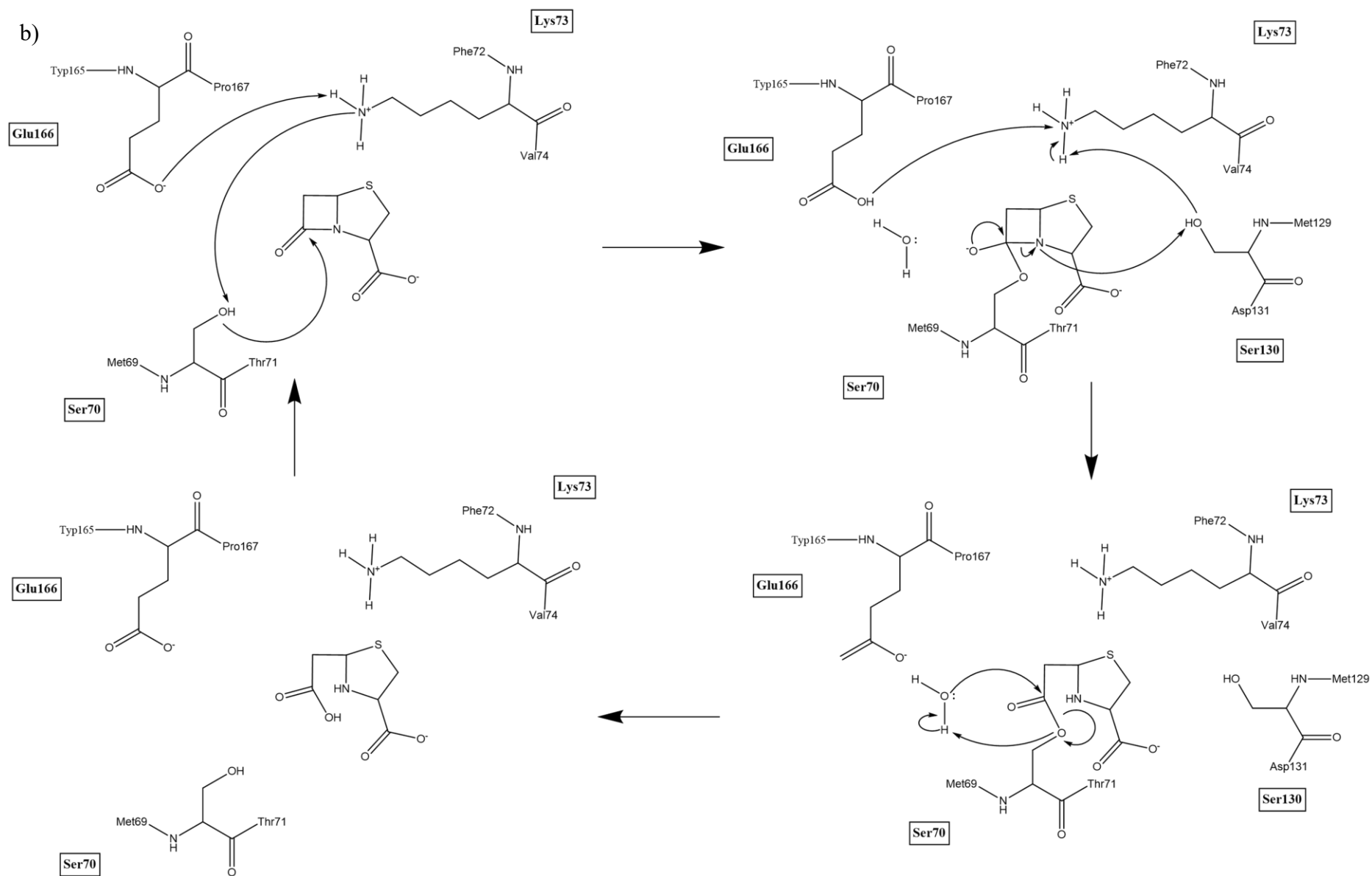


Figure 1.8 General reaction mechanism of TEM-type β -lactamases. (a) S70 is activated indirectly by the general base E166 through WAT297 (b) Glu166 deprotonates S70 through K73.

1.4.3. TEM-type extended spectrum β -lactamases (ESBLs)

As discussed previously, extensive use and misuse of β -lactam antibiotics have led to the development of TEM-1 variants. The sophisticated protein sequencing techniques facilitate the fast protein sequencing of these variants and hence the number, position and type of point mutations are easily acquired. Activity profiles of these variants were well studied. They present varied phenotypes compared to TEM-1, and many of them acquire the extra ability to hydrolyze later generation cephalosporins (e.g., cefotaxime and ceftazidime, and they are commonly used clinically^{56,57,77-81}), contrasting to the TEM-1 that possessing hydrolytic power limited to only penicillins, first and second generation of cephalosporins. These variants are regarded as extended spectrum β -lactamases (ESBLs). How these point mutations affect the hydrolytic profiles, kinetics and structures are therefore extensively investigated in the past few decades. A few common strategies are utilized to study these ESBLs. Site directed mutagenesis is utilized to systematically study the effect of different mutations or combinations of mutation at particular residue(s). These mutants are further analyzed by kinetic studies towards various types of antibiotics so as to understand changes in catalytic profiles and efficiencies towards different types of antibiotics. X-ray crystallography is the best way to study the structure of mutants, and kinetic information is often combined to explain the resistance mechanism of these mutants. Among the discovered point mutations, G238S, E104K and M182T are three of the common point mutations found in TEM-types ESBLs. Examples include TEM-19 (G238S), TEM-17 (E104K), TEM-15 (G238S/E104K), TEM-20 (G238S/M182T) and TEM-52 (G238S/E104K/M182T), which are discovered in clinical isolates, indicating the significance of understanding the properties of these point mutations.^{56,57,78}

1.4.3.1. Effect of G238S

Residue 238 is located exactly next to the β -3 strand, and it is the first amino acid in the β -3/ β -4 strand connecting loop, acting as the entrance of the binding pocket. Mutation in this residue is common, and serine is the most common amino acid for glycine to be mutated. G238S is known to be a key substitution conferring TEM-1 the extended hydrolytic power to later generation cephalosporins, especially cefotaxime and ceftazidime. Catalytic efficiencies of mutants possessing G238S towards these two antibiotics have been enhanced about 100 times.^{63,82} Interestingly, G238S containing mutants have lower hydrolytic power towards early generations of penams and cepheems. Two models were proposed to explain these observations. In the first model, an additional hydrogen bond was formed between S238 residue and oxime oxygen in third generation cephalosporins, which enhanced the binding affinity.⁸³ However, another model proposed that the position of β -3 strand or Ω -loop was shifted due to the steric conflict between the additional hydroxyl side chain of S238 and N170^{84,85}, resulting in the enlargement of the binding pocket. The second proposed model was consistent to the results of the site-directed mutagenesis and kinetic study.⁸⁵ There is still no consensus has been reached yet. The X-ray crystal structure of TEM-52 (PDB: 1HTZ) has suggested that the 2.8 Å displacement of loop consisting of residues 238-243 has been led to enlargement of the binding site so as to accommodate the bulky oxyimino side chains of the third generation cephalosporins. In addition, K104 also re-oriented its new positive side chain towards the active site.⁸⁶ However, no movement of the residue 238-243 loop was found and hence the active site expansion was not observed for TEM-72 (Q39K/M182T/G238S/E240K), although this might be due to the effect of E240K dominating the conformational changes in this loop.⁸⁷ Similar to TEM-52, the G238A structure also provided the observation of binding pocket expansion which attributed to the displacement of the Ω -loop due to the extra steric hindrance caused by

the additional methyl side chain.⁸⁸ Besides explaining the extended power to hydrolyze the third generation cephalosporins, active site expansion also provides a reason for the lower catalytic efficiency towards penicillin, which is possibly due to the enlarged cavity being unable to provide the best orientation for active site residues and antibiotics during binding and hydrolysis.

1.4.3.2. Effect of E104K

Similar to residue 238, residue 104 is also located at the entrance of active site opposite to the β -3 strand. E104K was found to enhance the hydrolyzing efficiency of third generation cephalosporins by about 10 times.⁸⁹ Unlike G238S substitution, the catalytic power of penicillin is not impaired after this point mutation. Usually, the effect becomes more drastic and obvious when E104K is presented with other key substitutions like G238S, R164H or R164S.⁸⁹ The binding affinity enhancement was explained by the direct interaction between K104 and carboxylic acid group of cephalosporins.⁹⁰ However, another explanation is that the SDN loop was perturbed by K104, which facilitated third generation cephalosporins to enter the active site.⁸⁹ A molecular modeling study suggested that the destabilizing effect of G238S could be compensated by E104K, but this effect was not observed in a later melting temperature study.⁸⁸

1.4.3.3. Effect of M182T

ESBL containing M182T only was also discovered and termed as TEM-135.⁹¹ However, TEM-135 is not clinically significant as it is not discovered in clinical isolation. It also did not confer any enhancement in antibiotics hydrolyzing efficiency. Nevertheless, incorporating M182T to other catalytically important point mutation(s) could enhance the hydrolytic power. This is reflected by the cefotaxime minimum inhibitory concentration (MIC), and G238S/E104K/M182T mutant (i.e., TEM-52) had even

greater MIC than the synergistically efficiency-enhanced G238S/E104K mutant (i.e., TEM-15).⁸⁶

However, even though residue 182 is positioned far away from the active site, the M182T point mutation was reported to be a global suppressor to compensate for the destabilizing effect due to missense mutations, surprisingly in both antibiotics- and inhibitor-resistant type of β -lactamases. A thermodynamic study revealed that ESBLs with M182T showed increased melting temperature when compared with those without M182T, indicating this point mutation could stabilize mutants. At the same time, the activities of M182T containing ESBLs towards third generation cephalosporins were retained or even enhanced, implying that the stability-activity trade-off has been overcome by M182T.^{88,92} In an X-ray crystallography and molecular modeling study, two new hydrogen bonds were formed between Thr182 and both the Glu63 and Glu64, which are located in the first hinge region connecting the two domains. These interactions limit the conformational flexibility of the two domains, leading to better orientation of catalytically important residues.^{86,93} Another X-ray crystallography, molecular modeling and site-directed mutagenesis study proposed that the hydrogen bond was actually formed between T182 and A185 and this formed a favorable N-cap and stabilized 183-195 helix.⁷⁴

1.5. Hydrogen/deuterium exchange-mass spectrometry (HDX-MS) for dynamic properties of protein

Currently, there is still no consensus about the effects of the three point mutations. Most of studies relied on X-ray crystallography, molecular modelling and site-directed mutagenesis. These methods could provide very important structural information, but only limited dynamic information of proteins could be obtained. It is believed that dynamic properties could affect the bioactivity of protein significantly. However, there is a lack of comprehensive study focusing on the dynamic properties of β -lactamases and it is essential to investigate the relationship between the dynamic properties of β -lactamases and their catalytic efficiencies in order to fully understand the reaction mechanism.

Hydrogen/deuterium exchange (HDX) is a widely used method to study the conformational dynamics of proteins. This method was introduced in 1954.⁹⁴ Considering the structure of peptide, three types of hydrogen can be observed: (1) alkyl hydrogens (or sometimes regarded as α -hydrogen); (2) side chain hydrogens, which are connected to the side chain functional groups; (3) amide hydrogen, which are located all along the polypeptide backbone, except for proline (Figure 1.9). Since alkyl hydrogen and side chain hydrogen have extremely long and short exchange time, it is impossible to observe their exchanges. However, the exchange time of amide hydrogen falls into the observable range of mass spectrometer. Therefore, during HDX analysis, only amide hydrogen (also called backbone hydrogen), is concerned. Amide hydrogens that have higher solvent accessibilities or that are involved in weak intramolecular interactions have higher HDX rate, and vice versa. By measuring the extent of HDX, flexibility of protein could be reflected.

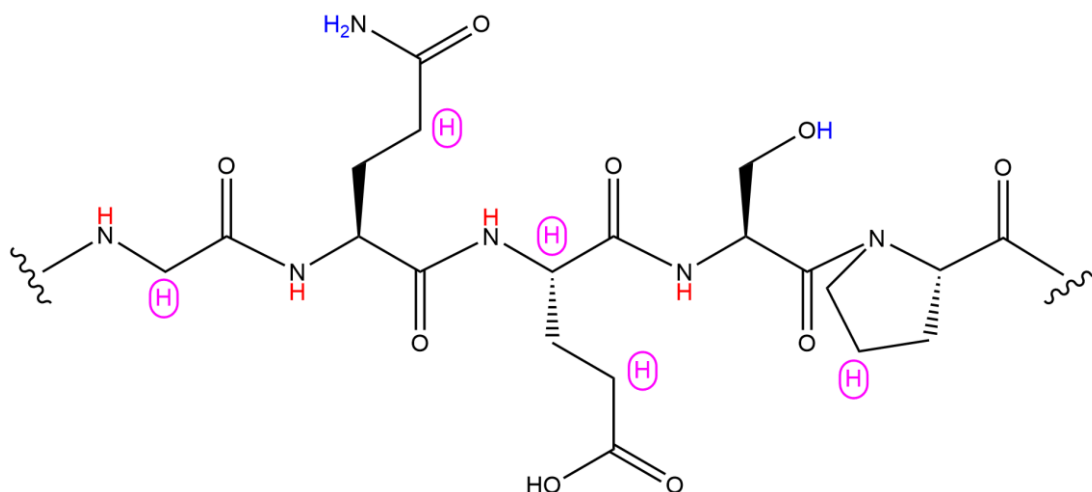


Figure 1.9 The structure of peptide Gly-Gln-Glu-Ser-Pro. Three types of hydrogen, exchangeable backbone amide hydrogens (red), side-chain hydrogens (blue), alkyl hydrogens (purple with rounded rectangle, indicated here only for part of the peptide backbone), are shown here.

Conventionally, nuclear magnetic resonance (NMR) is used for detection of HDX changes.⁹⁵ For example, the formation and stabilization of the secondary structure regions of BS3 class A β -lactamase from *Bacillus licheniformis* were investigated by HDX-NMR.⁹⁶ However, NMR requires relatively high concentration to generate observable signals and it works for relatively small protein only. The intrinsic property of mass spectrometry (MS) to distinguish isotopes, which have highly similar physical and chemical properties with the only obvious difference between them in mass, renders MS as a suitable detector for HDX. Besides, the high sensitivity and resolution and low sample consumption of MS are highly beneficial to HDX analysis. Moreover, by using time-of-flight as mass analyzer, the size limitation of protein for NMR is now overcome. More advanced mass analyzers, such as orbitrap, further enhance the mass accuracy and resolution, leading to more accurate and comprehensive analysis. Compared with X-ray crystallography, which requires qualified crystals to be grown and this is not an easy task, HDX-MS only requires samples in solution form. Also, it is not easy to solve

the X-ray crystal structure. Besides, HDX-MS provides in solution dynamic information which X-ray crystallography could only provide a static protein structure. However, it should be noted that HDX-MS and X-ray crystallography are complementary to each other as the visible information and interactions between residues provided by X-ray crystallography greatly assist the interpretation of HDX-MS results.

There are two types of labeling approaches: continuous labeling and pulse labeling.⁹⁷ Continuous labeling describes the protein or protein complexes undergoing deuterium labeling for a fixed period (e.g., 60 minutes). Since the D₂O concentration is very high (usually 90% or above), once a protein is labeled, deuterium labeled is not able to reverse to its protiated form theoretically. Therefore, the transition from its monoisotopic form to the deuterium form is almost unidirectional. Once enough time is given, proteins would be totally deuterated as a result of protein motions and protein breathing (Figure 1.10). This approach is useful in study of slow unfolding transitions.⁹⁸ Overall, continuous labeling is the mostly adopted approach as it is easier to perform.

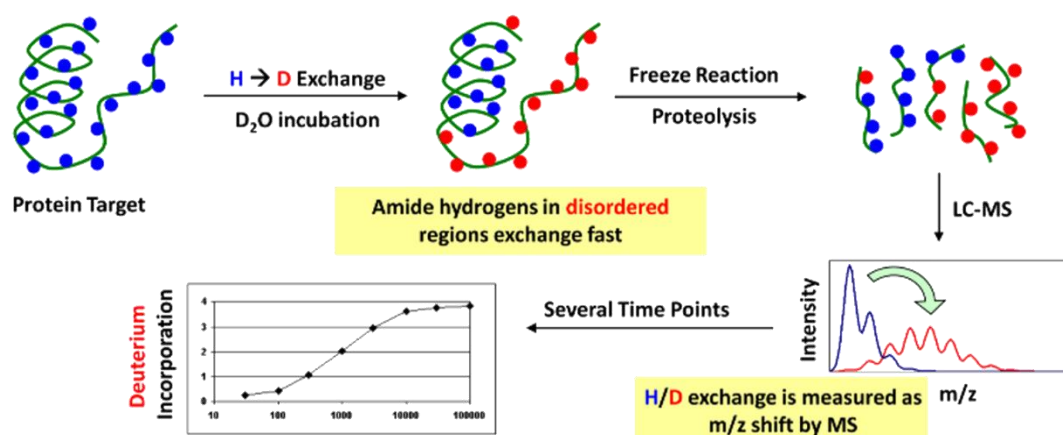


Figure 1.10 Procedures of continuous labeling local HDX-MS.

In contrast to continuous labeling, pulse labeling only exposes samples to deuterium for a very short time and such short period is called pulse. In this approach, proteins are usually induced to undergo conformational changes in response to a perturbing agent. Only the part of protein that is unfolded during the pulse could be labeled. It can be used to identify protein folding mechanisms⁹⁹ and probe significantly populated kinetic intermediate states in a folding reaction. Information obtained by pulse labeling can be complementary to information obtained by conventional spectroscopic and NMR experiments.¹⁰⁰⁻¹⁰⁶

Both local and global HDX can be performed after either continuous and pulse labeling. The main difference of these two approaches is whether the protein or protein complex undergoes proteolytic digestion or not. The protease used in HDX is usually pepsin, as it is acid-resistant and compatible with the HDX quench conditions, which are usually at 0 °C and pH 2.5. In short, global HDX provides a general picture for the conformational change under the influence of stimuli, while local HDX reveals the detail of exact exchange locations. Figure 1.11 summarizes the general procedures for HDX.

Several studies related to β -lactamases were performed using HDX-MS. A protein folding study about changes in solvent accessibility for the binding between AmpC β -lactamase and rottlerin was investigated by HDX-MS, so as to reveal the mechanism of aggregate-based inhibitors that the protein was partially unfolded when the protein was bound to the aggregate particle.¹⁰⁷ Pulse-labeling HDX-MS on a microfluidic chip was performed to monitor changes in the active site of TEM-1 upon the covalent binding with an inhibitor, and also subtle changes in regions far away from the active site.¹⁰⁸ Later, by using a similar approach to perform millisecond HDX-MS, dynamic modes

linked to individual hydrolytic steps of TEM-1 towards ampicillin, cephalexin and clavulanate were identified.¹⁰⁹ A recent study about the conformational changes upon the binding between the β -lactamase inhibitory protein and three typical class A β -lactamases, TEM-1, SHV-1 and PC-1, was performed by HDX-MS. H10 was found to be an allosteric site that could affect class A β -lactamase inhibition.¹¹⁰ Although these studies provide new information about the properties of β -lactamases, no in-solution conformational dynamics study related to the effect of point mutations or their combinations was done, which can combine with the previously known structural information to understand the properties of each point mutations more comprehensively.

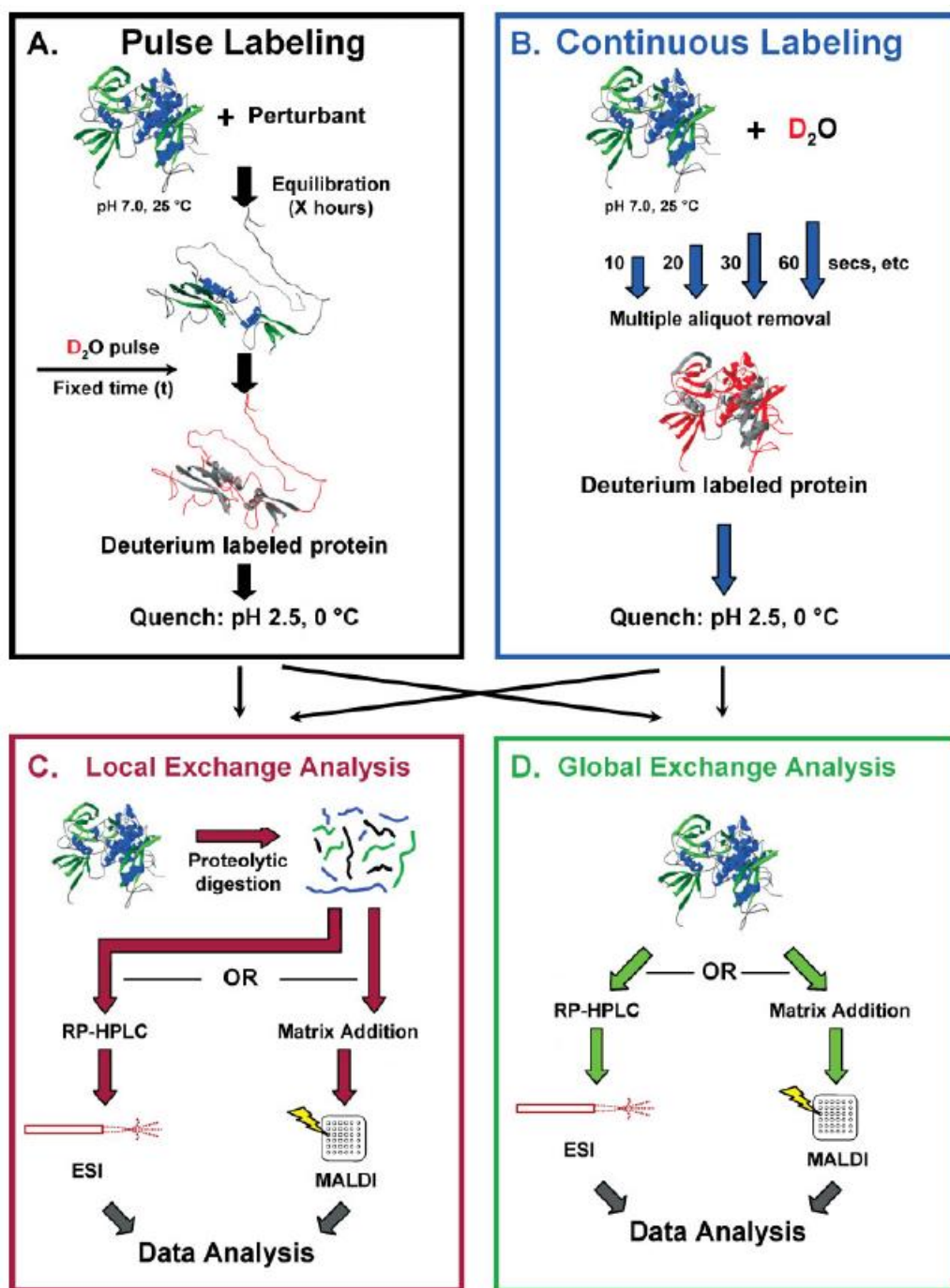


Figure 1.11 Normal process of HDX. Either (a) continuous labeling or (b) pulse labeling can be used for deuteration, followed by either (c) local exchange analysis or (d) global exchange analysis.¹¹¹

1.6. Objectives

Although there are already some extensive investigations of G238S, E104K and M182T on TEM-type ESBLs, there is still a lack of a comprehensive study on the conformational dynamics of ESBLs. Conformational dynamics are believed to be vital for biological functions of proteins. As β -lactamases are highly dynamic, and conformational dynamics are highly cooperative, especially during substrate binding, therefore, in this thesis, the individual and interactive effects of these three point mutations on the structural and dynamic properties of six TEM-type ESBLs, TEM-1, TEM-19 (G238S), TEM-17 (E104K), TEM-15 (G238S/E104K), TEM-20 (G238S/M182T) and TEM-52 (G238S/E104K/M182T), and their conformational changes induced by their binding with the third generation cephalosporin, cefotaxime, are investigated by using both global and local HDX-MS. The regional flexibility and conformational change information provide more insights into the structural/dynamics-function relationships of ESBLs to benefit the development of new antibiotics and β -lactamase inhibitors.

Chapter 2. Experimental

2.1. Protein expression

2.1.1. Expression and purification of TEM-type β -lactamases

All TEM-type β -lactamase used in this study were either purchased from GenScript or in-house expressed. The expression procedures were the same for both sources and listed below. The target DNA sequence of the six β -lactamases was optimized, synthesized and cloned into vector pET30a with His-tag and kanamycin resistance gene for protein expression. The *E. Coli* strain BL21 Star (DE3) was transformed with recombinant plasmid by heat shock at 42 °C in a water bath for 90 seconds. The transformed *E. coli* was pre-cultured on an agar plate with kanamycin at 37 °C for 16 hours. 10 mL of pre-cultured was added to 200 mL 2x yeast extract tryptone (2xTY) and 200 μ L isopropyl β -d-1-thiogalactopyranoside (IPTG) was then added to induce protein overexpression when OD₆₀₀ reached 0.8. After overnight incubation, the cells were collected by centrifugation and then resuspended in the lysis buffer (50 mM phosphate buffer, 300 mM NaCl, pH 7.4). Resuspended cells were homogenized and lysed by sonication for 1 minute. Cell debris was removed by centrifugation (10000 rpm, 60 minutes), followed by filtration using a 0.45 μ m PVDF filter.

Fast protein liquid chromatography was used for protein purification. In brief, the crude extract was loaded to a Ni²⁺ charged HisTrap™ Fast Flow Crude column (5 mL, GE Healthcare). 50 mM phosphate buffer (see section 2.3) was used to flow through the system which trigger the binding between the his-tagged protein and column. Elution of protein was performed using 1 M imidazole in 50 mM phosphate buffer and eluents were collected in fractions for further sodium dodecyl sulphate–polyacrylamide gel electrophoresis (SDS-PAGE) analysis. Imidazole was then removed by buffer exchange of the purified protein to 50 mM phosphate buffer.

2.1.2. Expression and purification of E166A TEM-type β -lactamases

Primers were specifically designed for the expression of the E166A mutants. Polymerase chain reaction (PCR) was performed between the designed primer and vector used for expressing the normal TEM-type β -lactamases in order to amplify excess amount of the vectors which encoded the targeted TEM-type β -lactamase sequences with E166A mutation. These mutated vectors were utilized for expression of E166A TEM-type β -lactamases, whose further expression followed the previous procedures of the vector transformation to the *E. Coli* strain BL21 Star (DE3).

2.2. Study of the completeness of binding between TEM-17 and cefotaxime

TEM-17 was reacted with cefotaxime at a fixed ratio 1:500 (mol/mol) for different durations, ranging from 5 seconds to 6 minutes. The reaction was quenched by using 1.5% formic acid (FA) to lower the pH of the reaction mixture to 2.5. The reaction was immediately subjected for liquid chromatography mass spectrometry (LC-MS) analysis. The LC separation method and MS detection method were similar to those of the global HDX-MS (see section 2.6), except the LC separation would be performed at room temperature. Data were analyzed and spectra were deconvoluted by MassLynx V4.2.

2.3. Preparation of phosphate buffer and deuterated phosphate buffer

The 50 mM phosphate buffer was prepared by mixing potassium phosphate monobasic and potassium phosphate powder, and then they were dissolved by Milli-Q water so that the concentrations of both components were 25 mM. The final pH of the buffer was tuned to 7.4. The 50 mM deuterium phosphate buffer was prepared similarly, but deuterium oxide (99.9%, Cambridge Isotope Laboratories) was used instead of Milli-Q water. Since the pD was 0.4 unit higher than pH, therefore pH of the deuterium buffer was tuned to 7.0 so that pD of the buffer was 7.4. All buffers were prepared under room

temperature.

2.4. Preparation of the E166A-TEM-type β -lactamase-cefotaxime complexes

To analyze the enzyme-substrate complex (ES*), E166A-incorporated apo-TEM-type β -lactamase was mixed with 100-fold molar excess of cefotaxime for 20 minutes to ensure the full conversion of apo-E166A-TEM-type β -lactamases to cefotaxime-bound acylated complexes. The completeness of the conversion was confirmed by intact protein UPLC-MS analysis.

2.5. Hydrogen/deuterium exchange (HDX) reaction

Either apo-TEM-type β -lactamases or E166A-TEM-type β -lactamases-cefotaxime complexes were diluted 10-fold (v/v) in 50 mM deuterated phosphate buffer (pD 7.4) and incubated from 10 seconds to 120 minutes (for global HDX-MS), or from 1 minute to 60 minutes (for local HDX-MS), at 25 °C. Control experiment was performed similarly but a non-deuterated buffer was used instead. At the desired time point, pre-cooled 1.5% FA was added to the reaction mixture at 1:1 ratio (v/v) to quench the HDX reaction by lowering the pH and temperature of the mixture to 2.5 and 0 °C, respectively. The quenched mixture was then immediately subjected to ultra-performance liquid chromatography mass spectrometry (UPLC-MS) for further analysis.

2.6. Global HDX-MS analysis

50 μ L of the quenched mixture was injected into a Waters nanoACQUITY UPLC system (Waters Corporation, Milford, MA, USA) and the whole column chamber was maintained at 0 °C during the whole chromatographic separation. Online desalting and preconcentration were performed utilizing the Waters ACQUITY UPLC Protein BEH C4 VanGuard Pre-column (300 Å, 1.7 μ m, 2.1 mm $\times$ 5 mm) by flowing 0.1% FA at 100

$\mu\text{L}/\text{min}$ for 3 minutes. The desalted analytes were then separated and eluted through a Waters ACQUITY UPLC Protein BEH C4 Column ($300 \text{ \AA}$, $1.7 \mu\text{m}$, $2.1 \text{ mm} \times 100 \text{ mm}$) in 10 minutes at $60 \mu\text{L}/\text{min}$ by the following gradient elution profile: A: 0.1% FA, B: acetonitrile (ACN) with 0.1 % FA; at 0 minute 20% B, at 5-8 minute 80% B, at 10 minute 20% B. Eluent was detected by the Waters Synapt G2-Si mass spectrometer (Waters Corporation, Milford, MA, USA) at the following conditions: capillary voltage 3.0 kV, source temperature 150°C , desolvation temperature 350°C . Mass spectra were acquired over a mass-to-charge ratio (m/z) range of 100 to 2000.

2.7. Optimization of digestion enzyme of local HDX-MS

The digestion efficiency of proteases type XVIII (Sigma, St. Louis, MO), pepsin column (Waters Enzymate BEH-Pepsin column, $300 \text{ \AA}$, $5 \mu\text{m}$, $2.1 \text{ mm} \times 30 \text{ mm}$), and combination of the protease XVIII and pepsin column was studied. TEM-20 was used as the model protein for optimization. For digestion involving proteases type XVIII, protease powder was dissolved in the local HDX-MS quenching solution at a TEM-20 to protease ratio of 1:20 (wt/wt). TEM-20 was digested for 2 minutes at 0°C . UPLC-MS analysis of TEM-20 was performed with different digestion methods (see section 2.8). For the combination digestion, protease type XVIII was used to digest TEM-20, followed by pepsin column. All data analyses were performed as described in section 2.9. Peptide lists were generated by Waters ProteinLynx Global Server (PLGS). They were further refined and raw mass spectra were processed by DynamX 3.0 (Milford, Massachusetts). The number of peptides, coverage and redundancy were calculated by DynamX 3.0. The average number of amino acids per peptide was calculated and qualities of spectra were reviewed manually.

2.8. Local HDX-MS analysis

50 μ L of the quenched mixture was injected into the Waters nanoACQUITY UPLC system and the whole column chamber was maintained at 0 °C during the whole chromatographic separation. Online pepsin digestion, desalting and preconcentration were performed utilizing a Water BEH Enzymate Protein Pepsin Column (300 Å, 5 μ m, 2.1 mm $\times$ 30 mm) and Waters ACQUITY UPLC BEH C18 VanGuard Pre-column (130 Å, 1.7 μ m, 2.1 mm $\times$ 5 mm) by flowing 0.1% FA at 100 μ L/min for 3 minutes. The desalted peptides were then separated and eluted through a Waters ACQUITY UPLC BEH C18 Column (130 Å, 1.7 μ m, 1 mm $\times$ 100 mm) in 12 minutes at 60 μ L/min by the following gradient elution profile: A: 0.1% FA, B: acetonitrile with 0.1% FA; at 0 minute 5% B, at 0.5 minute 8% B, at 6.5 minute 40% B, at 8-11 minute 95% B, at 11.5-12 minute 5% B. Eluent was detected by Waters Synapt G2-Si mass spectrometer at the following conditions: capillary voltage 3.0 kV, source temperature 150 °C, desolvation temperature 350 °C. Mass spectra were acquired over an m/z range of 100 to 2000 using MS^E mode.

2.9. Data analysis

For global HDX-MS data, spectra acquired were deconvoluted by MassLynx 4.2 to obtain masses of deuterated proteins at different time points. Deuterium uptake was calculated by subtracting the mass of non-deuterated protein from the deuterated protein of the time point of interest by Microsoft Excel. Deuterium uptake figures were also produced by Microsoft Excel.

For local HDX-MS data, ProteinLynx Global Server 3.0.2 was used to generate a peptide list from the MS^E spectra. The peptide list included sequence of identified peptides, masses of precursor ions, measured m/z , charge state, retention time,

fragmentation profile and PLGS searching score of each identified peptide. This list was imported to the DynamX 3.0 for further filtering and deuterium uptake analysis. The filtering criteria were minimum intensity: 5000; minimum products per amino acid: 0.3; minimum score: 7; maximum MH⁺ error: 5 ppm. The quality of the result analyzed by the software was reviewed manually. Sequence coverage, redundancy and number of peptide generated were also calculated automatically by the software. Figures of local HDX-MS data were produced by either DynamX or Microsoft Excel.

Chapter 3. Results and discussions

3.1. Mass spectrometric characterization of the TEM-type β -lactamases

The identities of all TEM-type β -lactamases investigated in this study were confirmed by UPLC-ESI-MS measurements. The mass spectrum (Figure 3.1a and g) showed that the measured mass of TEM-1 was 29860.9, which was in agreement with the theoretical molecular mass calculated based on the TEM-1 protein sequence (i.e., 29859.8). For other TEM-ESBLs, all measured masses were consistent with their theoretical molecular masses. It should be noted that small discrepancies between the theoretical and experimental masses could be attributed to the instrumental measurement errors which could be further corrected using leucine enkephalin as an internal mass calibrant. Nevertheless, besides experimental masses, the sequence coverage of all TEM-ESBLs had achieved 100% after PLGS database search (data not shown), which provided additional evidence on the identities of these six β -lactamases. Furthermore, the peak of TEM-type β -lactamase was shown as the dominant peak in the deconvoluted mass spectrum, and only one major peak was observed in the total ion chromatogram (TIC), indicating the high purities of the expressed TEM-type β -lactamases. In addition to the purities and the relatively rough molecular weights, the SDS-PAGE result also depicted the expression levels of the six β -lactamases, which were revealed by the darkness of bands. Two of them, TEM-19 (G238S) and TEM-15 (G238S/E104K) obviously had lower expression levels than the other four, possibly due to their low thermostabilities, and therefore they were susceptible to be expressed.⁸⁸ Table 3.1 summarized the comparison of the measured and theoretical molecular masses, and the point mutation(s) of TEM-type β -lactamases. All ESI-MS and deconvoluted spectra, TIC of TEM-1 as a representative, as well as the SDS-PAGE result, were shown in Figure 3.1, 3.2 and 3.3, respectively.

Table 3.1 Amino acid at position 104, 182 and 238, theoretical and measured molecular masses of the six TEM-type β -lactamases.

β -Lactamases	Amino acid at position			Theoretical	Measured
	104	182	238	molecular mass (Da)	molecular mass (Da)
TEM-1	E	M	G	29859.8	29860.9
TEM-15	K	M	S	29888.9	29888.1
TEM-17	K	M	G	29858.8	29859.8
TEM-19	E	M	S	29889.8	29888.8
TEM-20	E	T	S	29859.7	29860.9
TEM-52	K	T	S	29858.8	29859.8

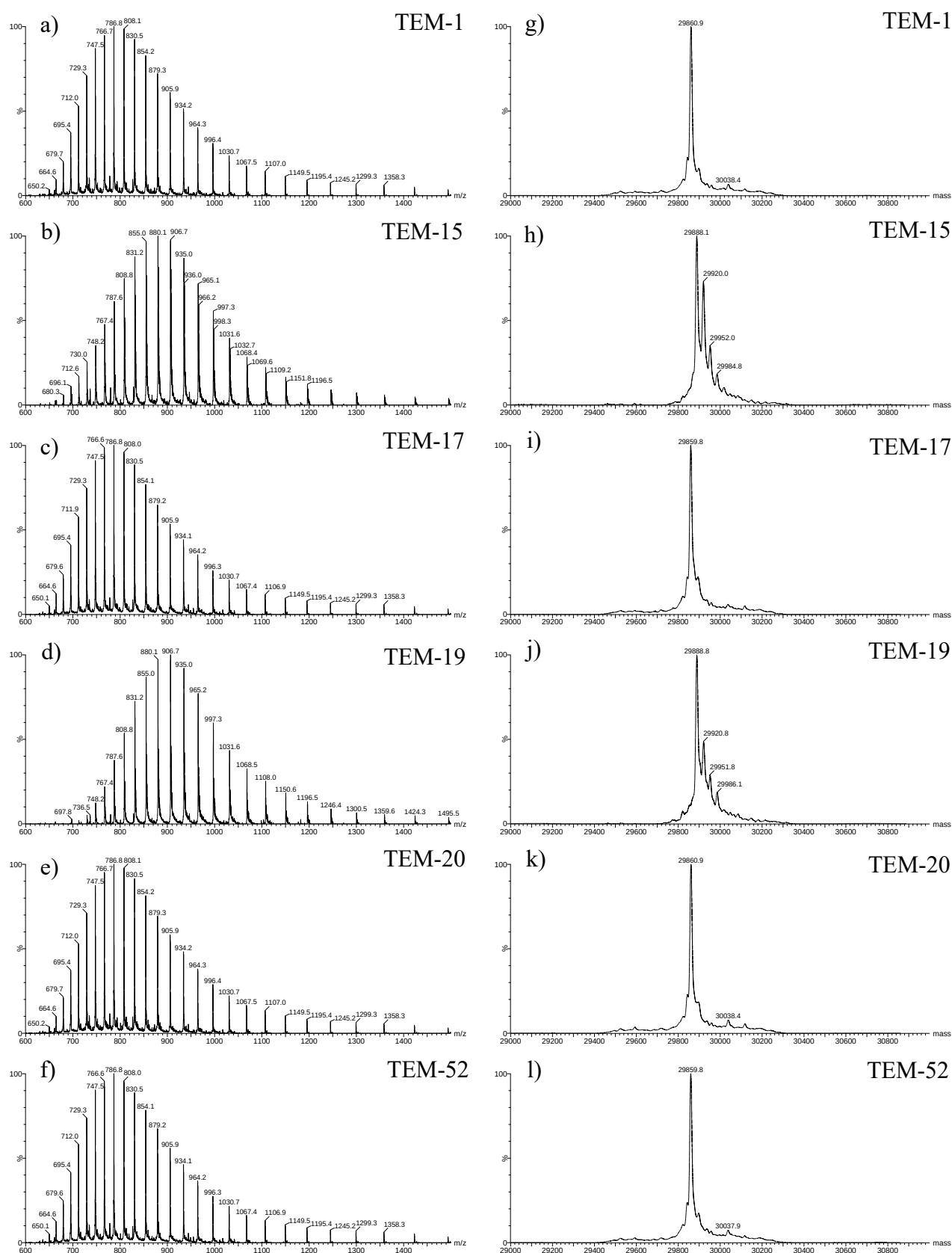


Figure 3.1 (a-g) ESI-MS spectra and (h-l) deconvoluted mass spectra of six TEM-type β -lactamases.



Figure 3.2 Total ion chromatogram of TEM-1 as a representative. The single dominant peak of TEM-1 indicated the high sample purity.

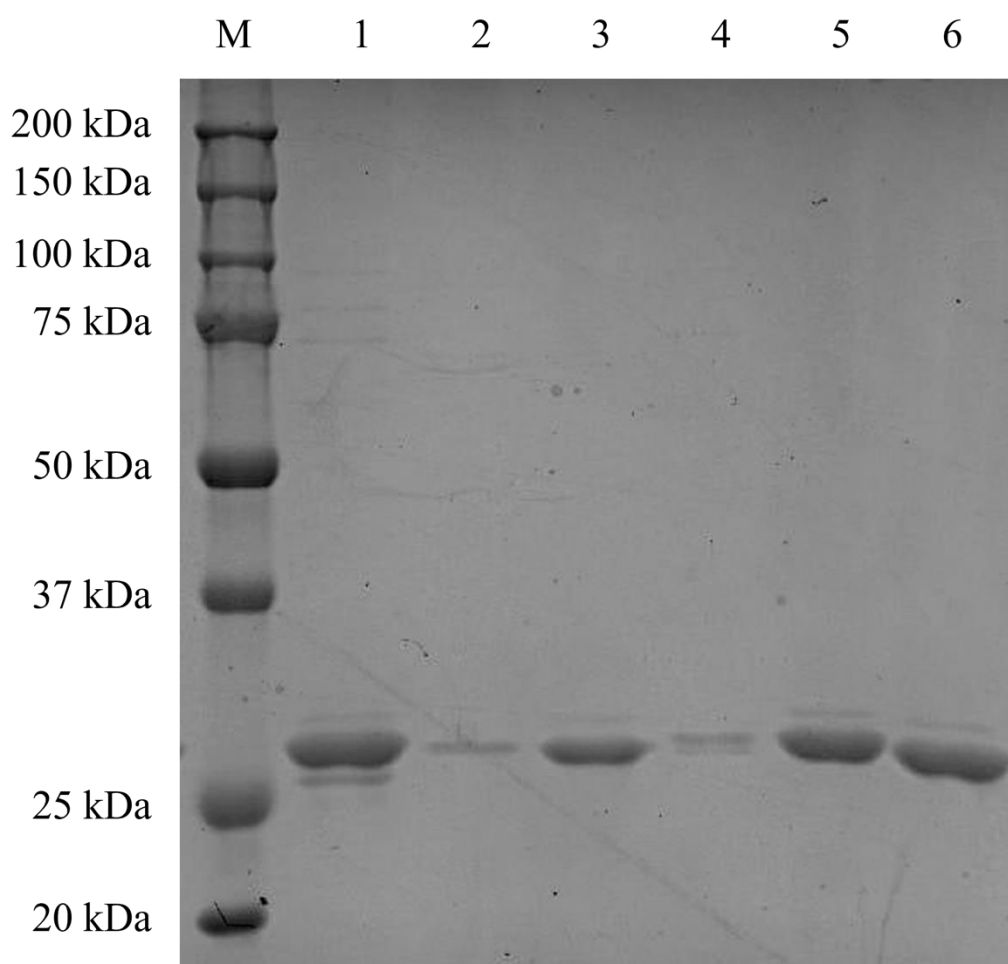


Figure 3.3 SDS-PAGE of the six TEM-type β -lactamases. Lane M: Ladder; Lane 1: TEM-1; Lane 2: TEM-15; Lane 3: TEM-17; Lane 4: TEM-19; Lane 5: TEM-20; Lane 6: TEM-52.

3.2. Global HDX-MS of apo-TEM-type β -lactamases

HDX-MS is a powerful technique to study changes in conformational dynamics upon environmental and structural changes such as pH, temperature and mutations.¹¹¹⁻¹¹⁴ In this study, a systematic study for conformational dynamics in TEM-type β -lactamases with individual or combination of point mutations at position at 104, 182 and 238 was conducted by HDX-MS. The global HDX of TEM-type β -lactamase was initiated by diluting the 1 volume part of protein with 9 volume parts of D₂O buffer. The exchange reaction was quenched at different times by reducing the pH and temperature to 2.5 and 0 °C respectively, and subjected to UPLC-MS analysis immediately. Generally, the isotopic distribution of the protein was shifted to higher m/z value upon the increase in HDX reaction time and Figure 3.4 showed the isotopic distribution shift of TEM-1 at various HDX time intervals. All TEM-type β -lactamases followed the EX2 regime as no bimodal isotopic distribution was observed, which showed the refolding process was significantly faster than the HDX process.¹¹⁵⁻¹¹⁸

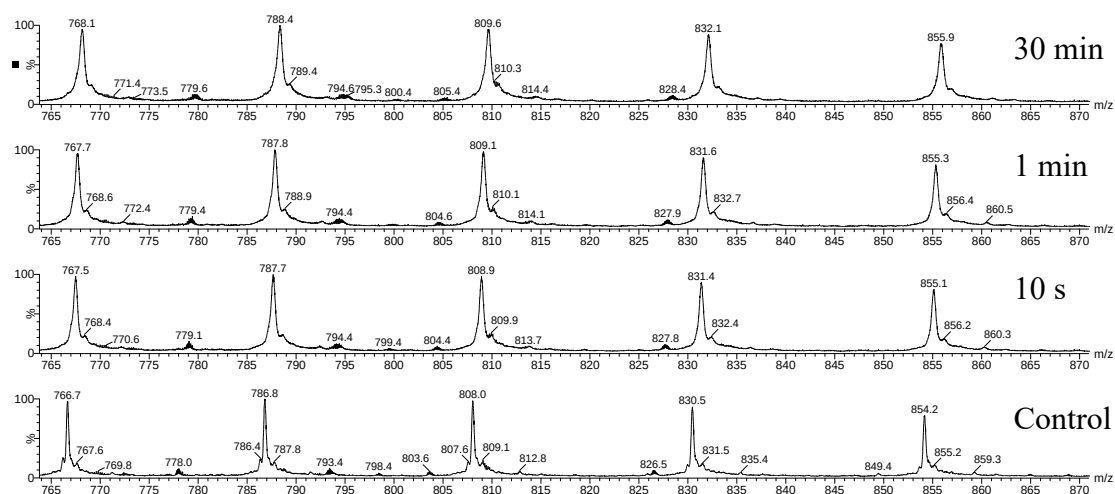


Figure 3.4 Isotopic distribution shift of TEM-1 at various HDX time intervals.

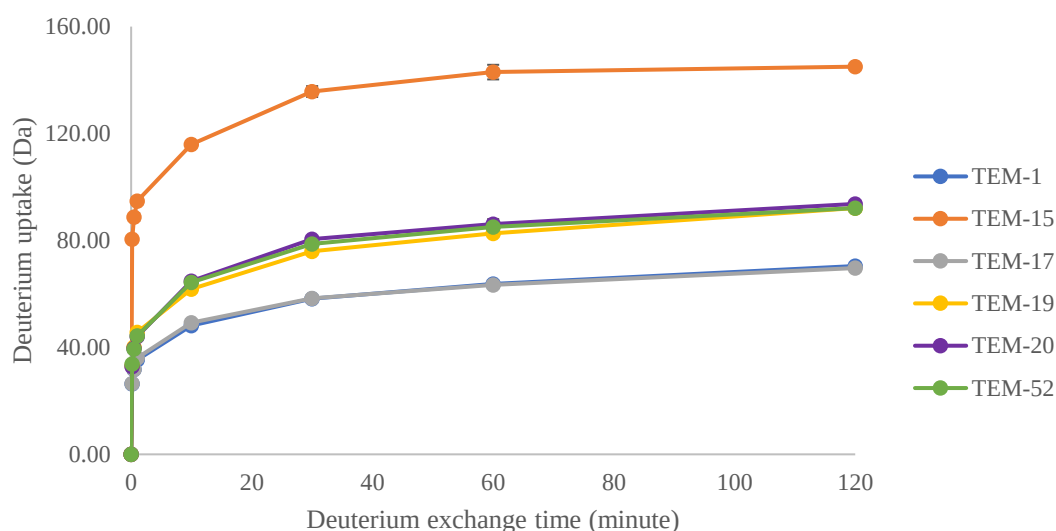


Figure 3.5 Deuterium uptake plot of the six apo-TEM-type β -lactamases.

The deuterium uptake against deuterium exchange time of six TEM-type β -lactamases was shown in Figure 3.5. Generally, mutants possessing G238S point mutation (i.e., TEM-15, TEM-19, TEM-20 and TEM-52) had their deuterium uptake significantly higher than that of TEM-1, indicating that flexibility was greatly enhanced. Surprisingly, E104K (i.e., TEM-17) had almost the same deuterium uptake with that of TEM-1. However, when E104K incorporated with G238S, the deuterium uptake obviously and significantly increased. ESBLs with G238S and/or E104K are known to be two key point mutations that confer ESBLs the enhanced hydrolytic power towards later generations of antibiotics, especially the third generation cephalosporins such as cefotaxime and ceftazidime, whose catalytic efficiency was $G238S/E104K > G238S > \text{wild type TEM-1}$.⁸⁸ The global HDX-MS result may provide insights about the correlation between the overall flexibility of β -lactamase and antibiotics hydrolytic power, and reflected that the increased flexibilities of ESBLs enhanced hydrolytic power towards third generation cephalosporins.

The third point mutation of interest, M182T, was also incorporated with G238S. The deuterium uptake of the ESBL that possessed this combination of point mutations, TEM-20 (G238S/M182T), was also investigated and compared with G238S only (TEM-19). Interestingly, both ESBLs had similar changes in deuterium uptake. However, obvious changes were observed for triple mutant TEM-52 (G238S/E104K/M182T) and TEM-15 (G238S/E104K). Incorporation of M182T into TEM-15 caused significant reduction in deuterium uptake, which was similar to those of TEM-19 and TEM-20. It was reported that M182T was not a point mutation that influenced antibiotics hydrolytic efficiency, i.e., secondary mutation, but a global suppressor that stabilized ESBLs.¹¹⁹ The global HDX-MS results were consistent with the proposed scenario that the M182T mutation could play a role in compensating the unfavorable structural and dynamic changes caused by the synergistic G238S and E104K mutations.^{56,58,88} However, since global HDX-MS results only revealed the overall flexibility of ESBLs, further investigation on the conformational changes of different localized regions in ESBLs was also necessary, particularly for TEM-19, TEM-20 and TEM-52 having their overall deuterium uptake overlapped, as these three mutants have demonstrated different hydrolytic power towards antibiotics.⁸⁸

3.3. Local HDX-MS analysis of free TEM-type β -lactamases

In order to acquire information of the regional conformational changes of ESBLs, local HDX-MS was performed. The procedures were similar to those of global HDX-MS, but with an additional enzymatic digestion step after quenching the HDX reaction and before UPLC-MS analysis.

3.3.1. Choice of digestion enzyme for local HDX-MS

For conventional proteomics study, trypsin is the most commonly used enzyme because of its specificity in the digestion position and optimal peptide length.¹²⁰⁻¹²² However, quenching of HDX reaction requires the pH to be reduced to 2.5, which causes the inactivation and denaturation of trypsin. This limits the choice of digestion enzyme, which must be acid-resistible. Pepsin is an acid-resistible enzyme commonly found in the stomach of many animals including humans. It is activated at pH below 3.0 but is irreversibly denatured at pH above 5-6, making it compatible for quenching HDX reaction and it is commonly used in HDX-MS studies. However, its non-specific digestion efficiency makes the prediction of resultant digestion peptides nearly impossible, and it may affect the spatial resolution of the analysis. For example, SDN loop is an important region for TEM-type β -lactamases, but it is only consisted of three amino acids only. If the length of digested peptide including this region is too long, the deuterium uptake of this region may not be able to precisely reflect the conformational dynamics of this important region. The use of alternative enzyme, or even a combination of enzymes, could mitigate this problem, as this will provide another digestion pattern that may be complementary to pepsin.¹²³⁻¹²⁵ Protease type XVIII (rhizopuspepsin) is another acid-resistible non-specific enzyme that is suitable for quenching HDX reaction and it can be utilized to compare with the digestion pattern of pepsin.^{123,125} Before performing local HDX-MS study, the performance of three

digestion methods, pepsin column, protease type XVIII and a combination of the former two enzymes, were studied. The digestion methods were compared according to various parameters, including the number of digested peptides generated, average number of amino acids for a peptide, reproducibility of spectra, sequence coverage and redundancy, and the results were summarized in Table 3.2.

The number of peptides listed in Table 3.2 was obtained by searching the PLGS and importing the result into DynamX for further data refining. All of the searched peptides were manually reviewed according to their spectra quality. Only peptides with high spectra quality, i.e, its isotopic pattern did not overlap with other peptides and with desired intensity, were counted as valid peptides. Among the three digestion methods, pepsin column provided the highest number of valid peptides. Normally, dual enzyme treatment (i.e., protease type XVIII with pepsin column) should be able to result more peptides. However, the cross digestion between pepsin and protease type XVIII possibly introduced interferences for LC-MS analysis. Even though more peptides were resulted, these interferences may have their isotopic patterns overlapping with those of peptides generated from β -lactamases and hence they would be rejected after manual review of data. Therefore, the final number of peptide generated by the dual enzyme treatment became less than using pepsin column only.

Theoretically, the more the peptides generated the more beneficial for the analysis as more information could be acquired. However, as aforementioned, peptide that consists of too many amino acids may not perfectly reflect the real conformational dynamics of certain important regions, and limit the spatial resolution of the analysis. As a result, the average number of amino acids for a peptide was also considered. This number was calculated by averaging the length of all valid digested peptides. Length of digested

peptides that lies between 5 and 20 is the most optimum value for HDX-MS analysis.¹²⁶ The combination of pepsin column and protease type XVIII provided the shortest average peptide length, and all three digestion methods provided suitable average peptide length for further analysis.

Reproducibility of spectra directly affects the accuracy of HDX-MS data analysis. Even though there are various softwares for data analysis, it is still worthwhile and necessary to spend time to check all spectra manually to confirm the reliability of raw data.¹²⁷ In general, pepsin column provided the highest reproducibility, while protease type XVIII provided satisfactory spectra reproducibility. However, reproducibility of the combination of pepsin column and protease type XVIII was quite poor. This might be due to the pepsin column also digested the solution form protease type XVIII during digestion of TEM-type β -lactamase and that might lead to very complex chromatograms and mass spectra with overlapping signals. In fact, the autolysis of protease XVIII might also generate unwanted peptides that might interfere with the data analysis, but spectral data might be further complicated if the protease XVIII was digested in the pepsin column.

Sequence coverage is defined as the percentage of amino acid covered by the digested peptides and redundancy is the number of detected peptides that cover a particular residue averaging over the whole amino acid sequence.¹²⁸ All three digestion methods showed satisfactory coverage and redundancy. Taking all parameters into account, pepsin column was chosen as the digestion method for this study, especially due to its excellent reproducibility and highest number of valid peptides resulted from digestion.

Table 3.2 Comparison of three digestion methods, pepsin column digestion, protease type XVIII digestion and combination of protease type XVIII and pepsin column.

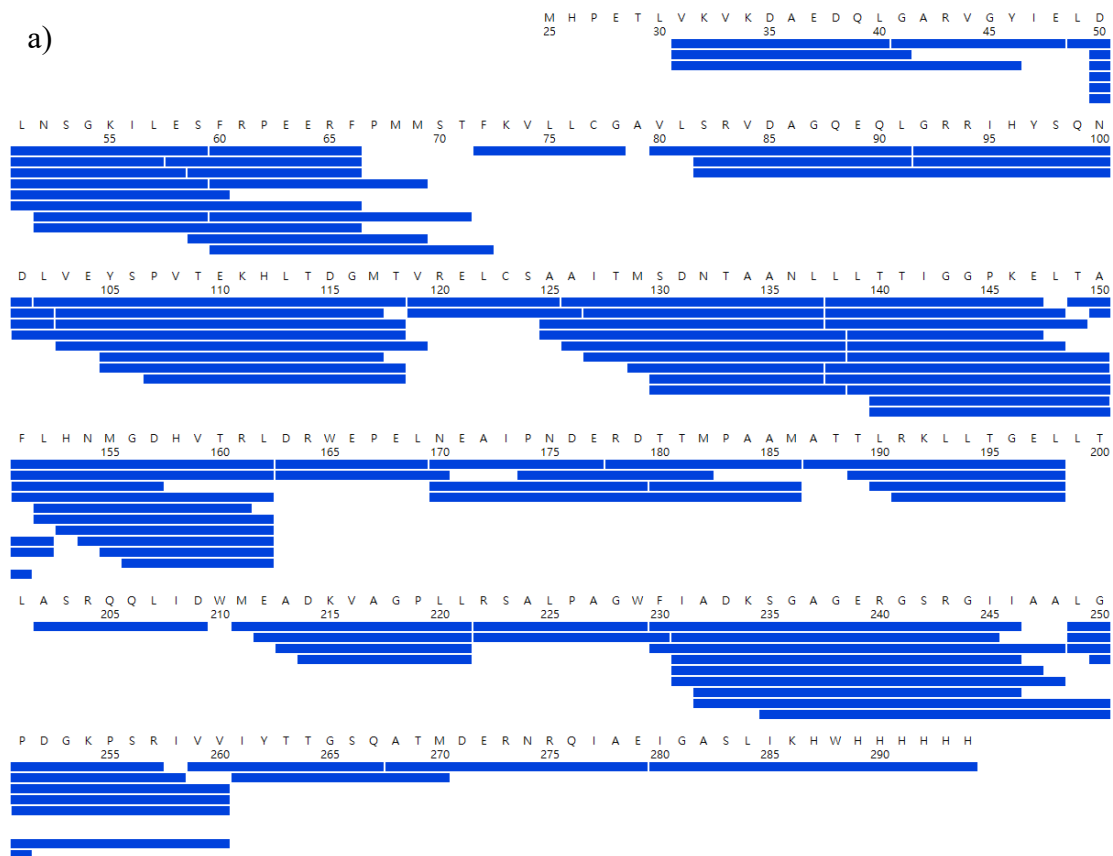
	Pepsin Column + XVIII	Pepsin Column	XVIII
Number of peptides	114	127	107
Average number of amino acids for a peptide	10.71	11.87	10.91
Reproducibility of spectra	Not satisfactory	Very good	Satisfactory
Sequence coverage	93.3%	98.1%	99.3%
Redundancy	4.85	5.69	4.35

3.3.2. Study of digestion patterns of apo-TEM-type β -lactamases

Before performing the local HDX-MS experiments, the digestion patterns, sequence coverage and redundancies of the six apo-TEM-type β -lactamases are essential to be studied, as adequate regional conformational information could only be obtained when high number of common peptides are found from the digests.

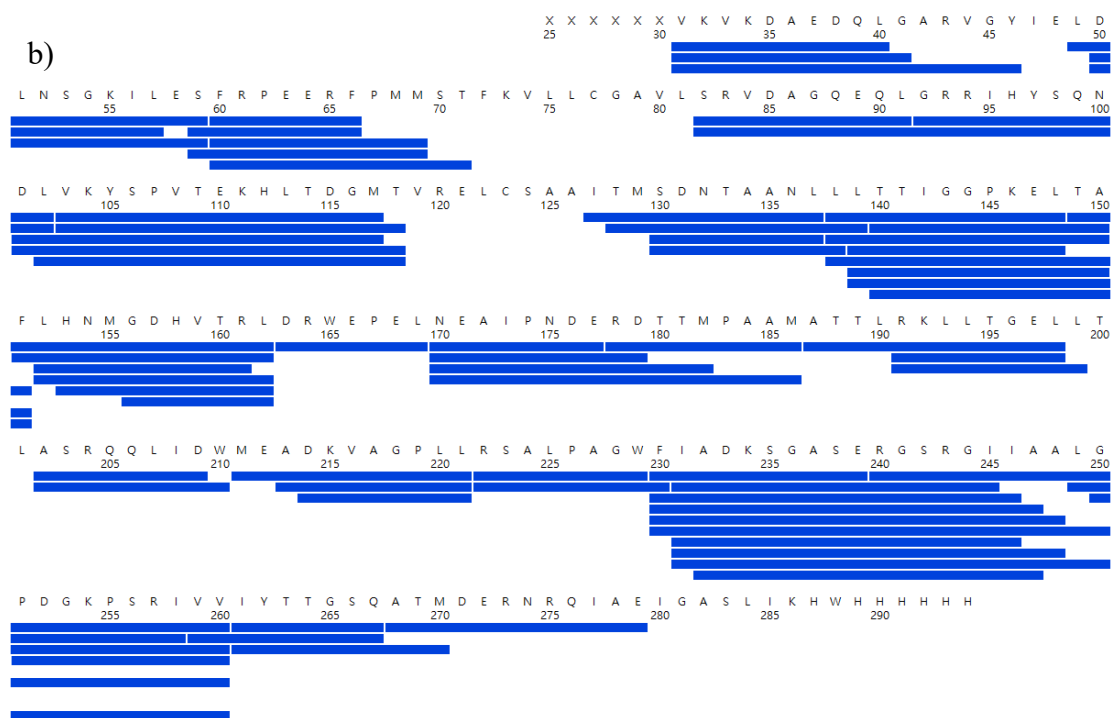
The coverage maps of all six apo-TEM-type β -lactamases were shown in Figure 3.6. The number of peptides (at least 56 peptides were found for each β -lactamases) and sequence coverage of all six TEM-type β -lactamases (ranging from 82.6% to 95.9%) were satisfactory. Generally, peptides that were commonly found in all six TEM-type β -lactamases would be used for comparison. In fact, the comparison between β -lactamases that only have one different point mutation could also provide insights into the effect of conformational changes after a particular mutation. Besides, since pepsin usually generates various peptides for the same region, and they have either the same length but with one or two residues shifted in sequence (e.g., residues 129-137 and 130-138), or one or two more consecutive residues (e.g., residues 139-150, 140-150 and 140-151); they could also provide insight into the conformational dynamics of a particular region, as their intrinsic HDX rates are assumed to be highly similar. This would be regarded as indirect comparison in this study. Usually, a peptide that is covered by more than three β -lactamases is useful for comparison, if necessary. Overall, the coverage of individual β -lactamases exceeded 80%. Overlapping peptides for all six apo-TEM-type β -lactamases also covered nearly 60% of the sequence and most of the important regions, therefore they were satisfactory for local HDX-MS analysis (Table 3.3).

a)

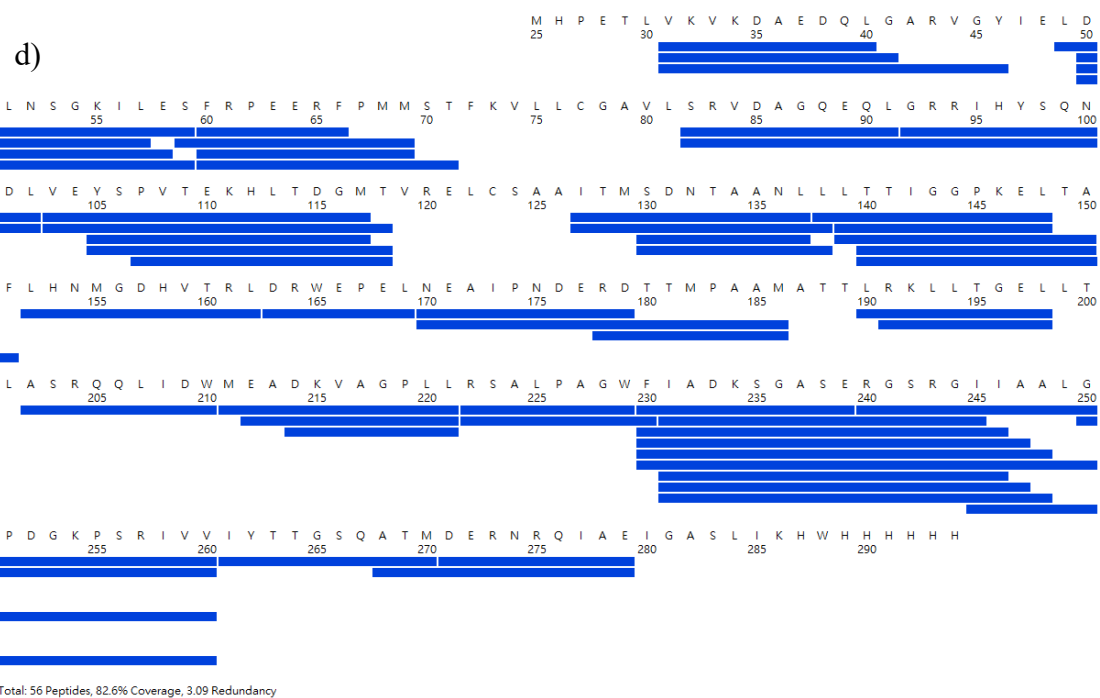


Total: 102 Peptides, 95.9% Coverage, 4.56 Redundancy

b)



Total: 71 Peptides, 84.1% Coverage, 3.85 Redundancy



e)



Total: 123 Peptides, 95.6% Coverage, 5.88 Redundancy



Figure 3.6 Coverage map of apo- (a) TEM-1, (b) TEM-15, (c) TEM-17, (d) TEM-19, (e) TEM-20 and (f) TEM-52.

Table 3.3 Number of peptides generated, sequence coverage, redundancies and number of common peptides resulted from pepsin digestion of the six apo-TEM-type β -lactamases.

β -Lactamase	Number of peptides	Sequence coverage	Redundancy
TEM-1	102	95.9%	4.56
TEM-15	71	84.1%	3.85
TEM-17	103	92.2%	4.89
TEM-19	56	82.6%	3.09
TEM-20	123	95.6%	5.88
TEM-52	93	90.7%	4.44
Common peptides	29	57.8%	2.23

3.3.3. Local HDX-MS results of apo-TEM-type β -lactamases

After data acquisition and data analysis, data were presented in either butterfly plots or difference plots in order to quickly and efficiently determine peptides that had significant deuterium uptake difference. Butterfly plots were constructed by plotting the deuterium uptake for different peptides at a particular HDX time, and each trace represented one β -lactamase. In this project, deuterium uptake difference more than 0.17 Da were defined as significant difference. This value was 3 times of the average standard deviation of deuterium uptake for all data points. Important regions could easily be recognized by observing if there were any meaningful trends there. However, this would require a large number of overlapping peptides and sequence coverage for all six β -lactamases in order to obtain comprehensive local conformation dynamics information.

The difference plot was constructed by plotting the uptake differences of overlapping peptides between two β -lactamases. Figure 3.7 depicts comparisons made for β -lactamases that had only one point mutation difference in this study. Each trace in the difference plot represented a particular HDX reaction time. Comparisons of apo-TEM-type- β -lactamases in the form of difference plot were shown in Figure 3.8. These plots might provide important insights into regions that were not directly involved in the hydrolysis, but indirectly affect the stability or catalytic efficiency of mutants. Instead of the butterfly plot discussed above, important regions that were covered by even one comparison only could still be noticed and investigated, so as to minimize loss of information. Two types of regions in β -lactamases were intentionally investigated, one was the known catalytically important regions, such as Ω -loop and SDN loop, while another one was the regions that possessing point mutations of concern in this study. Several allosteric regions, which might not be directly involved in the antibiotics

catalysis but might indirectly affect the conformation of those important regions were also investigated, and they could be spotted in difference plots.

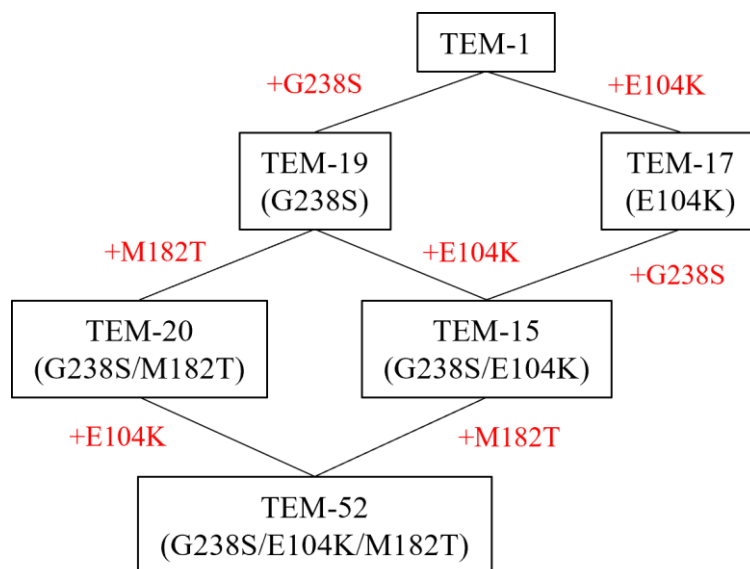
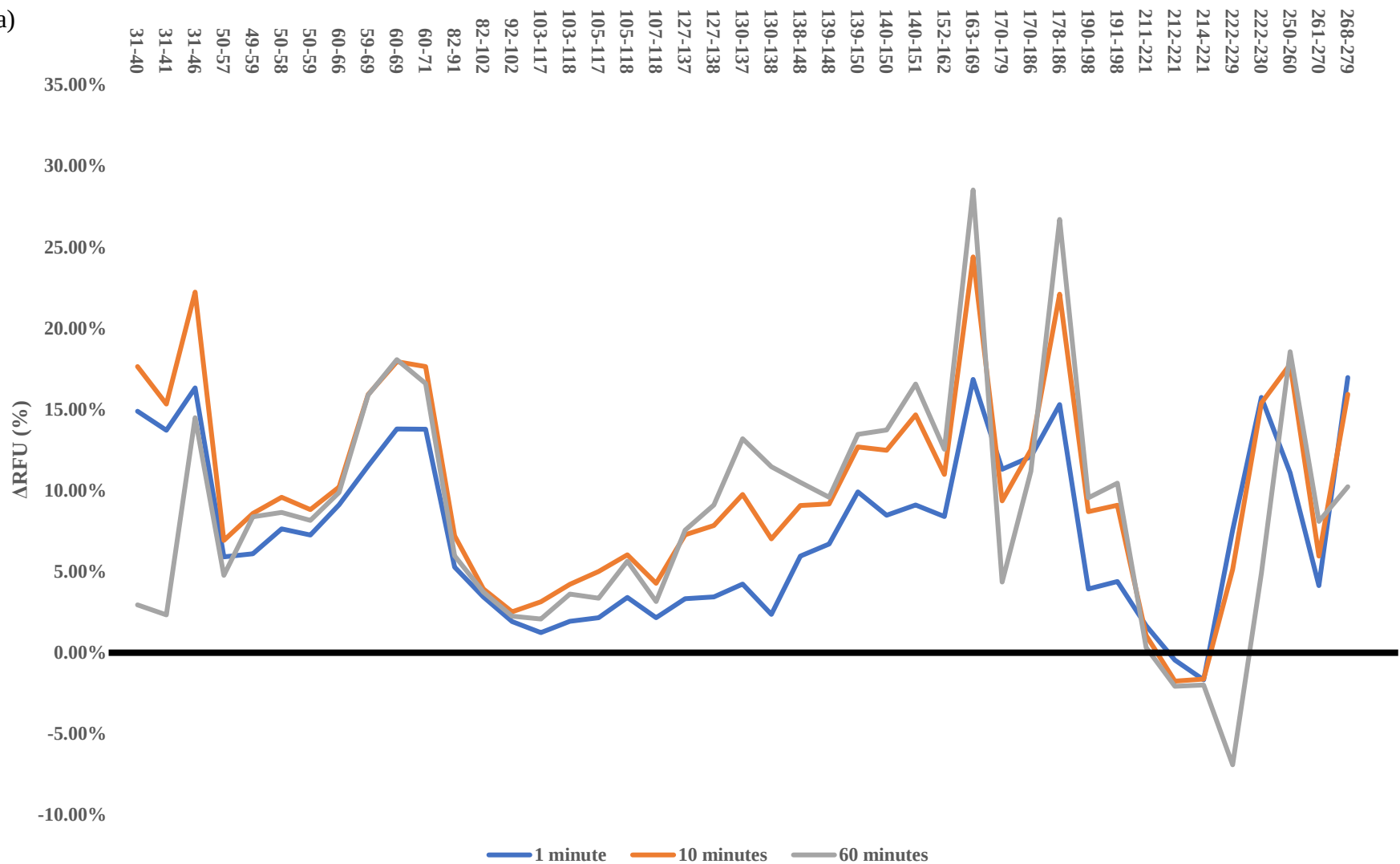
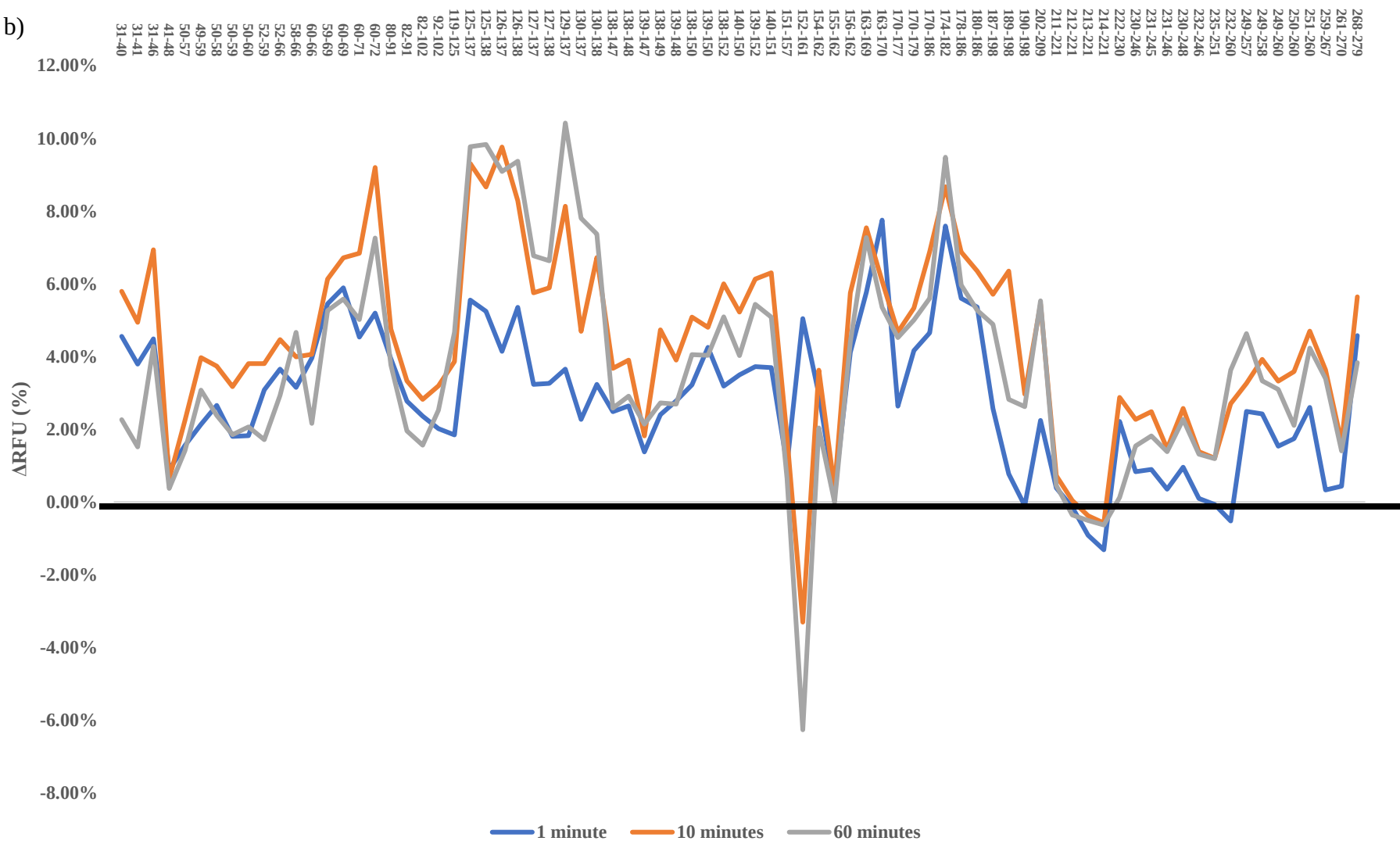


Figure 3.7 Comparisons of TEM-type β -lactamases that had one different point mutation.

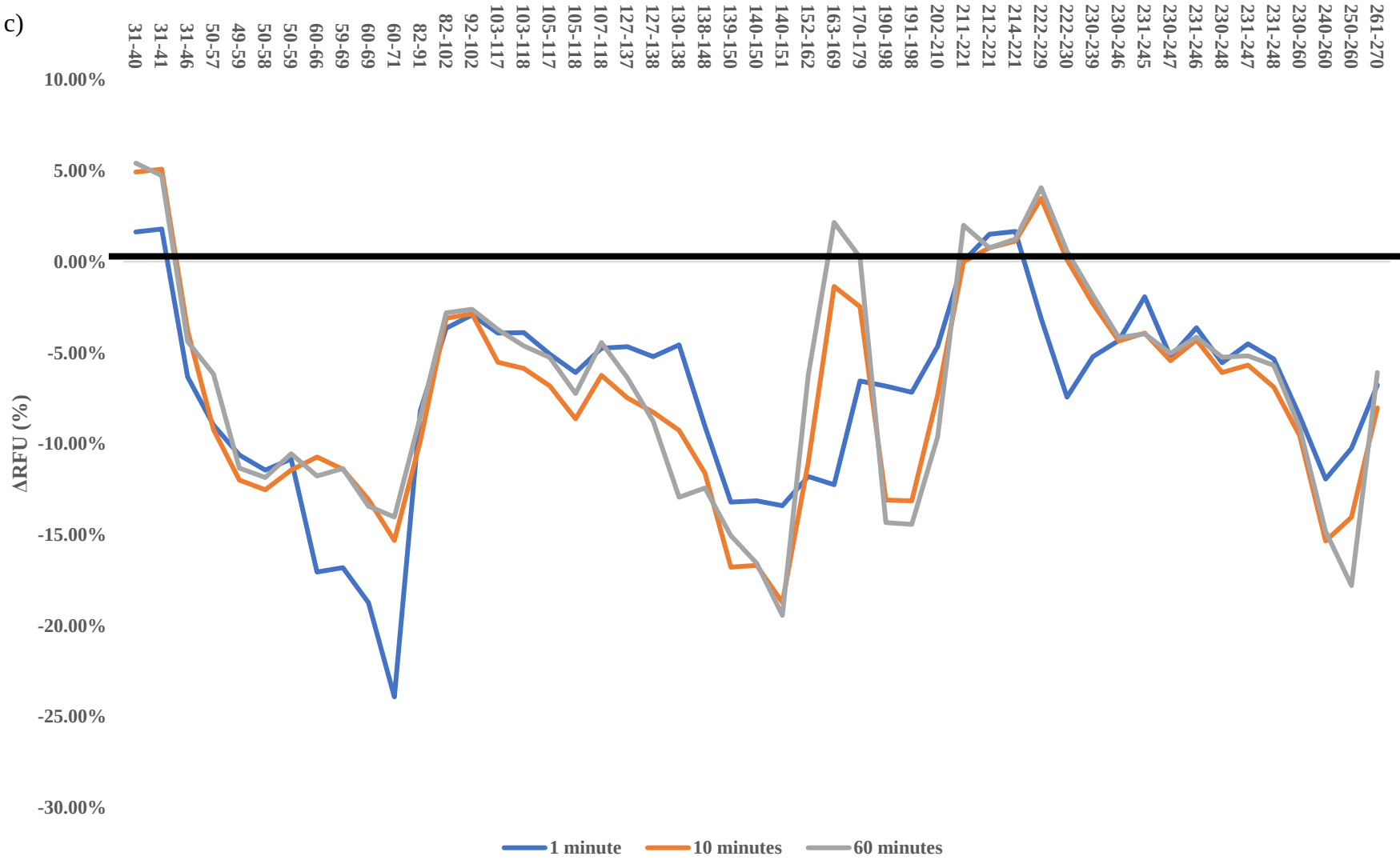
a)



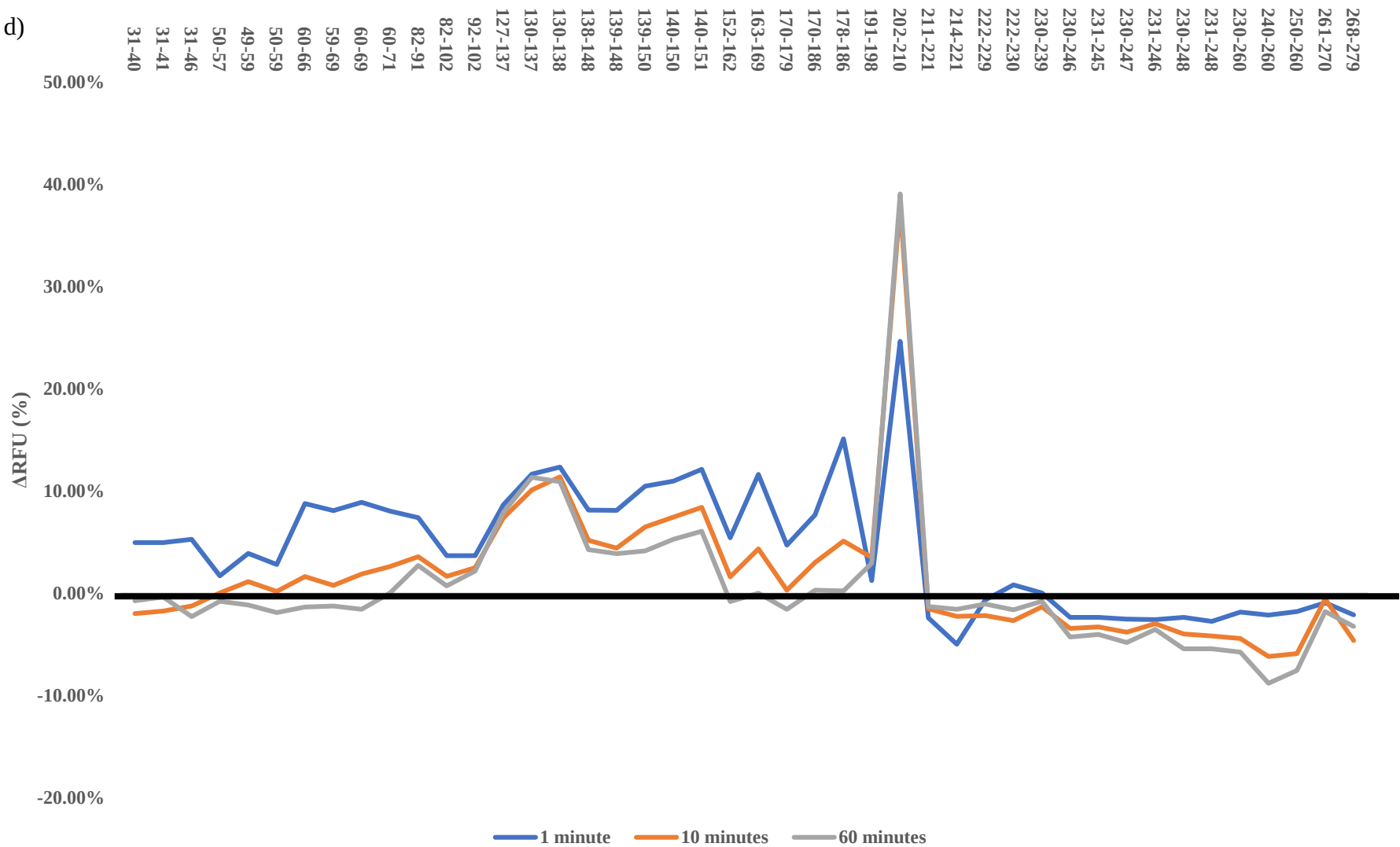
b)

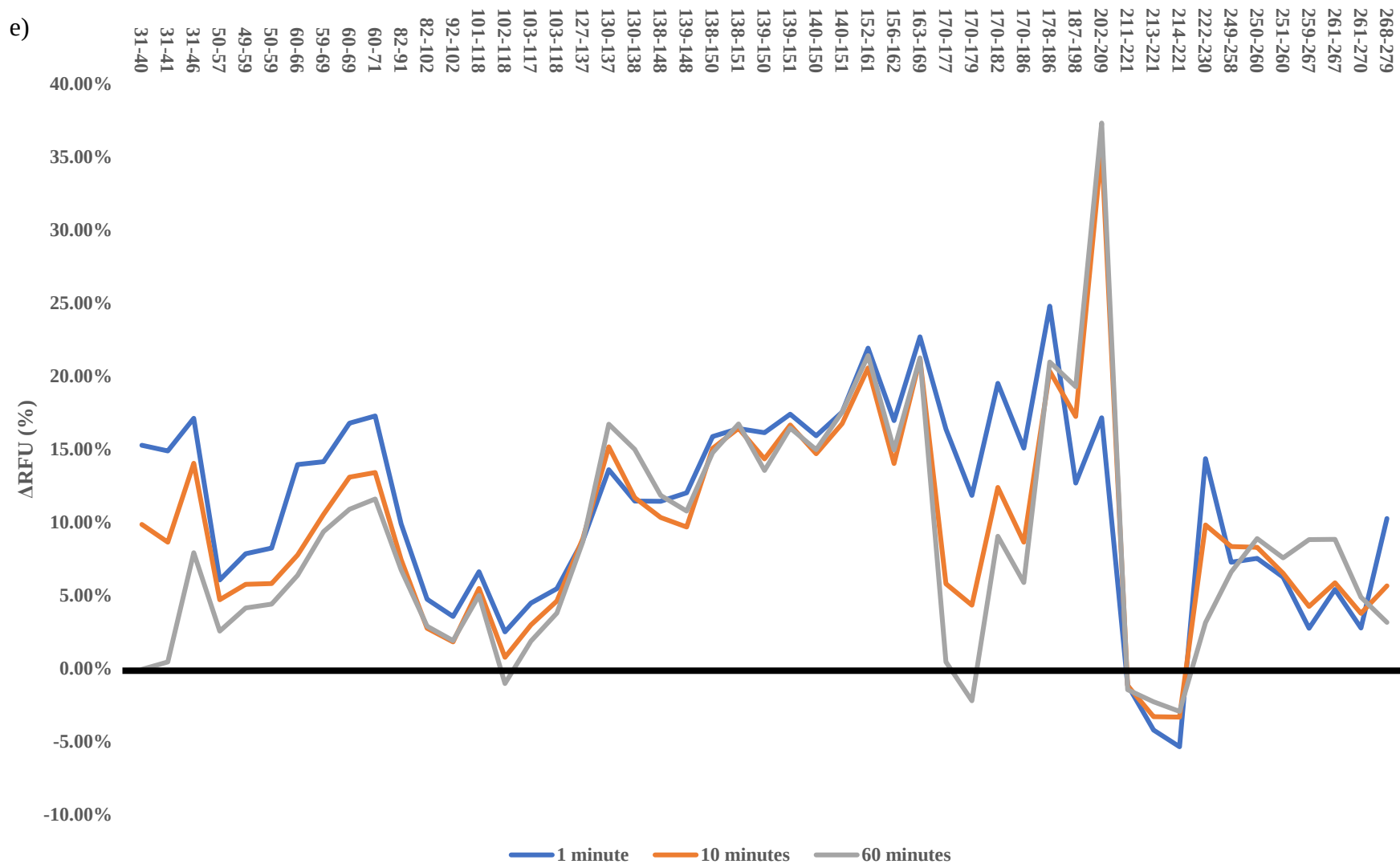


c)

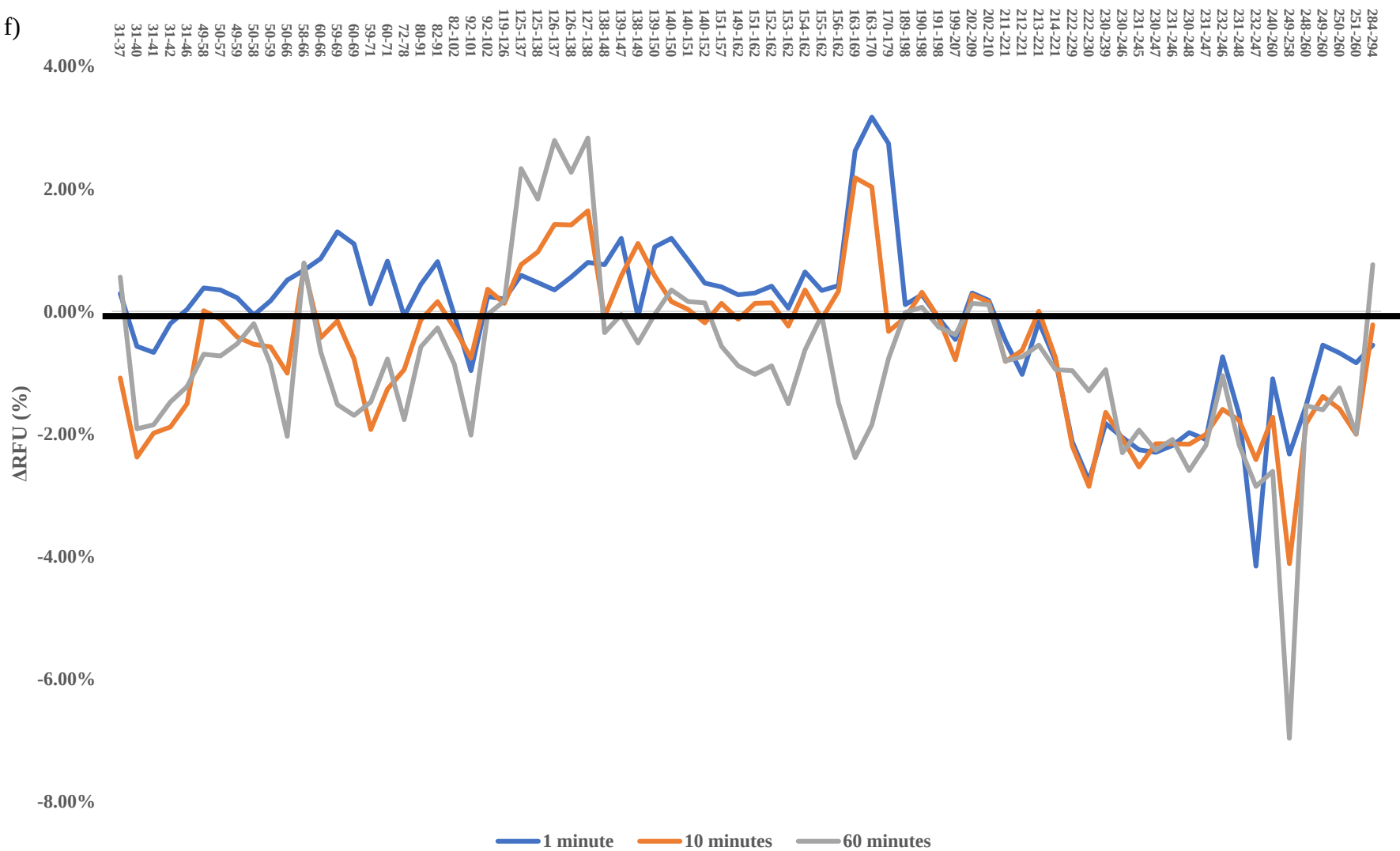


d)





f)



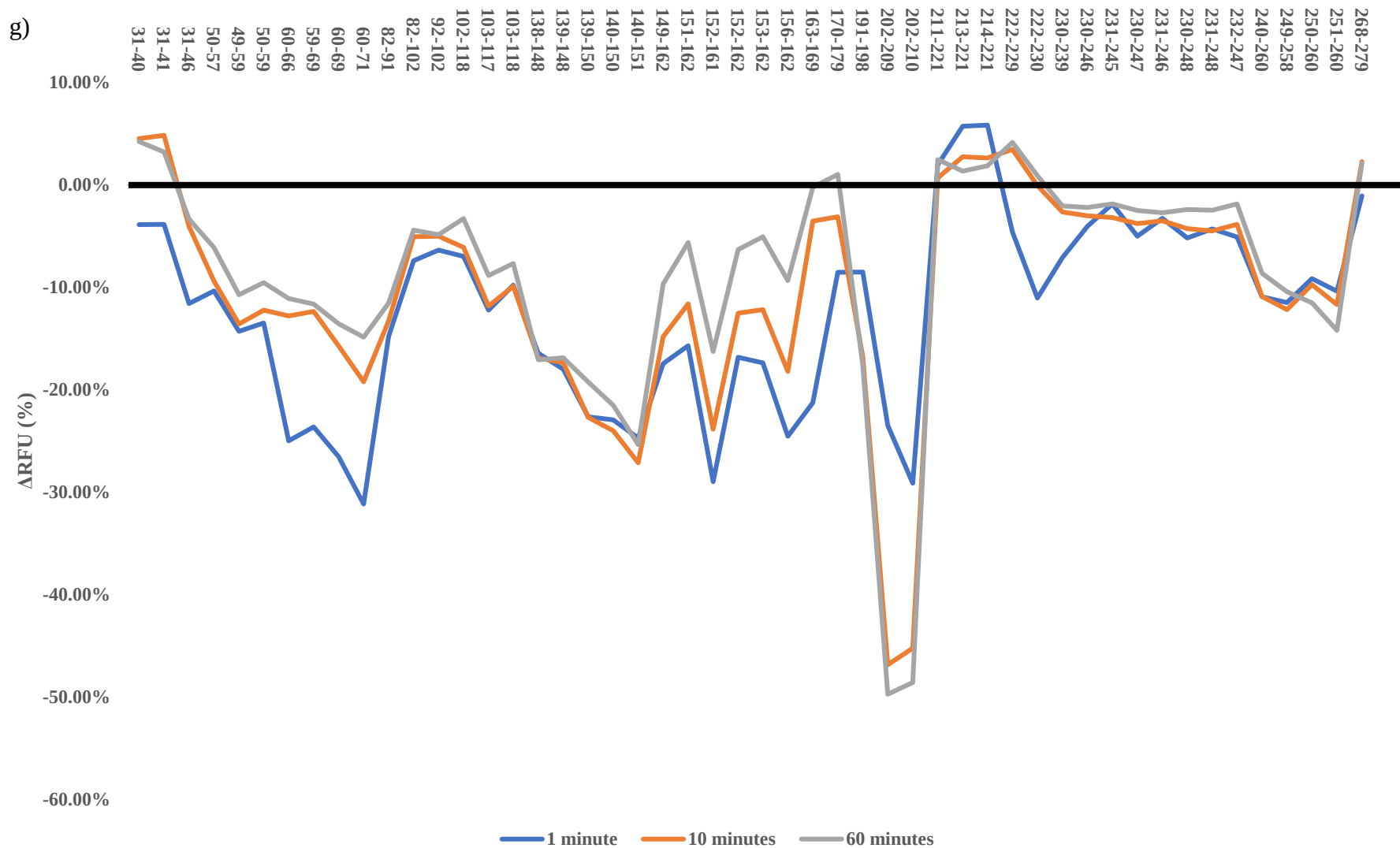


Figure 3.8 Different plot of comparison between (a) TEM-1 and TEM-19, (b) TEM-1 and TEM-17, (c) TEM-19 and TEM-20, (d) TEM-19 and TEM-15, (e) TEM-17 and TEM-15, (f) TEM-20 and TEM-52 and (g) TEM-15 and TEM-52.

Regions near active serine-70 (Residues 59-69, β 2/H2-loop)

Serine-70 (S70) is the most important residue particularly in class A, C and D β -lactamases. Its side chain hydroxyl group attacks the carbonyl group of the β -lactam ring, causing the ring opening and hence the deactivation of antibiotics. As a result, the conformational dynamics of S70-associated and/or nearby regions could significantly influence the catalytic properties. The local HDX-MS result for β 2/H2-loop (the region near the active site S70) showed that all mutants showed higher deuterium uptake than TEM-1. TEM-19 (G238S) showed greater flexibility enhancement in this region than TEM-17 (E104K). Interestingly, introduction of M182T to G238S (i.e., TEM-20) caused further flexibility enhancement, while this region became more compact when M182T was added to G238S/E104K (i.e., TEM-52) (Figure 3.9).

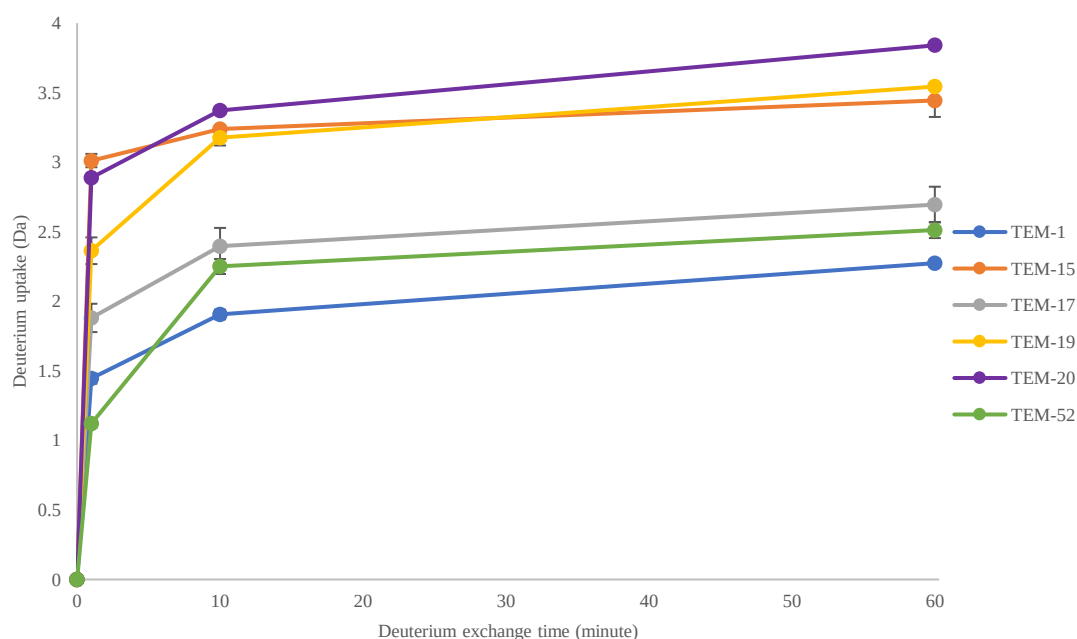


Figure 3.9 Deuterium uptake plot of residues 59-69 for apo-TEM-type β -lactamases.

Ω-loop (Residues 163-169 and 170-179)

Ω-loop is located at the entrance of active site and is involved in all individual steps of the hydrolysis of β-lactam antibiotics. The flexibility and orientation of it can significantly affect the properties of β-lactamases, including substrate specificity, binding affinity and catalytic efficiency.⁶⁸ The results revealed that this region became more flexible after mutation. G238S had greater effect to the flexibility than E104K. For TEM-19 (G238S) and TEM-15 (G238S/E104K), even though they had similar deuterium uptake after 60 minutes of HDX, their initial HDX rates were different. The synergistic effect G238S/E104K caused significantly faster initial exchange rate when compared with G238S and E104 alone. M182T stabilized this region by reducing the initial exchange rate. This may indicate that the flexibility of Ω-loop could be an important factor influencing the catalytic efficiency of ESBLs. Moreover, the N-terminal part of Ω-loop (residues 163-169, Figure 3.10a) was found to exhibit greater changes in deuterium uptake compared to the C-terminal part (residues 170-179, Figure 3.10b), revealing that the conformational dynamics of N-terminal part of this region might be more critical to the catalytical properties.

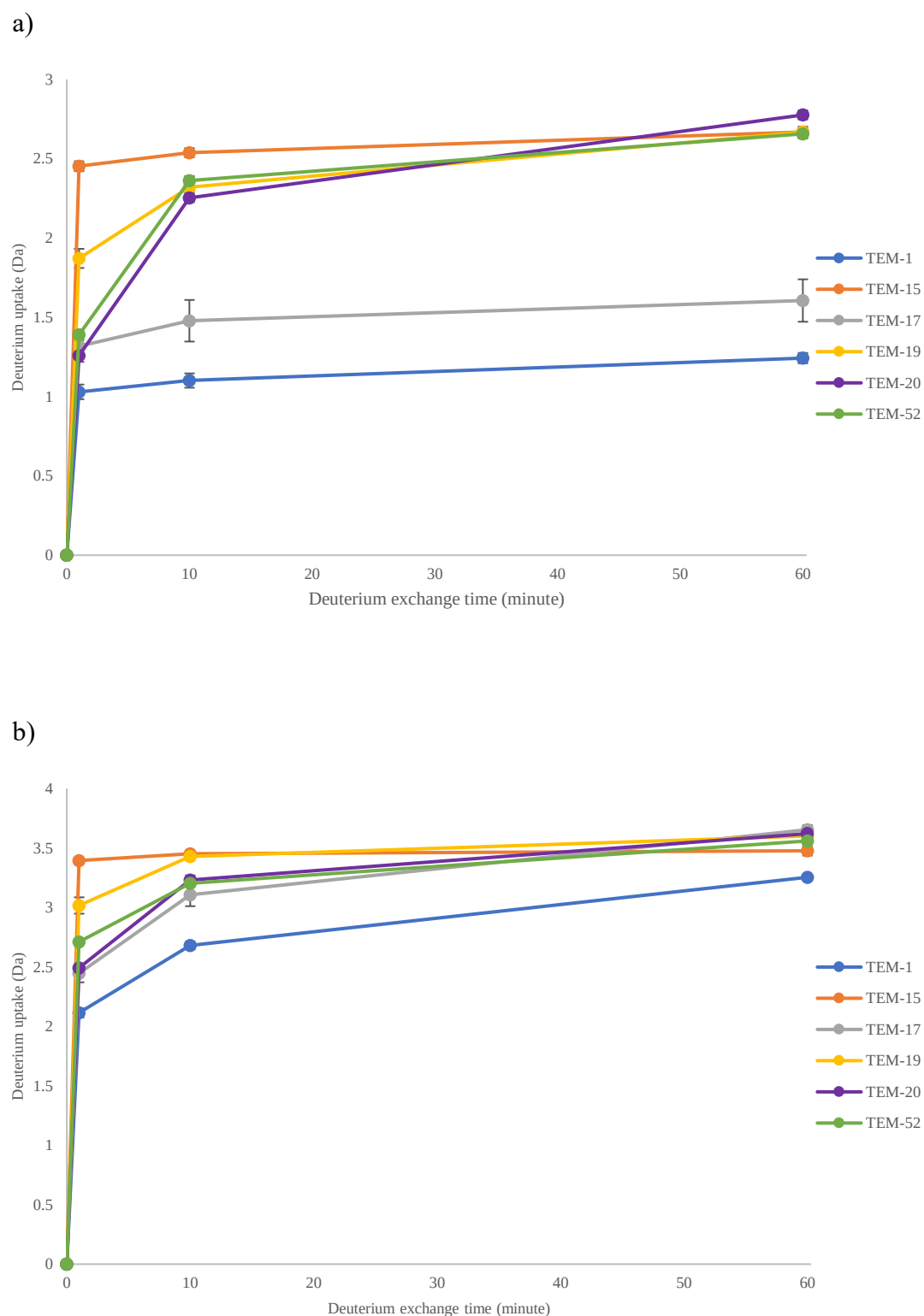
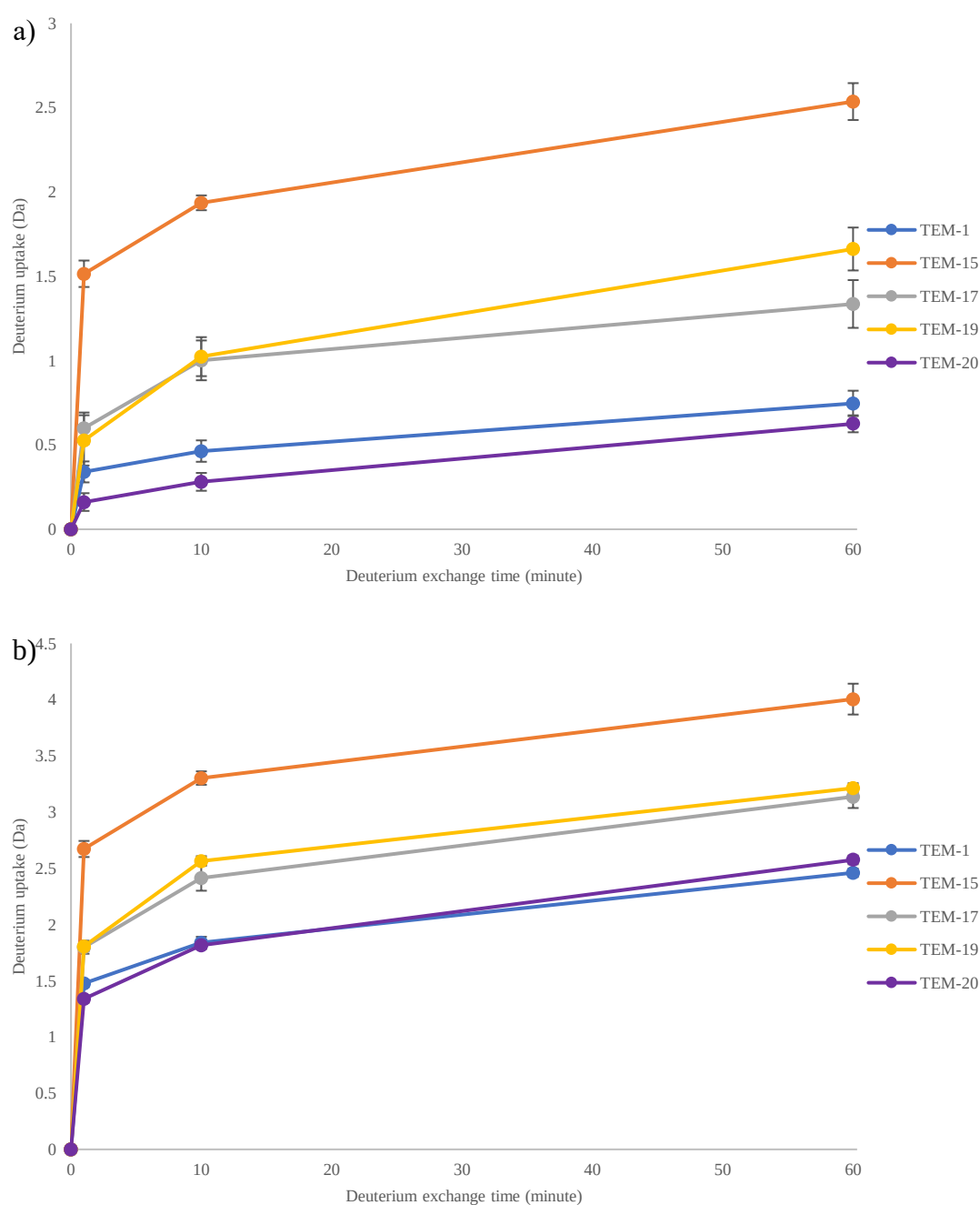


Figure 3.10 Deuterium uptake plot of residues (a) 163-169 and (b) 170-179 for apo-TEM-type β -lactamases.

SDN loop (Residues 130-138, 127-137 and 127-138)

This loop consists of three residues, S130, D131 and N132. This active site consisting region has been proven to be highly conserved. Mutations on this loop cause significant detrimental effects on the antibiotics hydrolyzing activities.¹²⁹ Similar to the previously discussed regions, as SDN loop is directly involved in the hydrolytic reaction, its proper orientation and flexibility are essential. Flexibility of SDN loop could be revealed by the local HDX-MS data, however, through an indirect uptake comparison among various digested peptides containing SDN loop, as discussed in section 3.3.2. This was due to the different intrinsic pepsin digestion patterns of different β -lactamases, even though they had only one different amino acid in their sequences. For residues 130-138, it was found in five β -lactamases except the triple mutant TEM-52 (G238S/E104K/M182T). Its deuterium uptake plot (Figure 3.11a) showed that TEM-19 (G238S) and TEM-17 (E104K) had higher flexibility than the wild type TEM-1. TEM-15 (G238S/E104K) showed a significantly higher flexibility than those of the two single mutants, indicating that the increased flexibility of this region might cause the expansion of the active pocket, which facilitated the binding between mutants and antibiotics. Also, SDN loop of TEM-19 and TEM-17 had similar flexibility, while flexibility of this part was enhanced in TEM-15, indicating that the synergistic effect of G238S and E104K might be demonstrated in SDN loop. However, after incorporating the M182T to the G238S single mutant (TEM-20), this region became more compact than TEM-1. Residues 127-137 also covered the SDN region but was also absent in the digestion pattern of TEM-52. The deuterium uptake of this peptide showed a similar flexibility trend to residues 130-138 (Figure 3.11b). Since residues 127-138 only had one extra amino acid at the C-terminal, although it was absent in the digestion pattern of G238/E104K, it was suitable to be indirectly compared with residues 127-137. The deuterium uptake plot of residues 127-138 revealed that both TEM-19 and TEM-17 had higher uptake than the TEM-1 (Figure 3.11c). TEM-20 (G238S/M182T) had similar

deuterium uptake to the TEM-1, while the TEM-52 (G238S/E104K/M182T) had a higher uptake than the TEM-1 but lower than both single mutants. Combining the uptake of the three peptides, SDN loop in TEM-15 was the most flexible one, followed by TEM-19 and then TEM-17. SDN loop in TEM-52 was more flexible than those in the TEM-1 and TEM-20, in which they had similar flexibilities of SDN loop, indicating that M182T controlled the flexibility of SDN loop after incorporating mutations and aided the effective hydrolysis of antibiotics.



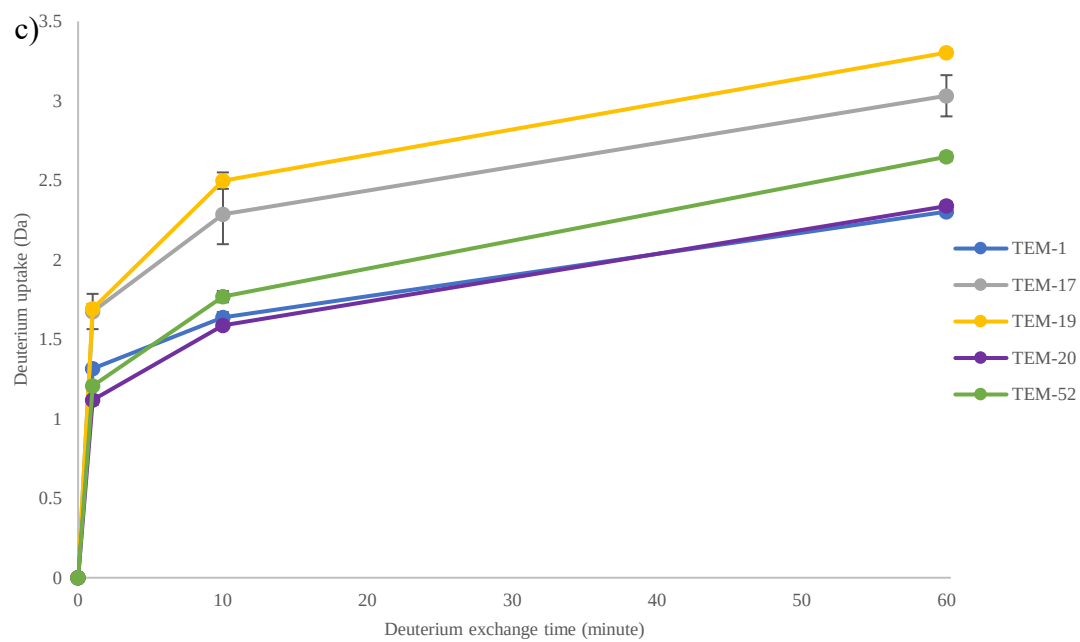


Figure 3.11 Deuterium uptake plot of residues (a) 130-138, (b) 127-137 and (c) 127-138 for apo-TEM-type β -lactamases.

β -3 strand (Residues 230-246)

The β -3 strand covering residue 238 and G238S is one of the key mutations that greatly affects the β -lactam antibiotics hydrolyzing ability of TEM-type β -lactamases. It is also one of the regions that constructs the active site, and the loop connecting β -3 and β -4 strand is located at the entrance of the binding pocket. Flexibility enhancement in this region is predicted to facilitate the entry and binding of antibiotics with bulky side chain. The results revealed that all mutants possessed a more flexible β -3 strand, while TEM-19 (G238S) had the most flexible one. TEM-17 (E104K) was slightly more flexible than the wild type TEM-1. After incorporation of M182T to TEM-19 and TEM-15, flexibility of β -3 strand in both TEM-20 (G238S/M182T) and TEM-52 (G238S/E104K/M182T) were reduced. The flexibility of TEM-20 was higher than that of TEM-52, which was consistent with the flexibility comparison between TEM-19 and TEM-15 (Figure 3.12).

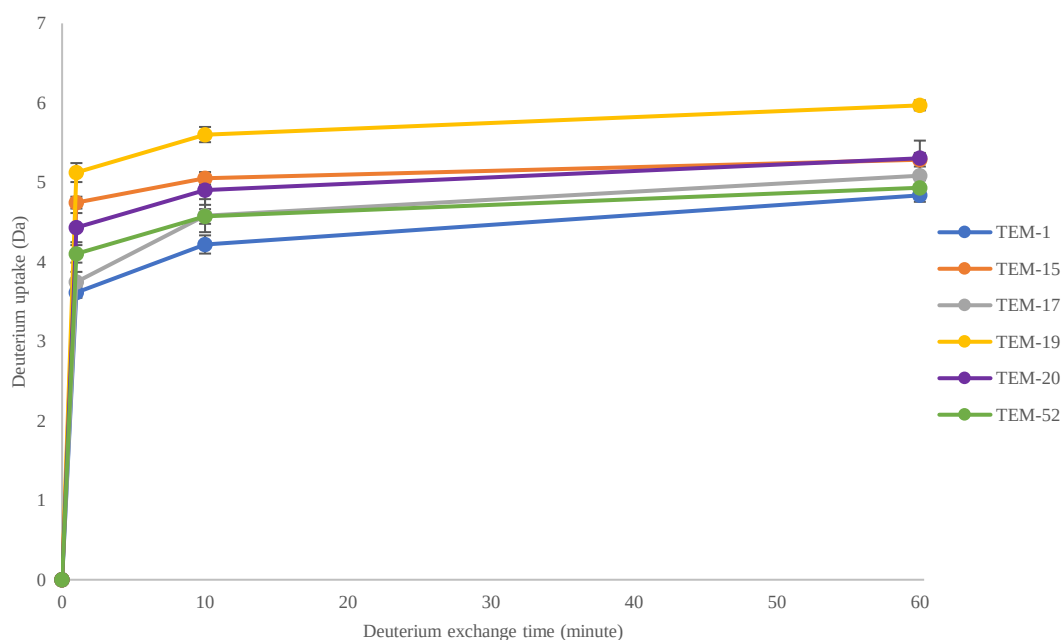


Figure 3.12 Deuterium uptake plot of residues 230-246 for apo-TEM-type β -lactamases.

Region containing residue 104 (Residues 103-117 and 103-118, H2/H3-loop)

Residue 104 is also a residue of our concern as the point mutation E104K is categorized as a key mutation that enhances the antibiotic catalytic efficiency. Residues 103-117 and 103-118, which is the H2/H3-loop, covered this residue, and this region involves several interactions with the catalytically important SDN loop such as V108-D131, T109-D131 and Y105-N132.⁶⁴ The results showed that flexibility of this region was enhanced for TEM-19 (G238S) and TEM-15 (G238S/E104K). TEM-17 (E104K) had a similar, or slightly lower flexibility than the wild type TEM-1. Both mutants containing M182T had become less flexible than TEM-1, while TEM-52 (G238S/E104K/M182T) had the most compact conformation and its flexibility was significantly lower than that of TEM-1. This might indicate that the compact conformation of this region in M182T-containing mutants might be used to compensate the destabilization effect to the SDN loop caused by key point mutations (Figure 3.13).

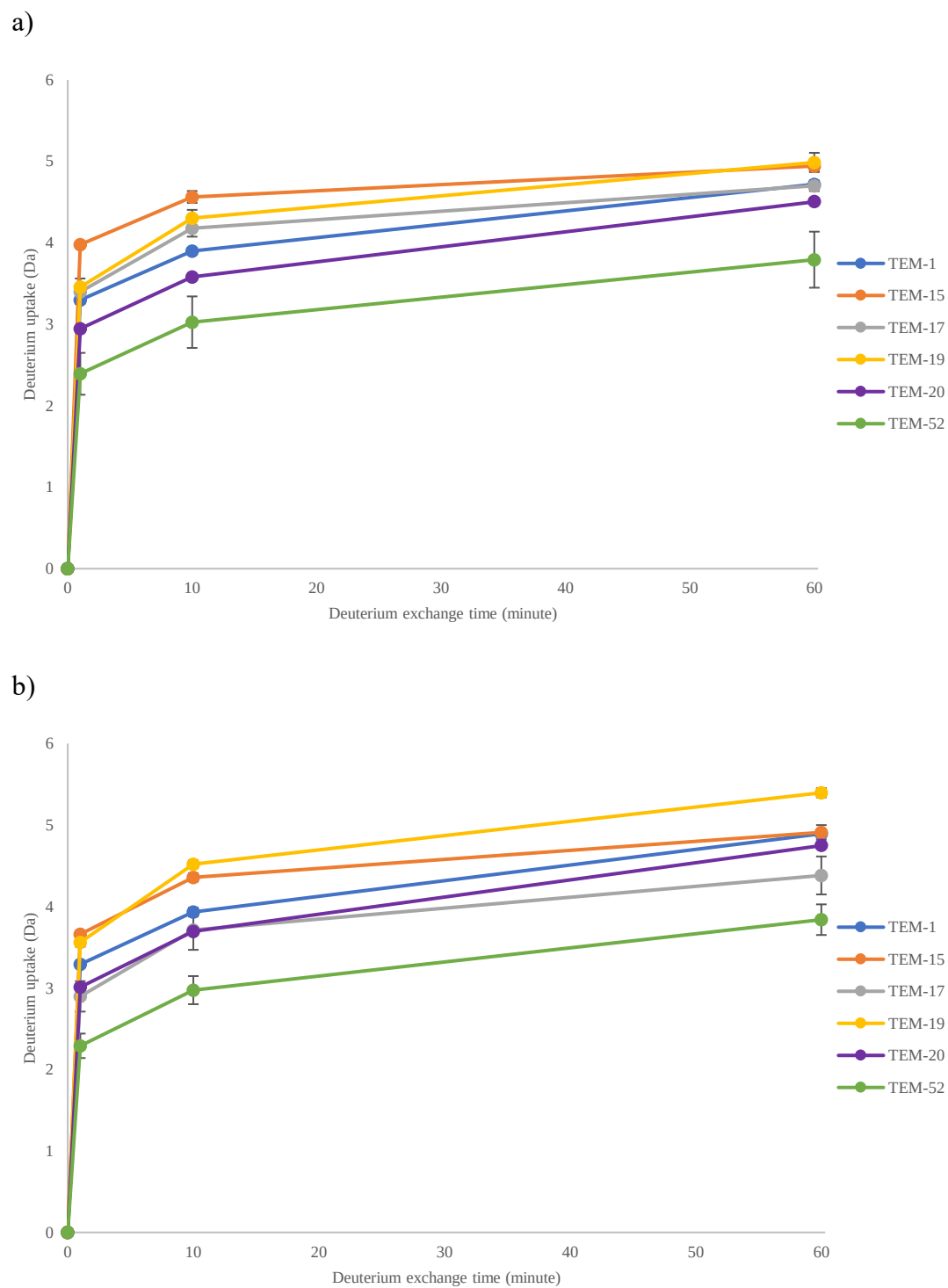


Figure 3.13 Deuterium uptake plot of residues (a) 103-117 and (b) 103-118 for apo-TEM-type β -lactamases.

Region containing residue 182 (Residues 178-186, H7/H8-loop and H8)

This residue is located near the catalytically important Ω -loop and only mutation found in this residue is from methionine to threonine. This mutation is believed to cause insignificant effect on the antibiotic catalytic efficiency of mutated TEM-type β -lactamases and hence it is regarded as a secondary mutation. Instead, the mutation on this residue was believed to alter the thermostability of β -lactamases.¹¹⁹ The results showed that residues 178-186, which is part of the H7/H8 loop and H8, covered this residue. However, pepsin digestion of the triple mutant TEM-52 did not generate this peptide. In fact, no peptide was found to cover residue 182 from the TEM-52 digested peptide pool. The flexibility of TEM-15 (G238S/E104K) was the highest, followed by TEM-19 (G238S) and TEM-17 (E104K). The synergistic effect was not obvious in this case as the flexibility enhancement in the double mutant was likely to be a result of simple accumulation of changes in flexibility of both single mutants. However, the effect of G238S on this region was found to be more significant than E104K, as the flexibility of both TEM-19 and TEM-15 was significantly higher than that of TEM-17. All three mutants containing only key point mutation(s), i.e., G238S and E104K, were more flexible than the wild type TEM-1. In contrast, the TEM-20 (G238S/M182T) was significantly more compact than the TEM-1, indicating that this secondary point mutation might tend to stabilize this region, which was near the Ω -loop, for mutants containing key point mutations (Figure 3.14).

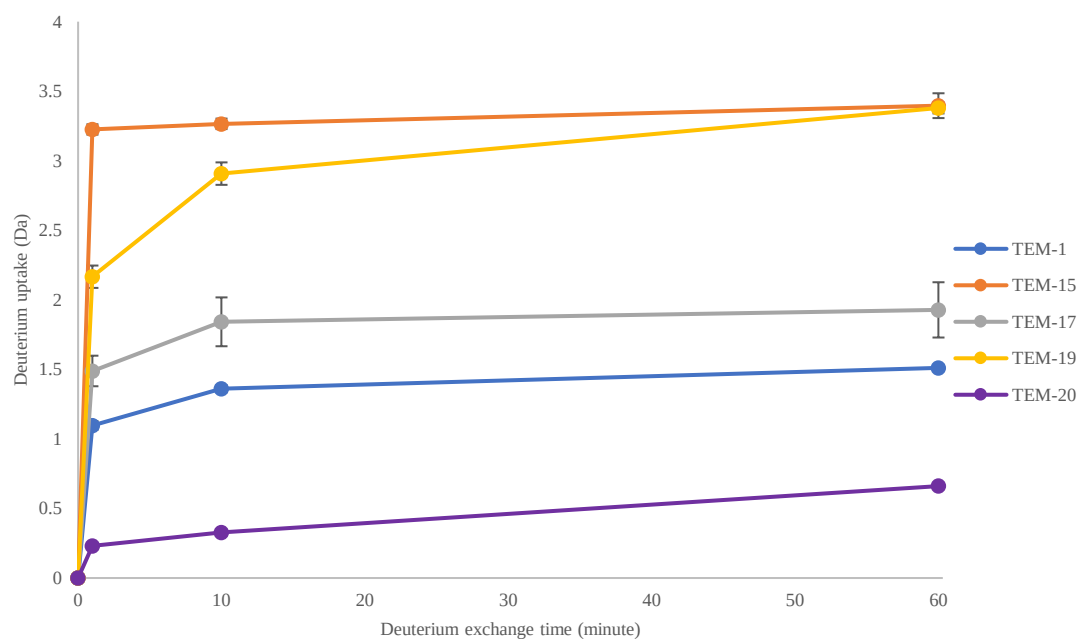


Figure 3.14 Deuterium uptake plot of residues 178-186 for apo-TEM-type β -lactamases.

H5-H6 region (Residues 140-150)

H5-H6 region is one of two allosteric regions first discovered in this study. This region is located outside the active site but near the Ω -loop, and it is connected by a two hydrogen bonds between T149 and L162 in the Ω -loop⁶⁴, as well as a hydrophobic interaction between P145 and W165 in the Ω -loop.⁶⁸ Flexibility enhancement was observed for mutants in the absence of M182T (i.e. TEM-19, TEM-17 and TEM-15), while incorporation of M182T caused the region to become even more compact than TEM-1. M182T might stabilize ESBLs by controlling the flexibility of this region so as to restrict the extensive movement of Ω -loop, which facilitated the optimized Ω -loop orientation for binding and hydrolyzing the β -lactam antibiotics (Figure 3.15).

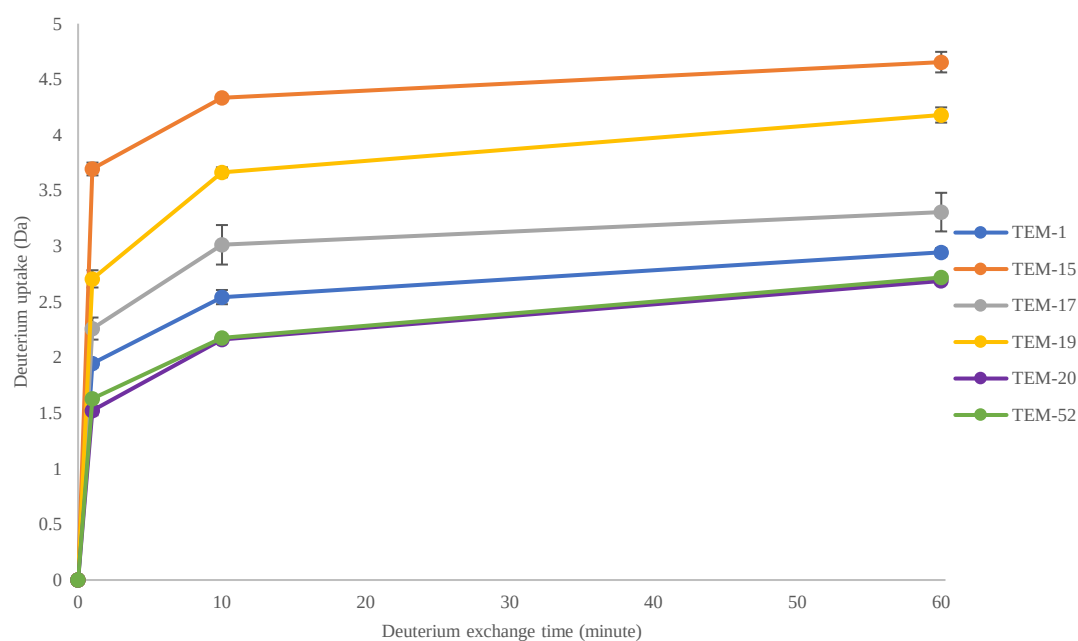
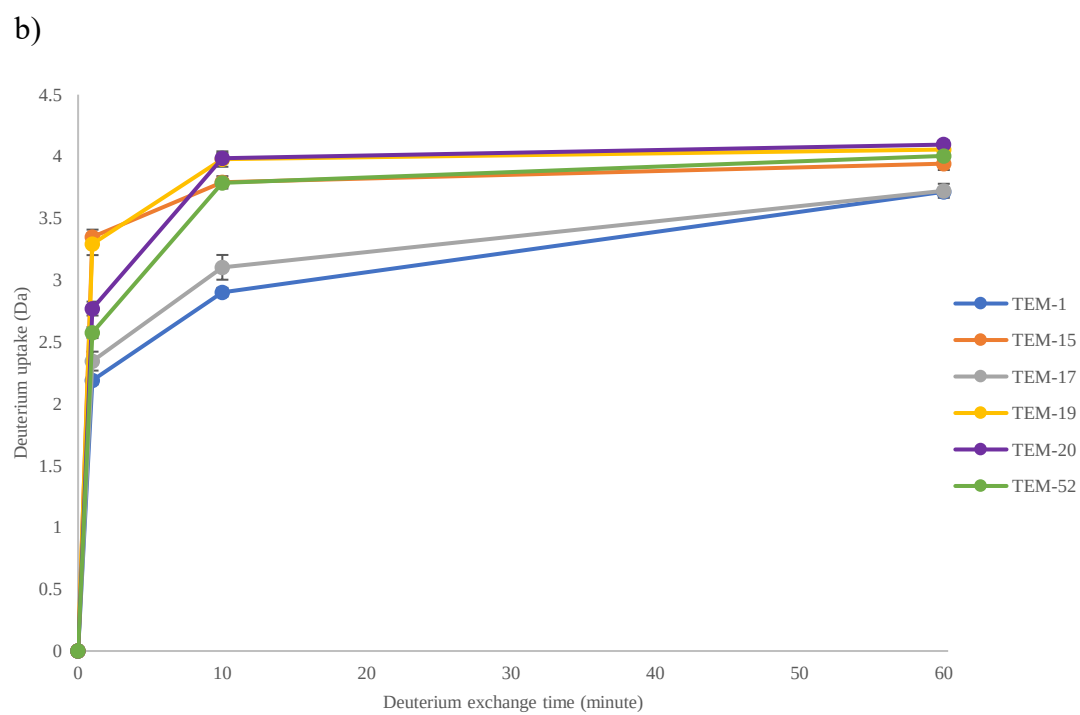
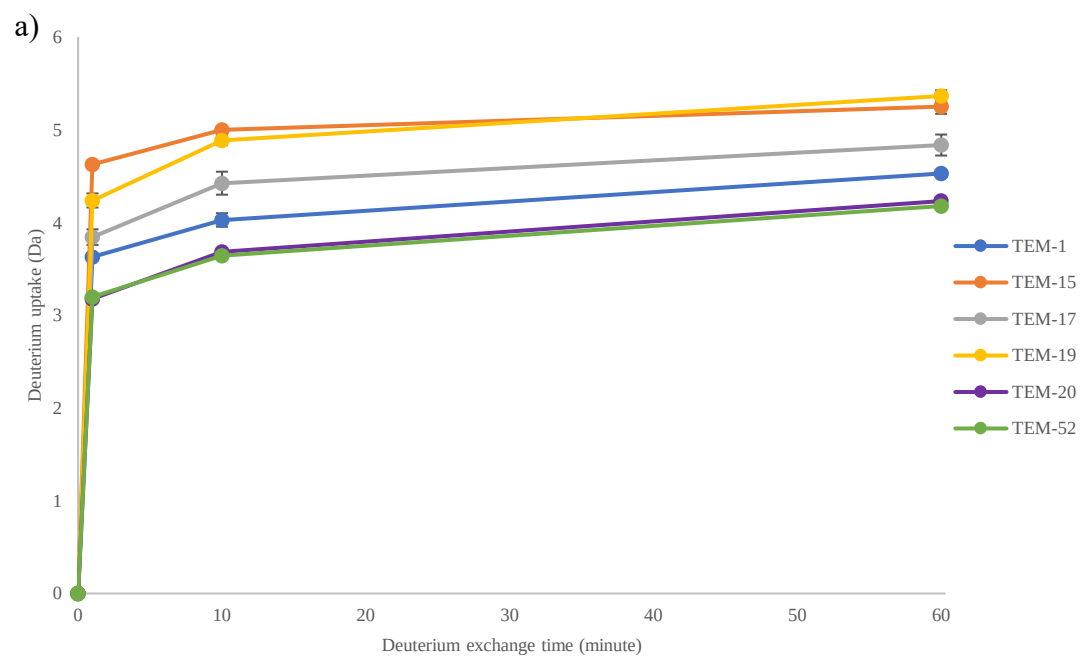


Figure 3.15 Deuterium uptake plot of residues 140-150 for apo-TEM-type β -lactamases.

An edge of β -sheet (Residues 49-59, 222-230 and 250-260)

This edge is another allosteric region first discovered in this study. There are three peptides comprising the edge of the β -sheet, including residues 49-59 (mainly the β 1/ β 2-loop and β -2 strand), 222-230 (mainly the H10/ β 3-loop) and 250-260 (mainly the β 4/ β 5-loop). This β -sheet edge is located at the opposite side of the active site. Our results showed all these three peptides had similar trends of deuterium uptake (Figure 3.16). TEM-19 (G238S) and TEM-15 (G238S/E104K) showed greater flexibility enhancement for the edge when compared with TEM-17 (E104K). M182T stabilized the edge by modulating the flexibility of the edge to an optimum level, i.e., similar or slightly more compact than TEM-1, as revealed by residues 49-59. Although residue 222-230 became more flexible for M182T-containing mutants, it is a loop-based peptide, therefore it tends to be more flexible. Since part of β -3 and β -4 strands in the other edge of the β -sheet were part of the active site, in order to accommodate the bulky side chain of the third generation cephalosporin, this edge must undergo several conformational rearrangements. Therefore, flexibility enhancement of this particular β -sheet edge was

necessary so as to reduce several unfavorable constraints after the active site expansion.



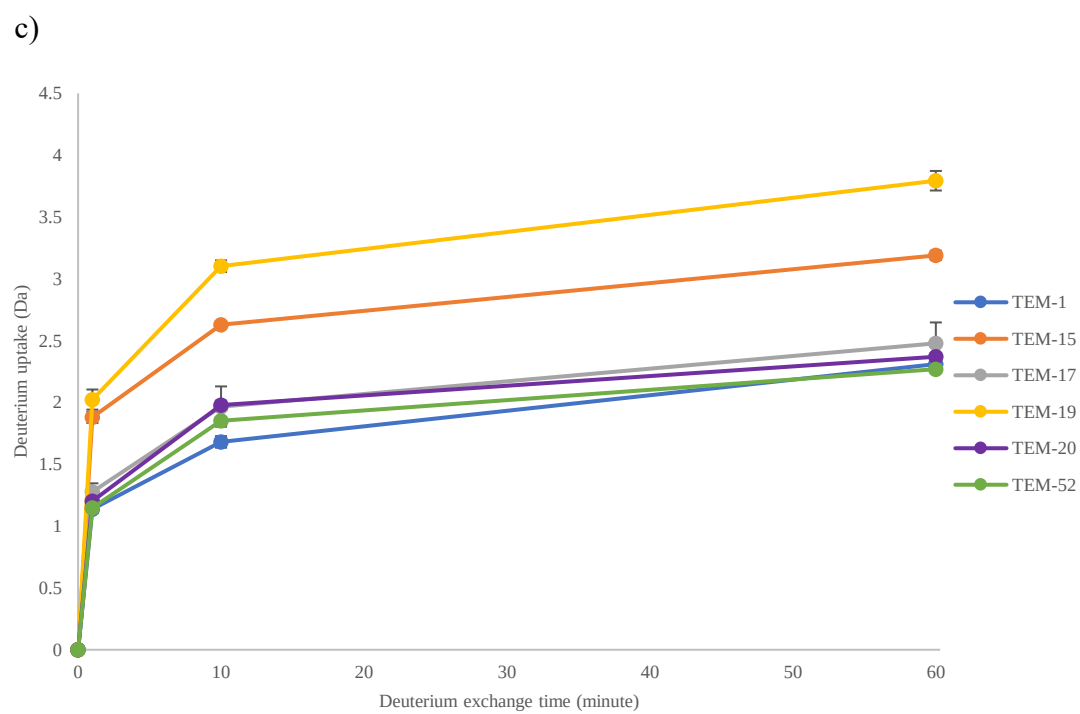


Figure 3.16 Deuterium uptake plot of residues (a) 49-59, (b) 222-230 and (c) 250-260 for apo-TEM-type β -lactamases.

3.4. Study of binding between the six β -lactamases with cefotaxime

After studying the apo-TEM-type β -lactamases with single or combination of point mutations, conformational dynamics of these β -lactamases upon binding with the third generation cephalosporin, cefotaxime, was also investigated. Cefotaxime-bound β -lactamases were therefore obtained for further conformational dynamics studies.

3.4.1. Completeness of binding between apo-TEM-type β -lactamases and cefotaxime

Before the HDX-MS analysis of cefotaxime-bound β -lactamases, conditions for achieving complete binding were investigated. Cefotaxime was mixed with ESBL at various ratios for different durations. The covalent acyl enzyme-substrate complex (ES*), could be detected by MS because its mass was different from that of the free ESBL. Figure 3.17 showed the ESI-MS spectrum of hydrolysis of cefotaxime by TEM-17. Two peaks that represented the TEM-17 and ES* were clearly observed. It was observed that population of ES* was increasing slowly as a function of time, indicating the ongoing process of cefotaxime hydrolysis (Figure 3.18). Interestingly, the mass of cefotaxime was 455 Da while the mass difference between TEM-17 and ES* was found to be 414 Da. This was attributed to the loss of $-\text{COOCH}_3$ and the hydrolysis of β -lactam ring.⁸⁴ However, the population of ES* started to decrease when the $[\text{ES}^*]/[\text{E}_{\text{total}}]$ reached about 20% (Figure 3.18), and the ratio could not reach a desired value of complete binding. This might be due to the extremely high catalytic efficiency of ESBLs towards cefotaxime.

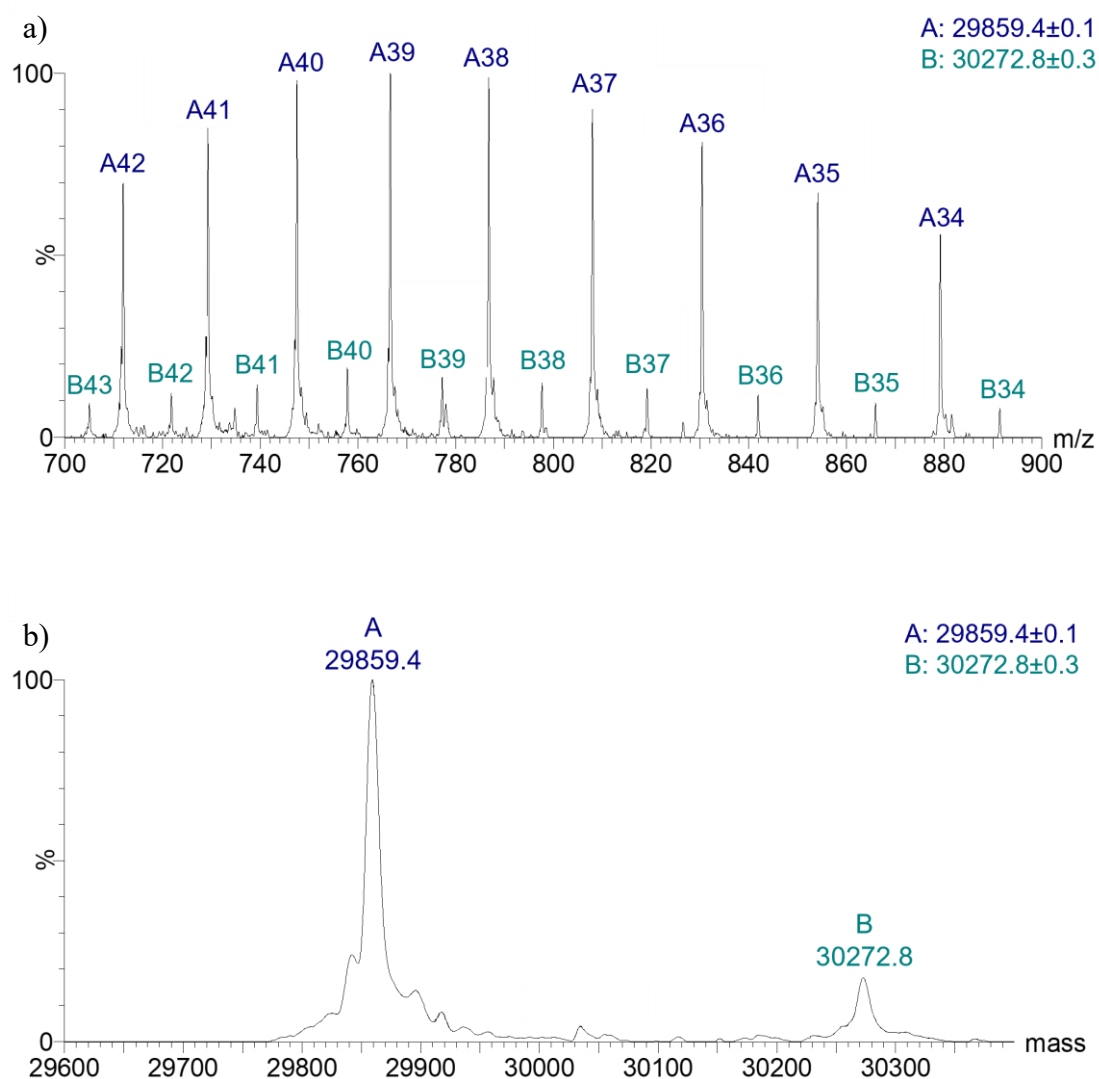


Figure 3.17 (a) ESI-MS and (b) deconvoluted mass spectra of reaction mixture of cefotaxime hydrolysis by TEM-17. Peak A and B represented free TEM-17 and the acylated complex of TEM-17 and cefotaxime, respectively.

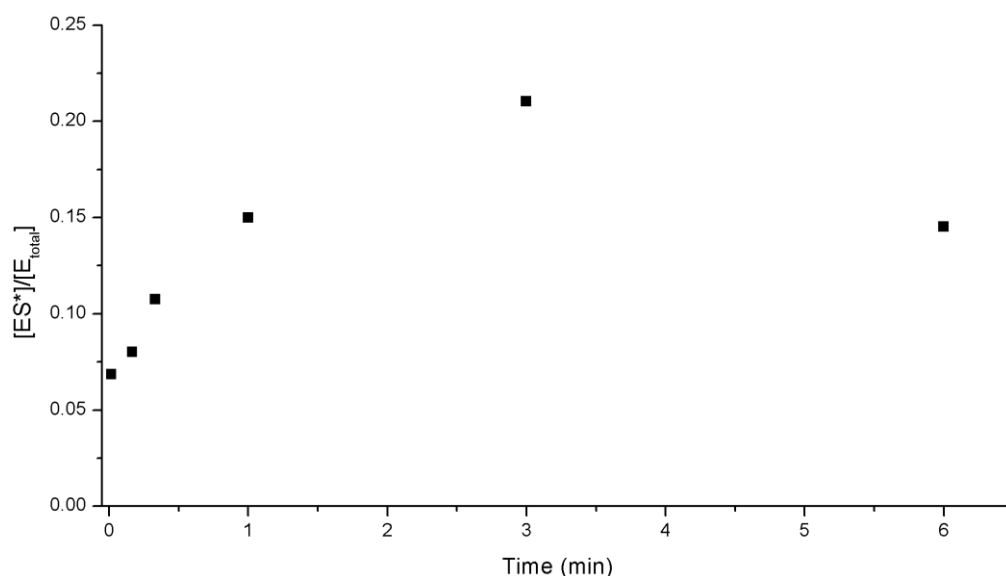


Figure 3.18 Plot of $[ES^*]/[E_{total}]$ against time of TEM-17 and cefotaxime reaction.

There are two approaches to achieve the complete cefotaxime binding of ESBLs. They include utilizing of excess amount of cefotaxime during reaction and incorporating a point mutation that could debilitate the deacylation efficiency, e.g. E166A, in ESBLs.^{130,131} However, injection of excess amount of cefotaxime to MS may cause signal saturation as well as signal suppression for proteins, which could be detrimental to the MS and the quality of spectra. The latter approach is common for forming adequate enzyme-substrate complex for X-ray crystallographic analysis and research studies that do not involve the E166 residue.¹³²⁻¹³⁶

3.4.2. Expression of the six TEM-type β -lactamases with E166A point mutations

The six TEM-type β -lactamases with E166A point mutations were successfully expressed by applying similar expression protocol for apo-TEM-type β -lactamases without E166A. Figure 3.19, 3.20 and 3.21 showed the SDS-PAGE result, mass spectra of the six E166A β -lactamases, and TIC of TEM-1A as a representative. Table 3.4 summarized the measured and theoretical masses of the six E166A β -lactamases, and all the measured masses had good agreement with their theoretical masses, confirming the correctness of expression. In this study, E166A β -lactamases would be labeled an “A” at the end of their name. For example, TEM-15A (G238S/E104K) is the E166A-mutated TEM-15, and E166A would not be included in the bracket to avoid redundancy.

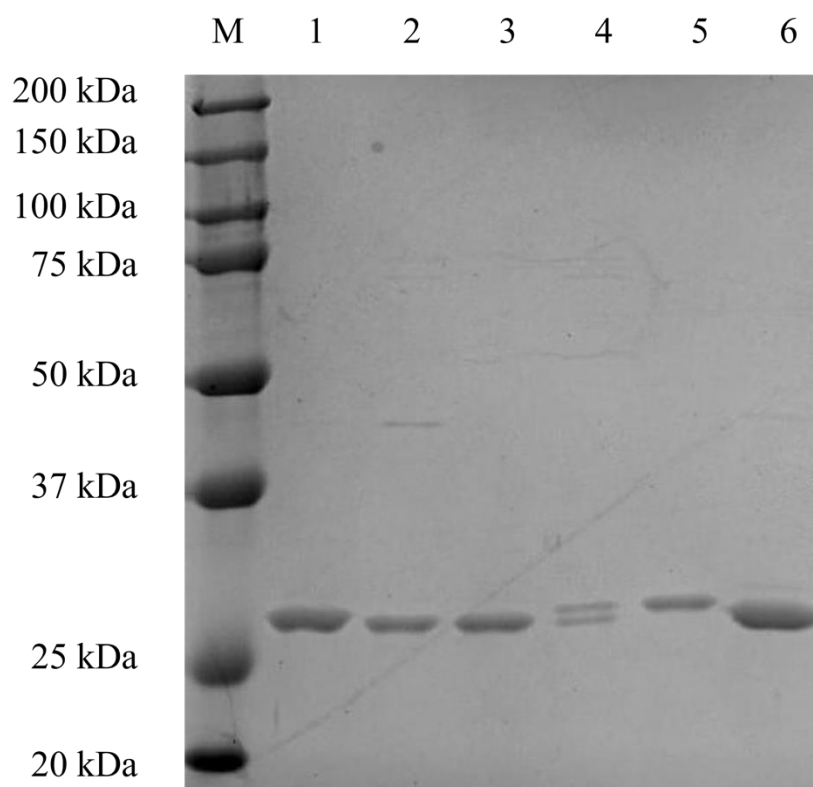


Figure 3.19 SDS-PAGE of the six E166A-TEM-type β -lactamases. Lane M: Ladder; Lane 1: TEM-1A; Lane 2: TEM-15A; Lane 3: TEM-17A; Lane 4: TEM-19A; Lane 5: TEM-20A; Lane 6: TEM-52A.

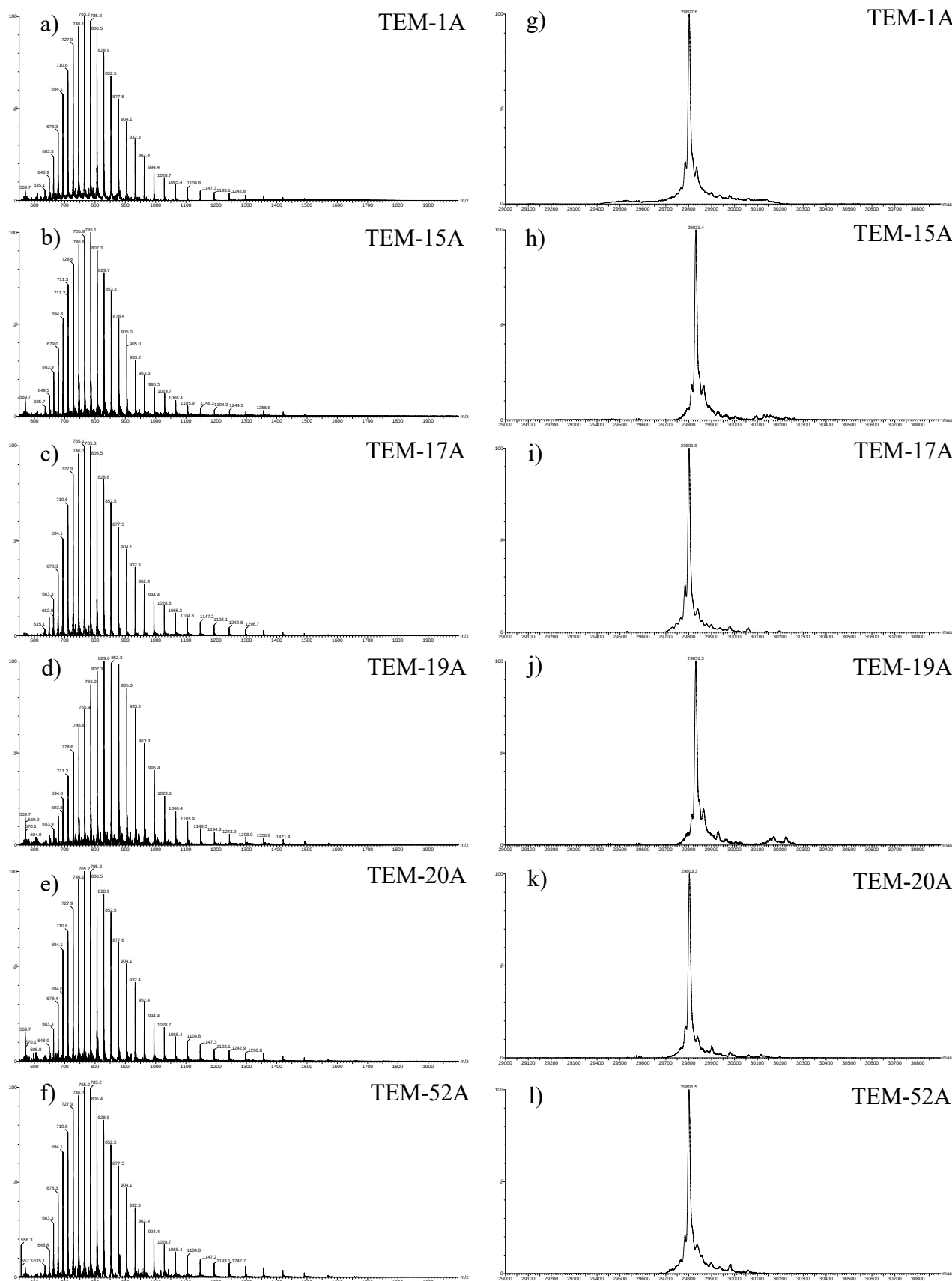


Figure 3.20 (a-g) ESI-MS spectra and (h-l) deconvoluted mass spectra of six E166A-TEM-type β -lactamases.

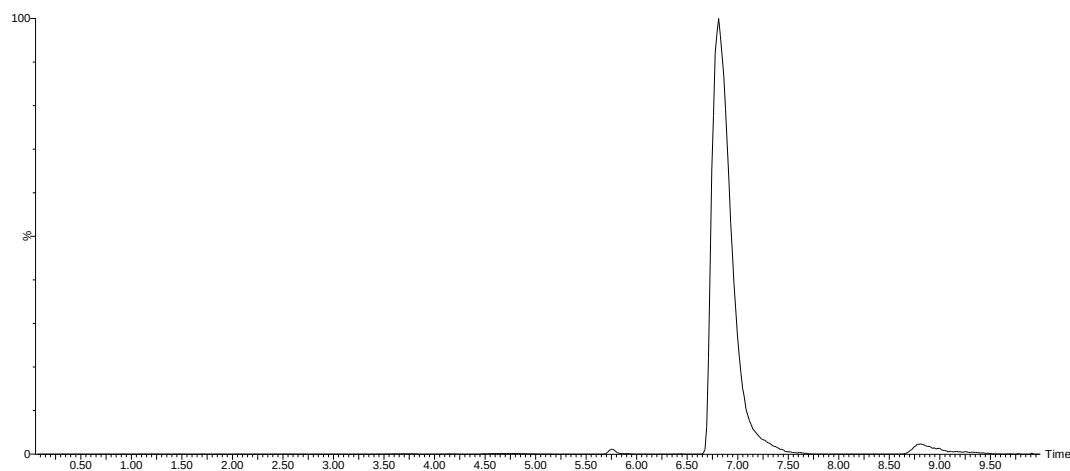


Figure 3.21 Total ion chromatogram of TEM-1A as a representative. The single dominant peak of TEM-1 indicated the high sample purity.

Table 3.4 Amino acid at position 104, 166, 182 and 238, theoretical and measured molecular masses of the six E166A mutated TEM-type β -lactamases.

β -Lactamases	Amino acid at position				Theoretical	Measured
	104	166	182	238	molecular mass (Da)	molecular mass (Da)
TEM-1A	E	A	M	G	29801.8	29802.6
TEM-15A	K	A	M	S	29830.8	29831.4
TEM-17A	K	A	M	G	29800.8	29801.9
TEM-19A	E	A	M	S	29831.8	29831.3
TEM-20A	E	A	T	S	29801.7	29803.3
TEM-52A	K	A	T	S	29800.8	29801.5

3.4.3. Complete binding between E166A-TEM-type β -lactamases and cefotaxime

Before the study of the conformational changes after binding, it was essential to convert all the apo-E166A-TEM-type β -lactamases to the ES*. This was because the co-existence of both apo-form and the bound-form of enzyme that indicated two different conformations of the same enzyme would be analyzed at the same time, hence the HDX changes of bound form would be diluted by that of the apo-form. In this study, 100-fold excess of cefotaxime was utilized to react with the E166A β -lactamases for 20 minutes to ensure the complete formation of the ES*. Only one protein multiply charged distribution was found in each mass spectrum, indicating the complete reaction between cefotaxime and E166A mutants (Figure 3.22). Moreover, the mass difference between the cefotaxime-bound E166A β -lactamase and apo-E166A β -lactamase was 396, indicating that the β -lactamase was bound to the -COOCH₃-cut cefotaxime (Table 3.5). Combining all evidence, the bioactivities of E166A β -lactamases and the completeness of the binding were confirmed.

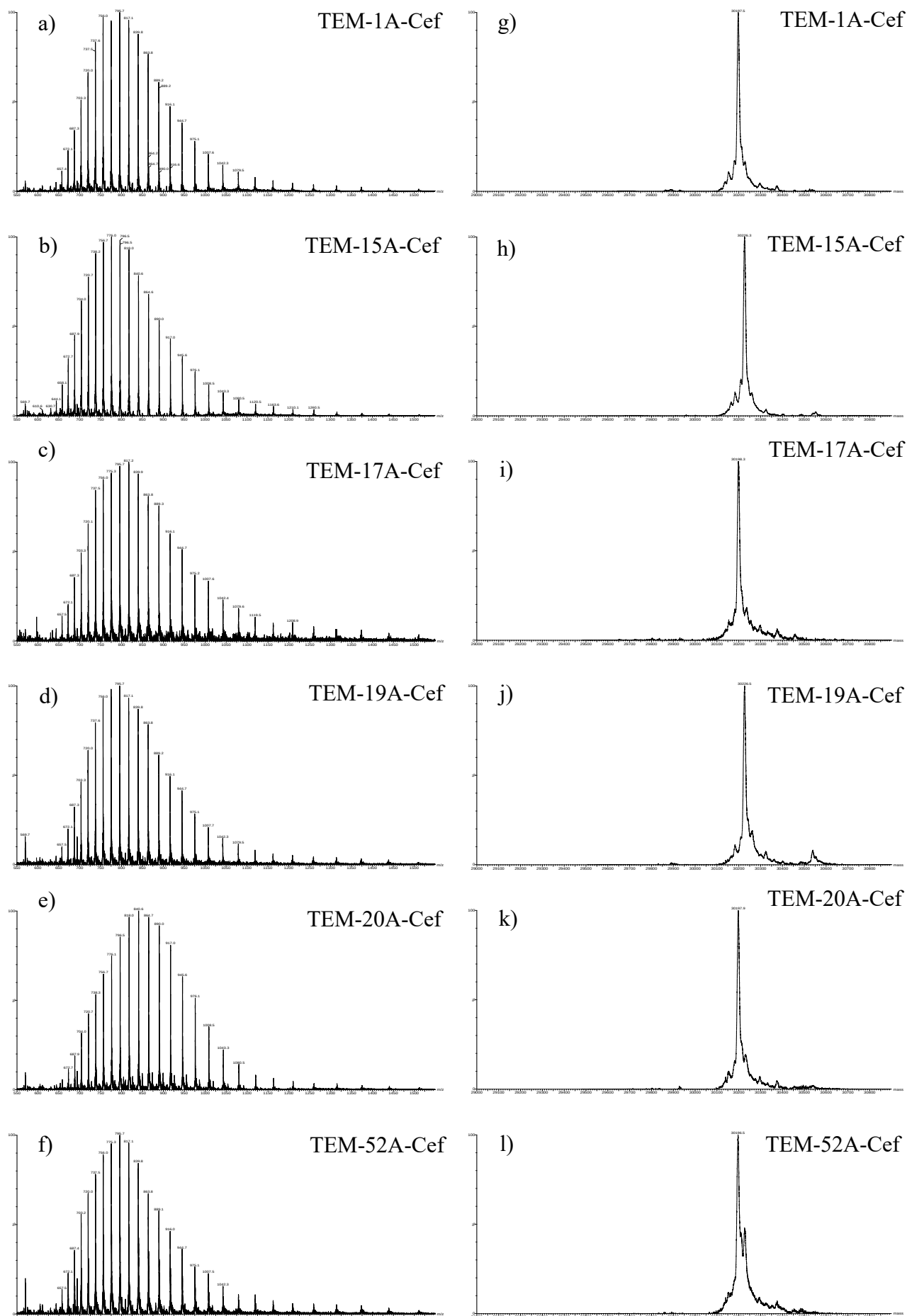


Figure 3.22 (a-g) ESI-MS spectra and (h-l) deconvoluted mass spectra of six cefotaxime-bound E166A-TEM-type β -lactamases.

Table 3.5 Mass differences between the cefotaxime-bound and the apo E166A-mutated β -lactamases.

Cefotaxime-bound E166A mutated β -lactamases	Measured molecular mass of the complex (Da)	Molecular mass of apo E166A mutated β -lactamase (Da)	Mass difference (Da)
TEM-1A-Cef	30197.5	29802.6	394.9
TEM-15A-Cef	30226.3	29831.4	394.9
TEM-17A-Cef	30198.3	29801.9	396.4
TEM-19A-Cef	30226.5	29831.3	395.2
TEM-20A-Cef	30197.9	29803.3	394.6
TEM-52A-Cef	30196.5	29801.5	395.0

3.4.4. Global HDX-MS of cefotaxime-bound E166A-TEM-type β -lactamases

The global HDX-MS result revealed that almost all cefotaxime-bound E166A-TEM-type β -lactamases had similar flexibility, but interestingly except TEM-17A-Cef (E104K) (Figure 3.23). Previously the apo-TEM-17 showed a similar conformation when compared with the wild type TEM-1, but the conformation of TEM-17A-Cef became much more compact than TEM-1A-Cef. Besides, it was revealed that the flexibility of apo-TEM-type β -lactamases had great variety, the similar bound state flexibility indicated that several mutants that had high flexibility in their apo-state may undergo great changes upon cefotaxime binding. Searching localized regions that underwent significant conformational changes could lead to a better understanding of the resistance mechanism.

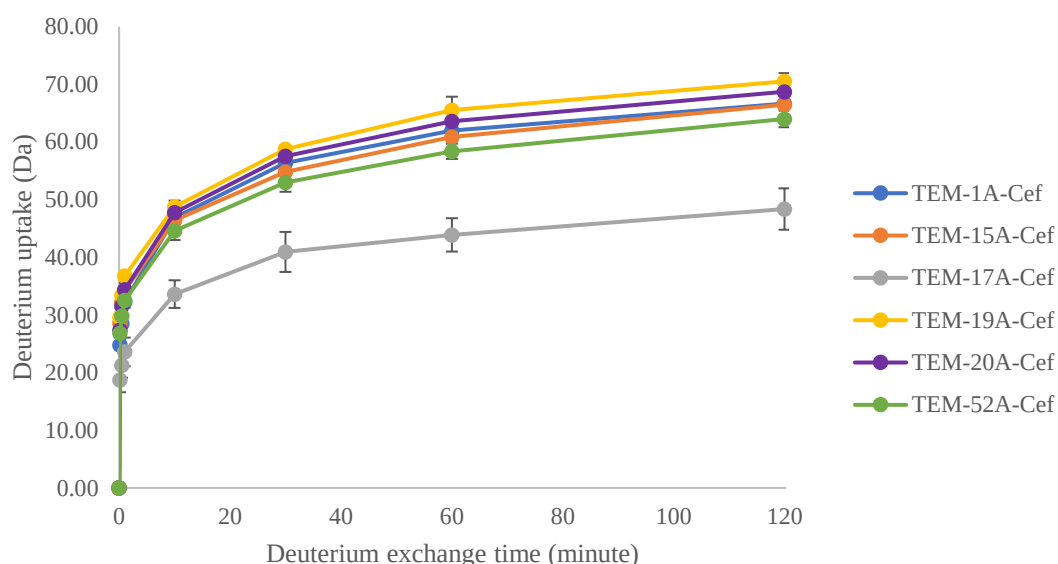
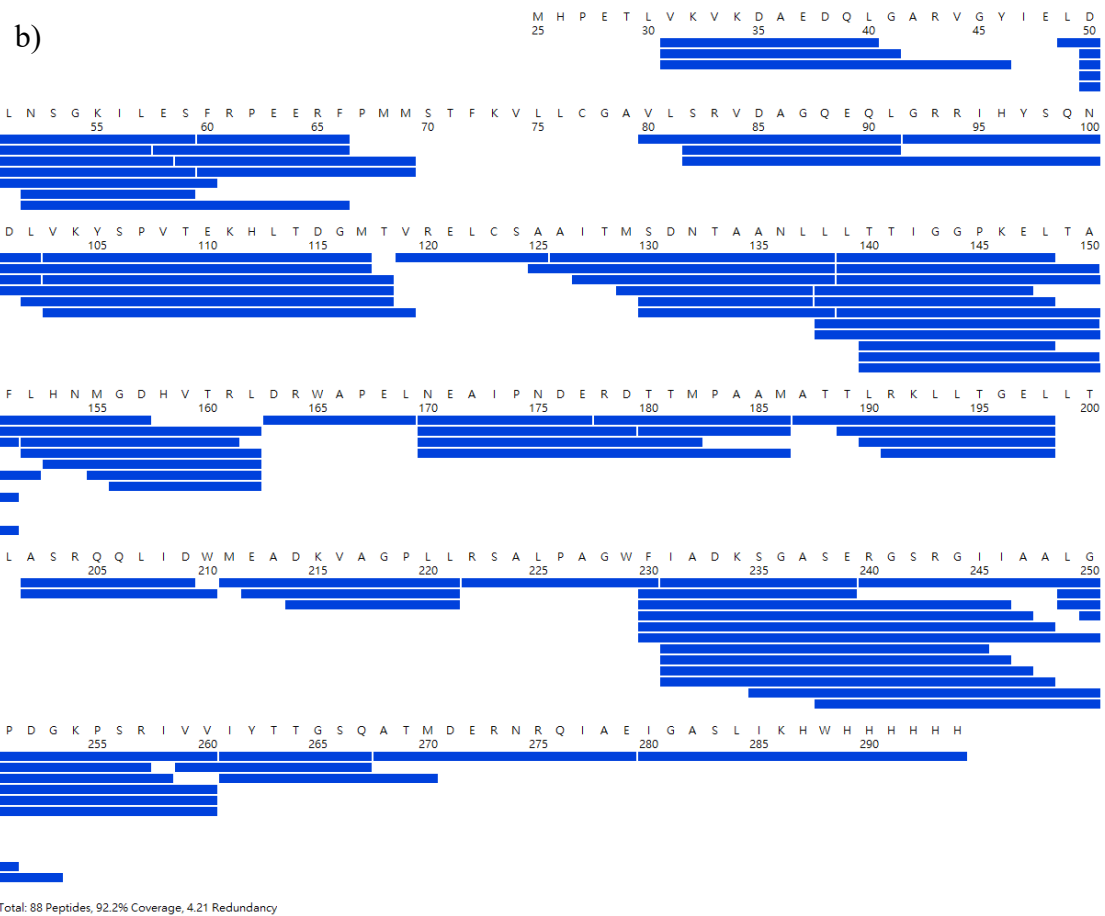


Figure 3.23 Deuterium uptake plot of the six cefotaxime-bound E166A-TEM-type β -lactamases.

3.4.5. Local HDX-MS of cefotaxime-bound E166A-TEM-type β -lactamase complexes

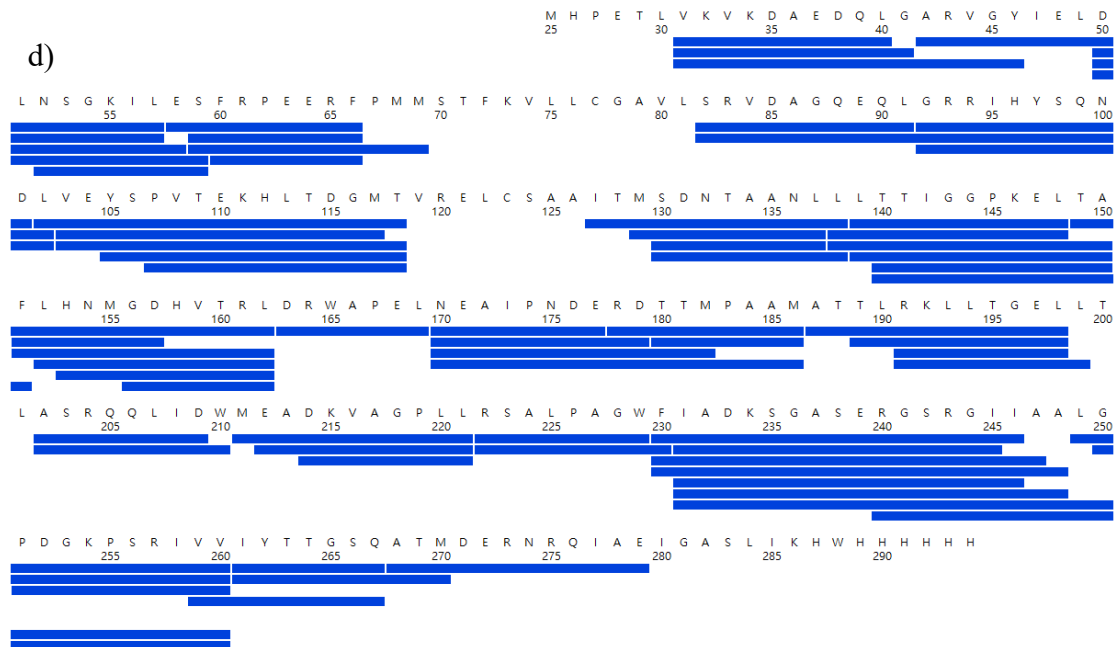
The coverage maps of all six cefotaxime-bound E166A TEM-type β -lactamase complexes were shown in Figure 3.24. The number of peptides and sequence coverage of all six complexes were satisfactory (Table 3.6). Although the number of overlapping peptides (24 peptides) and sequence coverage (61.1%) were relatively low, over 84% sequence coverage was achieved for each individual complex. More importantly, the digestion covered most of the important regions, such as region near S70, SDN and Ω -loop, of TEM-type β -lactamase. Moreover, as discussed previously, peptides that were covered by four or five complexes of our interest could also be applied for comparisons. As a result, the digestion patterns of the six complexes were satisfactory for further comparisons. Difference plots of comparisons of cefotaxime-bound E166A-TEM-type β -lactamase complexes having one different point mutations were shown in Figure 3.25.



c)



d)



e)



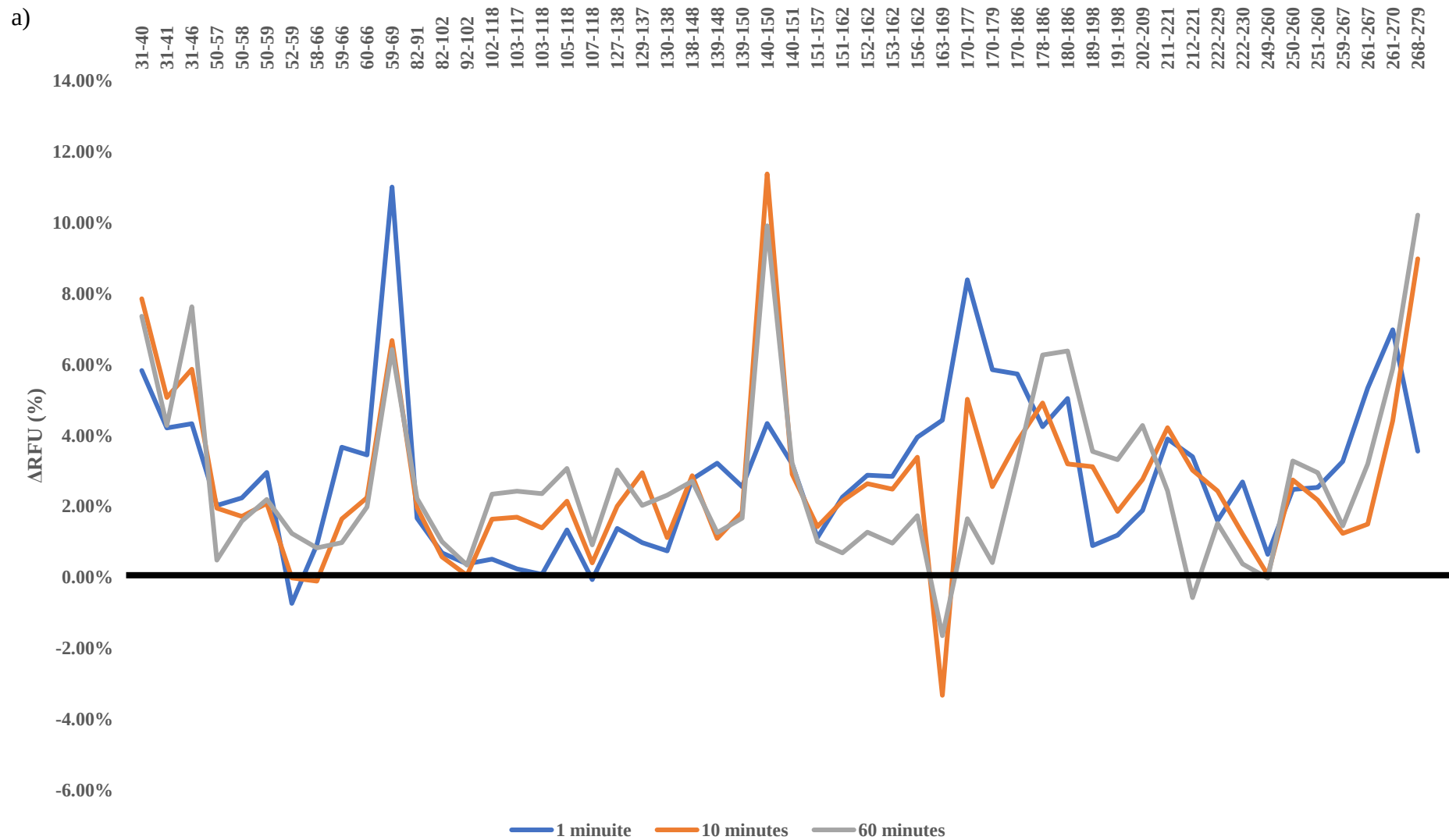
Total: 112 Peptides, 94.1% Coverage, 5.41 Redundancy



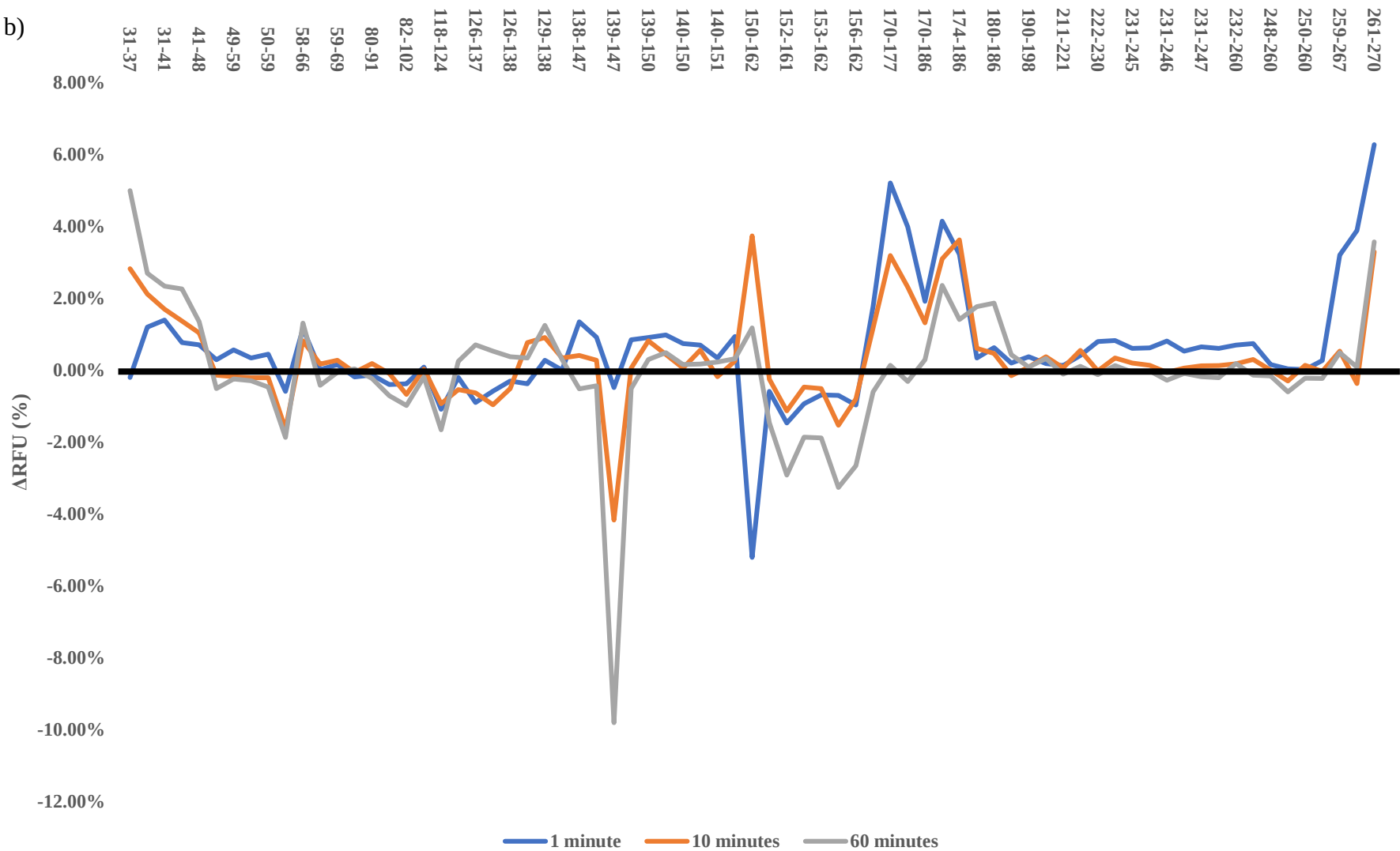
Figure 3.24 Coverage map of (a) TEM-1A-Cef, (b) TEM-15A-Cef, (c) TEM-17A-Cef, (d) TEM-19A-Cef, (e) TEM-20A-Cef and (f) TEM-52A-Cef.

Table 3.6 Number of peptides generated, sequence coverage, redundancies and number of common peptides resulted from pepsin digestion of the six cefotaxime-bound E166A-TEM-type β -lactamases.

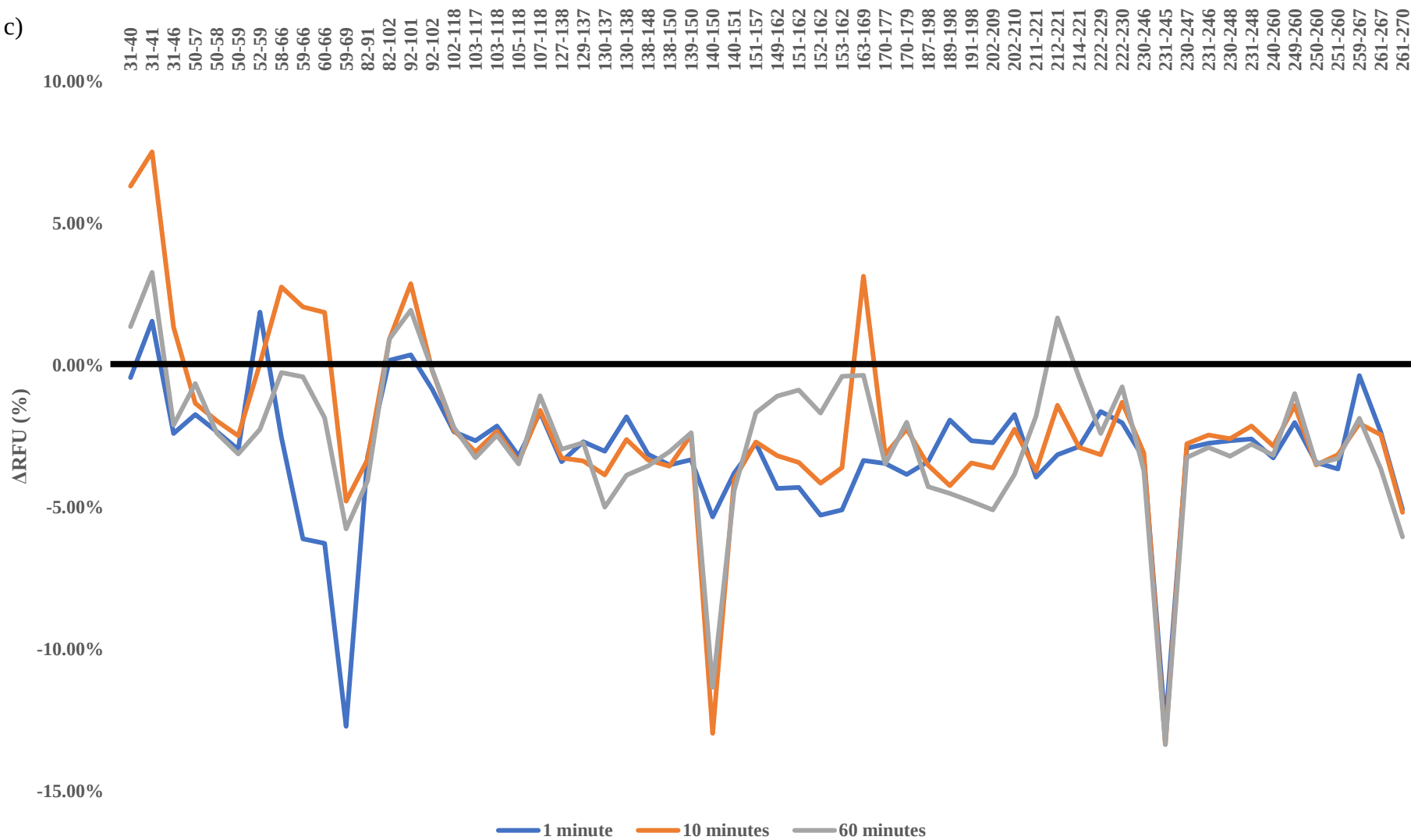
Cefotaxime-bound			
E166A mutated β -lactamases	Number of peptides	Sequence coverage	Redundancy
TEM-1A-Cef	95	90.7%	4.48
TEM-15A-Cef	88	92.2%	4.21
TEM-17A-Cef	107	89.3%	5.53
TEM-1A9-Cef	70	84.1%	3.59
TEM-20A-Cef	112	94.1%	5.41
TEM-52A-Cef	116	93.0%	6.71
Common peptides	24	61.1%	1.77

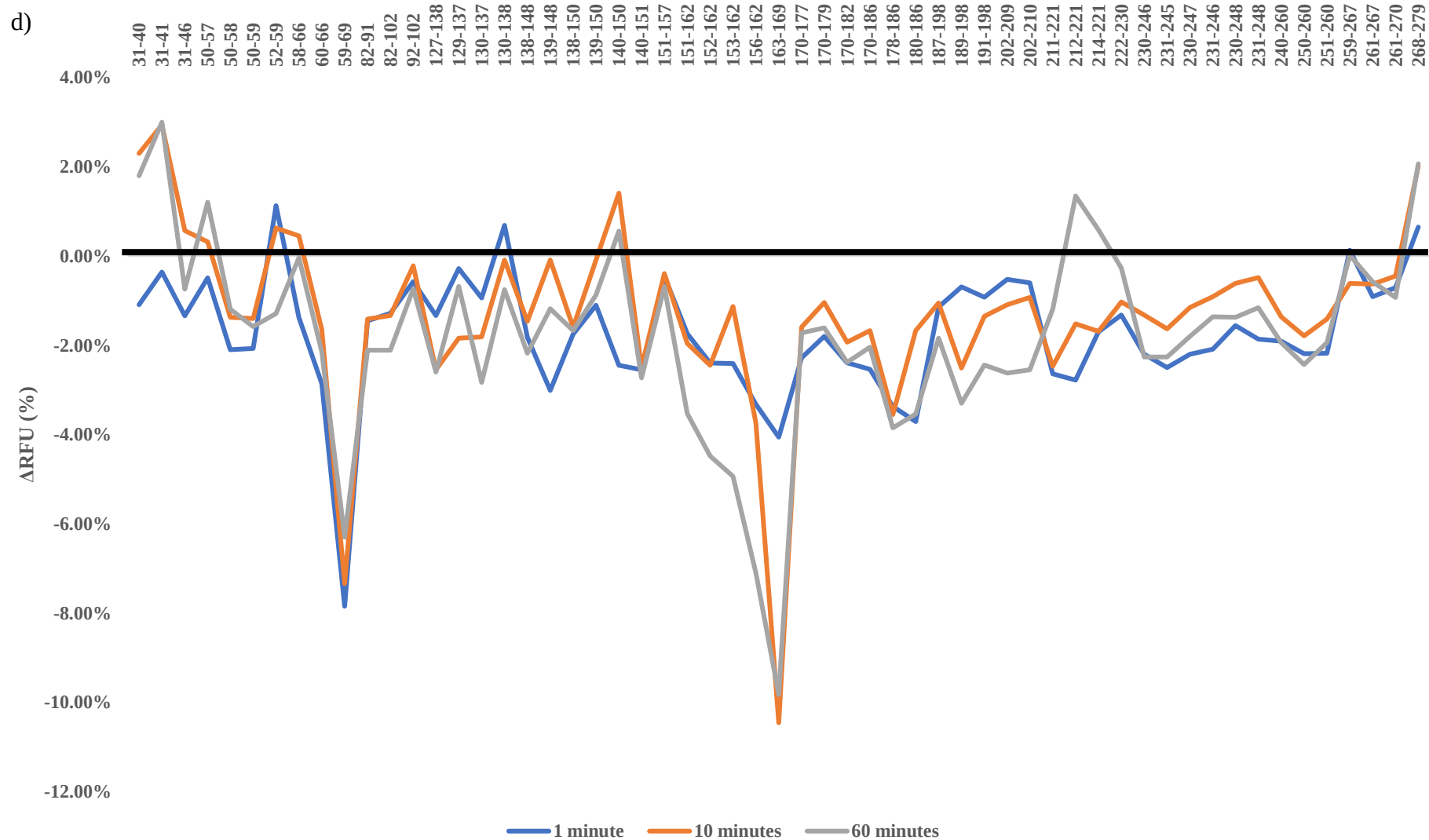


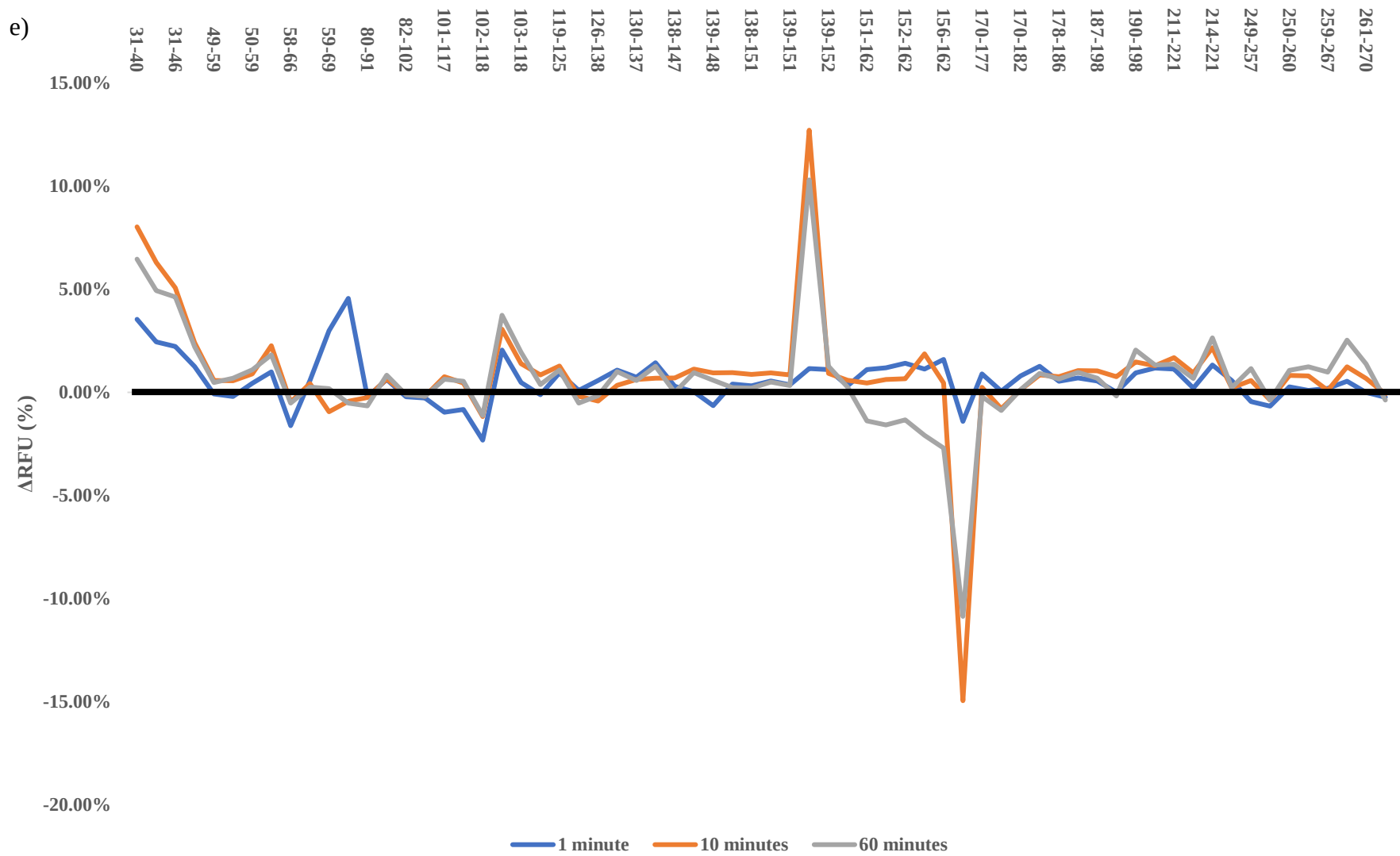
b)

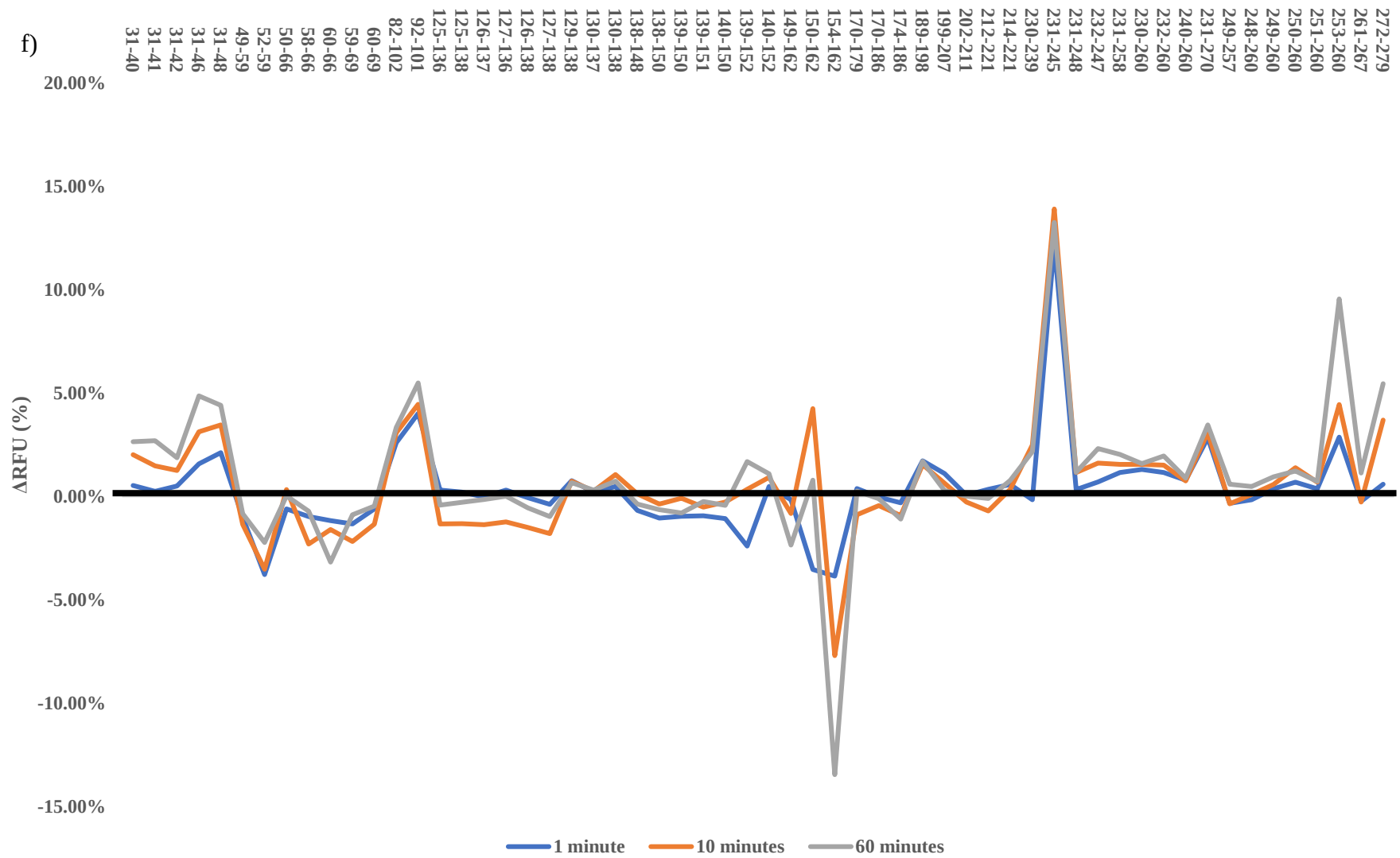


c)









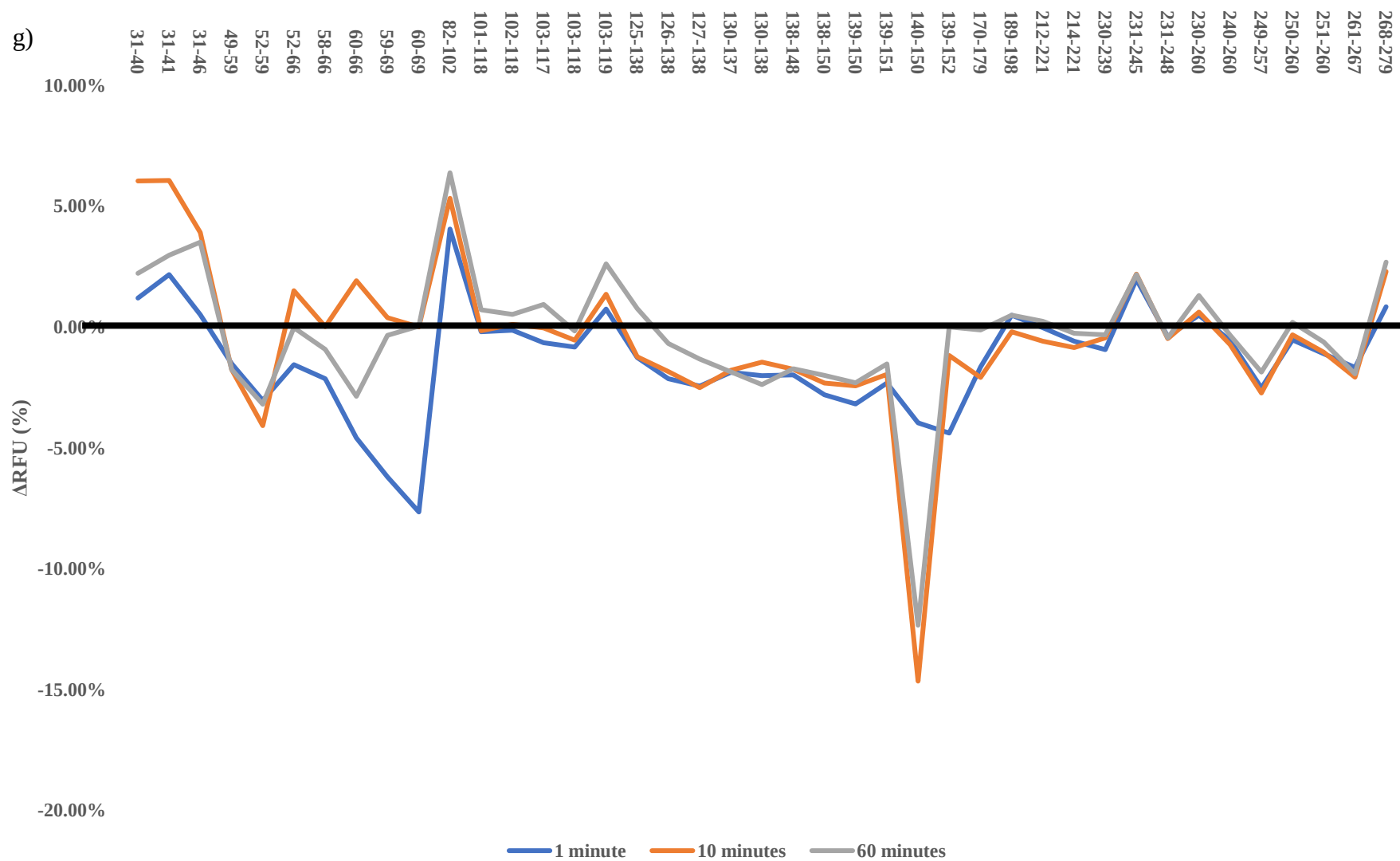


Figure 3.25 Different plot of comparison between (a) TEM-1A-Cef and TEM-19A-Cef, (b) TEM-1A-Cef and TEM-17A-Cef, (c) TEM-19A-Cef and TEM-20A-Cef, (d) TEM-19A-Cef and TEM-15A-Cef, (e) TEM-17A-Cef and TEM-15A-Cef, (f) TEM-20A-Cef and TEM-52A-Cef and (g) TEM-15A-Cef and TEM-52A-Cef.

Regions near active serine-70 (Residues 59-69, β 2/H2-loop)

Since S70 is directly involved in the binding and acylation of cefotaxime, the region near S70 was deemed an important region. The deuterium uptake plot showed almost all cefotaxime- β -lactamase complexes had similar flexibility, except TEM-19A-Cef (G238S) which had slightly more flexible conformation than the others (Figure 3.27a). The observation of similar flexibility in this region for complexes was reasonable, as the conformation of S70 in both TEM-1 (PDB: 1BTL) was highly similar to that of the triple mutant (i.e., TEM-52 with G238S/E104K/M182T, PDB: 1HTZ). Correct orientation of S70 was essential for the acylation, therefore the conformation of this region during acylation was expected to be similar for all TEM-type β -lactamases (Figure 3.26). The deuterium uptake of residues 60-69 also supported this explanation, as the flexibility of this region for all β -lactamases was similar, although this peptide was not found in TEM-19A-Cef (Figure 3.27b).

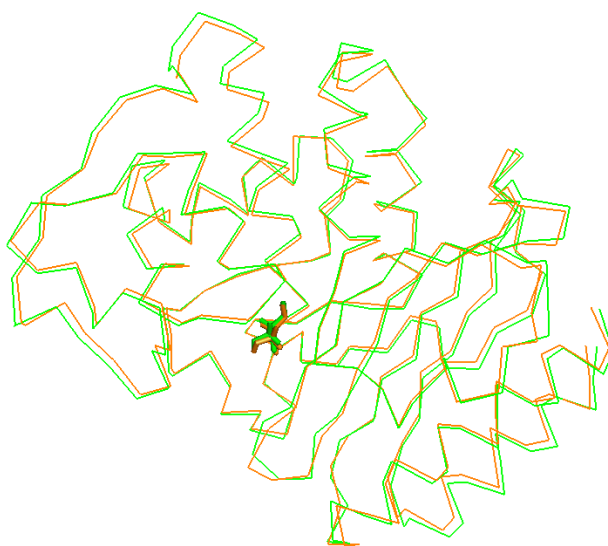


Figure 3.26 Aligned X-ray crystal structures of TEM-1 (green) and TEM-52 (orange). Sticks represent the S70 in TEM-1 (green) and TEM-52 (orange) respectively.

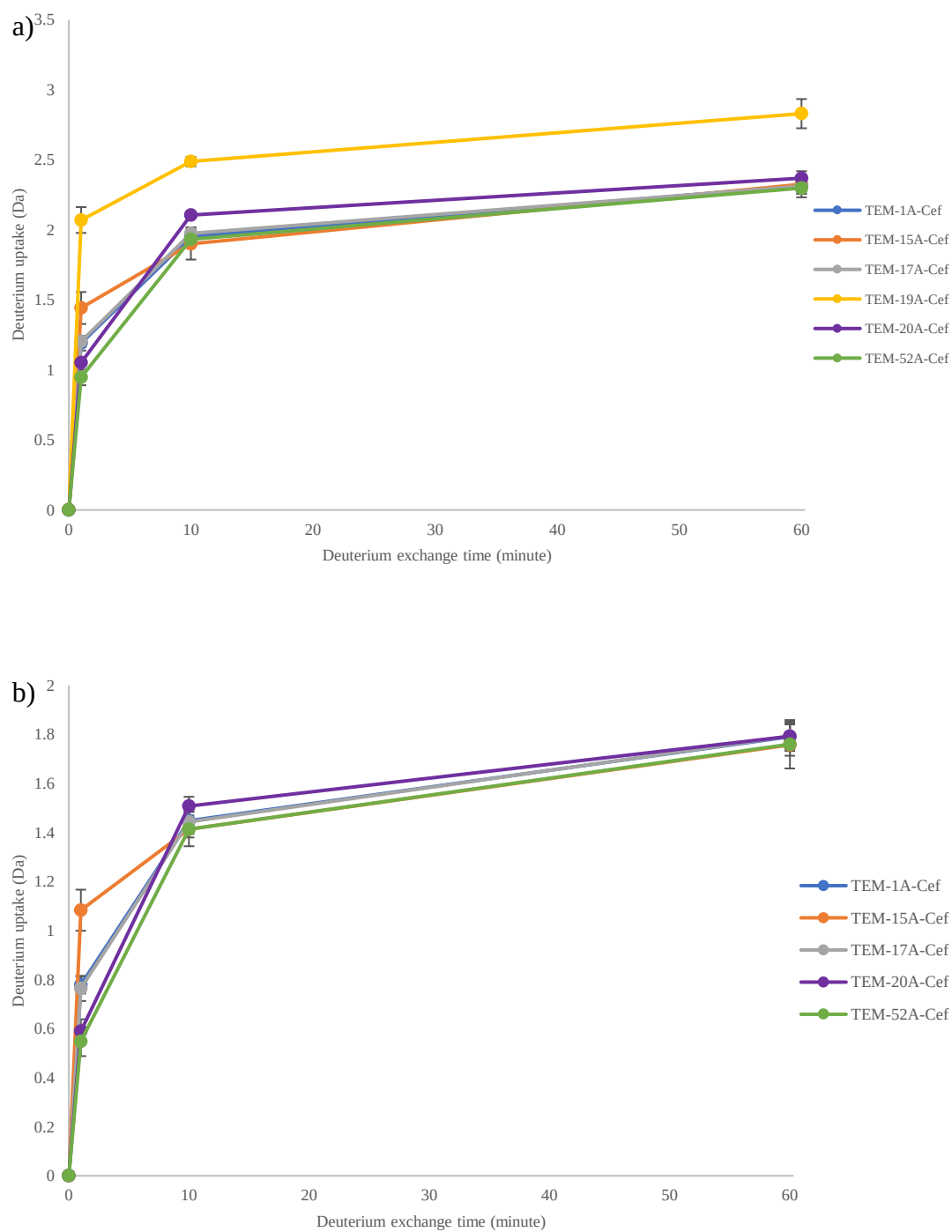


Figure 3.27 Deuterium uptake plot of residues (a) 59-69 and (b) 60-69 for cefotaxime-bound TEM-type β -lactamases.

Ω-loop (Residues 163-169 and 170-179)

Similar to the region near the S70, the catalytically important Ω-loop had similar flexibility for all the six cefotaxime-β-lactamases complexes, as reflected by the deuterium uptake of residues 170-179 (Figure 3.28a). The subtle flexibility differences were shown in the 1-minute HDX time point, where TEM-19A-Cef (G238S) and TEM-15A-Cef (G238S/E104K) were slightly more flexible than the others, followed by TEM-17A-Cef (E104K). Both complexes containing M182T (TEM-20A-Cef and TEM-52A-Cef) were more compact than those without M182T, but they were still more flexible than the wild type TEM-1A-Cef. The residues 163-169 were missed in the TEM-52A-Cef (G238S/E104K/M182T) (Figure 3.28b). In contrast to the C-terminal part of Ω-loop, the N-terminal part Ω-loop was shown to be either having similar flexibility to the wild type TEM-1, including TEM-17A-Cef and TEM-20A-Cef, or having more compact conformation for TEM-19A-Cef and TEM-15A-Cef, and the effect was more obvious for the TEM-15A-Cef. In contrast, this point mutation combination (G238S and E104K) caused a great flexibility enhancement of this region in the apo β-lactamase, which might indicate that the synergistic effect of G238S/E104K significantly altered the conformation of the Ω-loop.

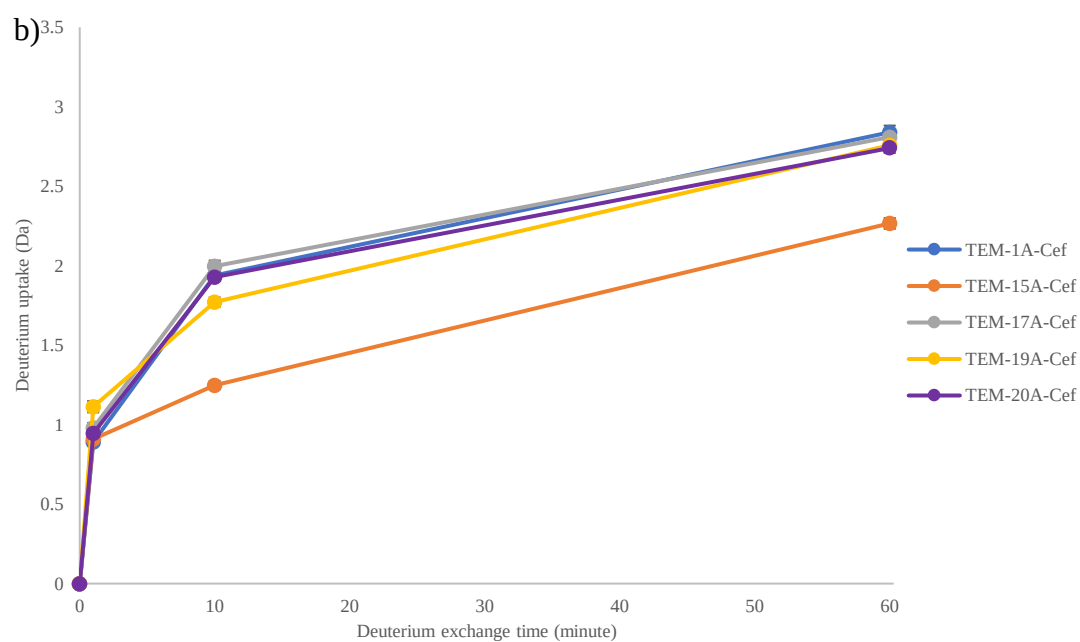
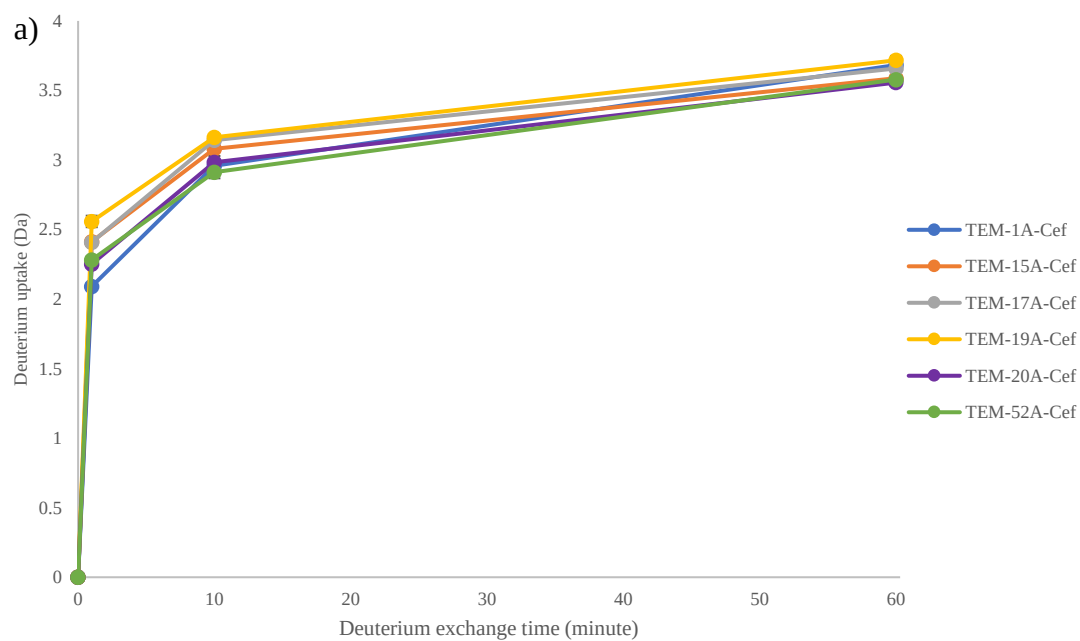


Figure 3.28 Deuterium uptake plot of residues (a) 170-179 and (b) 163-169 for cefotaxime-bound TEM-type β -lactamases.

SDN loop (Residues 130-138)

The SDN loop became extremely rigid and compact after binding with cefotaxime, which was reflected by the extremely low deuterium uptake value in residues 130-138 for all complexes (Figure 3.29a). As the standard deviations of almost all measurements were so large, it could only be concluded that deuterium uptake had occurred, as all deuterium uptake values exceeded the defined key value 0.17 Da, but their flexibility could not be confidently ranked from residues 130-138. In order to gain more insights into the flexibility differences of SDN loop among the complexes, residues 127-138 were also utilized for comparison, even though TEM-17A-Cef (E104K) did not cover this peptide (Figure 3.29b). It showed that TEM-19A-Cef (G238S) was slightly more flexible than the wild type TEM-1A-Cef and TEM-15A-Cef (G238S/E104K), which flexibilities of latter two complexes were similar. Two complexes containing the M182T, TEM-20A-Cef (G238S/M182T) and TEM-52A-Cef (G238S/E104K/M182T), had a slightly more compact SDN loop. The above observations showed that SDN loop was an extremely important region to tightly hold the cefotaxime, therefore the deuterium uptake efficiency of this region was hindered. Besides, the stabilizing effect of M182T might also act on the loop so that to compensate the effects caused by the key mutation, especially for G238S.

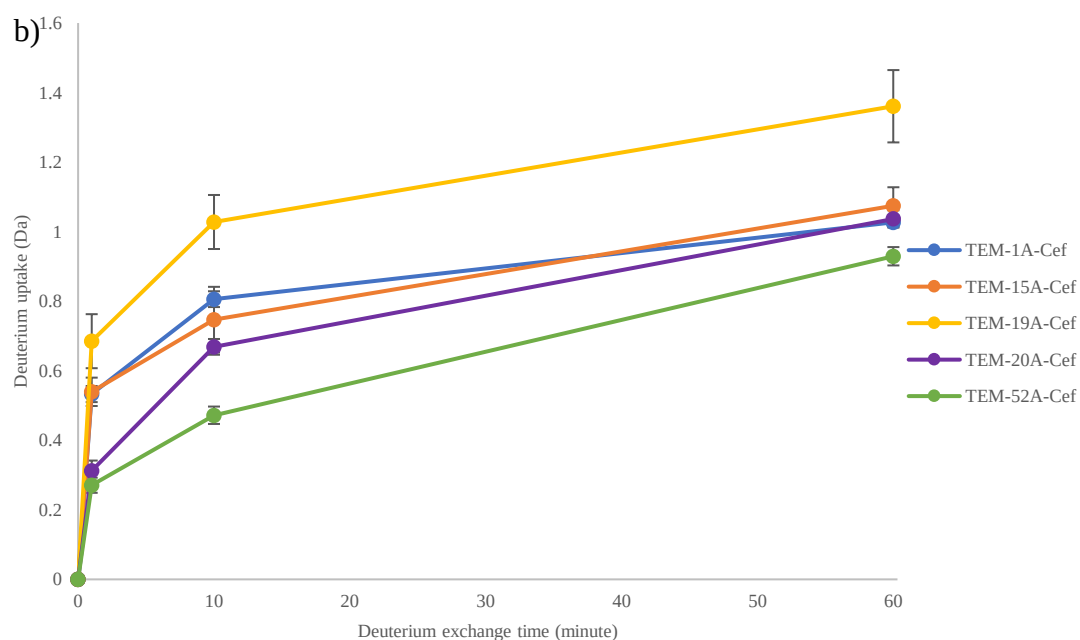
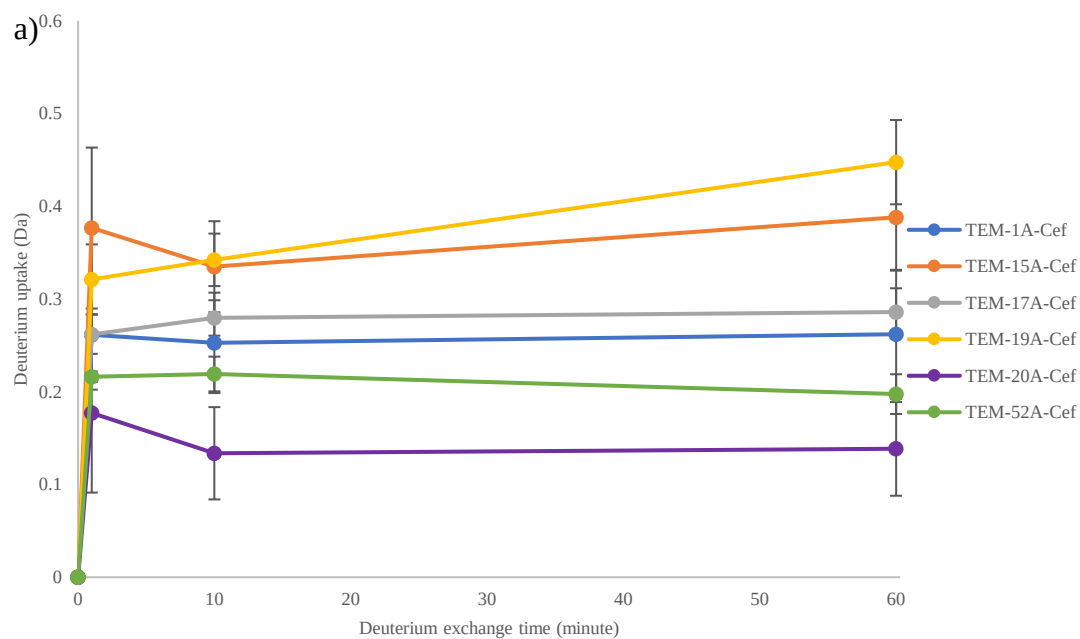


Figure 3.29 Deuterium uptake plot of residues (a) 130-138 and (b) 127-138 for cefotaxime-bound TEM-type β -lactamases.

β-3 strand (Residues 231-248)

β-3 strand in complexes was flexible, as the deuterium uptake of all complexes in this region reached a plateau after 1 min of HDX reaction (Figure 3.30). This might be due to the relatively long length of residue 231-248, which also covers part of the β-4 strand. However, all peptides covering the residue 238 and the β-3 strand had similar length, so the β-3 strand flexibility could only be determined by this peptide. The deuterium uptake showed that TEM-19A-Cef (G238S) was the most flexible, followed by TEM-15A-Cef (G238S/E104K) and TEM-52A-Cef (G238S/E104K/M182T). These three complexes were more flexible than the wild type TEM-1A-Cef and TEM-17A-Cef (E104K), which had similar flexibility to the TEM-1A-Cef. Only TEM-20A-Cef (G238S/M182T) showed a more compact conformation than TEM-1A-Cef. The flexibility of this region for these complexes indicated that the β-3 strand might be responsible for maintaining a larger cavity for antibiotics like cefotaxime which possessed a bulky side chain to effectively orient and fit in the active site. The reduced flexibility of TEM-20A-Cef might imply this region might be over-flexible for complexes containing key mutation G238S only, and therefore flexibility of β-3 strand in TEM-19A-Cef was required to be controlled by the M182T point mutation.

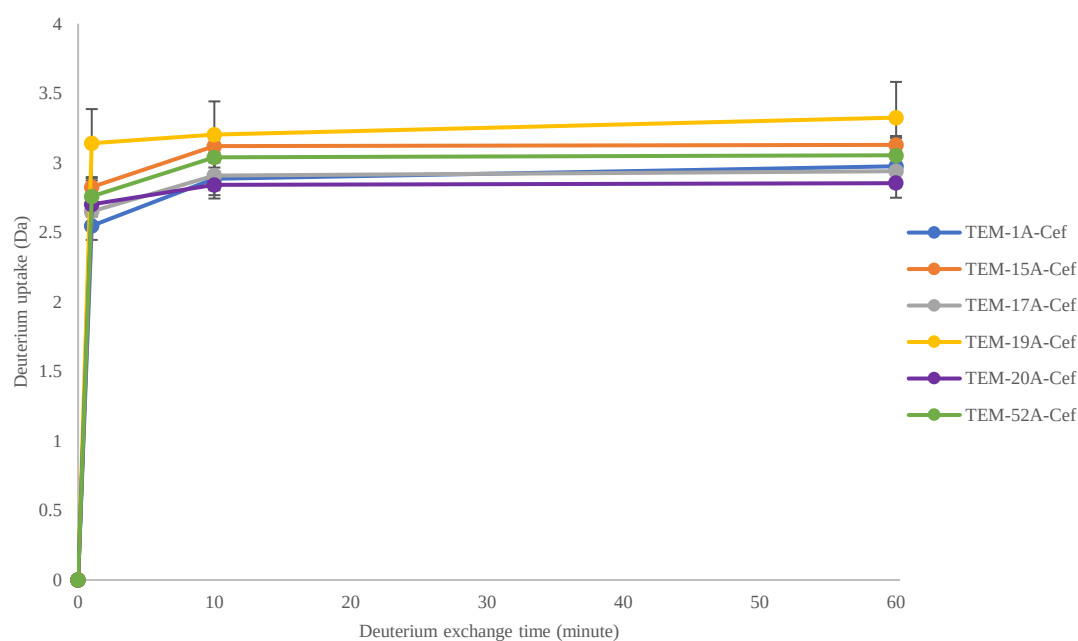


Figure 3.30 Deuterium uptake plot of residues 231-248 for cefotaxime-bound TEM-type β -lactamases.

Region containing residue 104 (Residues 103-117 and 103-118, H2/H3-loop)

Most of the β -lactamases complexes, except TEM-19A-Cef (G238S), had a more compact conformation after binding with the cefotaxime. TEM-17A-Cef (E104K) was the most compact one among complexes, followed by TEM-15A-Cef (G238S/E104K) and TEM-52A-Cef (G238S/E104K/M182T), both of which had similar flexibility. TEM-20A-Cef (G238S/M182T) was slightly more compact than TEM-1A-Cef (Figure 3.31). These observations indicated that the E104K point mutation reduced the flexibility of this region to ensure a proper conformation and orientation for effective antibiotics binding. Besides, M182T assisted to control the flexibility of this region in TEM-19A-Cef only, while no changes were found when M182T incorporated into TEM-15A-Cef, which might reflect the synergistic effect of G238S and E104K had already controlled the conformation of this region.

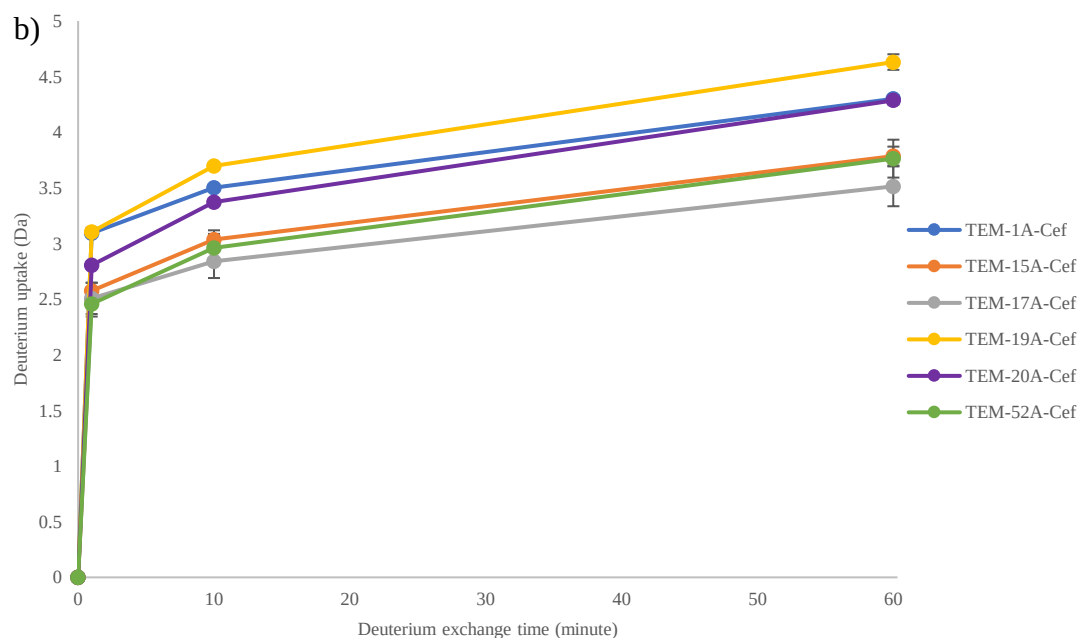
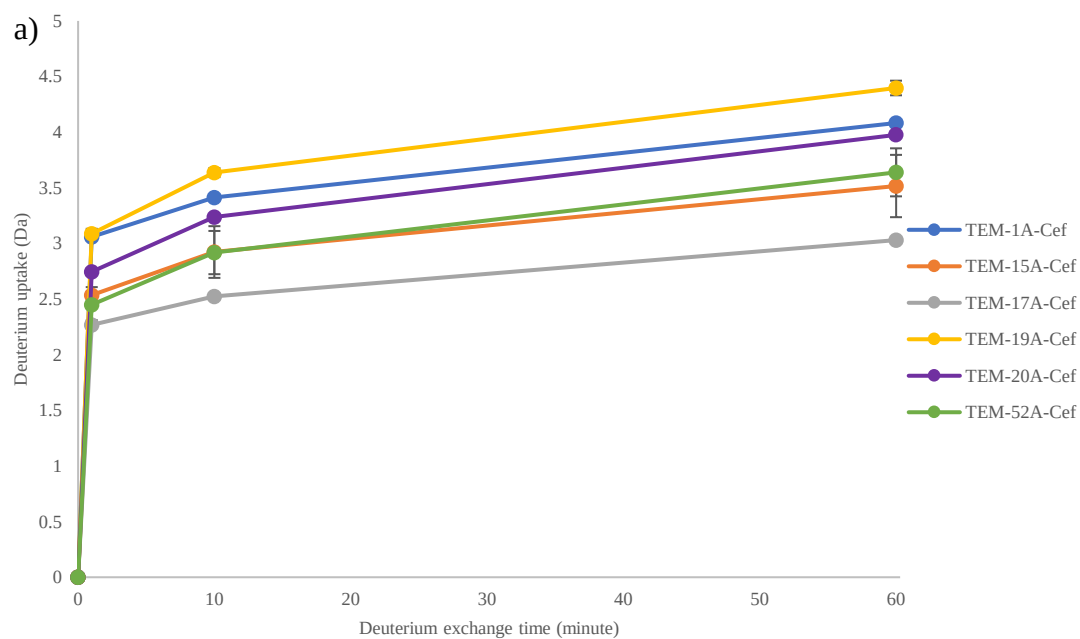
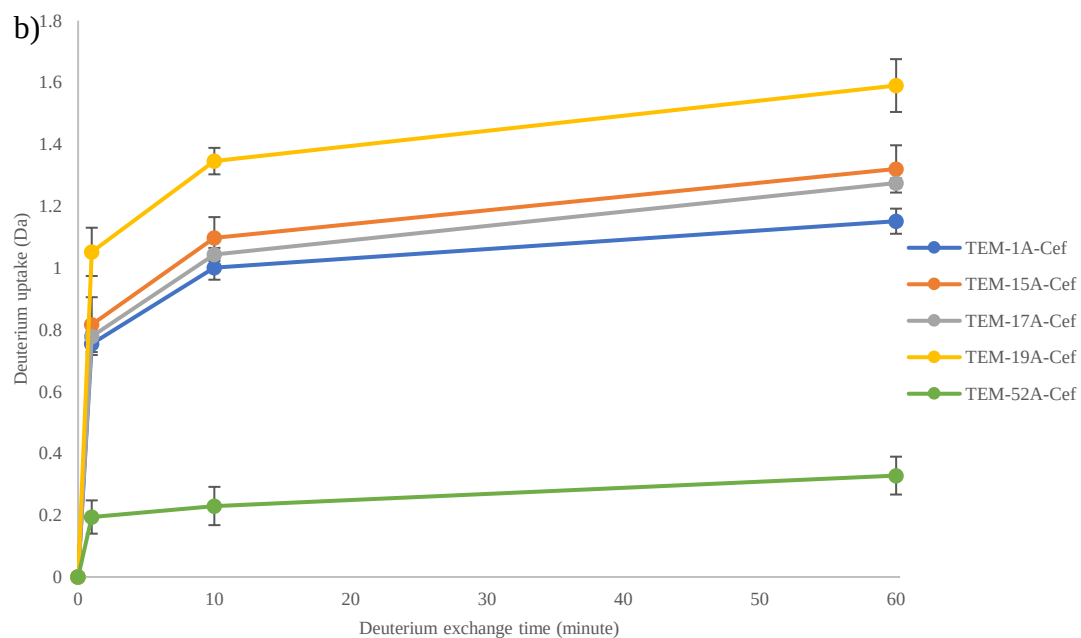
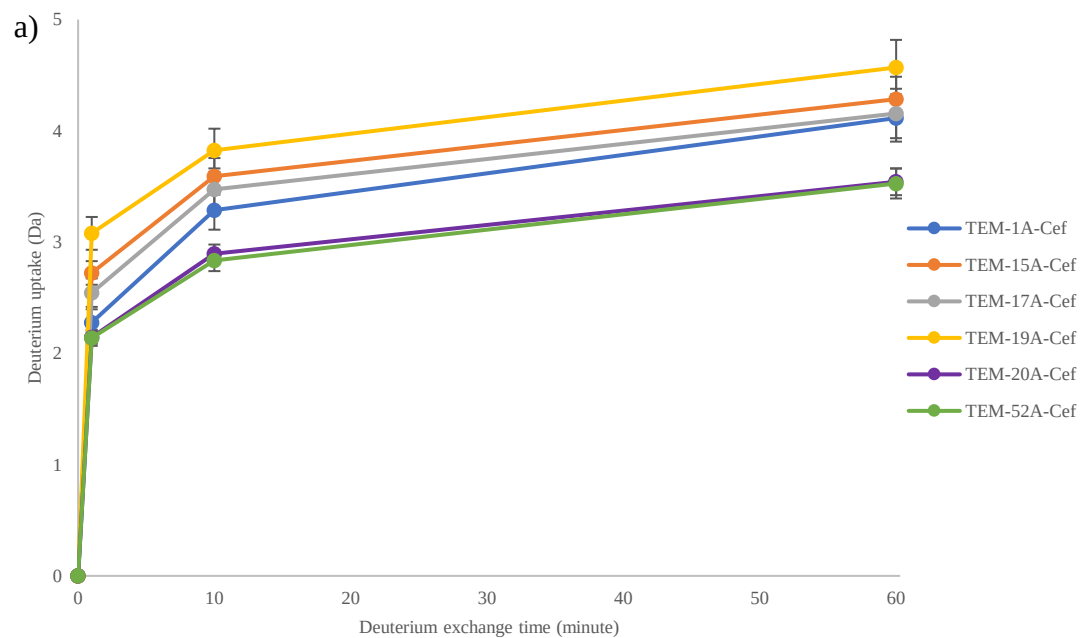


Figure 3.31 Deuterium uptake plot of residues (a) 103-117 and (b) 103-118 for cefotaxime-bound TEM-type β -lactamases.

Region containing residue 182 (Residues 178-186, H7/H8-loop and H8)

Only two peptides covering the residue 182 were found to be suitable for flexibility comparison. However, residues 178-186 was not found in TEM-20A-Cef and 170-186 had 17 amino acids in length, and also covered the previously discussed C-terminal Ω -loop. To facilitate the comparison, residues 170-179 (Figure 3.28a), which was the C-terminal Ω -loop, was also incorporated for comparison as combination of the deuterium uptake of residues 170-179 and 170-186 (Figure 3.32a) might provide insight into the deuterium uptake of residues 180-186, and residues 178-186 (Figure 3.32b) may serve as a cross-checking peptide to validate the result, even though TEM-20A-Cef result was missed. In fact, residues 180-186 was also found in complexes without M182T, therefore this peptide could be utilized as another cross-checking peptide (Figure 3.32c).

As previously discussed, the flexibility of the C-terminal Ω -loop was similar for all complexes (Figure 3.28a). Therefore, the theoretical trend of residues 180-186 was reasonably deduced from the trend of residues 170-186. It was found that TEM-19A-Cef (G238S) was the most flexible one, followed by TEM-15A-Cef (G238S/E104K). TEM-17A-Cef (E104K) was slightly more flexible than TEM-1A-Cef. However, for both mutants containing M182T, their flexibilities were lower than TEM-1A-Cef. Experimental trends in both residues 178-186 and 180-186 were consistent to the prediction. The triple mutant had its flexibility lower than the TEM-1A-Cef, and the order of flexibility for mutants containing only key mutation(s) followed the above prediction, i.e., the trend of residues 170-186.



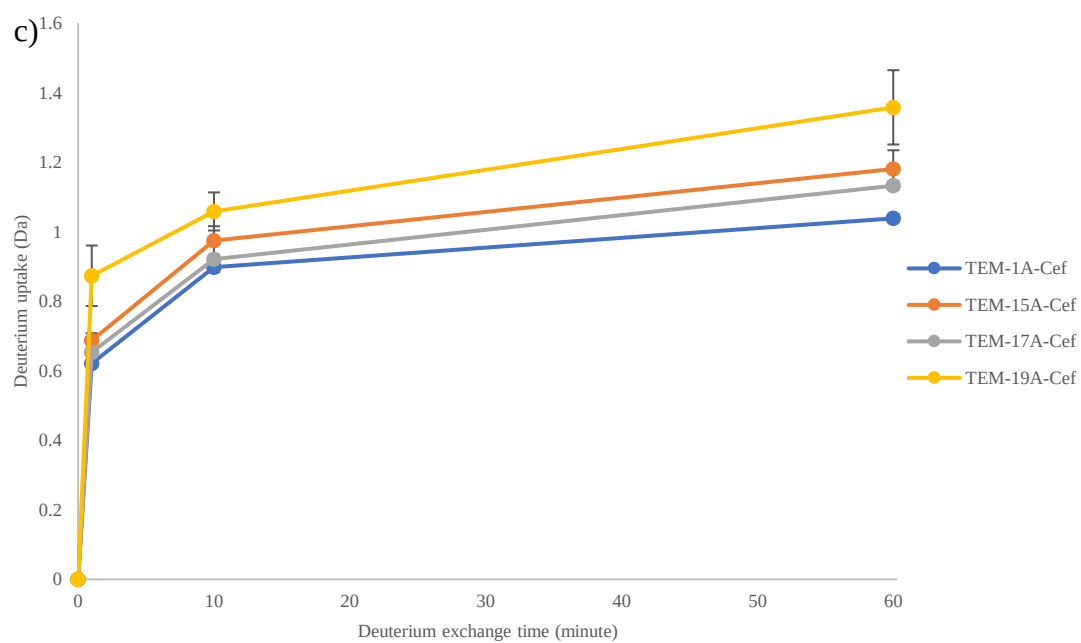


Figure 3.32 Deuterium uptake plot of residues (a) 170-186, (b) 178-186 and (c) 180-186 for cefotaxime-bound TEM-type β -lactamases.

H5-H6 region (Residues 140-150)

The flexibility of this region for the six complexes could clearly be separated into 3 groups. TEM-1A-Cef and TEM-17A-Cef (E104K) were in the same group with similar flexibility (Figure 3.33). TEM-19A-Cef (G238S) and TEM-15A-Cef (G238S/E104K) were another group, and they were significantly more flexible than the TEM-1A-Cef group. TEM-20A-Cef (G238S/M182T) and TEM-52A-Cef (G238S/E104K/M182T) were in the third group, and their flexibilities were slightly lower than that of the TEM-1A-Cef group. As H5-H6 region interacted with the N-terminal of Ω -loop, flexibility of this region might have to be optimized so as to facilitate the proper orientation of Ω -loop for cefotaxime binding. The two β -lactamase complexes containing only key mutation, TEM-19A-Cef and TEM-15A-Cef, had a very flexible H5-H6 region, and they were in contrast to the N-terminal Ω -loop in both complexes (Figure 3.28b), i.e., having a slightly compact conformation. On the other hand, both M182T containing complexes, TEM-20A-Cef and TEM-52A-Cef, had a more compact conformation, while Ω -loop of these two complexes did not have significant change in their flexibilities. This further confirmed the importance of the H5-H6 region that the flexibility change in the region compensated the effect of flexibility change in Ω -loop due to the incorporation of key mutations by M182T point mutation.

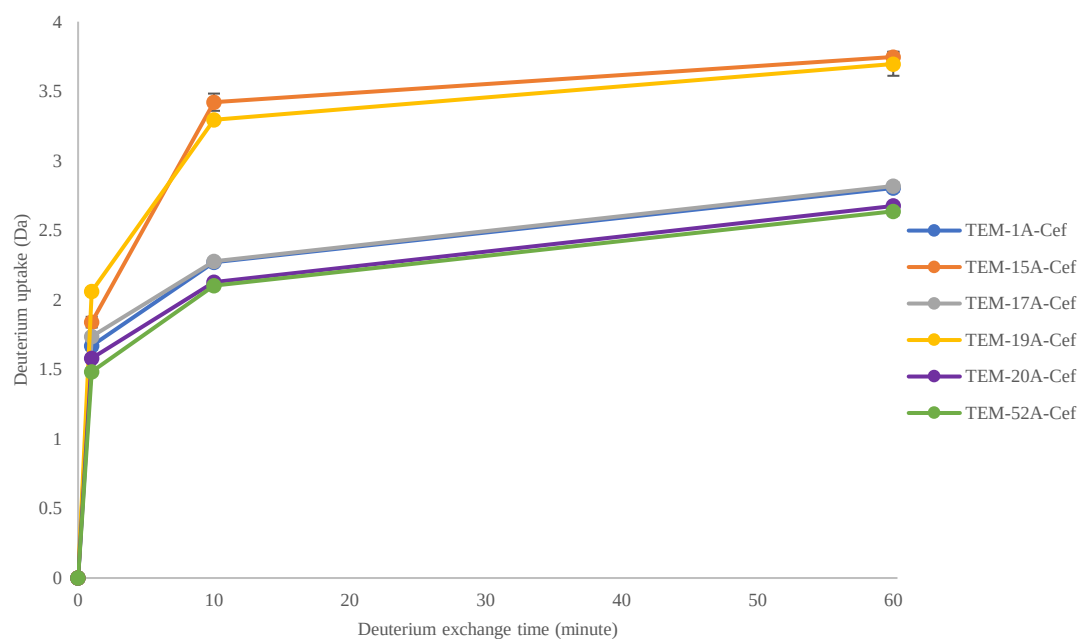
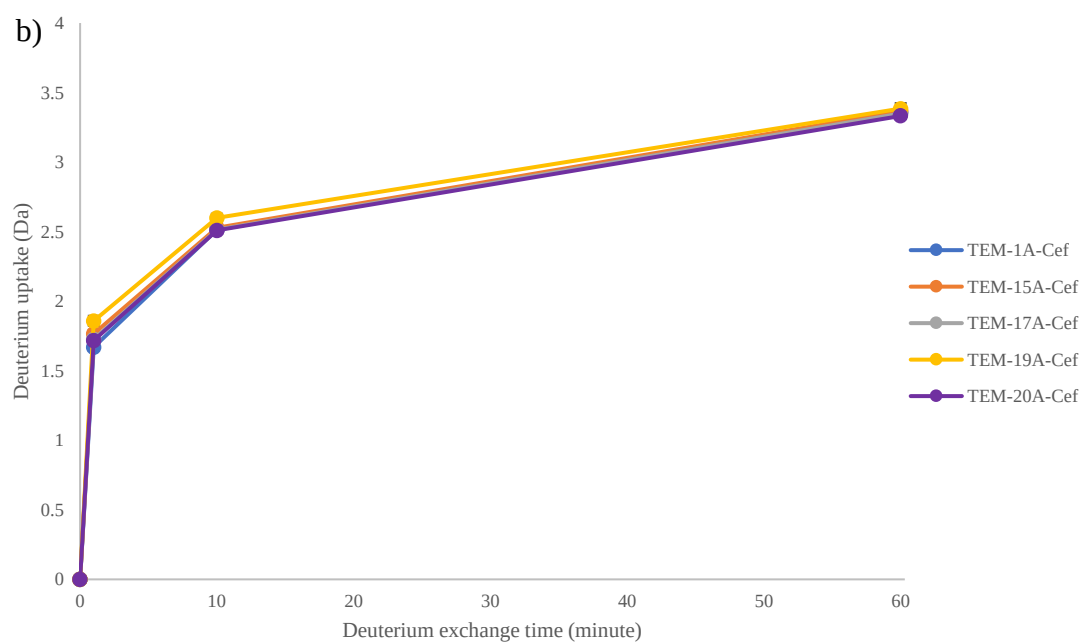
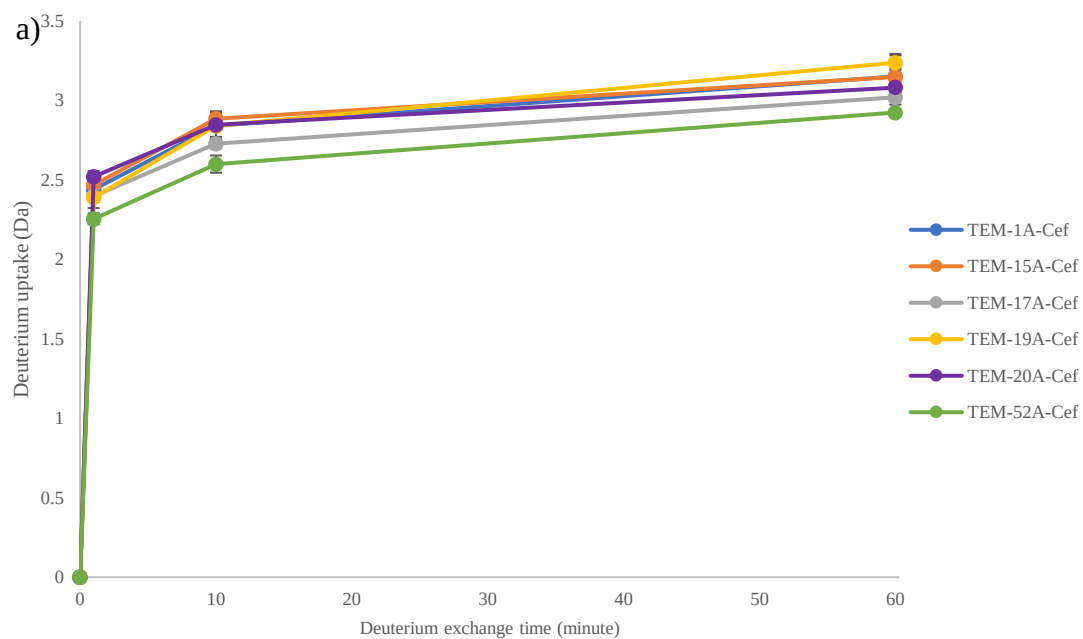


Figure 3.33 Deuterium uptake plot of residues 140-150 for cefotaxime-bound TEM-type β -lactamases.

An edge of β -sheet (Residues 52-59, 222-230 and 250-260)

The deuterium uptake plot of all the three peptides consisting of the edge of β -sheet showed that the flexibility of mutants was similar to that of TEM-1A-Cef. Only TEM-19A-Cef (G238S) was found to be slightly flexible in residues 250-260 (Figure 3.34c). This result indicated that the conformation of the edge was likely to be similar among complexes, which was similar to β -3 and β -4 strands (residue 231-248, Figure 3.30) in order to facilitate the cefotaxime binding.



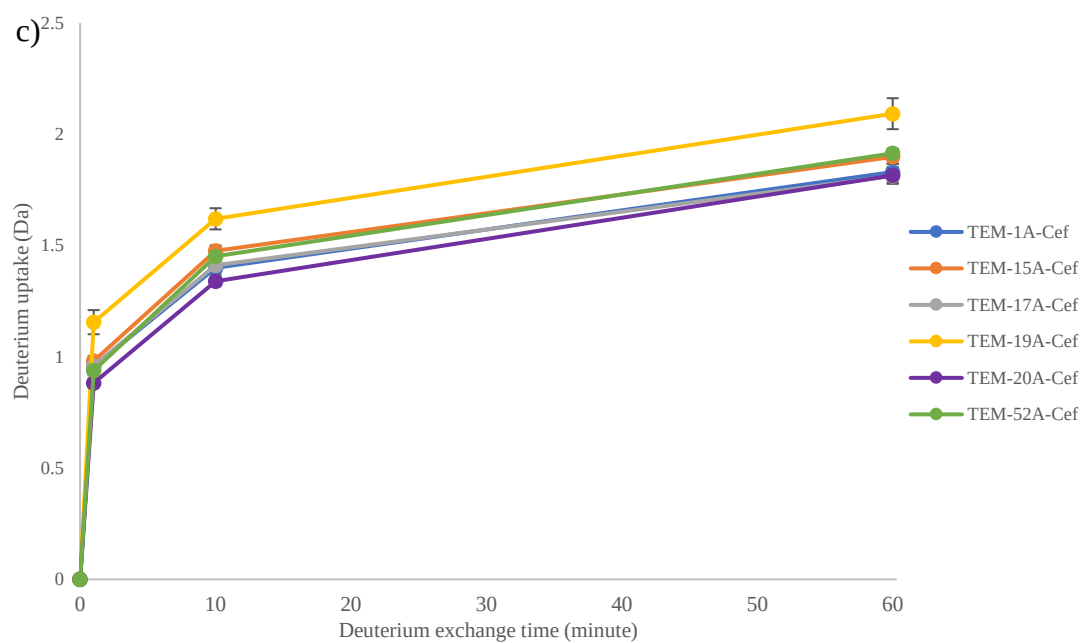
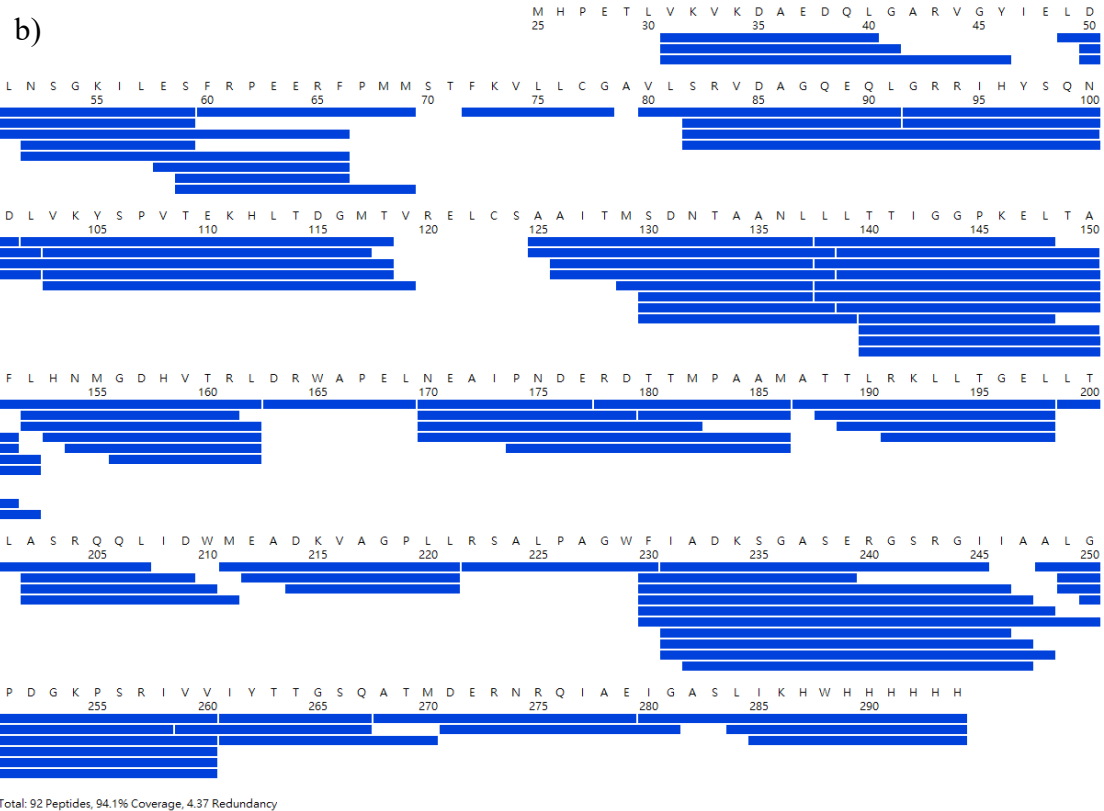


Figure 3.34 Deuterium uptake plot of residues (a) 52-59, (b) 222-230 and (c) 250-260 for cefotaxime-bound TEM-type β -lactamases.

3.4.6. Comparison of conformational changes in localized regions of six E166A-TEM-type β -lactamases upon cefotaxime binding

The coverage maps of the six comparisons between each E166A-TEM-type β -lactamase and its cefotaxime-bound complex were shown in Figure 3.35. The number of common peptides (ranging from 74 to 120 peptides) in each comparison and sequence coverage (ranging from 82.2% to 94.1%) were adequate and covered almost all important regions in TEM-type β -lactamase (Table 3.7). Similar to sections 3.3.3 and 3.4.5, if necessary, indirect comparison would be utilized to provide insights into the conformational changes upon cefotaxime binding for several important regions. Difference plots of comparisons of cefotaxime-bound E166A-TEM-type β -lactamases having one different point mutations were shown in Figure 3.36.





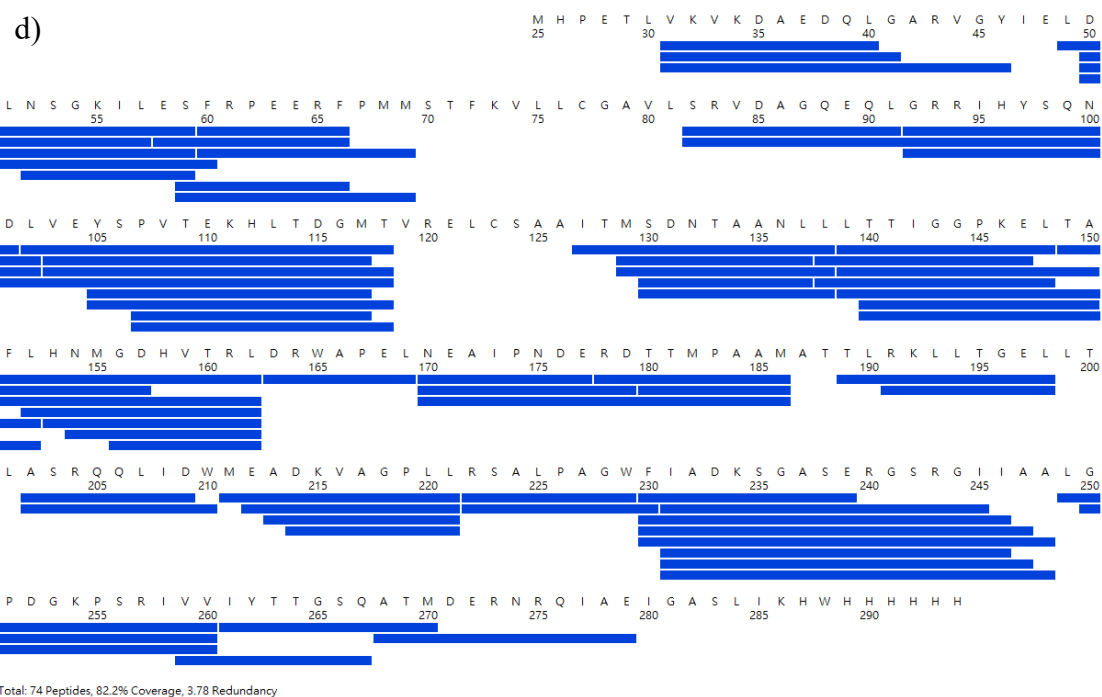
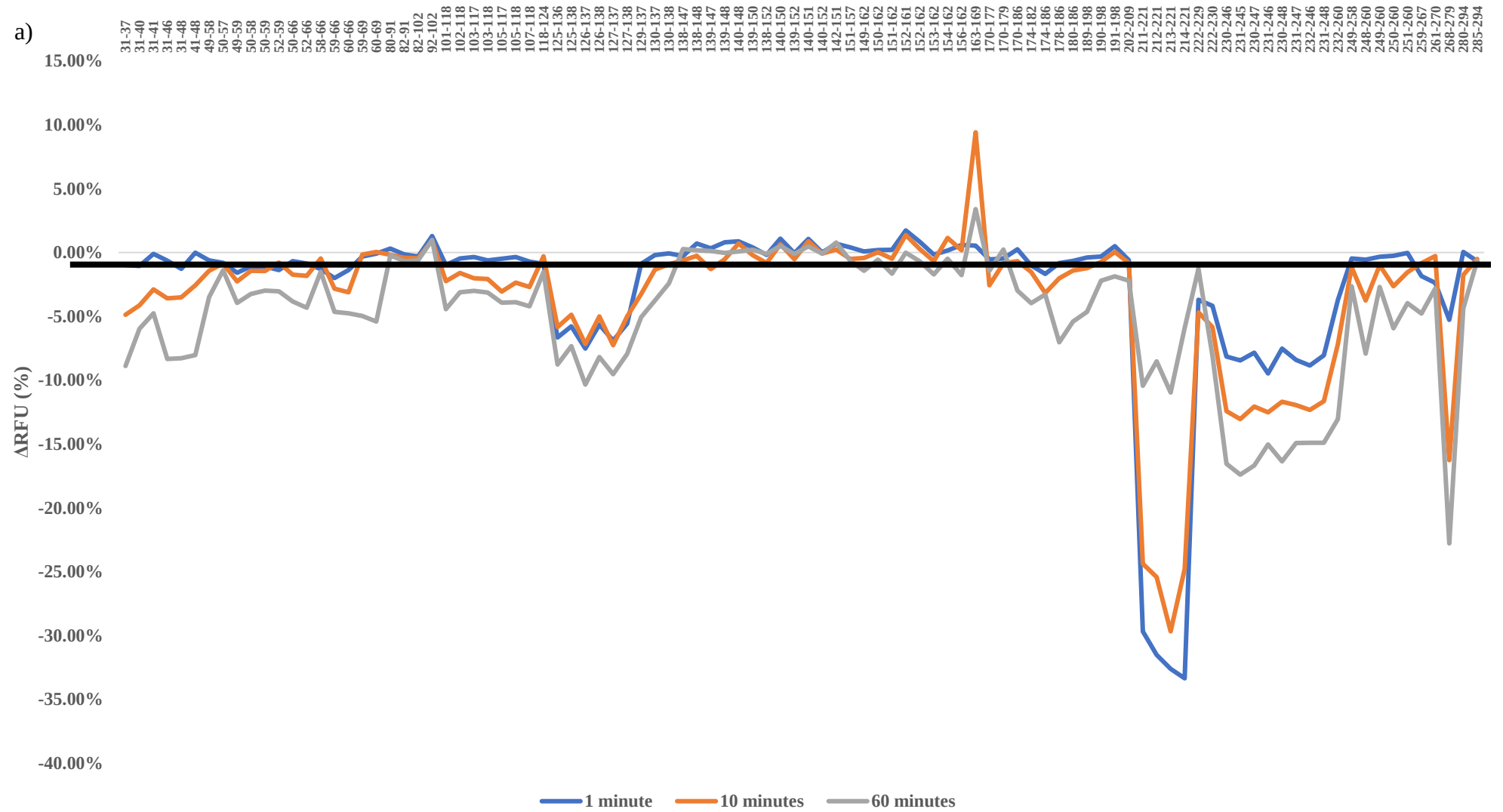


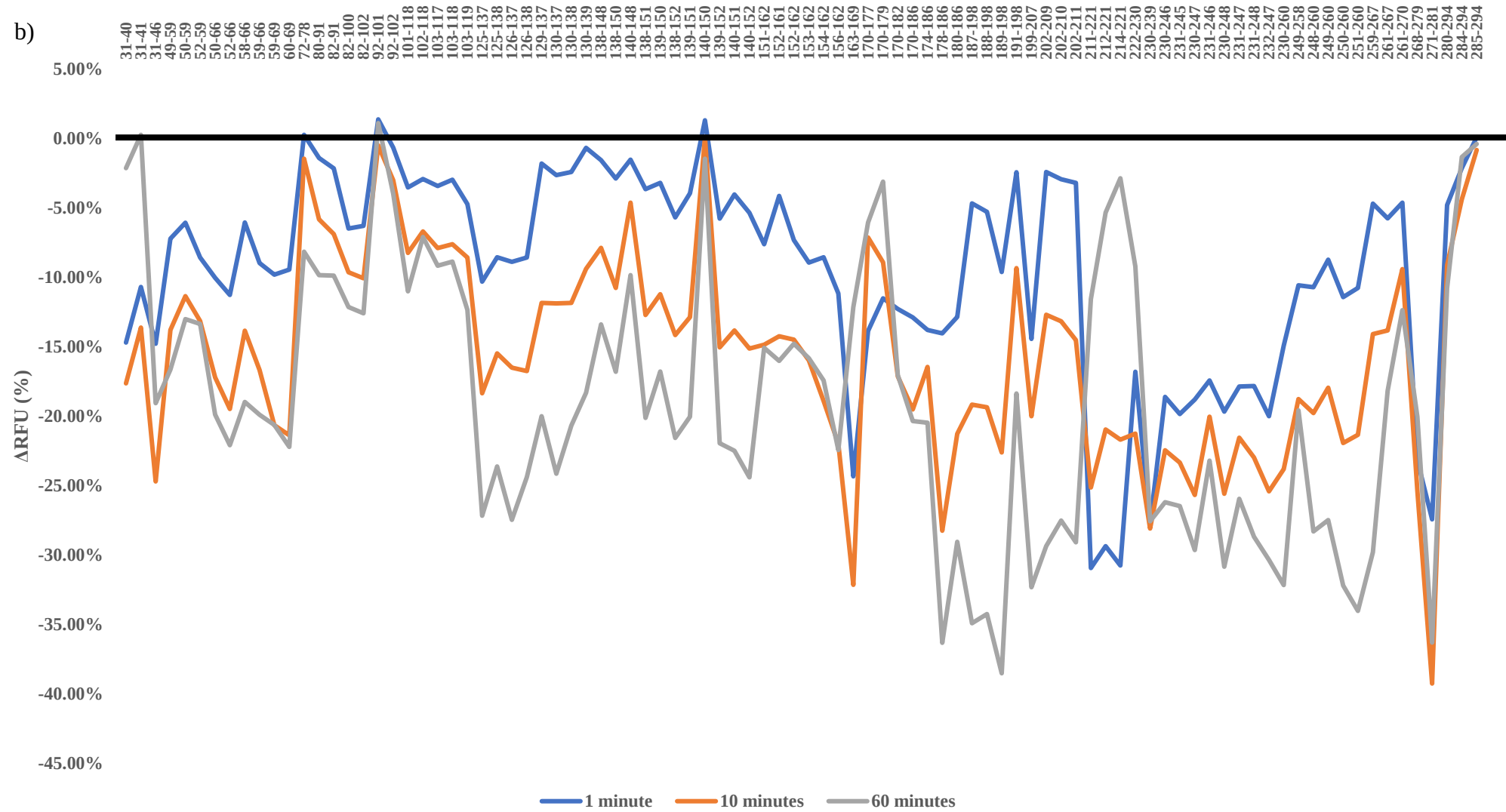


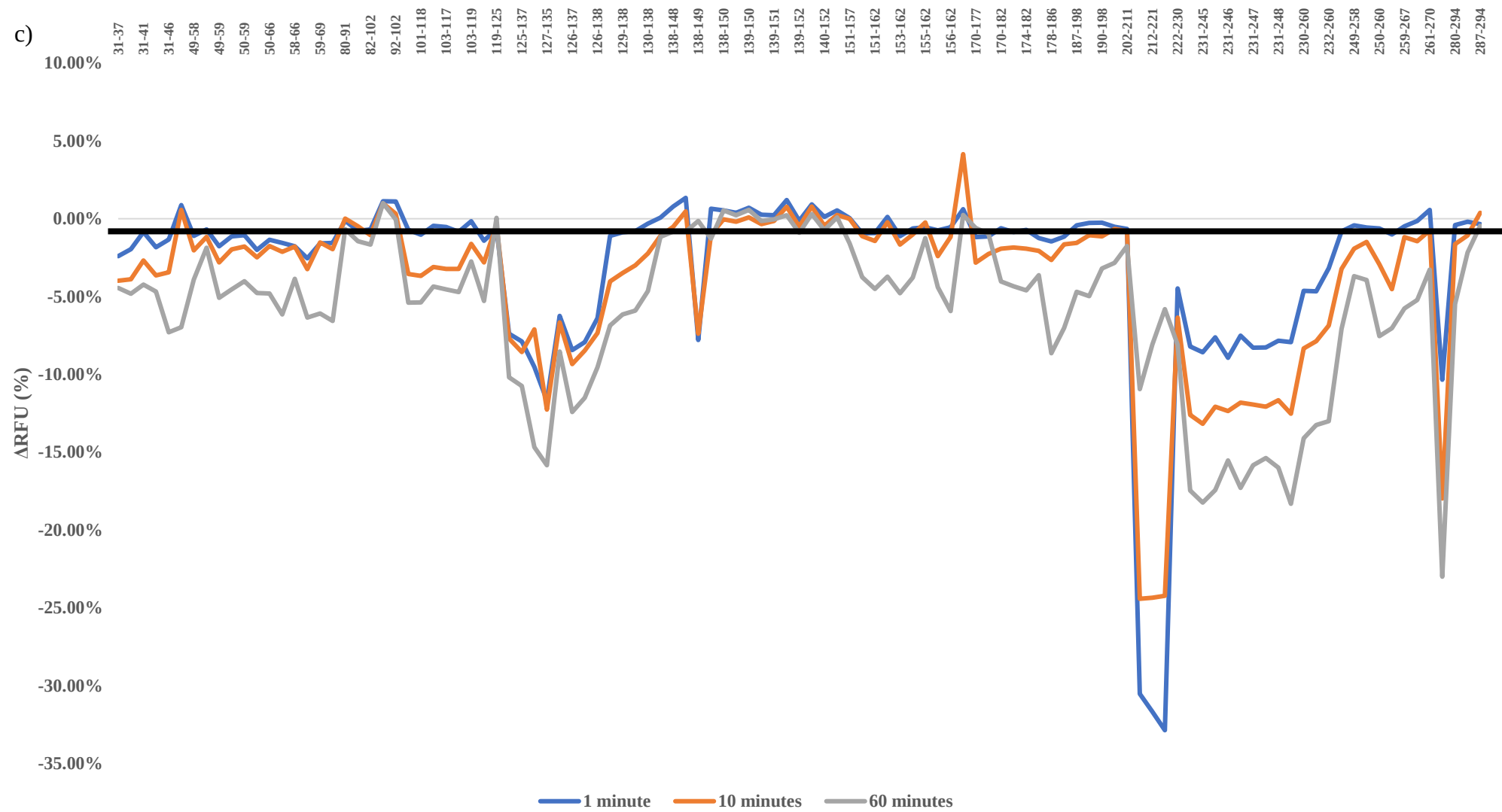
Figure 3.35 Coverage map of comparison between (a) TEM-1A and TEM-1A-Cef, (b) TEM-15A and TEM-15A-Cef, (c) TEM-17A and TEM-17A-Cef, (d) TEM-19A and TEM-19A-Cef, (e) TEM-20A and TEM-20A-Cef and (f) TEM-52A and TEM-52A-Cef.

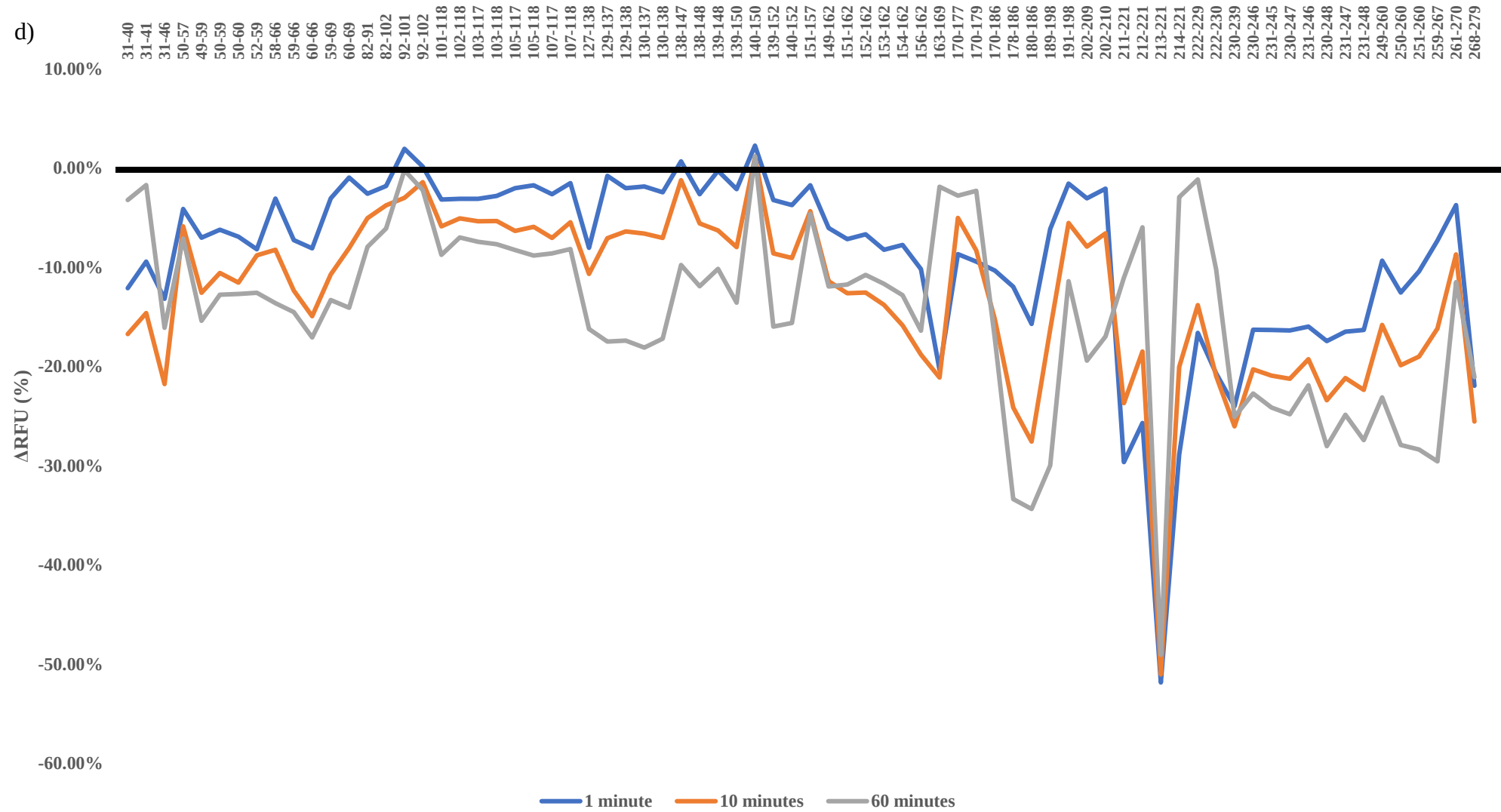
Table 3.7 Number of peptides generated, sequence coverage, redundancies and number of common peptides resulted from pepsin digestion of each comparison between E166A-TEM-type β -lactamases and their cefotaxime-bound complexes.

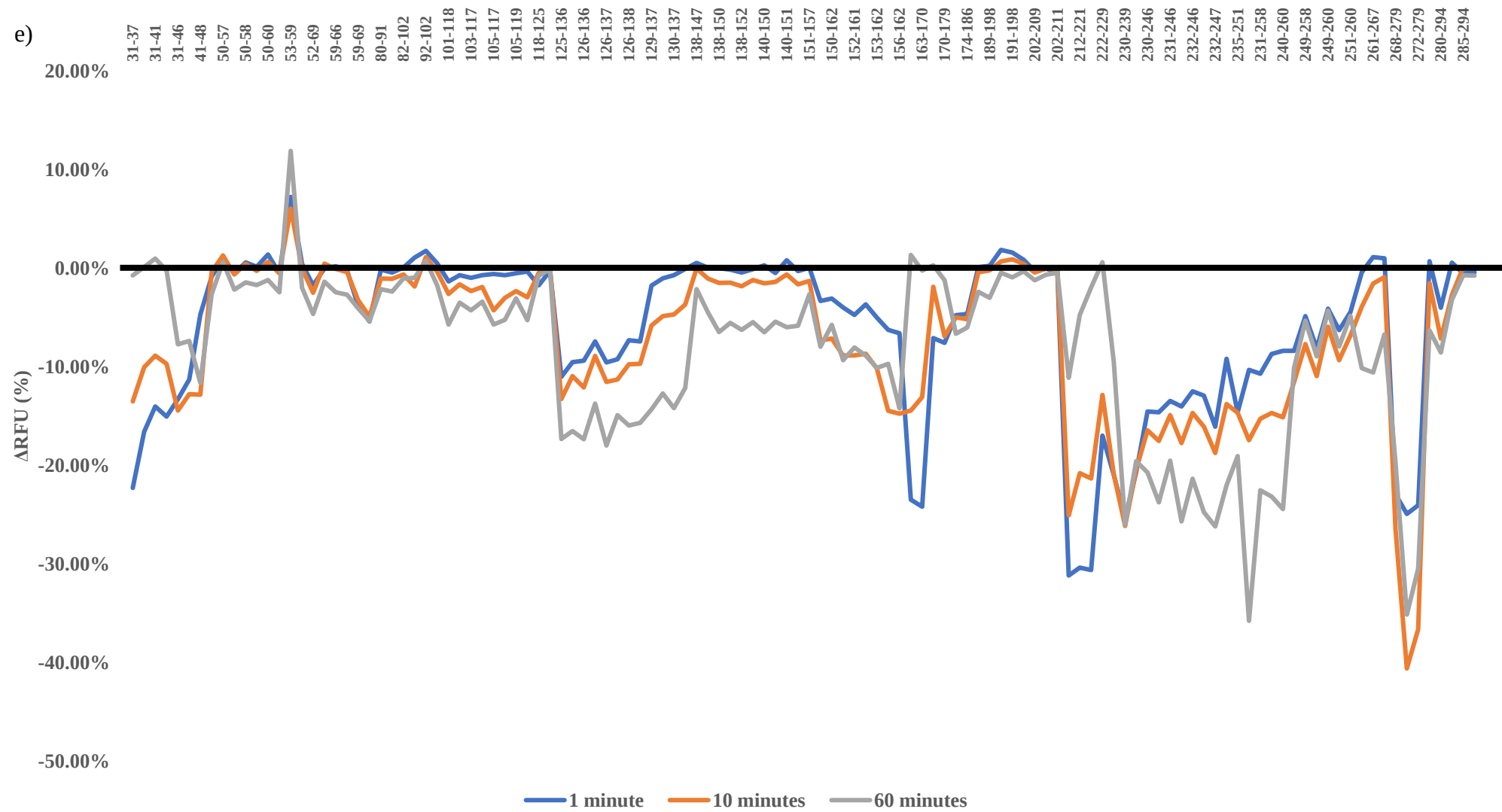
Comparison with its cefotaxime- bound complex	Number of common peptides	Sequence coverage	Redundancy
TEM-1A	98	91.9%	4.61
TEM-15A	92	94.1%	4.37
TEM-17A	109	93.0%	5.26
TEM-19A	74	82.2%	3.78
TEM-20A	120	94.1%	5.55
TEM-52A	100	91.1%	5.65
Common peptides	32	69.3%	2.14











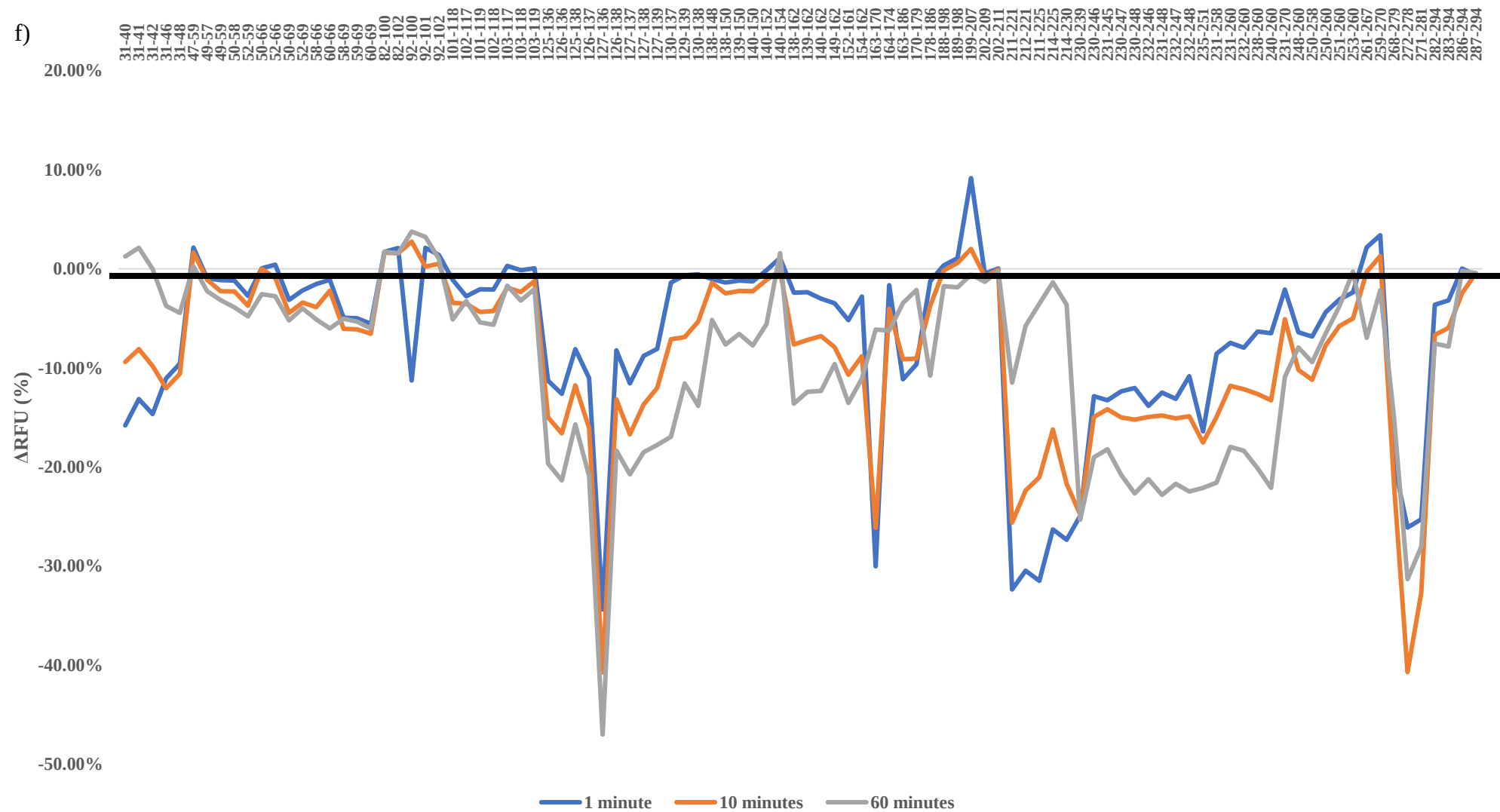


Figure 3.36 Different plot of comparison between (a) TEM-1A and TEM-1A-Cef (b) TEM-15A and TEM-15A-Cef, (c) TEM-17A and TEM-17A-Cef, (d) TEM-19A and TEM-19A-Cef, (e) TEM-20A and TEM-20A-Cef and (f) TEM-52A and TEM-52A-Cef.

Regions near active serine-70 (Residues 59-69, β 2/H2-loop)

The binding between cefotaxime and E166A-TEM-type β -lactamases showed that all complexes had flexibility reduction in this region (Figure 3.37). This was obviously due to S70 that was directly involved in the antibiotic hydrolysis, therefore this region must become more compact so as to interact with cefotaxime. Comparison of the conformational changes of the six β -lactamases showed that TEM-15A-Cef (G238S/E104K) and TEM-19A-Cef (G238S) had the greatest reduction in flexibility, followed by the two mutants having M182T, i.e., TEM-52A-Cef (G238S/E104K/M182T) and TEM-20A-Cef (G238S/M182T). TEM-17A-Cef (E104K) had a slightly greater change than the wild type TEM-1A-Cef, but their changes were relatively minor compared with the other four mutants. Overall, conformational changes found in this region were expected and reasonable.

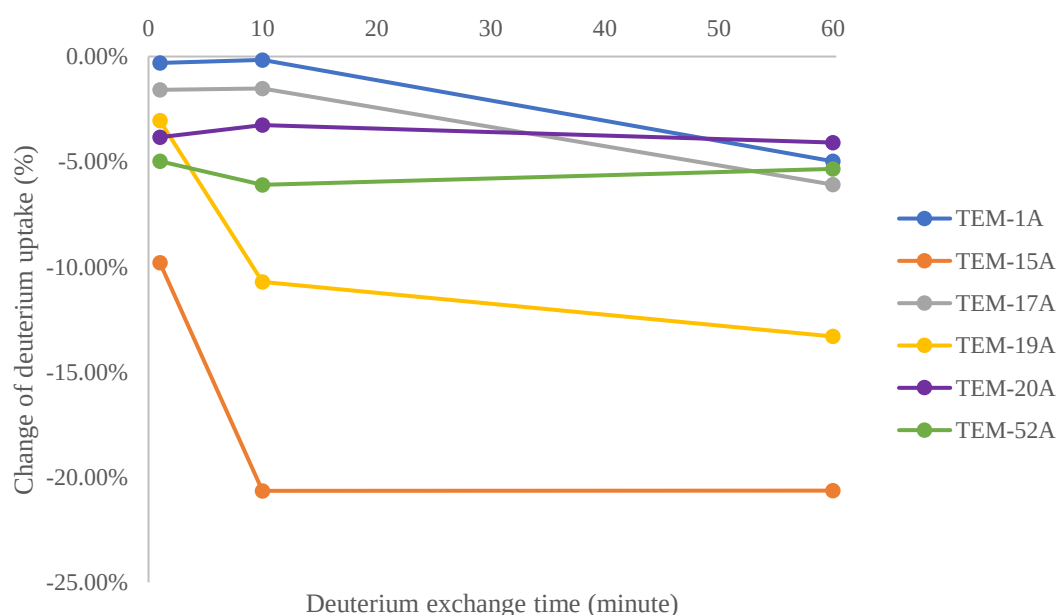
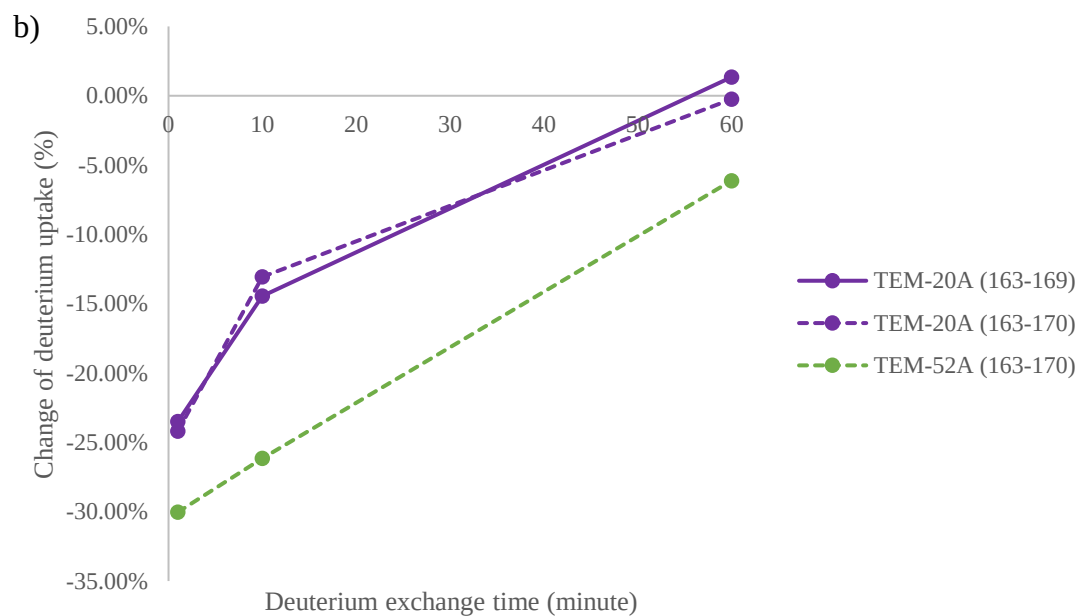
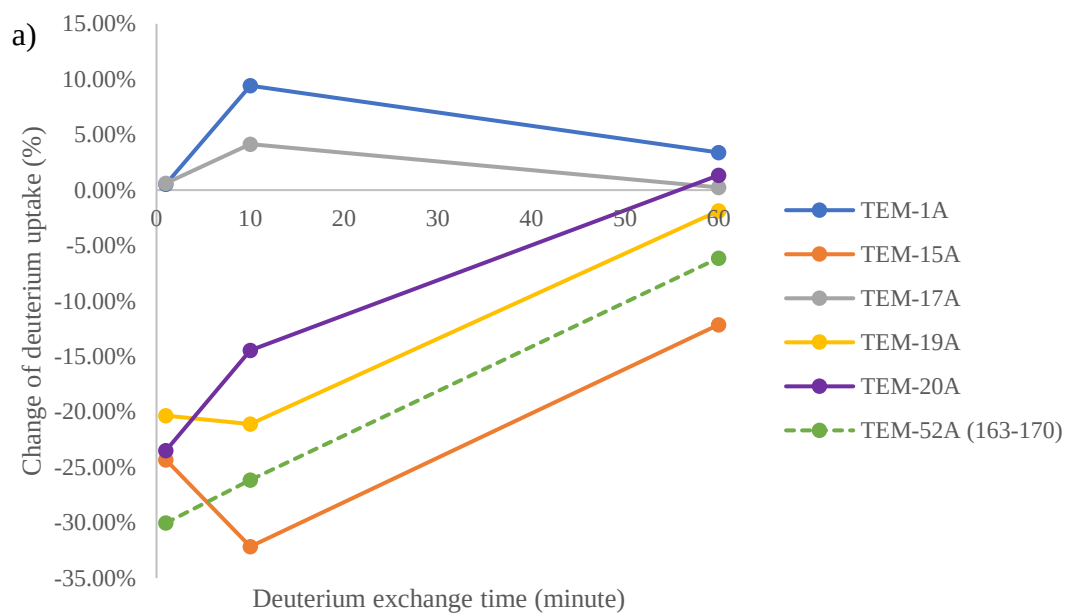


Figure 3.37 Plot of change of deuterium uptake of residues 59-69 upon cefotaxime binding.

Ω-loop (Residues 163-169 and 170-179)

The N-terminal Ω-loop, residues 163-169, covered five β-lactamases except TEM-52A-Cef (G238S/E104K/M182T) (Figure 3.38a). Since Ω-loop was a catalytically important region, in order to incorporate the triple mutant into the comparison, residues 163-170, which were covered by TEM-20A-Cef (G238S/M182T) and TEM-52A-Cef only, were also utilized. Trends of TEM-20A-Cef (G238S/M182T) for both residues 163-169 and 163-170, as well as their percentage changes of deuterium uptake were similar (Figure 3.38b), therefore it was reasonable to incorporate the trend of TEM-52A-Cef in residues 163-170 to residues 163-169 in order to make a more comprehensive comparison (Figure 3.38a). The comparison showed that TEM-15A-Cef (G238S/E104K) had the greatest reduction in flexibility, however incorporation of M182T into TEM-15A-Cef (i.e., TEM-52A-Cef) rendered less flexibility reduction after binding. A similar observation was also found in TEM-19A-Cef (G238S) and TEM-20A-Cef. Interestingly, TEM-17A-Cef (E104K) and TEM-1A-Cef only showed a minor change after binding, and they were even slightly more flexible than their apo-forms, in their N-terminal Ω-loop conformations. This might be due to the whole binding pockets in TEM-1 and TEM-17 were not flexible enough when compared with binding pockets in other mutants. Therefore, a flexible Ω-loop in these two β-lactamases was required to relieve the steric hinderance for accommodation of the bulky cefotaxime. Similar changes were also found in the C-terminal Ω-loop, and reflected by the residues 170-179 (Figure 3.38c). However, in general, the conformational changes were not as obvious and significant as the N-terminal Ω-loop.



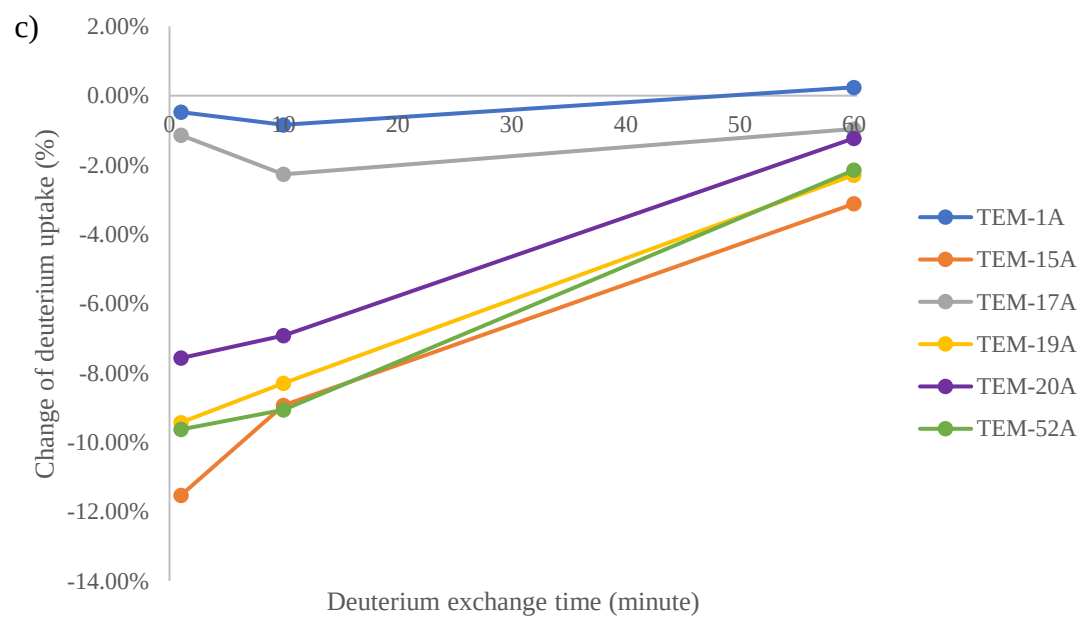


Figure 3.38 Plot of change of deuterium uptake of residues (a) 163-169 and 163-170 (TEM-52A only), (b) 163-169 (TEM-20A only) and 163-170 (TEM-20A and TEM-52A), (c) 170-179 upon cefotaxime binding.

SDN loop (Residues 130-138)

As discussed in the comparison for conformation of complexes, SDN loop region had a very compact conformation after binding with cefotaxime, and the flexibility was similar for all complexes. Therefore, the extent of conformational change mainly depended on the initial flexibility of β -lactamase. The comparison showed that extent of uptake reduction followed the order of TEM-15A-Cef (G238S/E104K) > TEM-19A-Cef (G238S) > TEM-52A-Cef (G238S/E104K/M182T) > TEM-20A-Cef (G238S/M182T) >> TEM-17A-Cef (E104K) > TEM-1A-Cef, which was consistent with the observation made in previous comparisons (Figure 3.39).

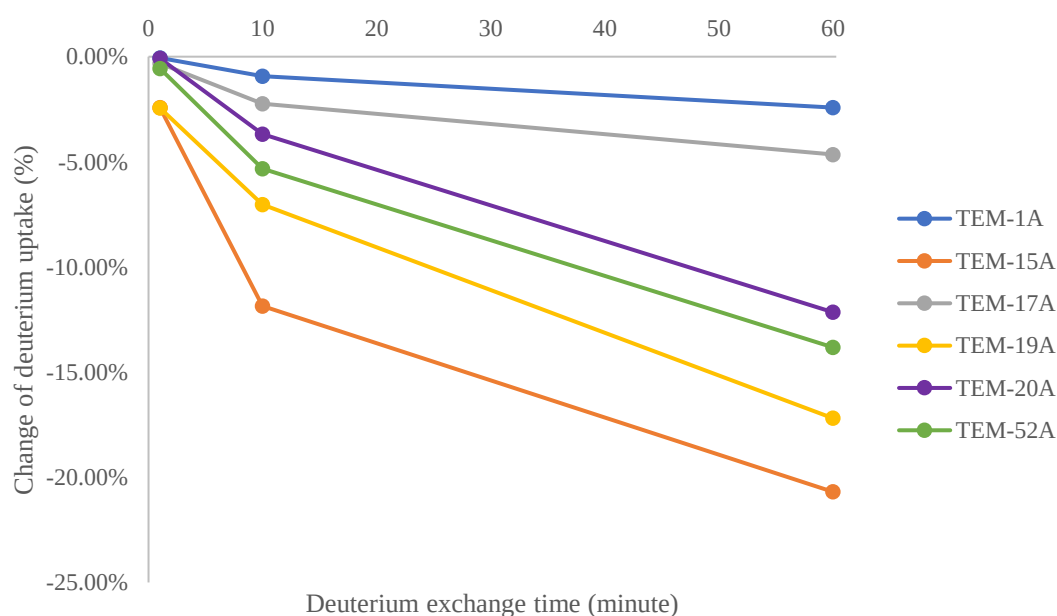


Figure 3.39 Plot of change of deuterium uptake of residues 130-138 upon cefotaxime binding.

β -3 strand (Residues 230-246)

The binding of cefotaxime with all β -lactamases showed an obvious and great reduction in flexibility for β -3 strand. Changes in β -3 strand were also expectedly consistent with that of the region near S70, as β -3 strand is also an active site consisting region and directly involves in the hydrolysis. Similar to several previous comparisons, TEM-15A-Cef (G238S/E104K) and TEM-19A-Cef (G238S) showed the greatest changes (Figure 3.40). Interestingly, after incorporation of M182T, flexibility reduction after binding in TEM-52A-was less than TEM-20A-Cef. TEM-17A-Cef (E104K) had similar flexibility reduction as TEM-1A-Cef, but changes for these two ESBLs were still obvious, implying the importance of this region for the antibiotic hydrolysis.

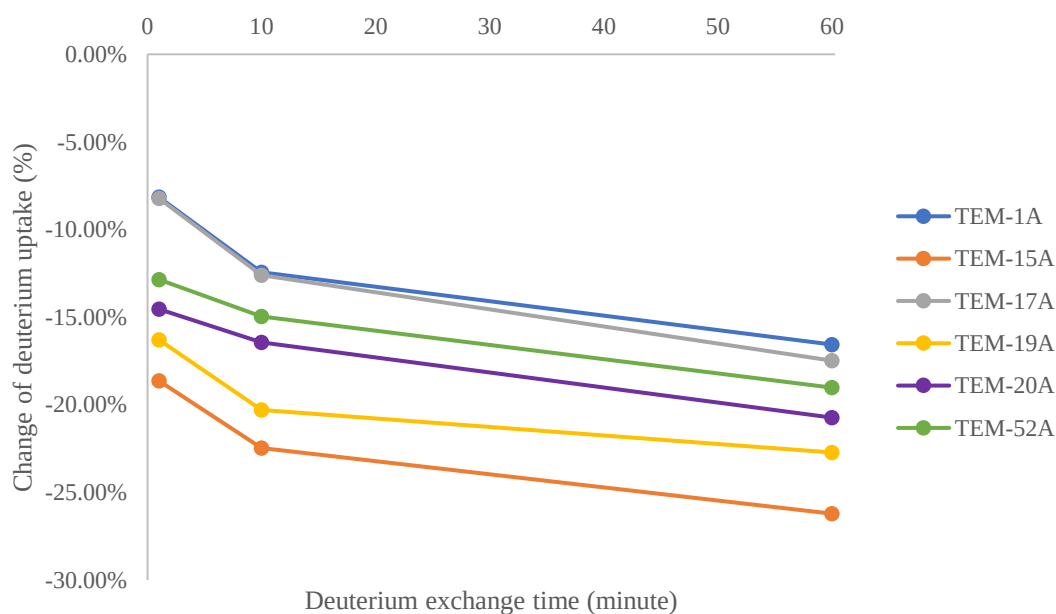
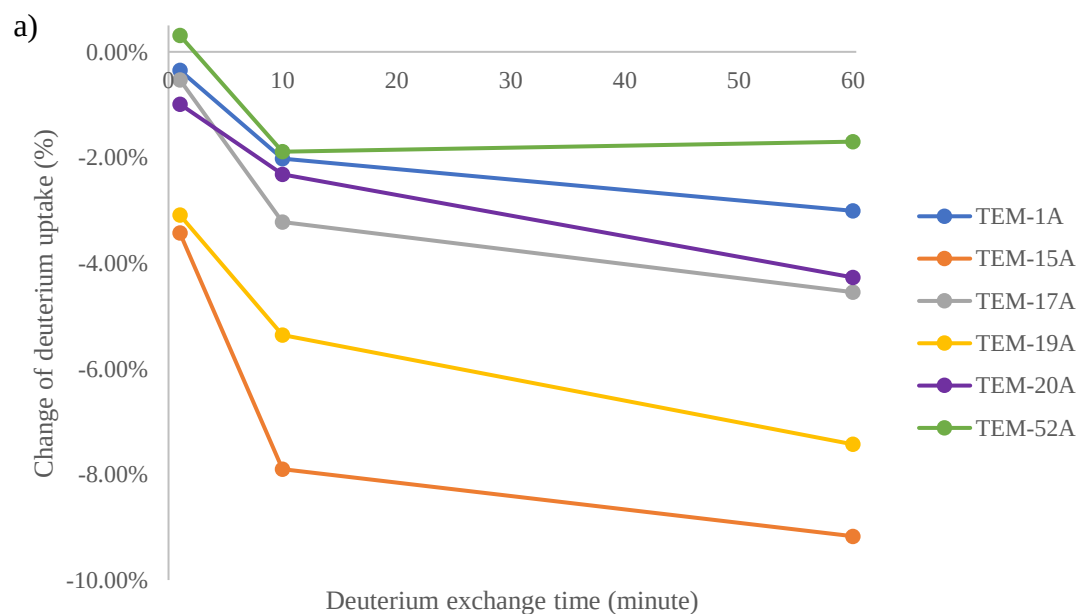


Figure 3.40 Plot of change of deuterium uptake of residues 230-246 upon cefotaxime binding.

Region containing residue 104 (Residues 103-117 and 103-118, H2/H3-loop)

Conformational changes of this region due to binding was not very significant for all six TEM-type β -lactamases, as less than 10% changes of deuterium uptake was observed. However, among the six ESBLs, changes were more obvious for TEM-15A-Cef (G238S/E104K) and TEM-19A-Cef (G238S). TEM-17A-Cef (E104K) also had an obvious change, which was reasonable as the mutation directly distorted its original conformation. Both M182T-containing mutant, TEM-52A-Cef (G238S/E104K/M182T) and TEM-20A-Cef (G238S/M182T), had similar and minor changes as the TEM-1A-Cef (Figure 3.41). This might suggest this region might be important towards cefotaxime binding for mutants possessing key point mutations



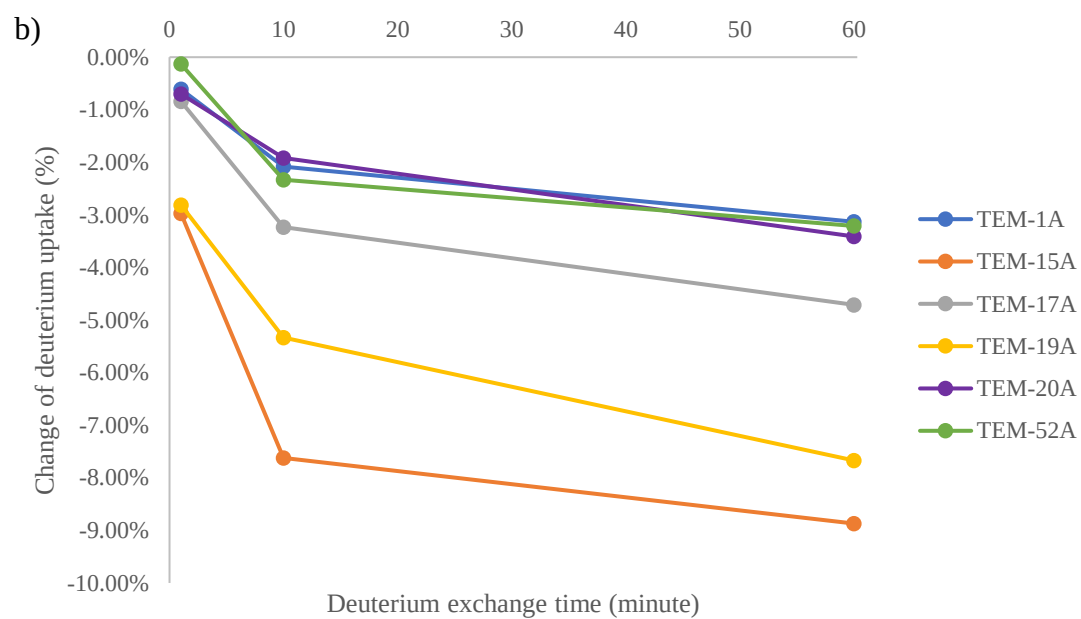
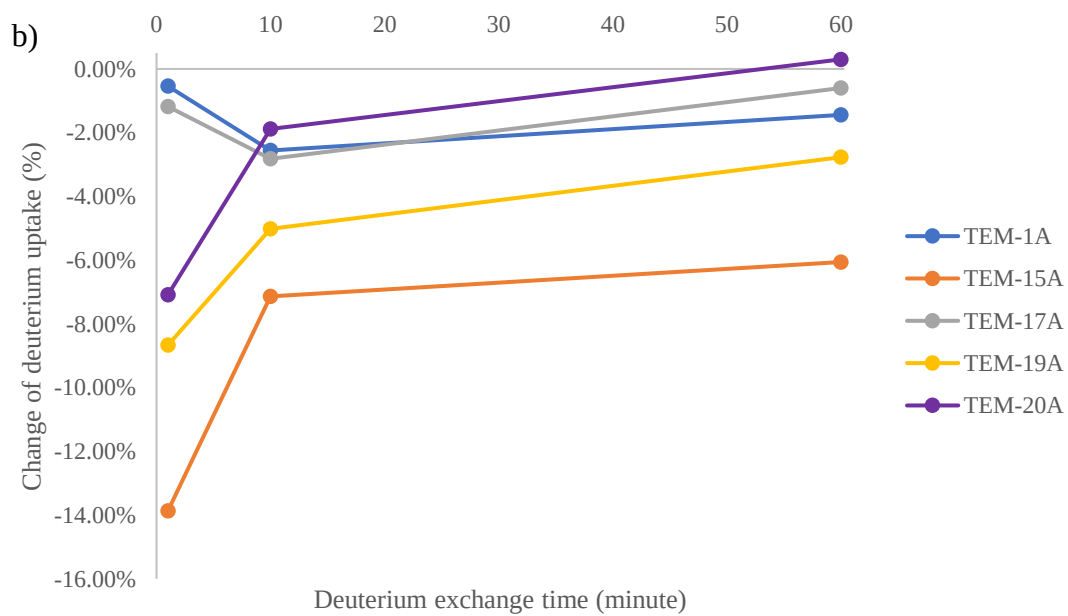
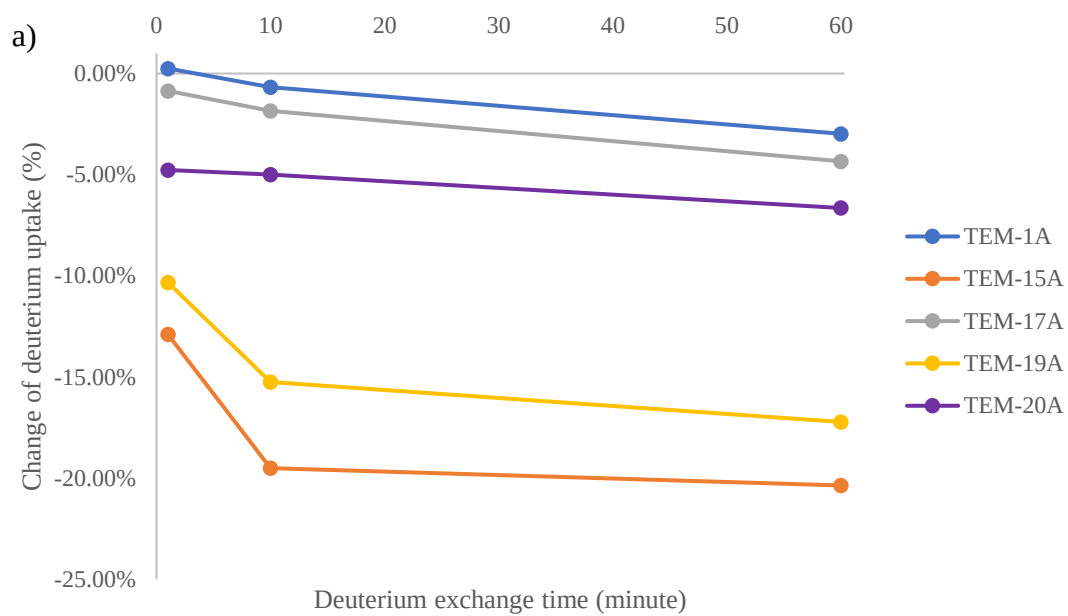


Figure 3.41 Plot of change of deuterium uptake of residues (a) 113-117 and (b) 113-118 upon cefotaxime binding.

Region containing residue 182 (170-186, 178-186 and 180-186, H7/H8-loop and H8)

Similar to the discussion of this region for the cefotaxime-bound ESBL complexes, an indirect comparison is required. Both residues 170-177 and 170-186 were not covered by TEM-52A-Cef (G238S/E104K/M182T), and combination of these two peptides could provide the theoretical trend of peptide 178-186. In the experiment, residues 178-186 was found but not covered by TEM-20A-Cef (G238S/M182T). A comprehensive comparison could be made by combining these results. Residues 180-186, which were not covered by TEM-20A-Cef and TEM-52A-Cef, could be used as the cross-checking peptide.

By combining the residues 170-186 and 170-177 (Figure 3.42a and b), which resulted in a theoretical trend of residues 178-186, it was clearly observed that TEM-15A-Cef (G238S/E104K) and TEM-19A-Cef (G238S) had the most significant reduction in flexibility after binding, and similar observations were also found in residues 180-186 (Figure 3.42c). However, the trend of the remaining mutants was not totally clear, but at least changes of deuterium uptake for TEM-1A-Cef, TEM-17A-Cef and TEM-20A-Cef were less than TEM-15A-Cef and TEM-19A-Cef (Figure 3.42a-c). The experimental trend of residues 178-186 was consistent to the above discussion, with an additional information that the triple mutant also had a similar change as the TEM-1A-Cef (Figure 3.42d). Therefore, conformational changes in this region were obvious for TEM-15A-Cef and then followed by TEM-19A-Cef, while others had a minor change and changes were similar to TEM-1A-Cef.



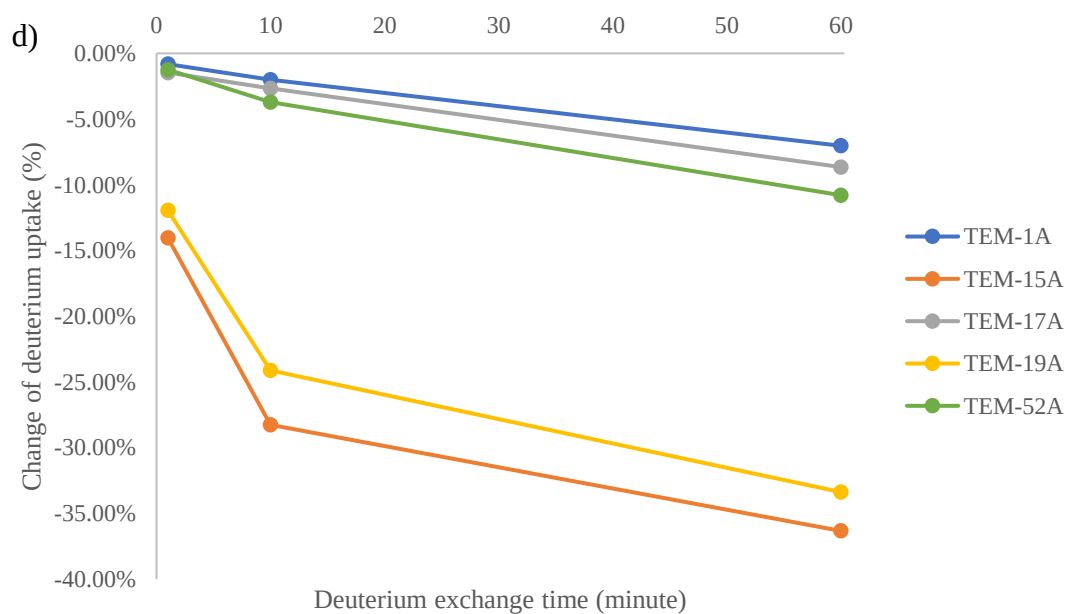
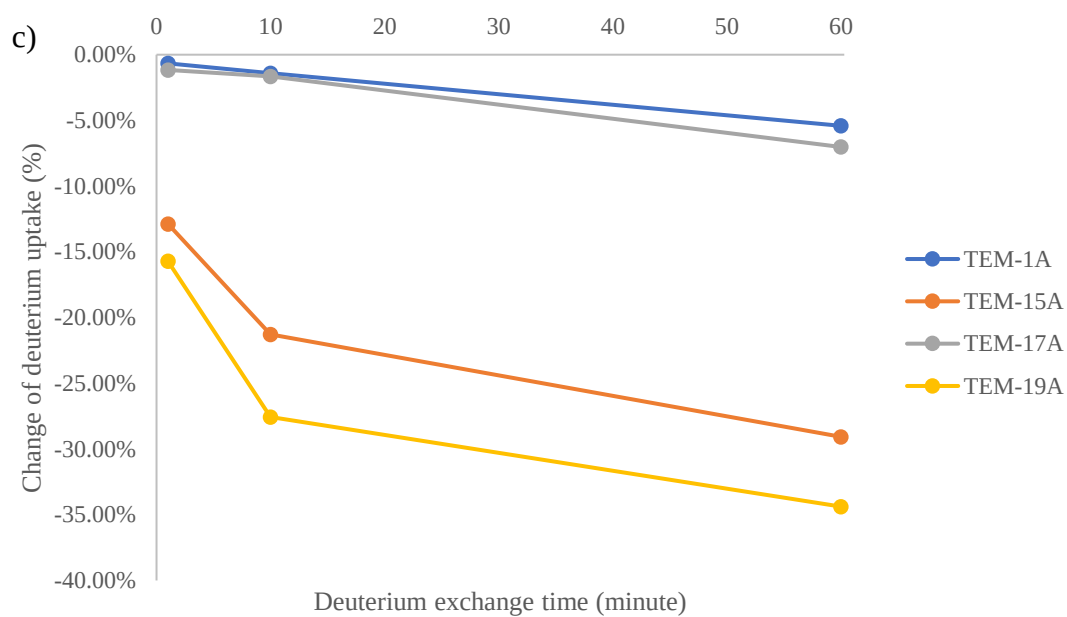


Figure 3.42 Plot of change of deuterium uptake of residues (a) 170-186, (b) 170-177, (c) 180-186 and (d) 178-186 upon cefotaxime binding.

H5-H6 region (Residues 140-150)

The conformational changes after cefotaxime binding in this region was clearly shown in the mutant containing M182T only. H5-H6 region of TEM-20A-Cef (G238S/M182T) and TEM-52A-Cef (G238S/E104K/M182T) became more compact (Figure 3.43). For other mutants and TEM-1A-Cef, only insignificant changes or no changes were observed. This implied that H5-H6 was one of the regions for M182T mutation to control or stabilize the overall β -lactamase conformation, especially for those TEM-type β -lactamases containing key mutation(s).

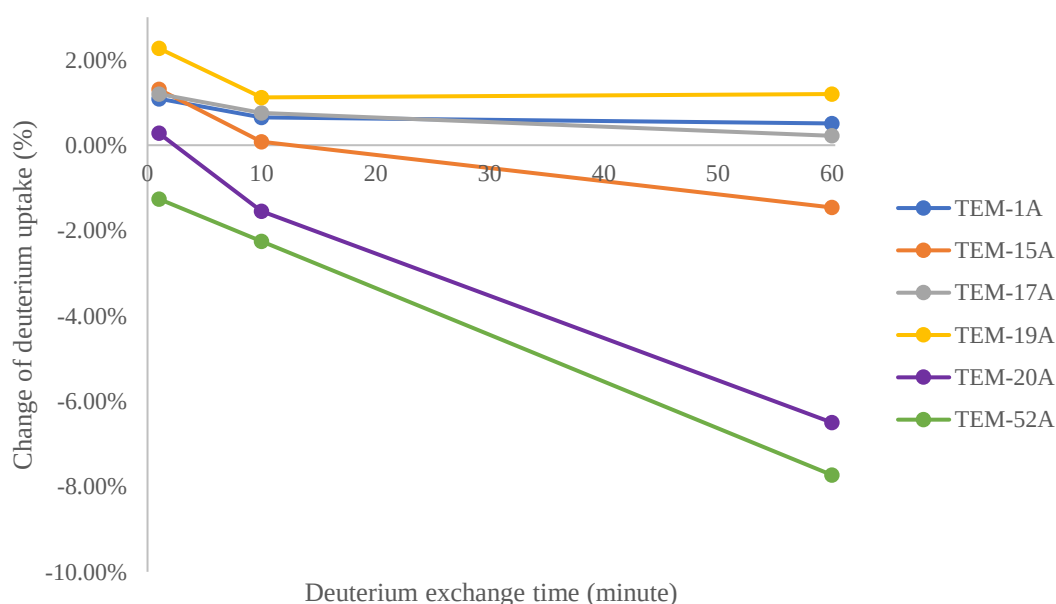
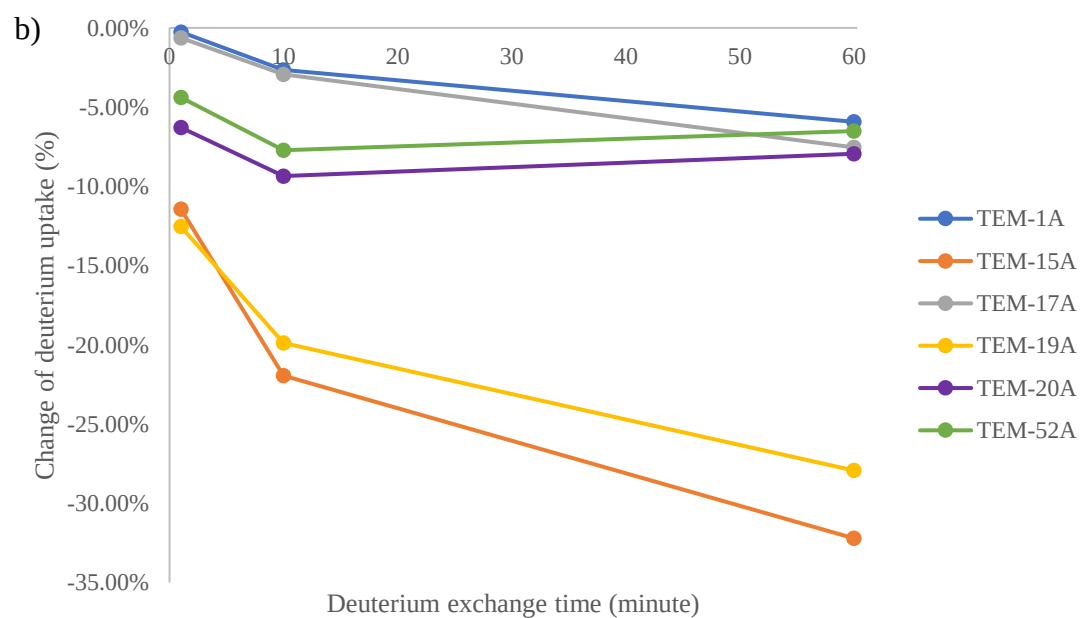
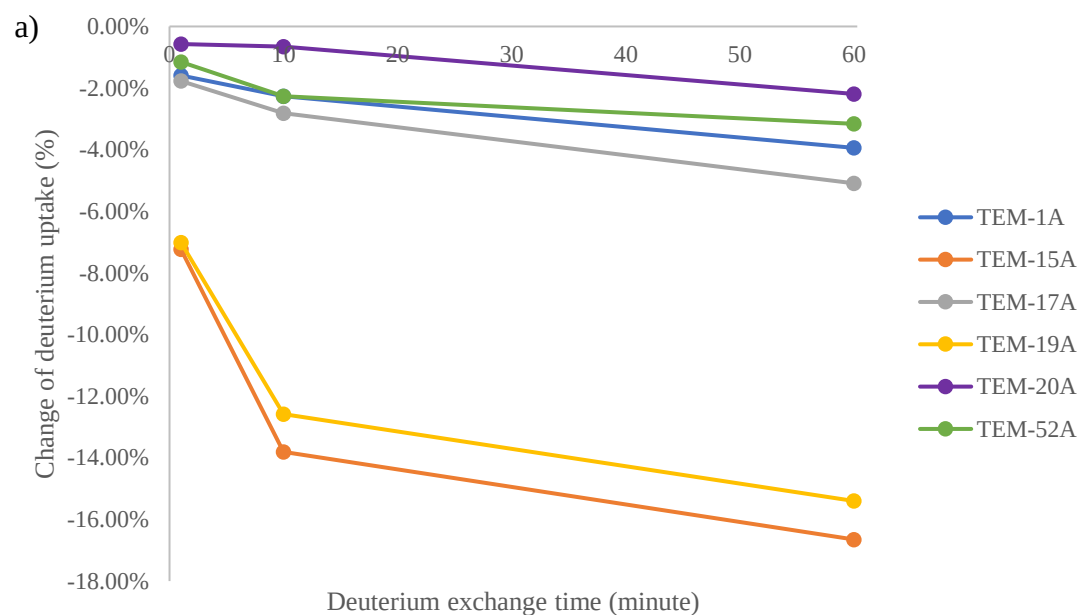


Figure 3.43 Plot of change of deuterium uptake of residues 140-150 upon cefotaxime binding.

An edge of β -sheet (Residues 49-59, 222-230 and 250-260)

For residues 49-59, obvious changes after cefotaxime binding were shown in TEM-15A-Cef (G238S/E104K) and TEM-19A-Cef (G238S) (Figure 3.44a). Other mutants, including TEM-1A-Cef, showed minor changes after binding. A more detailed comparison revealed that the two mutants containing M182T (TEM-20A-Cef and TEM-52A-Cef) underwent slightly smaller changes than the TEM-1A-Cef, while TEM-17A-Cef (E104K) had a slightly greater change. This observation was opposite for residues 250-260. M182T-containing mutants had slightly greater changes than TEM-1A-Cef and the change for TEM-17A-Cef was smaller (Figure 3.44b). However, the great reduction in flexibility of residues 250-260 after binding remained the same for TEM-15A-Cef and TEM-19A-Cef. Moreover, changes were less obvious for TEM-17A-Cef and TEM-1A-Cef than those M182T-containing mutants.

For residues 222-230, the great reduction in flexibility after binding could still be observed for TEM-15A-Cef and TEM-19A-Cef (Figure 3.44c). However, TEM-20A-Cef (G238S/M182T) also had a great flexibility reduction as the other two G238S-containing mutants. For TEM-52A-Cef (G238S/E104K/M182T), only two peptides, residues 214-225 and 214-230 which covered residues 220-230, could be used to predict its flexibility change of this region (Figure 3.44d). Both peptides showed similar flexibility changes upon the increment of the HDX time, and interestingly the values were also similar and the flexibility reduction was large and comparable to TEM-15A-Cef, TEM-19A-Cef and TEM-20A-Cef. By combining these results, this region might be relevant towards cefotaxime binding for G238S-containing mutants, including those mutants have been stabilized by M182T.



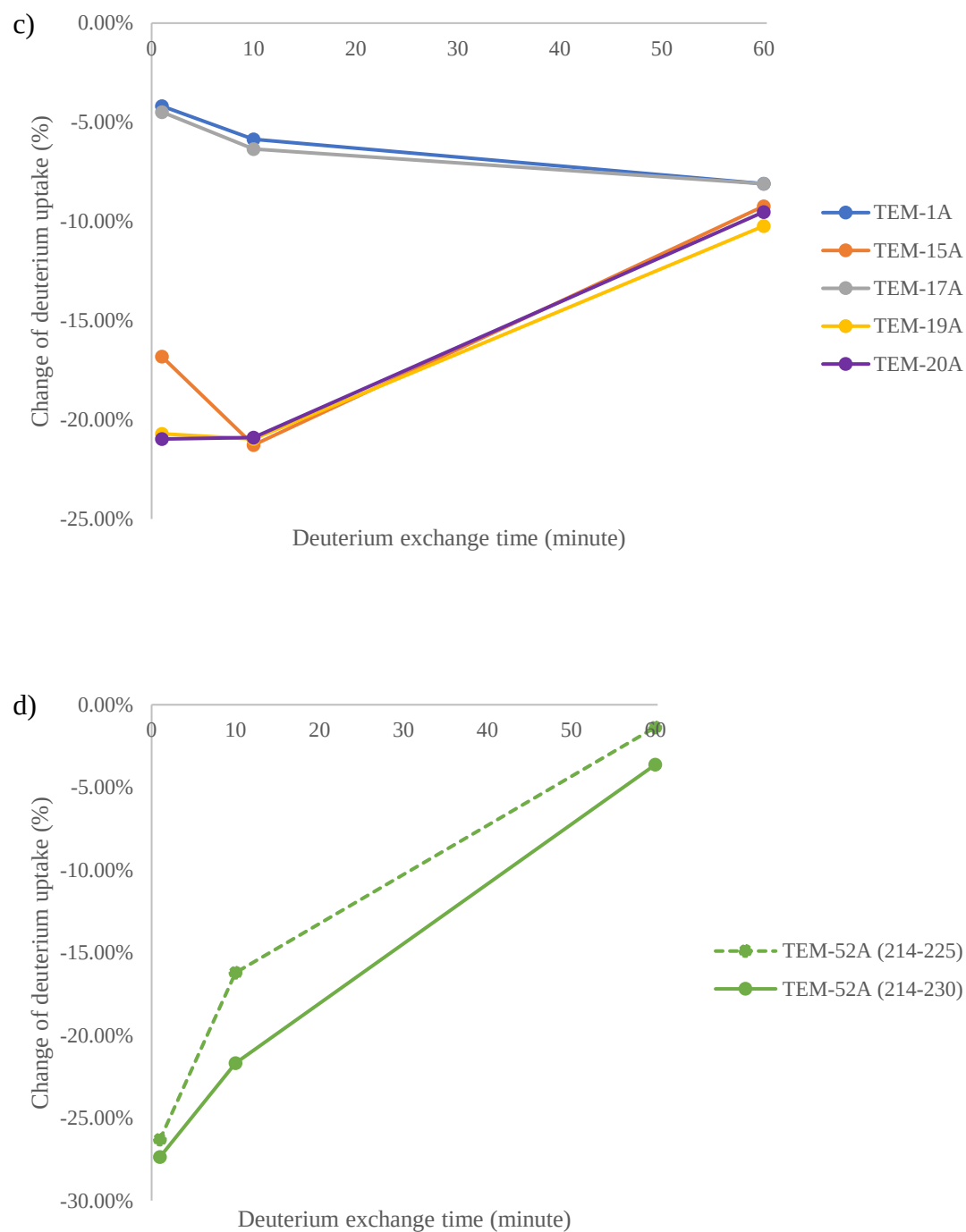


Figure 3.44 Plot of change of deuterium uptake of residues (a) 49-59, (b) 250-260, (c) 222-230 and (d) 214-225 and 214-230 (TEM-52A only) upon cefotaxime binding.

3.4.7. Insights into cefotaxime catalytic efficiencies of TEM-type ESBLs from conformational changes due to incorporation of single or combination of point mutations

In this study, conformational dynamics of wild type TEM-1 and five TEM-type ESBLs, as well as their cefotaxime-bound complexes, was comprehensively studied by HDX-MS. Global HDX-MS revealed that G238S-containing ESBLs possessed a more flexible conformation than TEM-1. Differences in cefotaxime hydrolytic power for these mutants indicated their conformational dynamics was most likely different, even though TEM-1 and TEM-17 shared a similar overall conformation. Regional conformational dynamics could be revealed by local HDX-MS. Regions of interest could be categorized into three groups: 1) catalytically important regions, including region near S70 (revealed by β 2/H2-loop in this study), Ω -loop, SDN loop and β -3 strand; 2) regions possessing point mutation, including residue 104 (E104K) and 182 (M182T), while residue 238 (G238S) was assigned to group 1 as β -3 strand is catalytically important; 3) allosteric regions that are not directly involved in the cefotaxime hydrolysis, including H5-H6 and an edge of β -sheet opposite to the active site, which were first discovered in the study..

Catalytically important regions for all mutants showed a more flexible conformation than the wild type TEM-1. This was consistent to the explanation that the active site was expanded in order to allow the bulky side chain of cefotaxime fitting in. At the same time, even though all mutants held a binding pocket with a larger cavity, a previous study reported that their cefotaxime catalytic efficiencies and thermostabilities were different. TEM-52 and TEM-15 were most catalytically efficient TEM-ESBLs among the six β -lactamases, followed by TEM-19, TEM-20 and then TEM-17. TEM-1 was expectedly the least efficient one.⁸⁸ However, in contrast to the hydrolytic power,

TEM-15 was the most thermally unstable ESBLs among the six β -lactamases, followed by TEM-19. TEM-17, possessing the key point mutation E104K, became slightly unstable compared with TEM-1. The two M182T-containing mutants, TEM-20 and TEM-52 were more stable than TEM-1, and the triple mutant was the most stable one.⁸⁸ In general, active site in TEM-19 was more flexible than that of TEM-17, which was similar to that of wild type TEM-1. The trend was consistent to the result from global HDX-MS, as well as their catalytic power that G238S had greater hydrolytic efficiency enhancement than E104K. Synergistic effect for G238S and E104K was obvious, as revealed in both results from global HDX-MS and local HDX-MS for catalytically important regions that its active site became highly flexible. Although a more flexible active site is beneficial for the hydrolysis, an excessive flexible structure renders lower protein stability. This explained the largely flexible TEM-15 active site had higher catalytic efficiency, however its thermal stability was sacrificed. As a global suppressor to compensate adverse effects to mutant stability, M182T altered the flexibility of these regions differently. At the region near S70, M182T compensated the effect of G238S by increasing its flexibility, while the synergistic effect of G238S/E104K were compensated by reducing the flexibility of this region. For Ω -loop, even though flexibility enhancement was greater in G238S/E104K than G238S, incorporation of M182T rendered both of their flexibilities reduced to a similar level. Similar observations were also found in SDN loop and β -3 strand. Flexibilities of these regions in their cefotaxime bound complexes were also investigated. Even though flexibilities of these regions among the six apo- β -lactamases could be ranked, they showed a highly similar flexibility in their complexes. As they were directly involved in the hydrolysis, their conformation of active site after binding should be similar. Surprisingly, among these four regions, SDN loop was the most compacted region, while the other three regions still retained certain flexibilities, indicating the importance of the orientation of

SDN loop during binding and hydrolysis. Conformation distortion, particularly due to point mutations in this loop is predicted to greatly affect the catalytic power of TEM-type β -lactamases.¹²⁹ For N-terminal part of Ω -loop (residues 163-169), TEM-15A-Cef (G238S/E104K) was shown to be more compact than other complexes, suggesting the effect of G238S/E104K might act on this region significantly and caused the large improvement in the cefotaxime catalytic power. Unfortunately, this peptide could not be found in the digest of TEM-52A-Cef (G238S/E104K/M182T).

Regions containing point mutations were investigated as it is reasonable to predict that point mutations could directly alter the regional conformation. Interestingly, only minor conformational rearrangement in regions containing residue 104 (revealed by mainly H2/H3-loop in this work) was observed for E104K single mutant (TEM-17) when compared with TEM-1. For mutants possessing G238S/E104K and G238S, flexibility of this region was enhanced, while M182T reduced their flexibilities lower than TEM-1. Therefore, instead of E104K, G238S and M182T were two mutations that mainly conferred conformational changes to this region. These observations suggested effects of mutations might not be necessary to directly act on the region near the mutation, but long range effects were propagated to several important regions so as to alter the catalytic properties of mutants. For region containing residue 182 (revealed by mainly H7/H8-loop and H8 in this work), the G238S/M182T mutant (i.e., TEM-20) had a more compact conformation than the TEM-1, in contrast to the G238S mutant having a more flexible conformation. Considering the conformational changes during cefotaxime binding, for H2/H3-loop, still the G238S/E104K and G238S mutants underwent a significant change. For M182T-containing mutants, stabilization effects of this point mutation were clearly observed, as conformational changes upon binding in both H2/H3-loop and H7/H8-loop and H8 were significantly reduced when compared with

mutants without M182T, indicating that after M182T stabilization, these catalytically irrelevant regions were not required to undergo great conformational change in order to facilitate the cefotaxime hydrolysis.

Two regions that are not involved in the cefotaxime hydrolysis, i.e., allosteric regions, were first found to relate to conformational changes of catalytically important region in this study. H5-H6 is located at the periphery of TEM-type β -lactamases and clearly far away from the active site. However, T149 in H5-H6 is connected to L162 in the Ω -loop by two hydrogen bonds⁶⁴, as well as P145 in H5-H6 interacts with W165 in the Ω -loop through a hydrophobic interaction.⁶⁸ Local HDX-MS results revealed that G238S/E104K and G238S caused significant enhancement of flexibility in H5-H6. However, more compacted H5-H6 than TEM-1 was found in M182T-containing mutants, in contrast to the flexibility in Ω -loop, which was more flexible than TEM-1. Moreover, in this region, only M182T-containing mutants underwent conformational changes during cefotaxime binding, while other four β -lactamases had insignificant changes. It is proposed that M182T stabilized the TEM ESBLs by modulating the flexibility of H5-H6, as it could control the conformational change and movement of Ω -loop through T149-L162 and P145-W165 interactions. As a result, it is proposed that H5-H6 could be a potential allosteric region for inhibition of highly stable TEM-type ESBLs with high antibiotic catalytic efficiency, as changing the conformation of H5-H6 may affect the conformational dynamics of Ω -loop, resulting significant reduction in thermostability and impairment of the hydrolytic power of TEM-type ESBLs.

The edge of β -sheet opposite to the active site is another region that does not directly involve in the cefotaxime hydrolytic reaction. Nevertheless, part of the β -3 and β -4 are the active site consisting regions, and as a well-ordered secondary structure,

conformation of other parts in the β -sheet could crucially affect the conformational dynamics of β -3 and β -4 strand. Particularly, the local HDX-MS results showed that β -3 strand became more flexible in order to accommodate the bulky side chain of cefotaxime, therefore it is hypothesized that the β -sheet edge of our concern should undergo certain conformational changes to facilitate changes from β -3 strands, and maintain the ordered secondary structure at the same time. Flexibility of edge (revealed by residues 49-59, 222-230 and 250-260) became higher for both G238S-containing mutants. However, the β -sheet might become over-flexible and hence their stability would be reduced. Incorporation of M182T might control the flexibility of this edge to a level similar to TEM-1, or become slightly compact, and hence they were more stable, and the conformational changes during cefotaxime binding became smaller when compared to the great changes for those G238S-containing mutants without M182T. It is believed that plasticity of these regions could also help to relieve the steric hindrance caused due to conformational changes of in particular β -3 strand.

Overall, our HDX-MS results provided important information about changes in conformational dynamics in different regions of TEM-type β -lactamases possessing different mutation(s) and explained how these mutations facilitated the accommodation of bulky side chain of cefotaxime. This information also helped to explain the thermal stability and cefotaxime catalytic efficiency of these mutants. Particularly, better understanding of the conformational dynamics of these mutants does not only assist for studying the antibiotics resistance mechanism, but also provides important insights for designing novel antibiotics or inhibitor, especially drugs working allosterically, to relieve the antibiotics resistance problem.

Chapter 4. Rapid detection of SARS-CoV-2 by mass spectrometry

4.1. Introduction

In December 2019, several unusual cases of pneumonia were reported in Wuhan, China. Soon after that, the number of cases rapidly escalated within a month in Wuhan. Even worse, this unknown pneumonia began to spread all over the world in the following few months. After confirming a novel coronavirus as the causative agent of this pneumonia, the World Health Organization (WHO) officially named this disease as “Coronavirus Disease 2019” (COVID-19) on 11 February 2020, replacing the former name “Wuhan Coronavirus” and “2019-nCoV”.¹³⁷ Since genetic studies of this novel coronavirus revealed its high similarity to the severe acute respiratory syndrome coronavirus (SARS-CoV), the causative agent of the severe acute respiratory syndrome in 2003, the International Committee on Taxonomy of Viruses (ICTV) announced “severe acute respiratory syndrome coronavirus 2” (SARS-CoV-2) as the official name of the causative agent of COVID-19.¹³⁸ On 12 March 2020, WHO announced COVID-19 as a pandemic.¹³⁹ Since mid-May 2021, more than 209 million confirmed cases and over 4.4 million deaths have been reported from more than 200 countries¹⁴⁰, and unfortunately, the number of confirmed and death case are still surging.

4.1.1. SARS-CoV-2 background

4.1.1.1. Features

SARS-CoV-2 has a circular or elliptic shape, with a diameter ranging from 60 – 140 nm^{141,142}, and it is categorized to the family Coronaviridae. Similar to other coronaviruses, a positive-sense single stranded RNA ((+)ssRNA) is wrapped up by nucleocapsid (N) protein, and surrounded by envelope (E) protein and membrane (M) glycoproteins. Spike (S) protein is attached to the surface of E protein and that results in the corona-like appearance. Recent studies have revealed that entry of SARS-CoV-2

relies on the binding between angiotensin-converting enzyme 2 (ACE2) of the host cell as the receptor and S proteins from the virus, which is similar to the endocytosis pathway of SARS-CoV.¹⁴³⁻¹⁴⁵ After entering the host cell, a series of events would be triggered including the virus replication and activation of the human intrinsic immune system. In a serious situation, patients may suffer from the life-threatening acute respiratory distress syndrome (ARDS). Even though patients could be recovered from ARDS, they may still suffer from some sequelae such as obstructive and restrictive ventilatory deficits, which significantly affect their quality of life.¹⁴⁶

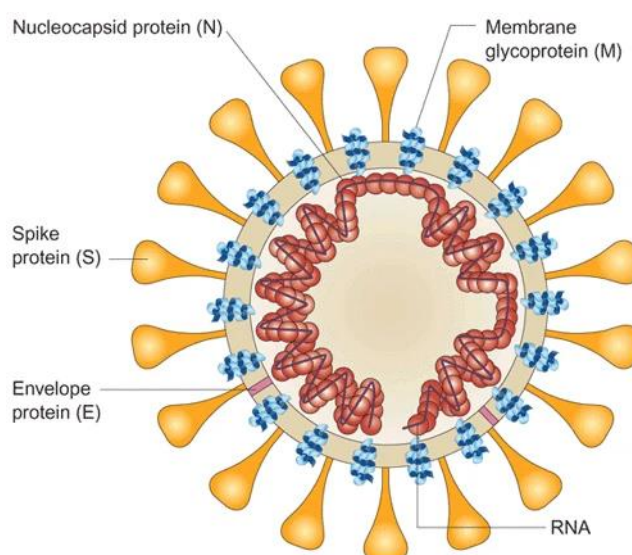


Figure 4.1 Schematic diagram of the SARS-CoV-2 structure.¹⁴⁷

4.1.1.2. Variants of SARS-CoV-2

It is reported that RNA viruses, to which SARS-CoV-2 belongs, possess higher mutation rate than that of the DNA viruses^{148,149}, which have five order of magnitude higher in mutability.¹⁵⁰ Recently various SARS-CoV-2 variants have been discovered. The reason of the occurrence of variants can be attributed to the selective pressure against the immune system of host, medicines used to treat infections, host genetic variety and diverse cellular microenvironments. These variants usually show enhanced virulence, transmission or infection power. Several variants are listed as the “Variants

of Concern” (VOCs) by the United States Centers for Disease Control and Prevention (CDC). The alpha variant, also known as 501Y.v1 or 20I, was first identified in UK. The transmission power was enhanced 70% compared with the native virus.¹⁵¹ The beta variant identified in South Africa, also known as 501Y.v2 or 20H, was found to evade the immune response, and cause false negative results in genetic testing.¹⁵² Other listed variants of concern including gamma and delta also pose serious threat to the current situation, especially the occurrence of the so-called “delta-plus” AY.1, AY.2 and AY.3 and omicron variants. All the current measures dealing with the pandemic will possibly become significantly less effective for these variants.¹⁵³

4.1.2. Detection of SARS-CoV-2

In order to confirm a definite COVID-19 diagnosis of a patient, several approaches are utilized by clinical physicians and scientists. For clinical physicians, diagnosis is mainly made based on the clinical symptoms of patients and monitoring of the hepatic functions by evaluating the level of several indicators particularly at the early stage of the pandemic via regular blood test and chest computed tomography imaging.^{154,155} In contrast, scientists working in the laboratories focus on the detection of the presence of SARS-CoV-2 mainly by nucleic acid and serological detection. It should be noted that nowadays these laboratory-based methods are extremely common and extensively utilized in both hospitals and the public.

4.1.2.1. Nucleic acid detection

It is always the first task for nucleic acid biologists to perform the genome sequencing of a new virus. Thanks to the polymerase chain reaction (PCR) and all developed contemporary nucleic acid sequencing methods, the genomic information can be easily acquired. With these valuable information, sequence alignments and comparison are

possible so as to understand the origin and even natural transmission pathway of the virus. More importantly, detection of the presence of the RNA of SARS-CoV-2 in suspected patients provides a more direct and simple approach for COVID-19 diagnosis. Currently, four nucleic acid detection methods for COVID-19 are utilized in laboratories, namely next generation sequencing, isothermal amplification, clustered regularly interspaced short palindromic repeats (CRISPR) and real-time reverse transcription polymerase chain reaction (qRT-PCR).

Next generation sequencing characterizes and identifies virus.¹⁵⁵ It also provides information of virus changes and transmission. Although the sequencing accuracy is high, the high instrument cost is a significant disadvantage. Also, sophisticated bioinformatic tools require experts to handle them, which hinders the widespread use of this technique.

Isothermal amplification is an alternative nucleic acid detection method for SARS-CoV-2. Huang et al. have developed a reverse transcription-loop-mediated isothermal amplification (RT-LAMP) method to detect SARS-CoV-2 within 30 minutes. Results are consistent with the traditional qRT-PCR.¹⁵⁶ This method is simple, as only one temperature is required for amplification. Besides, the detection sensitivity and specificity are comparable to the previous methods. Low instrumentation requirement is another advantage of this method so that this technique could be applied in medical resources inadequate areas. However, the primer design is critical for detection. Non-specific binding of primer could cause false positive results.

Various detection approaches have been developed based on clustered regularly interspaced short palindromic repeats (CRISPR) technique, which requires the

conversion of RNA to DNA. SHERLOCK (Specific High Sensitivity Enzymatic Reporter Unlocking), a CRISPR-Cas 13-based RNA detection technology developed by Broad Institute of MIT and Harvard, is one of the examples.¹⁵⁷ Zhang et al. have published a SHERLOCK-based protocol for SARS-CoV-2 detection, despite the fact that this protocol has not been fully validated.¹⁵⁸ Instead of Cas 13, another similar approach, namely HOLMES (an one-Hour Low-cost Multipurpose highly Efficient System) developed by Shanghai Institutes for Biological Sciences, utilizes Cas 12 for detection.¹⁵⁹ The CRISPR approaches are able to detect both DNA and RNA, but combination of nucleic acid amplification techniques is necessary to enhance the detection sensitivity.

Real-time reverse transcription polymerase chain reaction (qRT-PCR) is currently the major and gold standard detection method for SARS-CoV-2. Several qRT-PCR assays have already been published.^{160,161} By using additional fluorescent substances in the PCR and monitoring the real-time fluorescent signal, the amount of DNA can be evaluated. Common qRT-PCR methods including Taqman probe method and dual/multiplex fluorescent PCR can also be used in SARS-CoV-2 detection. qRT-PCR provides high sensitivity and specificity of detection. However, this method still suffers from high false negative rate, particularly for samples collected at the early stage of infection.¹⁶² Moreover, the cost of instrument is also high.

Generally, there are four steps in the workflow of nucleic acid detection for SARS-CoV-2, and they are sample collection, nucleic acid extraction, real-time PCR and data analysis.

Since infection of SARS-CoV-2 causes respiratory diseases and pneumonia, the first

sample collection targets are the upper respiratory tract, including nasal swab and oral swab, and the lower respiratory track, including sputum, bronchoalveolar fluid and respiratory tract fluid. Other targets involve invasive sampling, such as blood sample and conjunctiva swab.¹⁶³

The sensitivity and accuracy of the detection are highly dependent on the quality and integrity of the extracted RNA, which indicate the importance of the effective nucleic acid extraction methods. Sample collection is simple and most of the samples collected from suspected patients are oral swab or sputum. However, high viscosity and high protein or interference concentrations make the RNA extraction more challenging. Liquefying of sample by sonication is necessary in order to improve the extraction efficiency.

Another critical point for the success of nucleic acid detection is the viral loads in samples, and this varies significantly according to the number of days after symptom onset and the types of sample collected. For SARS-CoV and MERS-CoV, viral loads reach the maximum after 14 days of symptom onset and lower respiratory tract samples always present higher viral loads than upper respiratory tract samples samples.¹⁶⁴⁻¹⁶⁶ However, for SARS-CoV-2, the viral loads reach the peak in the first week and decrease afterward.¹⁶⁷⁻¹⁷⁰

In order to enhance the detection sensitivity, real-time PCR using Taqman probe-based assay is performed. Currently, various countries have developed and decided different primers for the assay.^{171,172} For example, according to National Institute For Viral Disease Control and Prevention of Chinese Center For Disease Control and Prevention (China CDC), positive infection is defined as the detection of both ORF1ab and N

genes¹⁷³, whereas Charité from Germany defines both RdRp and E genes as the targets. Isothermal amplification like reverse transcription loop-mediated isothermal amplification (RT-LAMP) assay is also able to detect SARS-CoV-2. The detection does not require any temperature change, and therefore no specific PCR instrument is required. Based on the different reported dyes used, detection results can be monitored either by naked eye or under UV light.

Even though nucleic acid detection is the golden standard for SARS-CoV-2 detection, it possesses several major limitations. First, high false negative rate of this approach has been reported. Second, positive result would only be given when virus is still present in patients. This means this method cannot reveal cases of patients who have already recovered from the disease. Besides, it is labor-intensive and only experienced personnel are suitable to process raw samples as extraction of pure RNA is not an easy task. The whole analysis typically takes several hours, and should be performed in high biosafety level (BSL) laboratories, and this limits the application of this approach. Moreover, as the current golden SARS-CoV-2 detection methods, numerous qRT-PCR tests are conducted every day all over the world, and this causes shortage of reagents required for qRT-PCR, particularly in some developing and resources-lacking countries.¹⁷⁴

4.1.2.2. Serological detection

Serological detection is an alternative approach for detection of SARS-CoV-2, and is even useful for the epidemiology of COVID-19. Serological tests aim at detecting antibodies and antigens in patients. Generally, serological assays include enzyme-linked immunosorbent assay (ELISA), chemiluminescence assay (CLIA), western blot (WB), protein microarray and neutralization.¹⁷⁵

Among all the aforementioned serological assays, ELISA is the most commonly used to detect antigens and antibodies. Antigens or anti-target antibodies are first immobilized in a 96-well plate, serum samples are then added to the plate and binding between antibodies and antigens occurs if antibodies are present. Functionalized secondary antibody conjugates or antigens are then added and bind to the target antibodies. Substrates or various stimulations are provided to induce physical or chemical properties changes such as color changes. For coronavirus, S and N proteins are commonly chosen as the antigens because of their importance and abundance. Also, producing recombinant proteins is only a routine task and renders the easy accessibility of antigens. Immunoglobulin G and M (IgG and IgM) are two common targets for diagnosis and epidemiology study.^{169,176,177} The whole process is simple and easy to be commercialized if the product is modified to strip-like one. However, immune response occurs only several days after virus infection, meaning serological tests may provide false negative results if samples are collected at the early stage of infection. For antigens, even though they are presented in samples of patients, but their quantities may not reach a detectable level by ELISA, particularly at the early stage of infection, rendering false negative results easily. Besides, the structures and sequences are highly conserved among coronavirus, especially for SARS-CoV and SARS-CoV-2^{178,179}, meaning there is a high possibility of cross-reactivity that renders test results inaccurate.¹⁸⁰ As SARS-CoV-2 is highly contagious and has high virulence, samples must be handled at BSL-3 laboratories, unless viruses in samples are properly and completely inactivated. It would be safe to process inactivated samples at BSL-2 and sometimes even BSL-1 laboratories. Both chemical approach such as guanidine isothiocyanate that could denature and dissolve viral proteins, and physical approach such as heating are utilized to inactivate viruses. However, these methods are too harsh to viral proteins that they denature them.

Denaturation of antibodies or antigens causes loss of immunogenicity and leads to the failure of serological tests.

In comparison, nucleic acid detection possesses advantages including high sensitivity and specificity, but suffers from false negative results and requires specific instrument, while serological detection is a fast detection method without specific instrumental requirement, on-site detection is possible, and it could be performed by individuals strictly following the instructions of the commercialized detection kit. However, its sensitivity and specificity are limited. As aforementioned, serological detection is not suitable for early-stage diagnosis. Cross-reactivity of antibodies and antigens may also induce false positive results. Overall, a new simple and rapid detection method for SARS-CoV-2 that can overcome these problems is urgently required.

4.1.2.3. Detection of SARS-CoV-2 by targeted proteomic approach using mass spectrometry

MS is widely applied for numerous areas, particularly in biological research studies. Besides the conformational dynamics of proteins previously discussed in chapters 1 - 3, proteomics is another field that many researchers focus on. Particularly, bottom-up proteomics is extensively utilized for both targeted and untargeted proteomics. In general, the proteomes of study targets, including commonly one control sample and one sample under certain treatments or stimulations, are subjected to proteolytic digestion, typically by trypsin. These digests are then separated and detected by LC-MS. Thanks to sophisticated databases such as Mascot, LC-MS results can be imported to these databases to search for proteins that are present in samples. By investigating both qualitative and quantitative differences, effects of treatments or stimulations could be revealed by changes in proteomes, and more importantly, biomarkers of certain

diseases could be discovered and applied clinically. Also, because of the continuous advancement made in the use of mass spectrometers, particularly in terms of their mass accuracies and resolutions, as well as the improving separation efficiencies of LC, confidence in the accuracies of results becomes higher, spurring a number of proteomics-related works in recent years.

A previous work demonstrated the proteolytic-mass spectrometric approach could be used to detect that the unfractionated intact microorganism, which enterobacteriophage MS2 was used as model. On-slide proteolytic digestion was performed on the unfractionated MS2 and the digest was directly detected by matrix-assisted laser desorption ionization mass spectrometry (MALDI-MS) in tandem mass spectrometry (MS/MS) mode. MS2 was identified by importing the MS/MS spectrum for database searching. The whole method is fast, simple and low sample consumption.^{181,182} Similar to the detection of the unique RNA sequence to identify SARS-CoV-2 by RT-PCR, it is also possible to detect the specific viral protein sequence, which is translated from the viral RNA so as to detect the presence of SARS-CoV-2. The RNA sequences similarities between SARS-CoV-2 and SARS-CoV or MERS-CoV are 79% and 50% respectively, which are distinct enough and support the proposed concept.¹⁸³ Besides, it is reasonable to predict specific peptides unique to SARS-CoV-2 that could be resulted after trypsin digestion.

LC-MS is another key to facilitate SARS-CoV-2 detection. The complexity of sample components severely complicates the detection, and hence a highly sensitive method is required. UPLC provides excellent separation efficiency with high resolution. Sensitivity and specificity could be further enhanced by using MS/MS in multiple reaction monitoring (MRM) mode. The two mass analyzers, which are commonly

quadrupole, only allow ion with specific m/z passing through while other unrelated ions or interferences are removed and discarded. Another quadrupole installed in between the two mass analyzers serves as a collision cell. Precursor ions selected in the first mass analyzer are fragmented through the collision induced dissociation, so as to generate product ions specific to their precursor ions. Such specific detection of masses of transitions greatly enhances identification reliability. Besides, because of the mass filtering process in the two mass analyzers, the detector can spend more time on the signal recording for target ions, and hence the sensitivity is significantly enhanced. Combining these advantages, the proteomics-mass spectrometric approach of detection of SARS-CoV-2 is believed to be feasible.

Several research studies have explored the possibility of detecting SARS-CoV-2 specific peptides. To detect the presence of S and N proteins, which show high potential as detection targets, approximately 2×10^{-5} viral particles are required.¹⁸⁴ Another study has demonstrated that one LC-MS can analyze 500 samples/day, with satisfactory validation of 84% of positive cases with 97% specificity achieved.¹⁸⁵

4.1.3. Objectives

Although effort has been made to prove the possibility of using MS with targeted proteomics for fast SARS-CoV-2 detection, several important concerns are aroused. First of all, the complicated procedures are rate-limiting. More than a half day, or even a day is required to complete the sample analysis. It would be better to improve the sample-to-result time to within an hour and increase the throughput of sample analysis. Second, the sensitivity should be able to meet the amount of virus in the common samples collecting for analysis nowadays, which are reported to be 1×10^4 - 1×10^{10} and 1×10^5 - 1×10^{11} copies/mL in nasal-derived swabs and saliva samples respectively.¹⁸⁶ Third, the persisting occurrence of new variants of SARS-CoV-2 should receive serious attention by researchers working on virus detection. Moreover, it is necessary to perform further validation using clinical samples.

This project has two objectives. First, a simple sample pretreatment method has to be developed, which includes sample clean-up steps and optimizes the digestion of viral protein. N protein of SARS-CoV-2 would be used as the model. Unique peptides of SARS-CoV-2, which are our main detection targets, are also investigated by analyzing the enzymatic digest. Then a rapid LC-MS analysis method in MRM mode is necessary for detection of the viral specific peptides, so as to confirm the presence of SARS-CoV-2 in samples.

4.2. Experimental

4.2.1. Nucleocapsid (N) protein

The His-tagged nucleocapsid protein without post-translational modification of SARS-CoV-2 was generously contributed by Guangdong Laboratory Animals Monitoring Institute. In brief, the gene of N-protein was encoded in the pET28a-N plasmid and the plasmid was transformed to *E. coli* for further induced expression. The expressed His-tagged N-proteins was about 46 kDa, which was consistent with the SDS-PAGE result (Figure 4.2).

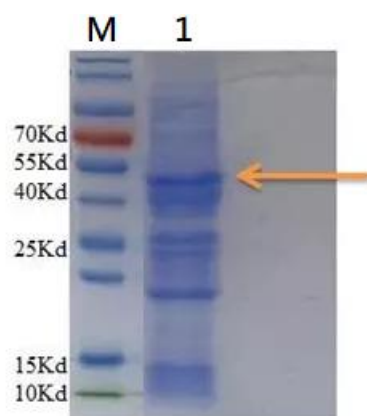


Figure 4.2 SDS-Page of the expressed N-protein. Lane M and 1 represent protein marker and induced strain respectively. This result is provided by the Guangdong Laboratory Animals Monitoring Institute.

4.2.2. Buffer exchange

20 μ L of sample (either pure N protein, N protein spiked saliva or blank saliva) was mixed with 190 μ L 20 mM ammonium bicarbonate in Amicon Ultra 0.5 mL Centrifugal Filters with 10 kDa cutoff. The mixture was centrifuged at 14800 rpm ($16152 \times g$) for 5 minutes (Sigma 1-14, Germany). The eluent was discarded and 200 μ L ammonium bicarbonate was added to the buffer exchange device for another round of centrifugation under the same condition. The eluent was discarded again and 200 μ L ammonium

bicarbonate was added for the third round of centrifugation, but the time was extended to 15 minutes at the same speed. The remaining solution in the device was collected by autopipette.

4.2.3. Trypsin digestion

Buffer exchanged samples were then subjected to trypsin digestion (Trypsin Gold, Mass Spectrometry Grade, Promega, U.S.). The digestion was performed at 60 °C in general. Digestion time and protein-to-trypsin ratio were varied for different parts in this project and they would be specified in the context if necessary. The trypsin digestion was quenched when the solution mixture contained about 2% FA.

4.2.4. LC-MS analysis for searching unique peptides of SARS-CoV-2 N-protein

The LC-MS analysis procedures were similar to that stated in section 2.8, except the chromatographic separation was performed at room temperature. The protein digest was injected into Waters nanoACQUITY UPLC system. Online desalting and preconcentration were performed utilizing the Waters ACQUITY UPLC BEH C18 VanGuard Pre-column (130 Å, 1.7 µm, 2.1 mm × 5 mm) by flowing 0.1% FA at 100 µL/min for 3 minutes at 100 µL/min. Desalted peptides were then separated and eluted through a Waters ACQUITY UPLC BEH C18 Column (130 Å, 1.7 µm, 1 mm × 100 mm) in 12 minutes at 60 µL/min by the following gradient elution profile: A: 0.1% FA, B: acetonitrile with 0.1% FA; at 0 minute 5% B, at 0.5 minute 8% B, at 6.5 minute 40% B, at 8-11 minute 95% B, at 11.5-12 minute 5% B. Eluent was detected by Waters Synapt G2-Si mass spectrometer at the following conditions: capillary voltage 3.0 kV, source temperature 150 °C, desolvation temperature 350 °C. Mass spectra were acquired over an m/z range of 100 to 2000 using MS^E mode.

4.2.5. Database search by PLGS and protein-protein basic local alignment search tool (blastp)

The database search procedures were similar to that stated in section 2.9. ProteinLynx Global Server 3.0.2 was used to generate the peptide list from the MS^E spectra. The peptide list includes sequence of identified peptides, masses of precursor ions, measured m/z , charge state, retention time, fragmentation profile and PLGS searching score of each identified peptide. The generated peptide list was subjected to protein-protein basic local alignment search tool (blastp) to investigate the specificity and uniqueness of peptides. Only peptides with 100% percent identity and all results related to SARS-CoV-2 were categorized as unique and specific peptides of N-protein of SARS-CoV-2.

4.2.6. MRM transitions of N-protein model peptides

Four peptides that passed the specificity check were selected as model peptides (GL Biochem, Shanghai), and were detected by the Agilent 6460 Triple Quad Mass Spectrometer. To establish MRM transitions, fragmentor was optimized so as to obtain the highest intensity in TIC. MS spectra were then analyzed and the charge state with the highest intensity would be selected as the mass of precursor ion. Further MS/MS analysis in product ion scan mode would be utilized to choose the fragments ions used in MRM. Generally, the highest intensity ion would be selected as the quantifier ion, while the second and third highest intensity ion were the qualifier ions, and all of their collision energies were optimized. Hence, three MRM transitions were established for one peptide.

4.2.7. LC-MS analysis of model peptides in MRM mode

Agilent 1290 Infinity LC System was used for the LC separation. Samples were separated and eluted through a Waters ACQUITY UPLC BEH C18 VanGuard Pre-column (130Å, 1.7 µm, 2.1 mm × 5 mm) connected with a Waters ACQUITY UPLC BEH C18 Column (130 Å, 1.7 µm, 2.1 mm × 50 mm) for 8 minutes at 0.4 mL/min by the following gradient elution profile: A: 0.1% FA, B: acetonitrile with 0.1% FA; at 0-0.5 minute 5% B, at 5 minute 95% B, at 5.1 minute 5% B, at 5.1-8 minute 95% B. Eluent was detected by Agilent 6460 Triple Quad Mass Spectrometer at the following conditions: Capillary voltage 4.0 kV, sheath gas flow 11 L/min, nebulizer 35 psi, sheath gas heater 300 °C, gas flow 8 L/min, gas temperature 300 °C. The MRM transition setting was summarized in Table 4.1.

Table 4.1 MRM transition setting for the four model N-protein peptides. The product ions in bold are the quantifier ions.

Precursor ion (<i>m/z</i>)	Product ion (<i>m/z</i>)	Fragmentor (V)	Collision energy (V)
MAGNGGDAALALLLDRLNQLQLESK			
836.1 [M+2H] ²⁺	742.5 (y ₆ ⁺)	130	26
	290.0 (y ₂ ⁺)		23
	813.0 (y ₇ ⁺)		25
MAGNGGDAALALLLDRLNQLQLESK			
828.9 [M+3H] ³⁺	202.8 (b ₂ ⁺)	130	25
	1141.8 (y ₂₂ ²⁺)		18
	813.5 (y ₁₄ ²⁺)		22
SRNSSRNSTPGSSRGTSPTARMAGNGGDAALALLLDRLNQLQLESK			
620.0 [M+6H] ⁶⁺	680.0 (b ₂₈ ⁴⁺)	140	13
	744.4 (b ₃₁ ⁴⁺ /y ₃₇ ⁵⁺)		10
	662.7 (b ₂₇ ⁴⁺ /b ₃₄ ⁵⁺)		19
SDNGPQNQRNAPRITFGGPSDSTGSGNQNGERSGARSK			
769.9 [M+5H] ⁵⁺	957.3 (b ₃₇ ⁴⁺)	110	19
	724.7 (y ₁₄ ²⁺ /b ₂₈ ⁴⁺)		19
	927.4 (b ₁₇ ²⁺)		24

4.3. Results and discussions

4.3.1. Strategy of detection of SARS-CoV-2-specific viral peptides

Currently, the nucleic acid detection approach is widely applied for COVID-19 test based on the detection of viral RNA of SARS-CoV-2 because of its specificity compared with other similar viral RNAs. Viral proteins, which are translated from this unique viral RNA, should also show their uniqueness and specificity in protein sequences. For the nucleocapsid (N) protein, even though it has about 90% similarity in sequence to the SARS-CoV N-protein, more than 30 differences could be found which are sufficient to generate adequate number of peptides unique to SARS-CoV-2 (Figure 4.3).¹⁸⁷ Besides, the high abundance of N protein in SARS-CoV-2 renders it an attractive target for virus detection.¹⁸⁸ Hence, specific and unique N protein peptides were selected as the detection targets and a targeted proteomic-mass spectrometric approach would be applied for the detection.

SARS-CoV	1	MSDNGPQ	S	NQRS	A	P	R	I	T	F	G	G	P	T	D	S	T	D	N	N	Q	N	G	R	N	G	A	R	P	K	Q	R	R	P	Q	G	L	P	N	N	T	A	S	W	F	T	A	L	T	Q	H			
SARS-CoV-2	1	MSDNGPQ	N	QRN	A	P	R	I	T	F	G	G	P	T	S	D	S	T	G	S	N	Q	N	G	E	R	S	A	R	S	K	Q	R	R	P	Q	G	L	P	N	N	T	A	S	W	F	T	A	L	T	Q	H		
SARS-CoV	61	GKEE	L	R	F	P	R	G	Q	G	V	P	I	N	T	N	S	G	P	D	D	Q	I	G	Y	R	R	A	T	R	R	V	R	G	D	G	K	M	K	E	L	S	P	R	W	F	Y	L	G	T	G	P	E	A
SARS-CoV-2	60	GKEDL	K	F	P	R	G	Q	G	V	P	I	N	T	N	S	G	P	D	D	Q	I	G	Y	R	R	A	T	R	R	I	R	G	D	G	K	M	K	D	L	S	P	R	W	F	Y	L	G	T	G	P	E	A	
SARS-CoV	121	SLPYGANKE	G	I	V	W	A	T	E	G	A	L	N	T	P	K	D	H	I	G	T	R	N	P	N	N	A	A	T	V	L	Q	L	P	Q	G	T	T	L	P	K	G	F	Y	A	E	G	S	R	G	G			
SARS-CoV-2	120	GLPYGANKDGI	I	W	V	A	T	E	G	A	L	N	T	P	K	D	H	I	G	T	R	N	P	A	N	N	A	A	T	V	L	Q	L	P	Q	G	T	T	L	P	K	G	F	Y	A	E	G	S	R	G	G			
SARS-CoV	181	SQASSRSSRSRGN	S	R	N	S	T	P	G	S	S	R	G	N	S	P	A	R	M	A	S	G	G	G	E	T	A	L	A	L	L	L	D	R	L	N	Q	L	E	S	K	V	S	G	K	G	Q							
SARS-CoV-2	180	SQASSRSSRSRNS	S	R	N	S	T	P	G	S	S	R	G	T	S	P	A	R	M	A	G	N	G	G	D	A	A	L	A	L	L	L	D	R	L	N	Q	L	E	S	K	M	S	G	K	G	Q							
SARS-CoV	241	QQQGQTVTKKSAAEASKKPRQKRTATKQ	Y	N	V	T	Q	A	F	G	R	R	G	P	E	Q	T	Q	G	N	F	G	D	Q	D	L	I	R	Q	G	T	D	Y	K																				
SARS-CoV-2	240	QQQGQTVTKKSAAEASKKPRQKRTATKA	Y	N	V	T	Q	A	F	G	R	R	G	P	E	Q	T	Q	G	N	F	G	D	Q	E	L	I	R	Q	G	T	D	Y	K																				
SARS-CoV	301	HWPQIAQFAPSASAFFGMSRIGMEVTPSGTWLTYH	G	A	I	K	L	D	D	K	D	P	Q	F	K	D	N	V	I	L	L	N	K	H	I	D	A																											
SARS-CoV-2	300	HWPQIAQFAPSASAFFGMSRIGMEVTPSGTWLTYT	G	A	I	K	L	D	D	K	D	P	N	F	K	D	V	I	L	L	N	K	H	I	D	A																												
SARS-CoV	361	YKTFPPTPEPKDKKKKTDEAQP	L	P	Q	R	Q	K	K	Q	P	T	V	T	L	L	P	A	A	D	M	D	D	F	S	R	Q	L	Q	N	S	M	S	G	A	S	A	D	S	T														
SARS-CoV-2	360	YKTFPPTPEPKDKKKKKADETAQ	L	P	Q	R	Q	K	K	Q	P	T	V	T	L	L	P	A	A	D	L	D	D	F	S	R	Q	L	Q	Q	S	M	S	--	S	A	D	S	T															
SARS-CoV	421	QA																																																				
SARS-CoV-2	418	QA																																																				

Figure 4.3 Sequences alignment of SARS-CoV-2 and SARS-CoV. Residues highlighted in yellow indicate differences between the protein sequences.

4.3.2. Sample pretreatment procedures and the searching of peptides specific to N protein of SARS-CoV-2

A typical proteomic analysis involves several standard steps such as protein denaturation, disulfide bond reduction and alkylation, proteolytic digestion, drying and reconstitution, which would take very long pretreatment time. Especially conventional trypsin digestion, which usually requires overnight digestion, does not fulfill the requirement of rapid detection. To achieve the rapid detection, digestion time must be reduced, but at the same time, the completeness of digestion may be sacrificed. It is also proposed that a buffer exchange step to ammonium bicarbonate, a mass spectrometric friendly buffer, should be included at the first step of sample pretreatment. As it can be imagined that the high complexity of components in the clinical sample could introduce great interferences for the analysis. A cleaner background and preconcentrated sample are also believed to be beneficial for trypsin digestion and further analysis.

As reduction of trypsin digestion time may cause incomplete digestion, two approaches were utilized to compensate for this drawback, and they are increasing the temperature of trypsin digestion to 60 °C during trypsin and the amount of trypsin used. Typically, trypsin digestion is performed at 37 °C to ensure the bioactivity of enzyme. Elevating the temperature may cause denaturation of trypsin, therefore the digestion time was limited to no more than 10 minutes in order to prevent the inactivation of trypsin, and to achieve the objective of fast detection. Initially, for the investigation of specific peptides, the trypsin-to-N protein ratios of 1:50 and 1:16 were used to explore if any robust peptides, i.e., peptides that could be generated easily at any conditions, could be resulted. In general, increasing the amount of trypsin used could result in higher intensities of peptides. However, the amount of N protein in clinical samples could be

highly varied, and it is impossible to optimize the amount of trypsin used for each sample. Therefore, 10 µg of trypsin was used for further spiked sample analysis. For the digestion time, 1, 3, 5, 10 and 60 minutes were tested. The minimum time for observing specific peptides was 3 minutes, and hence the digestion time was fixed at 3 minutes in the spiked sample analysis.

Detection of these high robustness specific peptides could enhance the accuracy of analysis. Besides, the specificity of peptides was confirmed by blastp searching. Here specific peptide was defined as all 100% percent of identity results belonging to SARS-CoV-2 (Figure 4.4). A peptide matched both SARS-CoV-2 and SARS-CoV was also shown in Figure 4.5. It was noteworthy that there were several peptides with almost all their blastp search results matching SARS-CoV-2 with 100% percent identity, but also accompanied with a result perfectly matched the bat coronavirus RaTG13 (Figure 4.6). The whole genome levels of these two viruses had 96% similarity.¹⁸⁹ However, to ensure the robustness and accuracy of our method, these special results were not counted in our specific peptide list. After screening, four peptides were selected as the target peptides and summarized in Table 4.2.

Peptide 1 was the most robust one that could be found in experiments at different conditions. Peptides 2 and 3 were missed cleaved and contained the sequence of peptide 1. Considering the possibility of incomplete digestion of trypsin within a short time, including several missed cleaved peptides could reduce the chance of false negative result.¹⁹⁰ Peptide 4 was independent of peptides 1-3, and including this peptide could enhance the variety of the detection targets.

Table 4.2 Four selected peptides used for SARS-CoV-2 detection. Peptides 2 and 3 were missed cleaved and contained the sequence of peptide 1, which are highlighted in red.

Sequence	Molecular weight (Da)	Alternative name of the peptide used in this thesis
MAGNGGDAALALLLDR	1670.9	Peptide 1
MAGNGGDAALALLLDRLNQLESK	2483.9	Peptide 2
SRNSSRNSTPGSSRGTSPPARMAGNGGDAALALLLDR	3715.1	Peptide 3
SDNGPQNQRNAPRITFGGPSDSTGSN	3846.0	Peptide 4
QNGERSGARSK		

	Description	Max Score	Total Score	Query Cover	E value	Per. Ident	Accession
✓	nucleocapsid phosphoprotein [Severe acute respiratory syndrome coronavirus 2]	74.4	74.4	100%	9e-14	100.00%	QIQ50129.1
✓	N protein [Severe acute respiratory syndrome coronavirus 2]	74.4	74.4	100%	9e-14	100.00%	QIH45050.1
✓	nucleocapsid phosphoprotein [Severe acute respiratory syndrome coronavirus 2]	74.4	74.4	100%	9e-14	100.00%	QIM47464.1
✓	nucleocapsid phosphoprotein [Severe acute respiratory syndrome coronavirus 2]	74.4	74.4	100%	9e-14	100.00%	QIS30122.1
✓	nucleocapsid phosphoprotein [Severe acute respiratory syndrome coronavirus 2]	74.4	74.4	100%	9e-14	100.00%	YP_009724397.2
✓	nucleocapsid phosphoprotein [Severe acute respiratory syndrome coronavirus 2]	74.4	74.4	100%	9e-14	100.00%	QIE07458.1
✓	nucleocapsid phosphoprotein [Severe acute respiratory syndrome coronavirus 2]	74.4	74.4	100%	9e-14	100.00%	QIS60879.1
✓	nucleocapsid phosphoprotein [Severe acute respiratory syndrome coronavirus 2]	74.4	74.4	100%	9e-14	100.00%	QIU78842.1
✓	nucleocapsid phosphoprotein [Severe acute respiratory syndrome coronavirus 2]	74.4	74.4	100%	9e-14	100.00%	QIS61476.1
✓	nucleocapsid phosphoprotein [Severe acute respiratory syndrome coronavirus 2]	74.4	74.4	100%	9e-14	100.00%	QIS30502.1
✓	nucleocapsid phosphoprotein [Severe acute respiratory syndrome coronavirus 2]	74.4	74.4	100%	9e-14	100.00%	QIQ49839.1
✓	nucleocapsid phosphoprotein [Severe acute respiratory syndrome coronavirus 2]	74.4	74.4	100%	9e-14	100.00%	QHW06046.1
✓	nucleocapsid phosphoprotein [Severe acute respiratory syndrome coronavirus 2]	74.4	74.4	100%	9e-14	100.00%	QIU80981.1
✓	nucleocapsid phosphoprotein [Severe acute respiratory syndrome coronavirus 2]	74.4	74.4	100%	9e-14	100.00%	QIU80993.1
✓	nucleocapsid phosphoprotein [Severe acute respiratory syndrome coronavirus 2]	74.4	74.4	100%	9e-14	100.00%	QII57305.1
✓	nucleocapsid phosphoprotein [Severe acute respiratory syndrome coronavirus 2]	74.4	74.4	100%	9e-14	100.00%	QIU78855.1
✓	nucleocapsid phosphoprotein [Severe acute respiratory syndrome coronavirus 2]	74.4	74.4	100%	9e-14	100.00%	QIU80969.1
✓	nucleocapsid phosphoprotein [Severe acute respiratory syndrome coronavirus 2]	74.4	74.4	100%	9e-14	100.00%	QIS30511.1
✓	nucleocapsid phosphoprotein [Severe acute respiratory syndrome coronavirus 2]	74.4	74.4	100%	9e-14	100.00%	QHO62884.1
✓	nucleocapsid phosphoprotein [Severe acute respiratory syndrome coronavirus 2]	74.4	74.4	100%	9e-14	100.00%	QIS60866.1
✓	nucleocapsid phosphoprotein [Severe acute respiratory syndrome coronavirus 2]	74.4	74.4	100%	9e-14	100.00%	QIQ50109.1
✓	nucleocapsid phosphoprotein [Severe acute respiratory syndrome coronavirus 2]	74.4	74.4	100%	9e-14	100.00%	QIS30062.1
✓	nucleocapsid phosphoprotein [Severe acute respiratory syndrome coronavirus 2]	74.4	74.4	100%	9e-14	100.00%	QIS61010.1

Figure 4.4 An example of blastp search result of a peptide specific to N protein of SARS-CoV-2. The red rectangular box indicates all results belonging to SARS-CoV-2 have 100% percent identity.

	Description	Max Score	Total Score	Query Cover	E value	Per. Ident	Accession
✓	nucleocapsid protein, partial [Severe acute respiratory syndrome coronavirus 2]	68.9	68.9	100%	5e-12	100.00%	QII87776.1
✓	nucleocapsid protein [Bat SARS-like coronavirus]	68.9	68.9	100%	6e-12	100.00%	ATO98154.1
✓	nucleocapsid protein N [SARS coronavirus WH20]	68.9	68.9	100%	6e-12	100.00%	AAK16200.1
✓	putative nucleocapsid protein [SARS coronavirus CUHK-Su10]	68.9	68.9	100%	6e-12	100.00%	AAP30714.1
✓	nucleocapsid protein [Bat SARS-like coronavirus]	68.9	68.9	100%	6e-12	100.00%	ATO98166.1
✓	nucleocapsid protein [Severe acute respiratory syndrome-related coronavirus]	68.9	68.9	100%	6e-12	100.00%	NP_828858.1
✓	nucleocapsid protein [Bat SARS-like coronavirus]	68.9	68.9	100%	6e-12	100.00%	ATO98190.1
✓	nucleocapsid protein [SARS coronavirus civet020]	68.9	68.9	100%	6e-12	100.00%	AAU04673.1
✓	nucleocapsid protein [Bat SARS-like coronavirus]	68.9	68.9	100%	6e-12	100.00%	ATO98142.1
✓	nucleocapsid protein [Bat SARS-like coronavirus]	68.9	68.9	100%	6e-12	100.00%	ATO98178.1
✓	nucleocapsid protein [Bat SARS-like coronavirus RsSHC014]	68.9	68.9	100%	6e-12	100.00%	AGZ48815.1
✓	nucleocapsid protein [SARS coronavirus ExoN1]	68.9	68.9	100%	6e-12	100.00%	ACZ72117.1
✓	nucleocapsid protein N [SARS coronavirus Frankfurt.1]	68.9	68.9	100%	6e-12	100.00%	AAP33707.1
✓	nucleocapsid protein [Bat SARS-like coronavirus]	68.9	68.9	100%	6e-12	100.00%	ATO98129.1
✓	nucleocapsid protein [Bat SARS-like coronavirus]	68.9	68.9	100%	6e-12	100.00%	ATO98117.1
✓	nucleocapsid protein [Bat SARS-like coronavirus]	68.9	68.9	100%	6e-12	100.00%	ATO98228.1
✓	nucleocapsid protein [Coronavirus BtRS-BetaCoV/YN2018C]	68.9	68.9	100%	6e-12	100.00%	QDF43833.1
✓	nucleocapsid protein [SARS coronavirus BJ182a]	68.9	68.9	100%	6e-12	100.00%	ACB69868.1
✓	nucleocapsid protein [Coronavirus BtRS-BetaCoV/YN2018A]	68.9	68.9	100%	6e-12	100.00%	QDF43823.1
✓	nucleocapsid protein [SARS coronavirus B039]	68.9	68.9	100%	6e-12	100.00%	AAV49739.1
✓	nucleocapsid protein [SARS coronavirus ExoN1]	68.9	68.9	100%	6e-12	100.00%	ACZ71986.1
✓	nucleocapsid protein [SARS coronavirus civet010]	68.9	68.9	100%	6e-12	100.00%	AAU04655.1
✓	nucleocapsid protein [Coronavirus BtRS-BetaCoV/YN2018B]	68.9	68.9	100%	6e-12	100.00%	QDF43828.1

SARS-CoV-2

SARS or other proteins not related to SARS-CoV-2

Figure 4.5 An example of blastp search result of a peptide not specific to N protein of SARS-CoV-2. The peptide sequence subjected to blastp searching also matched the sequence of SARS-CoV.

	Description	Max Score	Total Score	Query Cover	E value	Per. Ident	Accession
✓	nucleocapsid phosphoprotein, partial [Severe acute respiratory syndrome coronavirus 2]	103	103	100%	4e-27	100.00%	BCA37476.1
✓	nucleocapsid phosphoprotein, partial [Severe acute respiratory syndrome coronavirus 2]	103	103	100%	6e-27	100.00%	QIK02784.1
✓	nucleocapsid phosphoprotein, partial [Severe acute respiratory syndrome coronavirus 2]	103	103	100%	7e-27	100.00%	QIK02783.1
✓	Chain A, SARS-CoV-2 nucleocapsid protein [Severe acute respiratory syndrome coronavirus 2]	103	103	100%	9e-27	100.00%	6M3M_A
✓	nucleocapsid phosphoprotein [Severe acute respiratory syndrome coronavirus 2]	106	106	100%	1e-25	100.00%	QIS61476.1
✓	nucleocapsid phosphoprotein [Severe acute respiratory syndrome coronavirus 2]	105	105	100%	2e-25	100.00%	QIS30122.1
✓	nucleocapsid phosphoprotein [Severe acute respiratory syndrome coronavirus 2]	105	105	100%	2e-25	100.00%	QIU78842.1
✓	nucleocapsid phosphoprotein [Severe acute respiratory syndrome coronavirus 2]	105	105	100%	2e-25	100.00%	QIU81197.1
✓	nucleocapsid phosphoprotein [Severe acute respiratory syndrome coronavirus 2]	105	105	100%	2e-25	100.00%	QIU78855.1
✓	nucleocapsid phosphoprotein [Severe acute respiratory syndrome coronavirus 2]	105	105	100%	2e-25	100.00%	QIE07458.1
✓	nucleocapsid phosphoprotein [Severe acute respiratory syndrome coronavirus 2]	105	105	100%	2e-25	100.00%	QIU96530.1
✓	nucleocapsid phosphoprotein [Severe acute respiratory syndrome coronavirus 2]	105	105	100%	2e-25	100.00%	QIS30511.1
✓	nucleocapsid phosphoprotein [Severe acute respiratory syndrome coronavirus 2]	105	105	100%	2e-25	100.00%	QIQ08827.1
✓	nucleocapsid phosphoprotein [Severe acute respiratory syndrome coronavirus 2]	105	105	100%	2e-25	100.00%	QIU80993.1
✓	nucleocapsid protein [Bat coronavirus RaTG13]	105	105	100%	2e-25	100.00%	QHR63308.1
✓	N protein [Severe acute respiratory syndrome coronavirus 2]	105	105	100%	2e-25	100.00%	QIH45050.1
✓	nucleocapsid phosphoprotein [Severe acute respiratory syndrome coronavirus 2]	105	105	100%	2e-25	100.00%	QIS60878.1
✓	nucleocapsid phosphoprotein [Severe acute respiratory syndrome coronavirus 2]	105	105	100%	2e-25	100.00%	QIU80969.1
✓	nucleocapsid phosphoprotein [Severe acute respiratory syndrome coronavirus 2]	105	105	100%	2e-25	100.00%	QHO62884.1
✓	nucleocapsid phosphoprotein [Severe acute respiratory syndrome coronavirus 2]	105	105	100%	2e-25	100.00%	QIS60866.1

Figure 4.6 An example of blastp search result of a peptide having almost all results belonging to SARS-CoV-2 with 100% percent identity, but one result also perfectly matched the bat coronavirus RaTG13. This is not included in the specific peptide list.

4.3.3. MRM transitions development

To establish MRM transitions for peptides, suitable precursor ions and product ions are required, and parameters including fragmentor, collision energies for each MRM pair are required to be optimized. Fragmentor was unique for each peptide and optimizing this parameter could maximize the MS signal. Since peptides could be multiply charged, usually the m/z of the charge state with highest intensity would be chosen, except for peptide 3. It was because no fragment ion of the highest signal charge state was observed from the further MS/MS analysis, but replacement by the second highest signal charge state solved the problem.

Product ion scan was performed for each selected precursor ion. Generally, few collision energies were varied to fragment the precursor ion, so as to observe the generated pattern of fragment ion. This could provide sufficient information to find fragment ions with high signals. Several criteria were set for the selection of the quantifier and qualifier ion. For the quantifier ion, fragment ion with the highest signal would be chosen. However, if that ion represented the mass of a single amino acid, it would be discarded as this type of ion would be easily interfered by contaminants or background noise. Besides, fragment ions with different charge states may have similar m/z , which could not be separated due to the limited resolution of quadrupole. For example, in peptide 3, one of its MRM transitions is m/z 620.0 > 744.4. The fragment ion with m/z 744.4 could represent b_{31}^{4+} or y_{37}^{5+} . In order to ensure the quantification accuracy, these confused ions would not be selected as the quantifier ion, but they could be qualifier ions. For qualifier ions, two fragment ions with the second and third highest intensity would be selected in this project. Like the criteria for quantifier ion, fragments representing single amino acid were not chosen as they could not be used to confirm the identity of a peptide. Collision energies were optimized for each MRM transition

and finally three transitions were established for each peptide. The optimized parameters for each MRM transition were listed in section 4.2.7 (Table 4.1).

4.3.4. LC separation

An 8-minute LC separation profile was developed for this analysis. It was found that all the four peptides were eluted within 3 minutes, indicating that the established method could perform rapid instrumental analysis (Figure 4.7). The remaining 5 minutes were intended for carrying out the clean-up and this time could be further reduced if necessary. Peptides 1 and 2 were eluted at the similar elution time but this did not affect the detection, as the signal in each MRM pair would be recorded in its individual channel. Furthermore, for the three MRM transitions of a peptide they must be eluted at the same time so as to confirm the detection of the viral peptide (Figure 4.8).

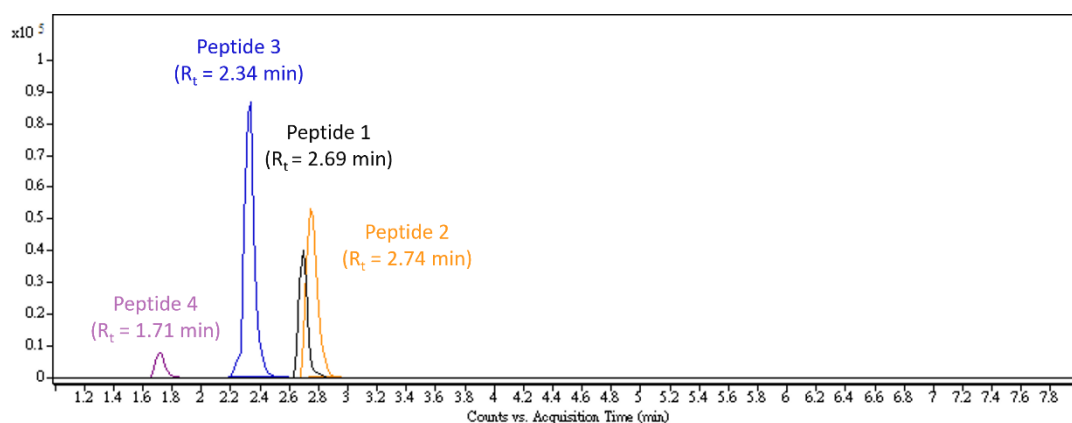
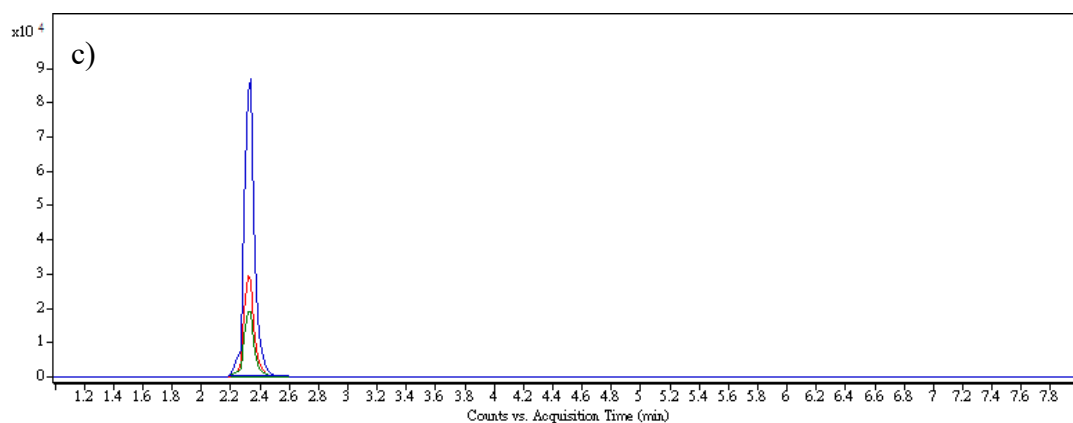
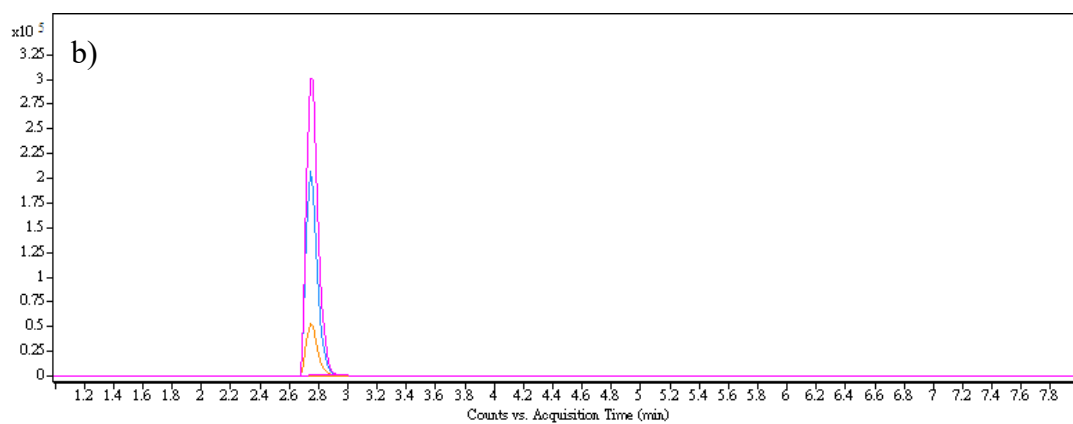
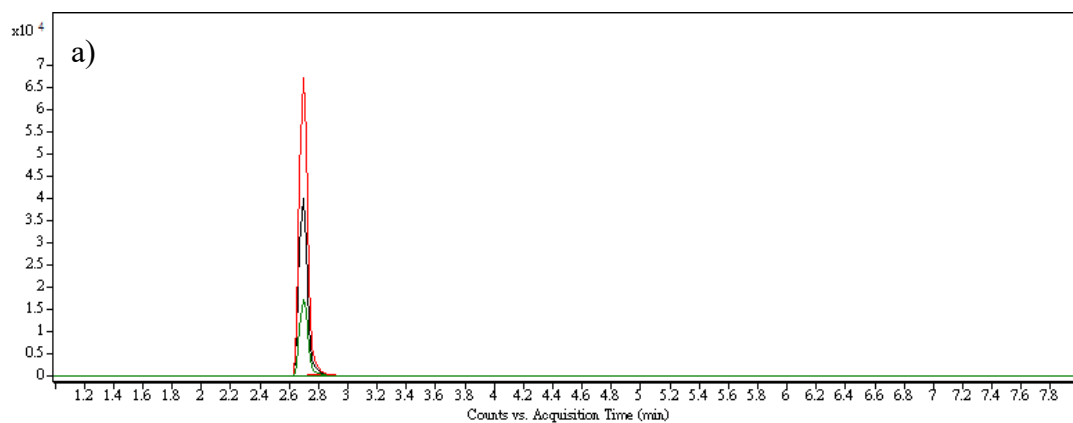


Figure 4.7 A chromatogram of the LC separation of the four selected peptides. Only one MRM transition from each peptide was selected for demonstration of the separation efficiency.



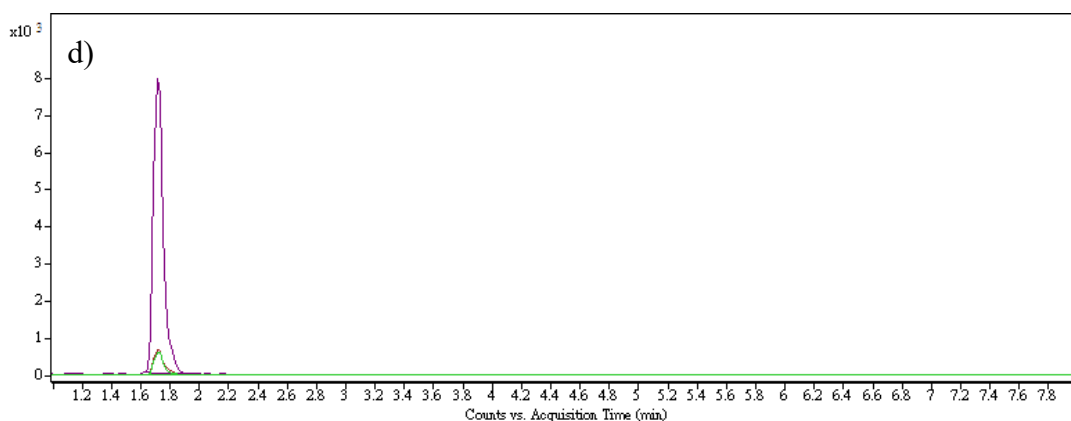


Figure 4.8 All three MRM transitions of (a-d) peptide 1-4 were eluted at the same time which indicated the positive detection of these peptides.

4.3.5. Linearities of the four selected peptides

MRM is advantageous for performing quantification and it could be applied for both small organic molecules and peptides. Linearities and linear ranges of the quantifier ions of the four peptides were evaluated. Generally, these quantifier ions showed excellent linearities ($R^2 > 0.99$) and covered satisfactory linear ranges. Linearities and linear ranges of the four peptides were summarized in Table 4.3.

Table 4.3 Linearities and linear ranges of the four selected peptides.

	MRM pair	Linearity (R^2)	Linear range (ppb)
Peptide 1	836.1 → 742.5	0.9985	5 – 5000
Peptide 2	828.9 → 202.8	0.9998	1 – 1000
Peptide 3	620.0 → 680.0	0.9950	50 – 10000
Peptide 4	769.9 → 957.3	0.9971	100 – 10000

4.3.6. Limit of detection of the four peptides in saliva matrix

In order to evaluate the limit of detection (LOD) of the four peptides, peptides were spiked into the saliva collected from a healthy person to resemble the real situation that our detection targets existed in a highly complex background. The LOD was defined as two out of three MRM channels showing their peaks with signal-to-noise ratio (S/N) greater than 3. This established method could detect target peptides to down to 1 ppb (Table 4.4).

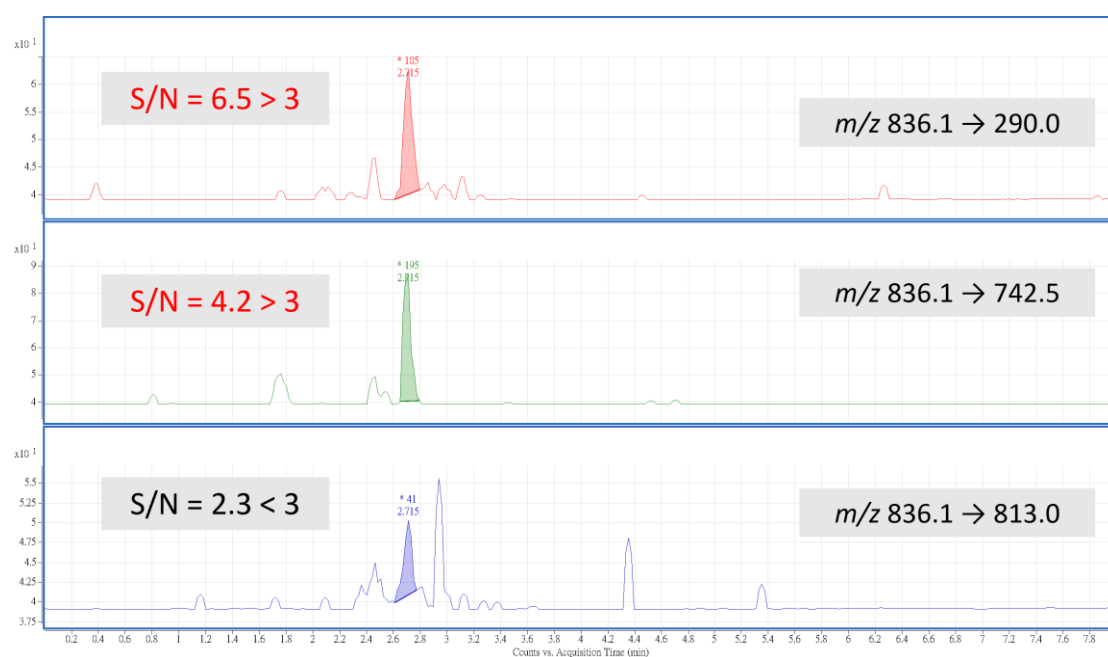


Figure 4.9 An example of the confirmation of detection for peptide 1. Two out of three channels had their S/N greater than 3.

Table 4.4 LOD of the four selected peptides in saliva matrix.

	Sequence	LOD (ppb)
Peptide 1	MAGNGGDAALALLLDR	5
Peptide 2	MAGNGGDAALALLLDRLNQLESK	1
Peptide 3	SRNSSRNSTPGSSRGTSARMAGNG GDAALALLLDR	30
Peptide 4	SDNGPQNQRNAPRITFGGPSDSTGSN QNGERSGARSK	21

4.3.7. Limitations and improvements of this approach

In this project, it has been demonstrated the possibility of targeted proteomics approach for rapid detection of SARS-CoV-2 by detecting the viral peptides specific to N protein. Four peptides unique for SARS-CoV-2 were selected as the detection targets. The criteria used in this study were relatively straightforward compared with other studies^{190,191}. But it should be noted that several peptides that were eliminated from the list might also be utilized as evidence to support the positive detection of the four specific peptides. Effort had been made for the optimization of the amount of trypsin required for the analysis, but this amount could never be confirmed that unless clinical sample were used to validate the method, as it was reasonable to predict that real-life samples could be much more complex than our laboratory samples. Besides, the applicability of this approach towards SARS-CoV-2 variants is required to be investigated. Nevertheless, it is reasonable to predict this approach could be rapidly adopted for variant detection by simply changing the m/z of MRM transitions, if sequences of specific peptides are found to be mutated. In short, this simple and rapid method with no complicated sample pretreatment was demonstrated to be applicable in detection and quantification of SARS-CoV-2 virus.

Chapter 5. Conclusions and future prospects

The conformational dynamics of the effects of the single, or combinations of the three common point mutations, G238S, E104K and M182T, on TEM-type β -lactamases was successfully investigated by HDX-MS. X-ray crystallography and molecular dynamic simulation are two conventional methods to study the properties of β -lactamases. X-ray crystallography provides visible protein structures and elucidates the complicated intramolecular interaction network, however it mainly reveals information at static protein state, with limited dynamic information. Molecular dynamic simulation provides dynamic information at different conditions easily, particularly at nanosecond timescale, but the data resulted are sensitive to the initial settings of various parameters which could easily cause great deviations from the real situation, as well as inconsistencies with experimental results. Thus, using HDX-MS, unique and complementary information to results produced by X-ray crystallography and molecular dynamics simulation could be obtained, including experimental in-solution protein conformation and dynamic information, which could be easily generated by this technique. Six TEM-type β -lactamases with different combinations of point mutations were analyzed by HDX-MS. The results revealed that generally G238S caused greater enhancement in flexibility than E104K, and that the G238S-containing mutants were more flexible than the wild type TEM-1. E104K, a point mutation also conferred extended hydrolytic efficacy towards the new generation antibiotics, did not cause any significant overall changes in the conformation. However, the synergistic effect of G238S and E104K could contribute to greater changes in flexibility than their single point mutations. These findings were consistent with the explanation of new generation antibiotics hydrolytic efficiencies enhancement of these ESBLs, i.e., expansion of active site in order to accommodate the bulky side chain of new generation antibiotics. M182T, the reported global suppressor for the TEM-type ESBLs, modulated the

flexibility of various localized regions differently in order to compensate the excessive conformational changes caused by the key mutations G238S and/or E104K.

Changes in conformational dynamics in the six TEM-type β -lactamases upon their binding to the third generation cephalosporin, cefotaxime, were also studied. TEM-type β -lactamases used for the binding studies were incorporated with the extra E166A point mutation, as the deacylation rate of the antibiotics hydrolysis was greatly reduced in order to facilitate the study of the conformation in the acyl-enzyme form. Conformational changes in several catalytically important and point mutation-containing regions were investigated, as well as two allosteric regions which were first discovered and discussed in this study. Catalytically important regions, including the region near S70, Ω -loop, SDN loop and β -3 strand, were found to have significantly conformational changes as they directly interacted with the antibiotics. Two allosteric regions that were far away from the active site, H5-H6 and the edge of β -sheet opposite to the active site, were found to undergo meaningful conformational changes upon the incorporation of single or combination of point mutations. H5-H6 was suggested to be related to the stabilization caused by M182T, which was particularly reflected by the HDX changes before and after cefotaxime binding. Only M182T containing mutants showing a greater uptake reduction in this region, indicating that H5-H6 could be a key region for TEM-type ESBLs stabilization, particularly for mutants containing key mutations like G238S and E104K. On the other hand, the edge of the β -sheet opposite to the active site involving antibiotics interaction was found to be more flexible particularly in their cefotaxime-bound form of G238S containing mutants. This could be attributed to the balance of effective binding between β -3 strand and cefotaxime, and the sufficient room for the bulky cefotaxime staying in the active site, which rendered another edge of the β -sheet more flexible to compensate the essential flexibility changes

for binding. However, the β -sheet would become less ordered and thermostabilities of mutants would be reduced. Incorporation of M182T into mutants could maintain the flexibility of this edge similar to the wild type TEM-1. As the result, thermostabilities of mutants were increased, and the enhanced cefotaxime catalytic efficiency could still be retained at the same time. Overall, the conformational dynamics revealed by this study provides better understanding to the antibiotics resistance mechanism and also insights into the design of new antibiotics or inhibitors. Apart from the active site related regions of TEM-type β -lactamases targeted by conventional antibiotics and inhibitors, allosteric regions, e.g., H5-H6 and the previously discussed edge of the β -sheet, also serve as attractive sites for inhibition of the TEM-type β -lactamases. In the future, molecular dynamic simulation could be performed in order to further validate the conclusion, and provide complementary dynamic information at nanoscale scale, facilitating better understanding on the dynamic properties of TEM-type ESBLs at different time scales. Moreover, a detailed kinetic study could be conducted, so that kinetic parameters of each individual steps (i.e., K_d , k_2 and k_3) could yield more insights on the effects of different combination of point mutations on the antibiotic hydrolysis mechanism.

The recent outbreak of COVID-19 is a serious worldwide health threat. Besides vaccine and drug development, prompt virus detection is another important area to combat the pandemic. In chapter 4, the possibility of identification of SARS-CoV-2 by combining the concept of targeted proteomics and mass spectrometric approach based on the detection of peptides specific and unique to SARS-CoV-2 N protein was demonstrated. Four specific viral peptides were successfully selected. Feasibility of this method was demonstrated by 1) its excellent specificity and sensitivity because MRM was used, 2) its excellent linearity and broad linear range; 3) the low limit of detection of specific peptides in the complex saliva matrix; and 4) no special reagent required for this method.

This simple and rapid method also has great potential to be applied in common laboratories with low BSL because maintaining native conformation and structures of antigens, in contrast to qRT-PCR and ELISA, were not essential. In the future, this method could be further improved and refined after analyzing the clinical samples, particularly with mutated viral proteins from SARS-CoV-2 VOCs. Besides, the digestion method could be further improved to perform complete digestion within a short time, and this could facilitate a more accurate quantification results for clinical use.

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